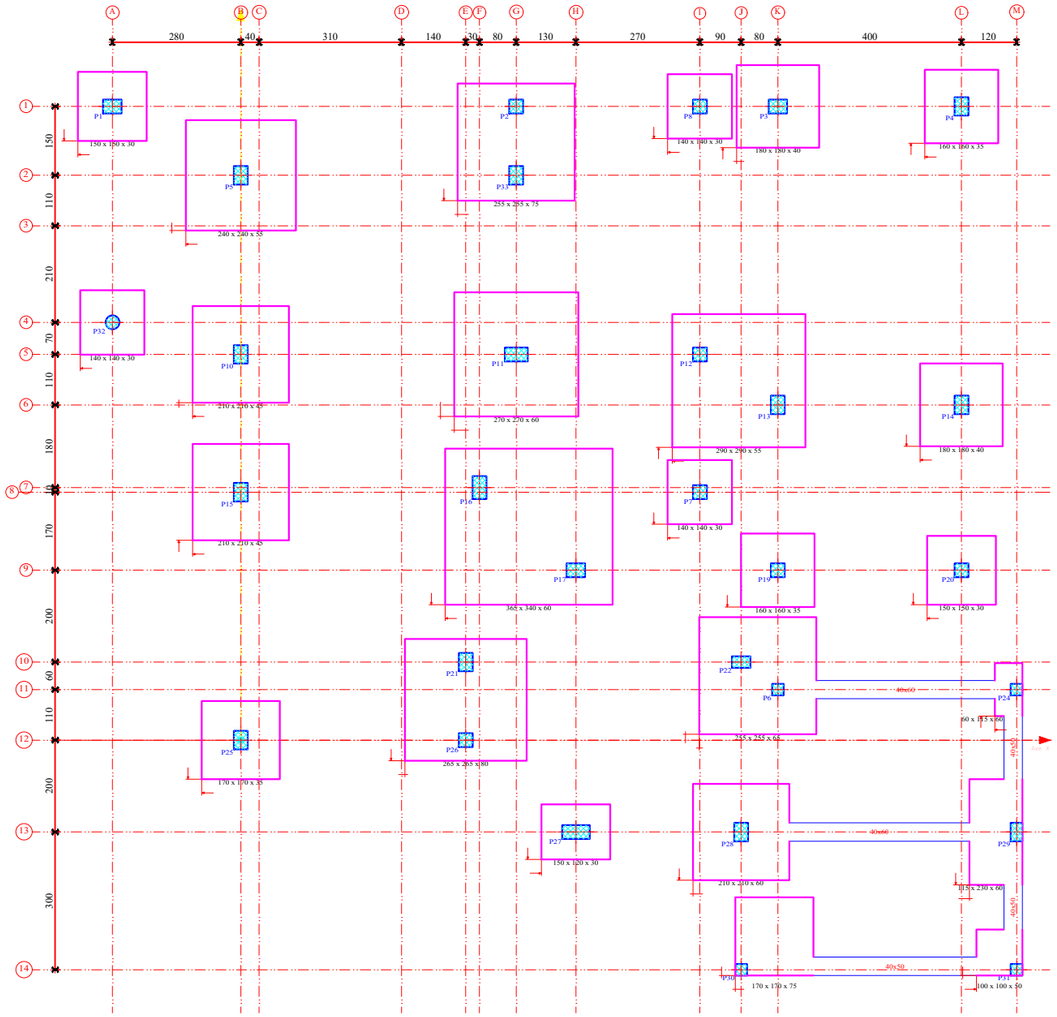

AFRIK
STUDIO
& partners

DOSSIER D'EXECUTION

PLAN DE COFFRAGE



Fondation
Coffrage
Béton: B25
Echelle: 1:100

Implantation des poteaux - Fondation			
Poteau	Dimension (cm)	Coordonnées du centre	
		Coordonnée X (cm)	Coordonnée Y (cm)
P1	40x30	-280	1380
P2	30x30	600	1380
P3	40x30	1170	1380
P4	30x40	1570	1380
P5	30x40	0	1230
P6	25x25	1170	110
P7	30x30	1000	540
P8	30x30	1000	1380
P10	30x40	0	840
P11	50x30	600	840
P12	30x30	1000	840
P13	30x40	1170	730
P14	30x40	1570	730
P15	30x40	0	540
P16	30x50	520	550
P17	40x30	730	370
P19	30x30	1170	370
P20	30x30	1570	370
P21	30x40	490	170
P22	40x25	1090	170
P24	25x25	1690	110
P25	30x40	0	0
P26	30x30	490	0
P27	60x30	730	-200
P28	30x40	1090	-200
P29	25x40	1690	-200
P30	25x25	1090	-500
P31	25x25	1690	-500
P32	Diamètre 30	-280	910
P33	30x40	600	1230
Cote de départ des poteaux: -1.50 m			

TABLE DES ÉLÉMENTS DE FONDATION						
Références	Dimensions (cm)	Épaisseur (cm)	Armature inf. X	Armature inf. Y	Armature sup. X	Armature sup. Y
P1 et P20	150x150	30	10HA12e=15	10HA12e=15		
P3 et P14	180x180	40	12HA12e=15	12HA12e=15		
P4 et P19	160x160	35	10HA12e=15	10HA12e=15		
P5	240x240	55	16HA12e=15	16HA12e=15		
P7, P8 et P32	140x140	30	9HA12e=15	9HA12e=15		
P10 et P15	210x210	45	14HA12e=15	14HA12e=15		
P11	270x270	60	19HA12e=14	20HA12e=13		
P24	60x115	60	7HA12e=15	4HA12e=15		
P25	170x170	35	11HA12e=15	11HA12e=15		
P27	150x120	30	8HA12e=15	10HA12e=15		
P28	210x210	60	14HA12e=15	14HA12e=15		
P29	115x230	60	15HA12e=15	9HA12e=12		
P30	170x170	75	11HA12e=15	11HA12e=15	11HA12e=15	11HA12e=15
P31	100x100	50	6HA12e=15	6HA12e=15		
(P22-P6)	255x255	65	17HA12e=15	17HA12e=15	17HA12e=15	17HA12e=15
(P2-P33)	255x255	75	17HA12e=15	17HA12e=15	17HA12e=15	17HA12e=15
(P12-P13)	290x290	55	19HA12e=15	19HA12e=15	19HA12e=15	19HA12e=15
(P16-P17)	365x340	60	22HA12e=15	24HA12e=15	22HA12e=15	24HA12e=15
(P21-P26)	265x265	80	18HA12e=14	18HA12e=14	18HA12e=14	18HA12e=14

Table de longrines de redressement



-Aciers HA400
-Diamètre du plus gros granulat 15 mm
-Béton fondation: fc28= 25 Mpa
-Béton poutres, poteaux: fc28= 20 Mpa
-Enrobage poteaux: 2,5cm
-Tenue au feu: 1h
-Contrainte du sol: 0.15 Mpa

Légende

- Poteaux naissants
- Raidisseurs naissants
- PC=Poteaux circulaires
- Poteaux continuants
- Raidisseurs continuants

BURKINA - FASO
LA PATRIE OU LA MORT, NOUS VAINCRONS.

DATE: JANVIER 2026

PROJET DE CONSTRUCTION D'UN BATIMENT R+2 A OUAGADOUGOU

ETUDES TECHNIQUES

MAITRE D'OUVRAGE :

MAITRE D'OEUVRE :

BUREAU DE CONTROLE

ENTREPRISE DE CONSTRUCTION

Etudes:

COMPAORE D. P. Jean Landry

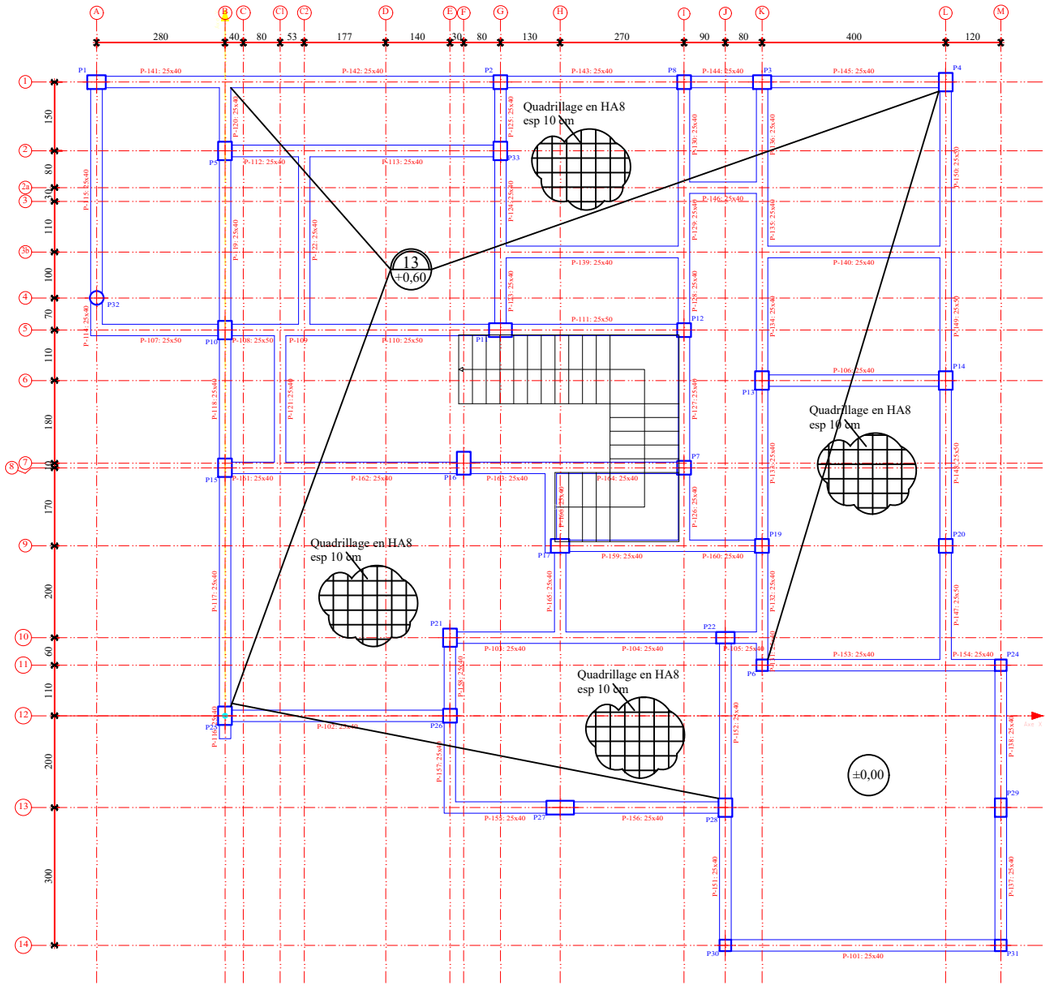
N° Planche: N°

Echèlle: Ajuster au format

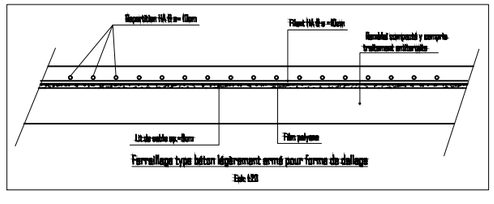
PLAN DE COFFRAGE FONDATION

Observations :

PLANCHE 01



P8 RDC
Coffrage
Béton: B20
Echelle: 1:100



-Aciers HA400
-Diamètre du plus gros granulat 15 mm
-Béton poutres, poteaux: fc28= 20 Mpa
-Enrobage poteaux: 2,5cm
-Tenue au feu: 1h

Légende

- Poteaux naissants
- Raidisseurs naissants
- PC=Poteaux circulaires
- Poteaux continuants
- Raidisseurs continuants

Implantation des poteaux - PB RDC			
Poteau	Dimension (cm)	Coordonnées du centre	
		Coordonnée X (cm)	Coordonnée Y (cm)
P1	40x30	-280	1380
P2	30x30	600	1380
P3	40x30	1170	1380
P4	30x40	1570	1380
P5	30x40	0	1230
P6	25x25	1170	110
P7	30x30	1000	540
P8	30x30	1000	1380
P10	30x40	0	840
P11	50x30	600	840
P12	30x30	1000	840
P13	30x40	1170	730
P14	30x40	1570	730
P15	30x40	0	540
P16	30x50	520	550
P17	40x30	730	370
P19	30x30	1170	370
P20	30x30	1570	370
P21	30x40	490	170
P22	40x25	1090	170
P24	25x25	1690	110
P25	30x40	0	0
P26	30x30	490	0
P27	60x30	730	-200
P28	30x40	1090	-200
P29	25x40	1690	-200
P30	25x25	1090	-500
P31	25x25	1690	-500
P32	Diamètre 30	-280	910
P33	30x40	600	1230
Cote de départ des poteaux: +0.60 m			

BURKINA - FASO
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ETUDES TECHNIQUES

MAITRE D'OUVRAGE :	
MAITRE D'OEUVRE :	
BUREAU DE CONTROLE	
ENTREPRISE DE CONSTRUCTION	

Etudes:

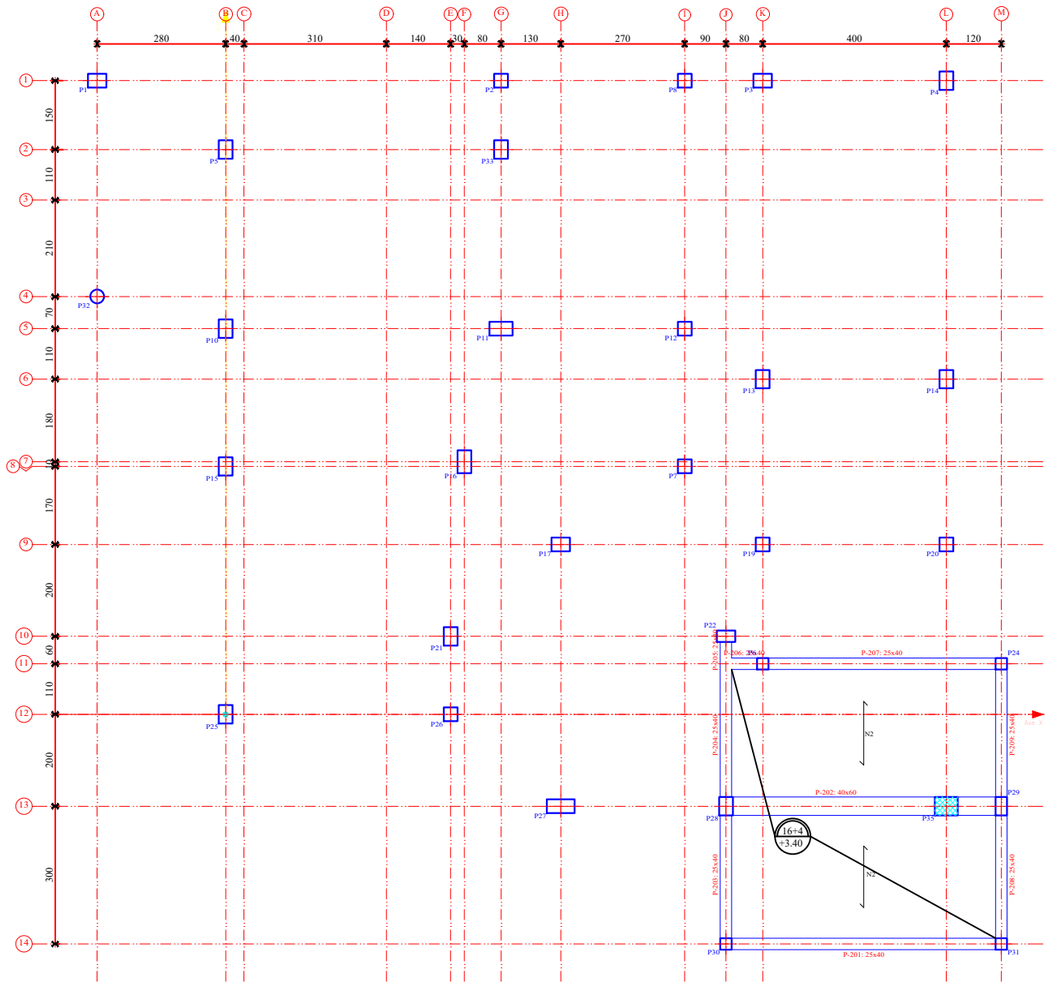
COMPAORE D. P. Jean Landry

N° Planche:	N°
Echèlle:	Ajuster au format

PLAN DE COFFRAGE LONGRINE

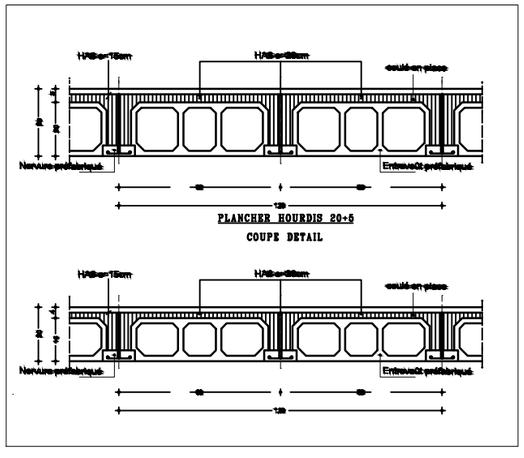
Observations :

PLANCHE 02

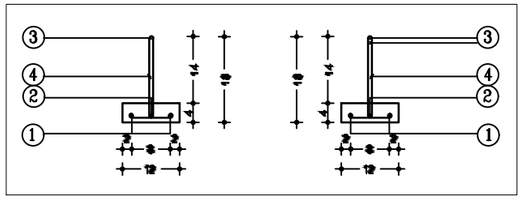


Implantation des poteaux - PH GARAGE			
Poteau	Dimension (cm)	Coordonnées du centre	
		Coordonnée X (cm)	Coordonnée Y (cm)
P1	40x30	-280	1380
P2	30x30	600	1380
P3	40x30	1170	1380
P4	30x40	1570	1380
P5	30x40	0	1230
P7	30x30	1000	540
P8	30x30	1000	1380
P10	30x40	0	840
P11	50x30	600	840
P12	30x30	1000	840
P13	30x40	1170	730
P14	30x40	1570	730
P15	30x40	0	540
P16	30x50	520	550
P17	40x30	730	370
P19	30x30	1170	370
P20	30x30	1570	370
P21	30x40	490	170
P22	40x25	1090	170
P25	30x40	0	0
P26	30x30	490	0
P27	60x30	730	-200
P28	30x40	1090	-200
P32	Diamètre 30	-280	910
P33	30x40	600	1230
P35	50x40	1570	-200
Cote de départ des poteaux: +3.40 m			

PH GARAGE
Coffrage
Béton: B20
Echelle: 1:100



DESIGNATION NERVURES	NOMENCLATURE DES ACIERS			
NERVURES	①	②	③	④
N1 [0-2[	2HA10	1HA10	1HA8	HA 6 ep=15cm
N2 [2-4[	2HA12	1HA12	1HA8	HA 6 ep=15cm
N3 [4-6[	2HA14	1HA14	1HA8	HA 6 ep=15cm
N4 [6-8[	2HA14	1HA14	1HA10 1HA10	HA 6 ep=15cm



-Aciers HA400
-Diamètre du plus gros granulat 15 mm
-Béton poutres, poteaux: fc28= 20 Mpa
-Enrobage poteaux: 2,5cm
-Tenue au feu: 1h

Légende

- Poteaux naissants
- Raidisseurs naissants
- PC=Poteaux circulaires
- Poteaux continuants
- Raidisseurs continuants

BURKINA - FASO
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PROJET DE CONSTRUCTION D'UN BATIMENT R+2 A OUAGADOUGOU

ETUDES TECHNIQUES

MAITRE D'OUVRAGE :	
MAITRE D'OEUVRE :	
BUREAU DE CONTROLE	
ENTREPRISE DE CONSTRUCTION	

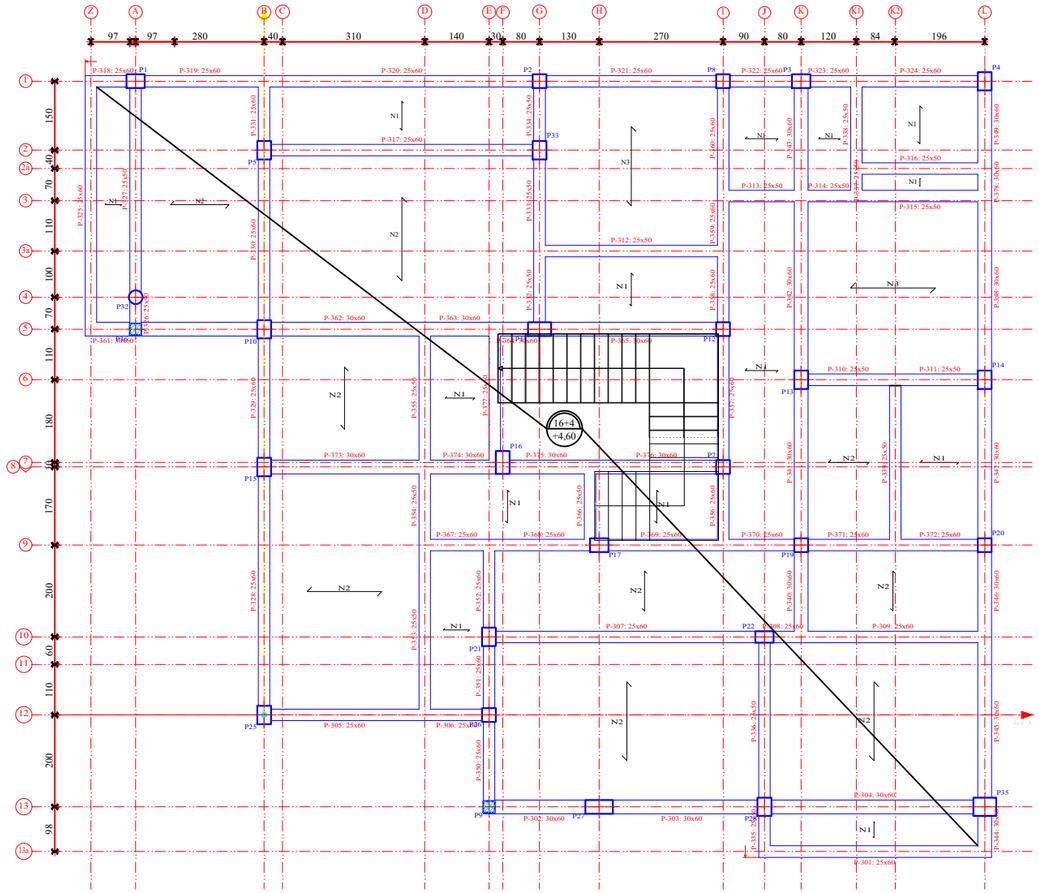
Etudes:

COMPAORE D. P. Jean Landry

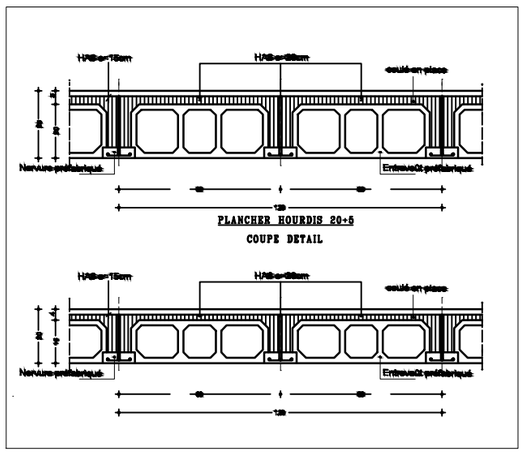
N° Planche:	N°	PLAN DE COFFRAGE PH GARAGE
Echèlle:	Ajuster au format	

Observations :

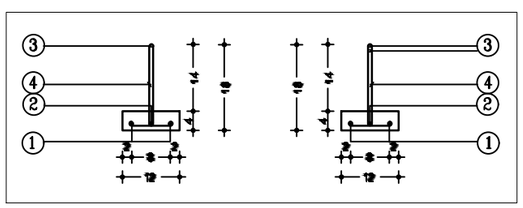
PLANCHE 03



Implantation des poteaux - PH RDC			
Poteau	Dimension (cm)	Coordonnées du centre	
		Coordonnée X (cm)	Coordonnée Y (cm)
P1	40x30	-280	1380
P2	30x30	600	1380
P3	40x30	1170	1380
P4	30x40	1570	1380
P5	30x40	0	1230
P7	30x30	1000	540
P8	30x30	1000	1380
P9	25x25	490	-200
P10	30x40	0	840
P11	50x30	600	840
P12	30x30	1000	840
P13	30x40	1170	730
P14	30x40	1570	730
P15	30x40	0	540
P16	30x50	520	550
P17	40x30	730	370
P19	30x30	1170	370
P20	30x30	1570	370
P21	30x40	490	170
P22	40x25	1090	170
P25	30x40	0	0
P26	30x30	490	0
P28	30x40	1090	-200
P33	30x40	600	1230
P35	30x40	1570	-200
P36	25x25	-280	840
Cote de départ des poteaux: +4.60 m			



DESIGNATION NERVURES	NOMENCLATURE DES ACIERS			
NERVURES	①	②	③	④
N1 [0-2[	2HA10	1HA10	1HA8	HA 6 ep=15cm
N2 [2-4[	2HA12	1HA12	1HA8	HA 6 ep=15cm
N3 [4-6[	2HA14	1HA14	1HA8	HA 6 ep=15cm
N4 [6-8[	2HA14	1HA14	1HA10 1HA10	HA 6 ep=15cm



-Aciers HA400
-Diamètre du plus gros granulat 15 mm
-Béton poutres, poteaux: fc28= 20 Mpa
-Enrobage poteaux: 2,5cm
-Tenue au feu: 1h

Légende

- Poteaux naissants
- Raidisseurs naissants
- PC=Poteaux circulaires
- Poteaux continuants
- Raidisseurs continuants

BURKINA - FASO
LA PATRIE OU LA MORT, NOUS VAINCRONS.

DATE: JANVIER 2026

PROJET DE CONSTRUCTION D'UN BATIMENT R+2 A OUAGADOUGOU

ETUDES TECHNIQUES

MAITRE D'OUVRAGE :	
MAITRE D'OEUVRE :	
BUREAU DE CONTROLE	
ENTREPRISE DE CONSTRUCTION	

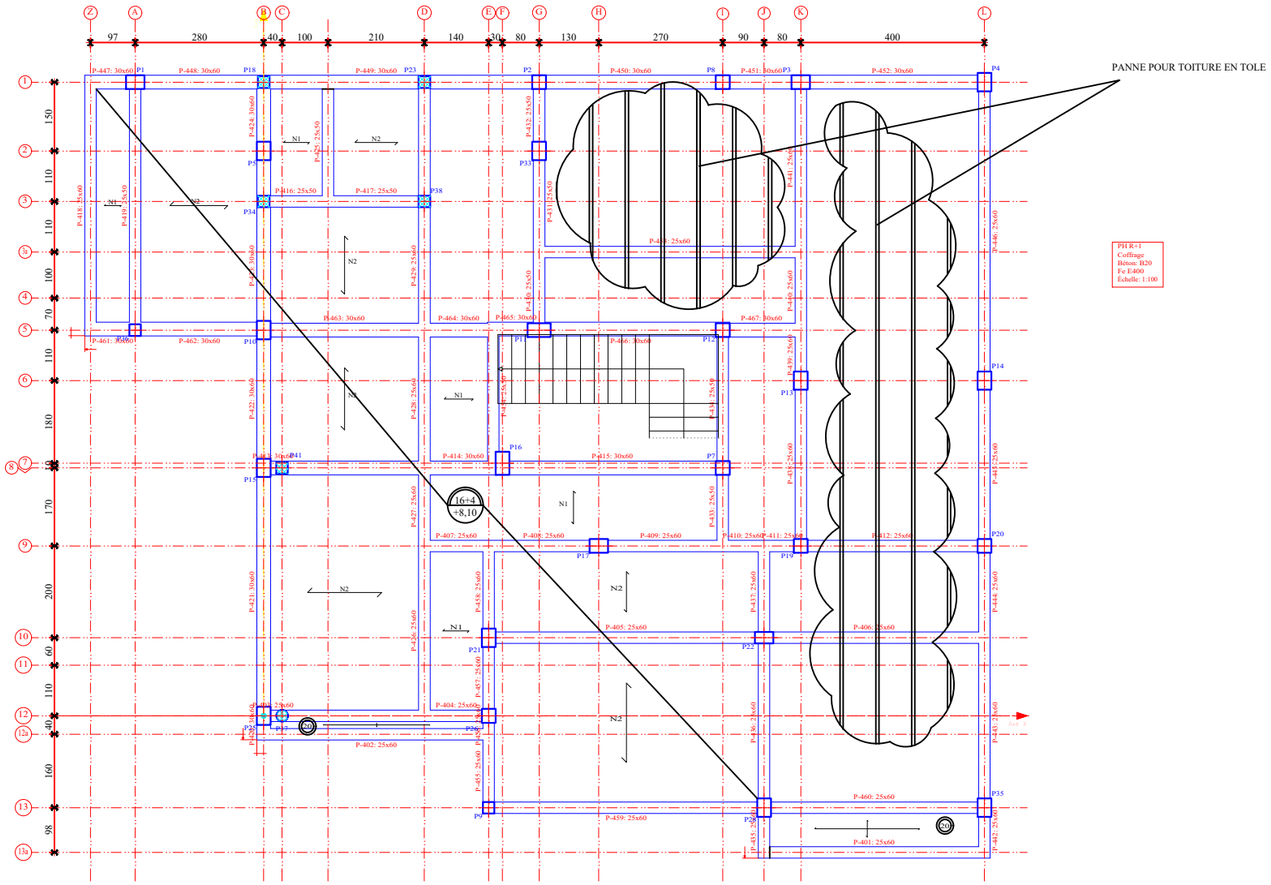
Etudes:

COMPAORE D. P. Jean Landry

N° Planche:	N°	**PLAN DE COFFRAGE PH RDC**
Echèlle:	Ajuster au format	

Observations :

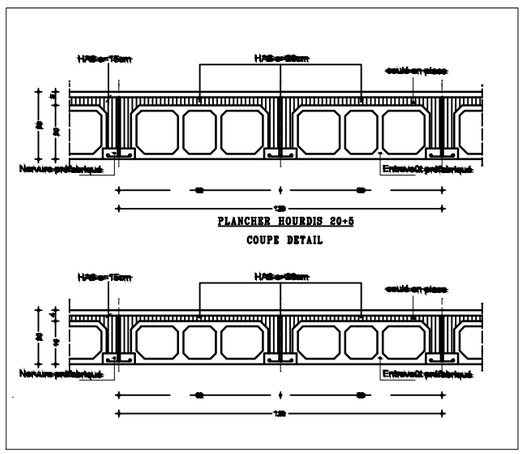
PLANCHE 04



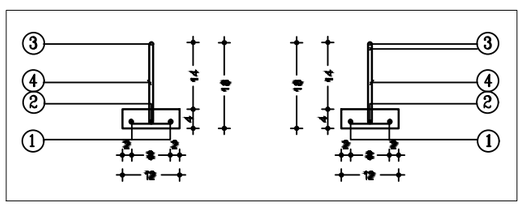
Implantation des poteaux - PH R+1			
Poteau	Dimension (cm)	Coordonnées du centre	
		Coordonnée X (cm)	Coordonnée Y (cm)
P7	30x30	1000	540
P11	50x30	600	840
P12	30x30	1000	840
P17	40x30	730	370
P18	25x25	0	1380
P22	40x25	1090	170
P23	25x25	350	1380
P26	30x30	490	0
P34	25x25	0	1120
P37	Diamètre 25	40	0
P38	25x25	350	1120
P41	25x25	40	540
Cote de départ des poteaux: +8.10 m			



DETAILS DE LA DALLE PLEINE
20 cm



DESIGNATION NERVURES	NOMENCLATURE DES ACIERS			
NERVURES	①	②	③	④
N1 [0-2[	2HA10	1HA10	1HA8	HA 6 ep=15cm
N2 [2-4[	2HA12	1HA12	1HA8	HA 6 ep=15cm
N3 [4-6[	2HA14	1HA14	1HA8	HA 6 ep=15cm
N4 [6-8[	2HA14	1HA14	1HA10 1HA10	HA 6 ep=15cm



-Aciers HA400
-Diamètre du plus gros granulat 15 mm
-Béton poutres, poteaux: fc28= 20 Mpa
-Enrobage poteaux: 2,5cm
-Tenue au feu: 1h

Légende

- Poteaux naissants
- Raidisseurs naissants
- PC=Poteaux circulaires
- Poteaux continuants
- Raidisseurs continuants

BURKINA - FASO
LA PATRIE OU LA MORT, NOUS VAINCRONS.

DATE: JANVIER 2026

PROJET DE CONSTRUCTION D'UN BATIMENT R+2 A OUAGADOUGOU

ETUDES TECHNIQUES

MAITRE D'OUVRAGE :	
MAITRE D'OEUVRE :	
BUREAU DE CONTROLE	
ENTREPRISE DE CONSTRUCTION	

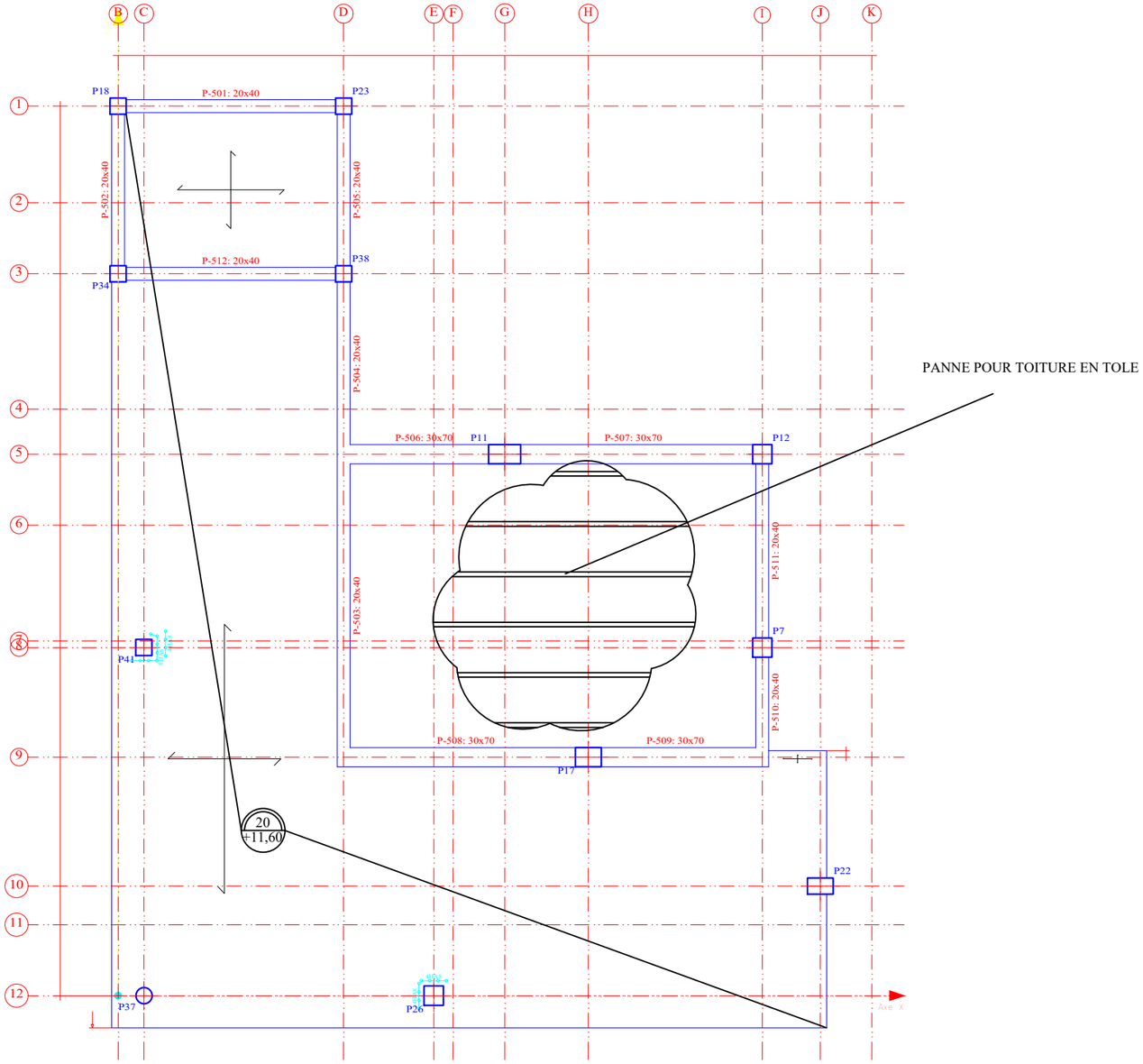
Etudes:

COMPAORE D. P. Jean Landry

N° Planche:	N°	**PLAN DE COFFRAGE PH R+1**
Echèlle:	Ajuster au format	

Observations :

PLANCHE 05

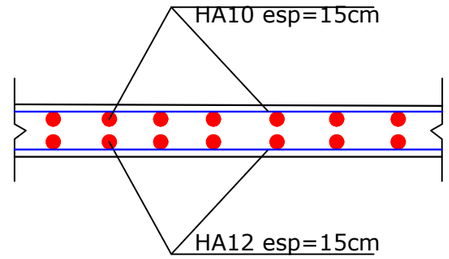


PH R+2
Coffrage
Béton: B20
Fe E400
Échelle: 1:100

-Aciers HA400
-Diamètre du plus gros granulat 15 mm
-Béton poutres, poteaux: fc28= 20 Mpa
-Enrobage poteaux: 2,5cm
-Tenue au feu: 1h

Légende

- Poteaux naissants
- Raidisseurs naissants
- PC=Poteaux circulaires
- Poteaux continuants
- Raidisseurs continuants



DETAILS DE LA DALLE PLEINE
20 cm

BURKINA - FASO LA PATRIE OU LA MORT, NOUS VAINCRONS.		DATE	JANVIER 2026
PROJET DE CONSTRUCTION D'UN BATIMENT R+2 A OUAGADOUGOU		ETUDES TECHNIQUES	
MAITRE D'OUVRAGE :		Etudes: COMPAORE D. P. Jean Landry	
MAITRE D'OEUVRE :			
BUREAU DE CONTROLE			
ENTREPRISE DE CONSTRUCTION			
N° Planche:	N°	PLAN DE COFFRAGE PH R+2	
Echèlle:	Ajuster au format		
Observations :		PLANCHE 06	

ESCALIERS

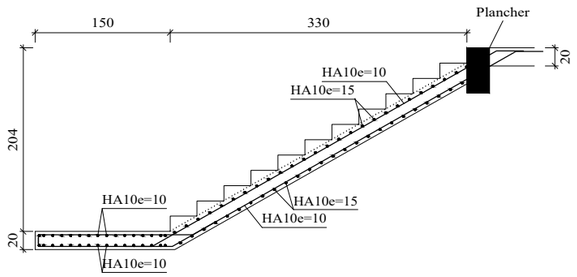
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
Escalier 1-Tronçon 1	1	HA10	16	352	5632	34.7
	2	HA10	16	264	4224	26.0
	3	HA10	32	175	5600	34.5
	4	HA10	32	212	6784	41.8
	5	HA10	16	386	6176	38.1
	6	HA10	16	420	6720	41.4
	7	HA10	16	500	8000	49.3
	8	HA10	16	628	10048	61.9
	9	HA10	100	163	16300	100.5
					Total+10%:	471.0
					HA10:	471.0
					Total:	471.0

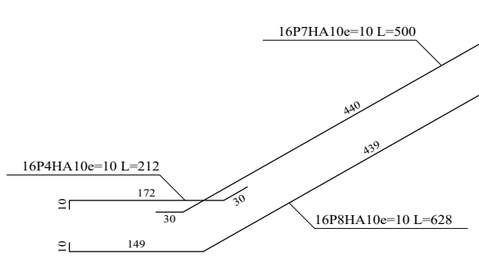
Récapitulatif Acier Escalier 1		Long. total (m)	Poids+10% (kg)
Fe E400	HA10	694.8	471

Escalier 1

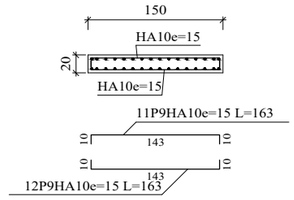
Tronçon 1		
Géométrie	Emmarchement	1.500 m
	Épaisseur	0.20 m
	Marche	0.300 m
	Contremarche	0.170 m
	Franchissement de dénivelé	3.91 m
	Nb de marches	23
	Niveau final	PH RDC
	Niveau initial	PB RDC
Charges	Poids propre	4.91 kN/m2
	Marches (Bétonné avec la dalle)	1.81 kN/m2
	Revêtement	1.00 kN/m2
	Rampes	3.00 kN/m
	Charge d'exploitation	3.00 kN/m2
Matériaux	Béton	B20
	Acier	Fe E400
	Enr. géométrique	3.0 cm

Section C-C

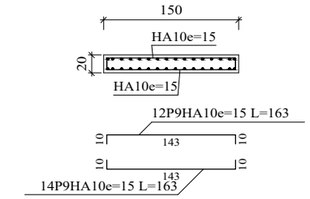




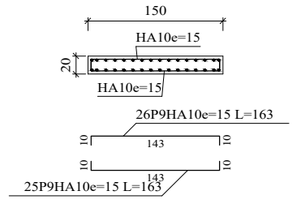
Section D-D



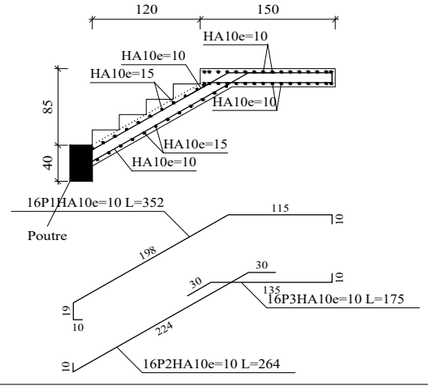
Section E-E



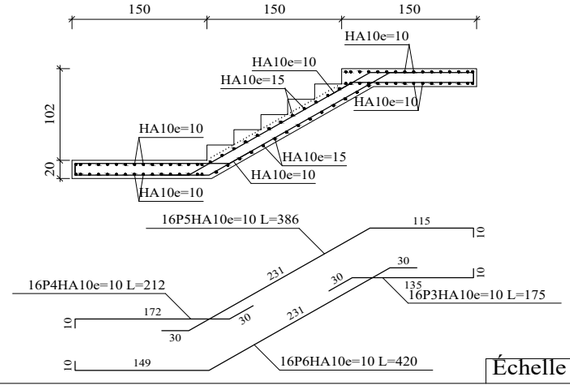
Section F-F



Section A-A



Section B-B



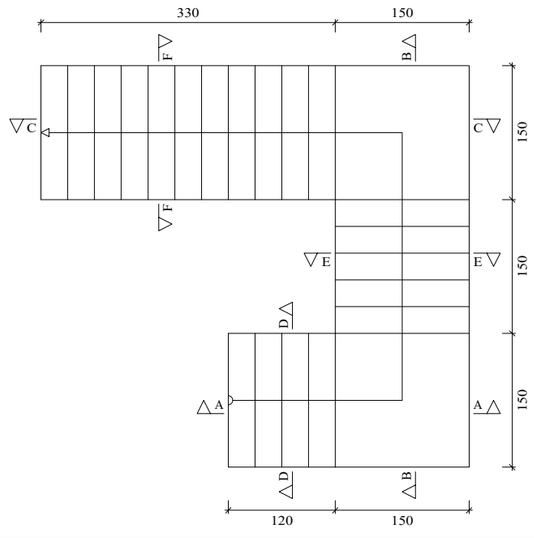
Échelle 1:50

Récapitulatif Acier Escalier 2		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA12	345.2	337	
	HA14	183.7	244	581

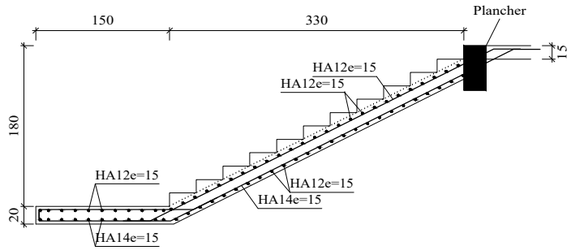
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
Escalier 2-Tronçon 1	1	HA12	11	347	3817	33.9
	2	HA14	11	268	2948	35.6
	3	HA14	22	180	3960	47.9
	4	HA12	22	215	4730	42.0
	5	HA12	11	382	4202	37.3
	6	HA14	11	422	4642	56.1
	7	HA12	11	491	5401	48.0
	8	HA14	11	620	6820	82.4
	9	HA12	98	167	16366	145.3
					Total+10%:	581.4
					HA12:	337.2
					HA14:	244.2
					Total:	581.4

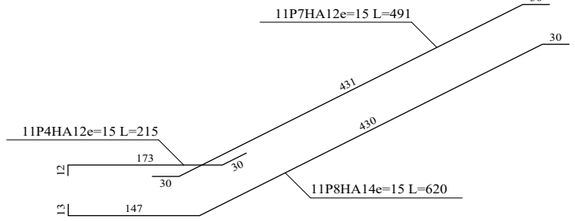
Escalier 2

Tronçon 1		
Géométrie	Emmarchement	1.500 m
	Épaisseur	0.20 m
	Marche	0.300 m
	Contremarche	0.150 m
	Franchissement de dénivelé	3.45 m
	Nb de marches	23
	Niveau final	PH R+1
	Niveau initial	PH RDC
Charges	Poids propre	4.91 kN/m2
	Marches (Bétonné avec la dalle)	1.65 kN/m2
	Revêtement	1.00 kN/m2
	Rampes	3.00 kN/m
	Charge d'exploitation	3.00 kN/m2
Matériaux	Béton	B20
	Acier	Fe E400
	Enr. géométrique	3.0 cm

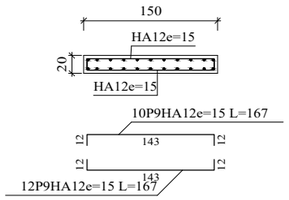


Section C-C

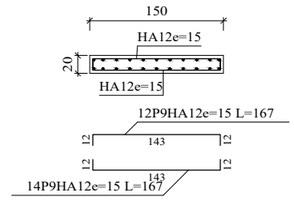




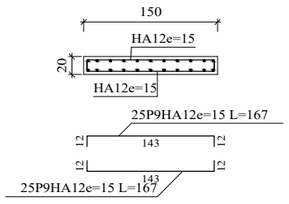
Section D-D



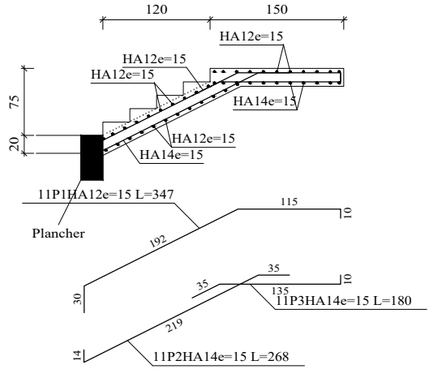
Section E-E



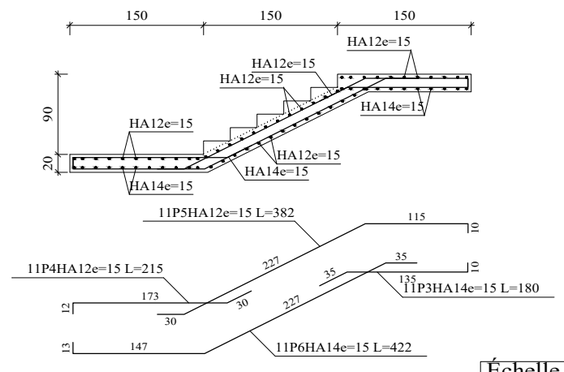
Section F-F



Section A-A



Section B-B

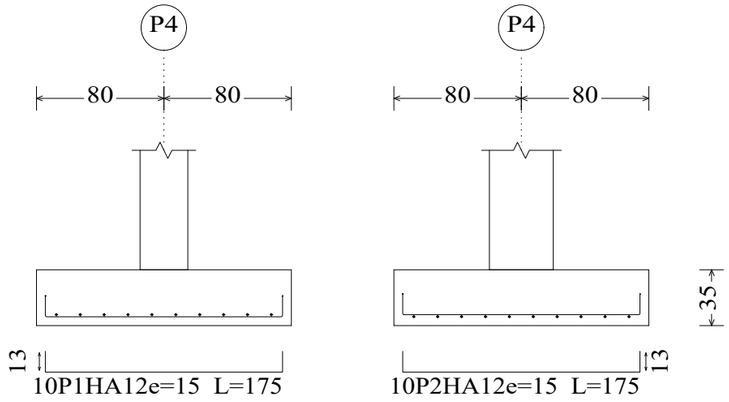


Échelle 1:50

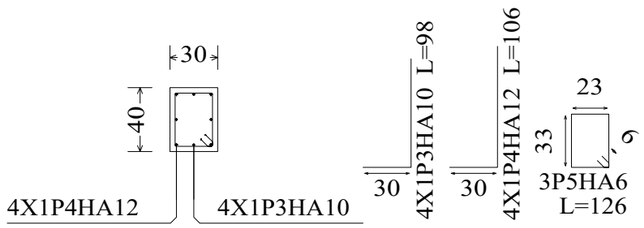
SE ELLES LONGRINES DE REDRESSE ENT

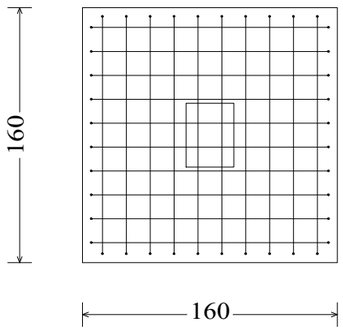
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P4	1	HA12	10	175	1750	15.5
	2	HA12	10	175	1750	15.5
	3	HA10	4	98	392	2.4
	4	HA12	4	106	424	3.8
	5	HA6	3	126	378	0.8
					Total+10%:	41.8
					HA6:	0.9
					HA10:	2.6
					HA12:	38.3
					Total:	41.8

P4



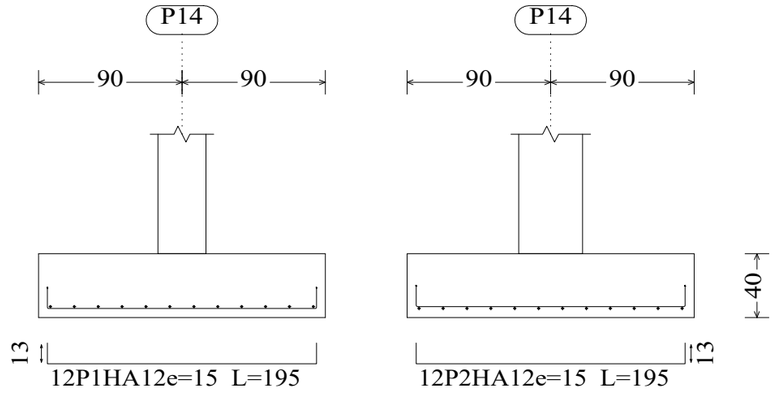
P4

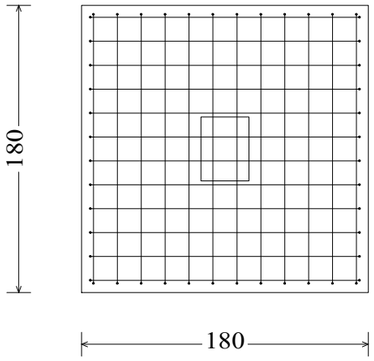




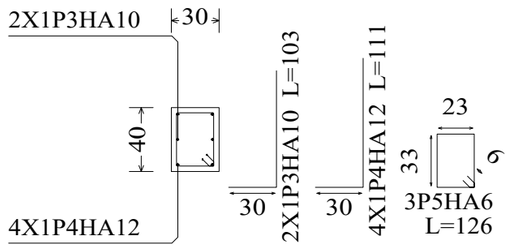
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

P14





P14

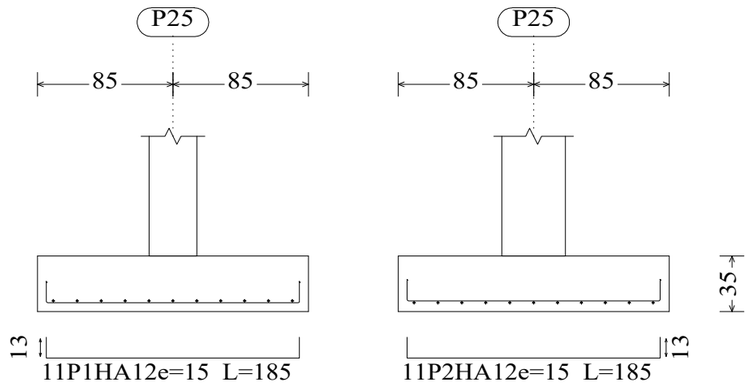


Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P14	1	HA12	12	195	2340	20.8
	2	HA12	12	195	2340	20.8
	3	HA10	2	103	206	1.3
	4	HA12	4	111	444	3.9
	5	HA6	3	126	378	0.8
					Total+10%:	52.4
					HA6:	0.9
					HA10:	1.4
					HA12:	50.1
					Total:	52.4

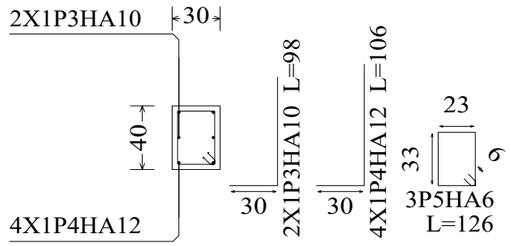
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

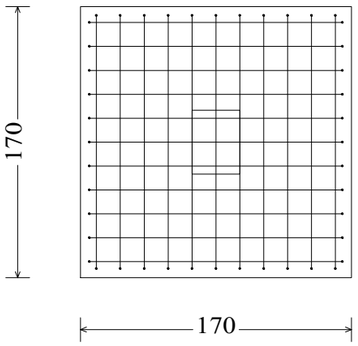
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P25	1	HA12	11	185	2035	18.1
	2	HA12	11	185	2035	18.1
	3	HA10	2	98	196	1.2
	4	HA12	4	106	424	3.8
	5	HA6	3	126	378	0.8
					Total+10%:	46.2
					HA6:	0.9
					HA10:	1.3
					HA12:	44.0
					Total:	46.2

P25



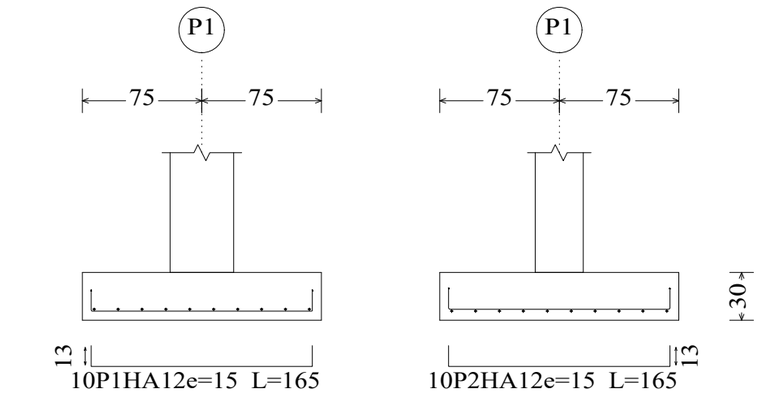
P25

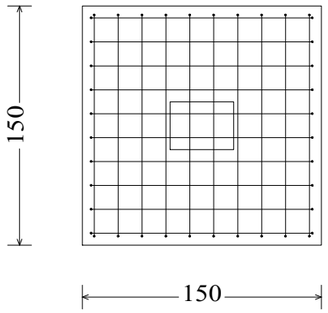




Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

P1





P1

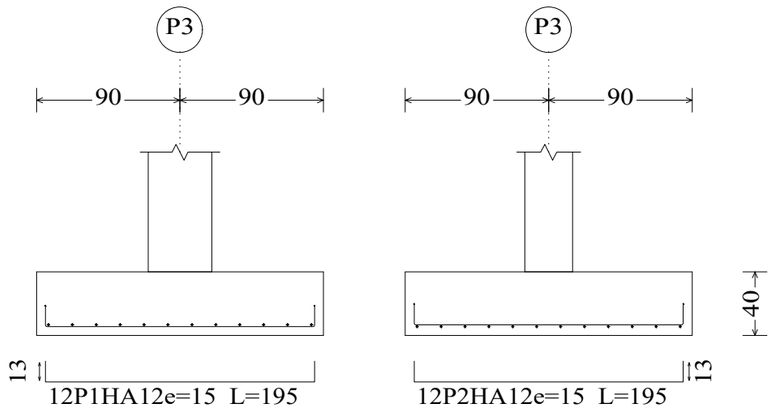


Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P1	1	HA12	10	165	1650	14.6
	2	HA12	10	165	1650	14.6
	3	HA10	2	93	186	1.1
	4	HA12	4	101	404	3.6
	5	HA6	3	126	378	0.8
					Total+10%:	38.2
					HA6:	0.9
					HA10:	1.2
					HA12:	36.1
					Total:	38.2

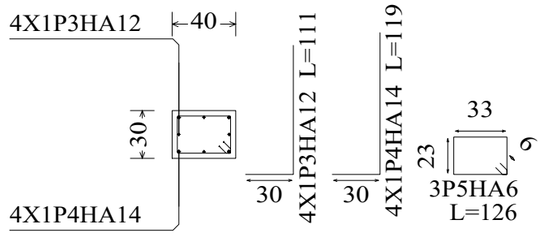
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

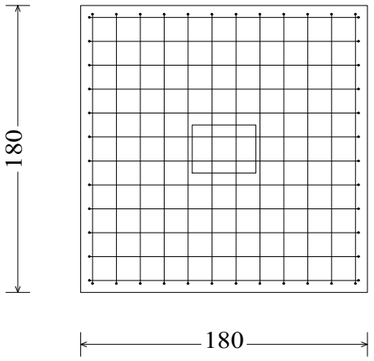
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P3	1	HA12	12	195	2340	20.8
	2	HA12	12	195	2340	20.8
	3	HA12	4	111	444	3.9
	4	HA14	4	119	476	5.8
	5	HA6	3	126	378	0.8
					Total+10%:	57.3
					HA6:	0.9
					HA12:	50.1
					HA14:	6.3
					Total:	57.3

P3



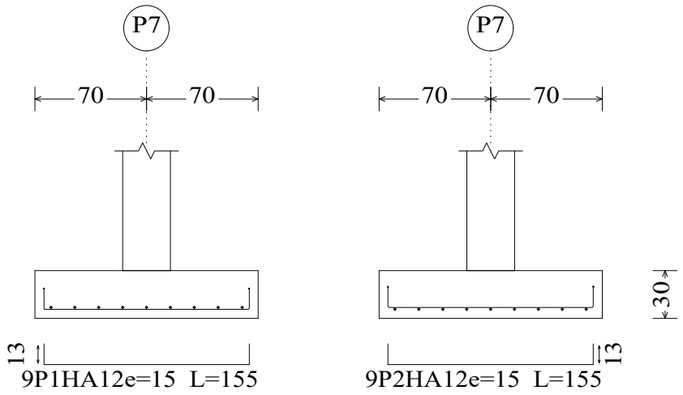
P3

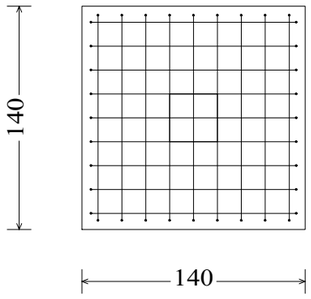




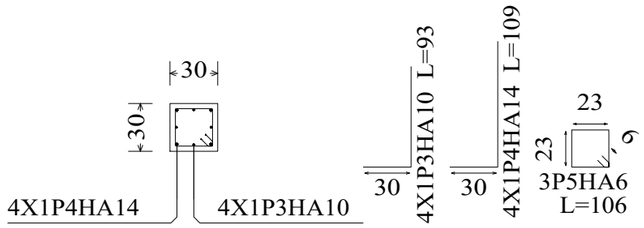
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

P7





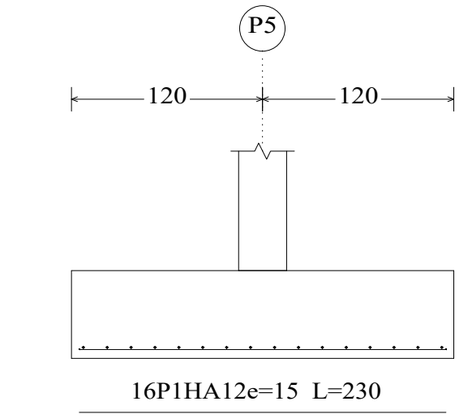
P7

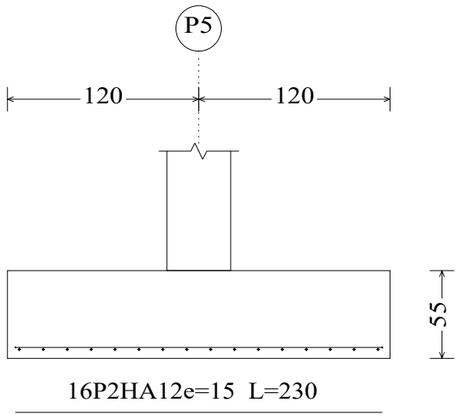


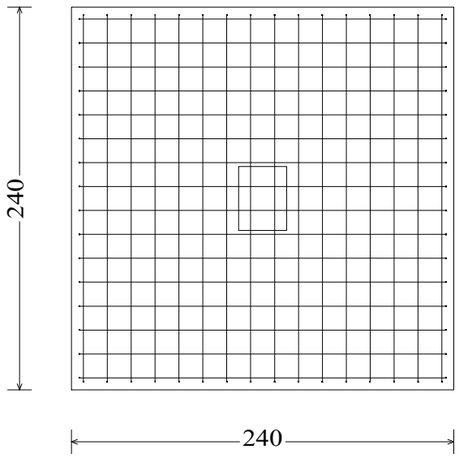
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P7	1	HA12	9	155	1395	12.4
	2	HA12	9	155	1395	12.4
	3	HA10	4	93	372	2.3
	4	HA14	4	109	436	5.3
	5	HA6	3	106	318	0.7
					Total+10%:	36.4
					HA6:	0.8
					HA10:	2.5
					HA12:	27.3
					HA14:	5.8
					Total:	36.4

Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

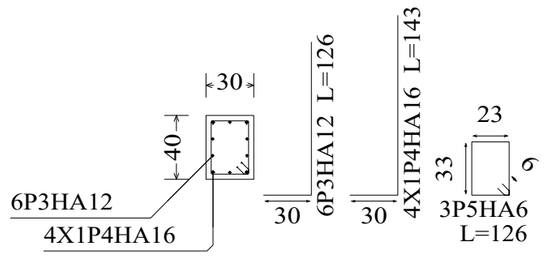
P5







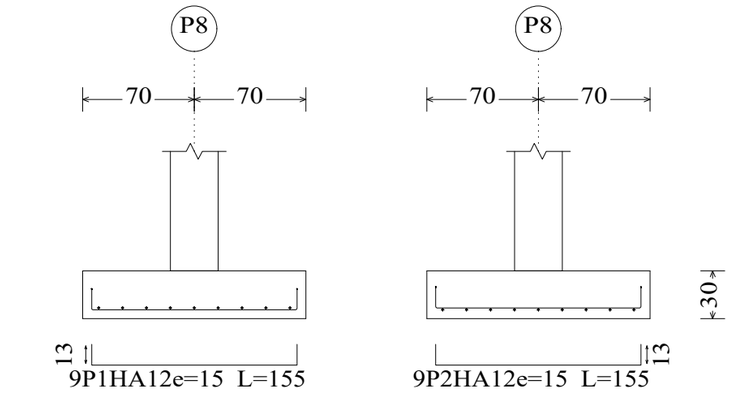
P5

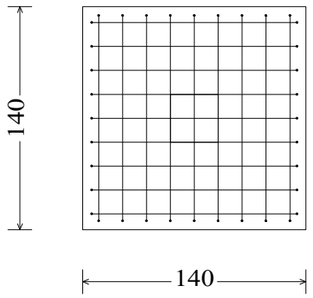


Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P5	1	HA12	16	230	3680	32.7
	2	HA12	16	230	3680	32.7
	3	HA12	6	126	756	6.7
	4	HA16	4	143	572	9.0
	5	HA6	3	126	378	0.8
					Total+10%:	90.1
					HA6:	0.9
					HA12:	79.3
					HA16:	9.9
					Total:	90.1

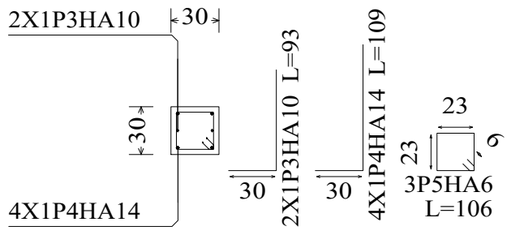
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

P8





P8

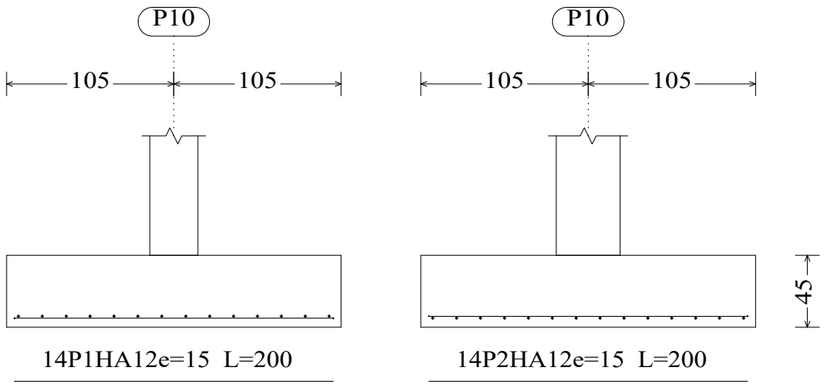


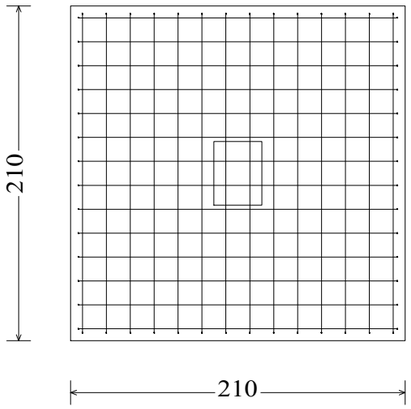
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P8	1	HA12	9	155	1395	12.4
	2	HA12	9	155	1395	12.4
	3	HA10	2	93	186	1.1
	4	HA14	4	109	436	5.3
	5	HA6	3	106	318	0.7
					Total+10%:	35.1
					HA6:	0.8
					HA10:	1.2
					HA12:	27.3
					HA14:	5.8
					Total:	35.1

Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

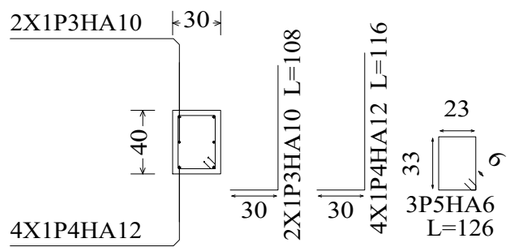
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P10	1	HA12	14	200	2800	24.9
	2	HA12	14	200	2800	24.9
	3	HA10	2	108	216	1.3
	4	HA12	4	116	464	4.1
	5	HA6	3	126	378	0.8
					Total+10%:	61.6
					HA6:	0.9
					HA10:	1.4
					HA12:	59.3
					Total:	61.6

P10





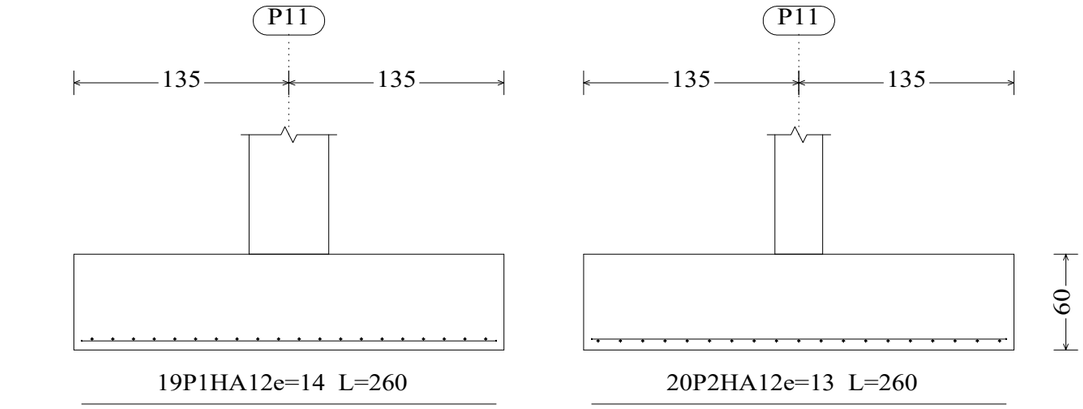
P10

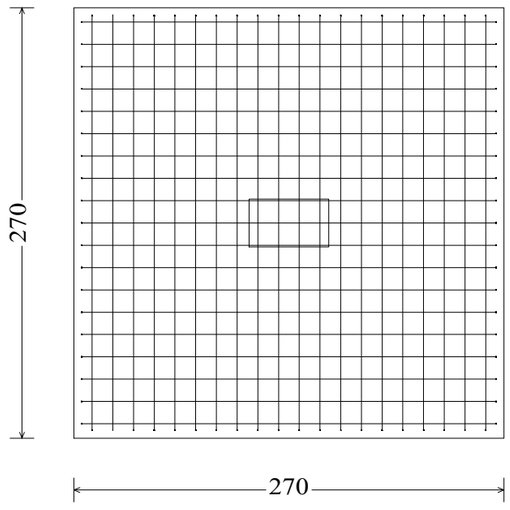


Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

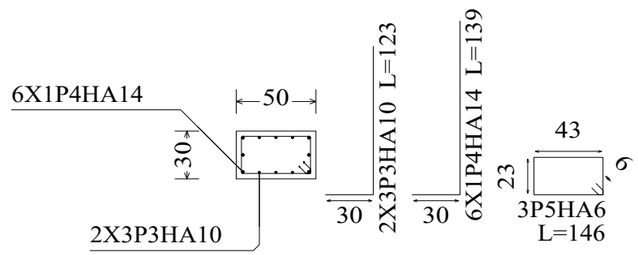
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P11	1	HA12	19	260	4940	43.9
	2	HA12	20	260	5200	46.2
	3	HA10	6	123	738	4.6
	4	HA14	6	139	834	10.1
	5	HA6	3	146	438	1.0
					Total+10%:	116.4
					HA6:	1.1
					HA10:	5.1
					HA12:	99.1
					HA14:	11.1
					Total:	116.4

P11





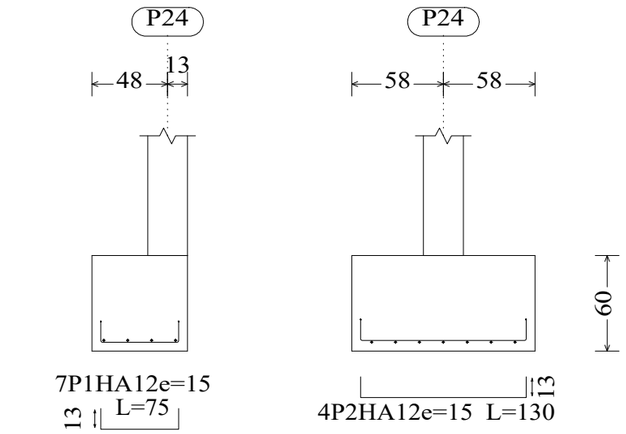
P11



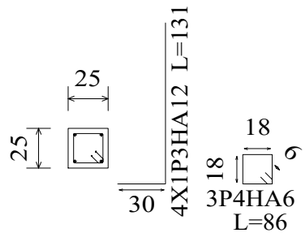
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

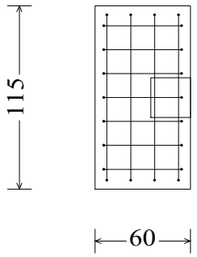
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P24	1	HA12	7	75	525	4.7
	2	HA12	4	130	520	4.6
	3	HA12	4	131	524	4.7
	4	HA6	3	86	258	0.6
					Total+10%:	16.1
					HA6:	0.7
					HA12:	15.4
					Total:	16.1

P24



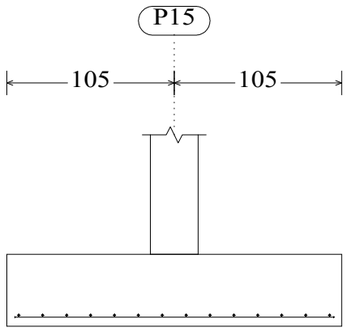
P24



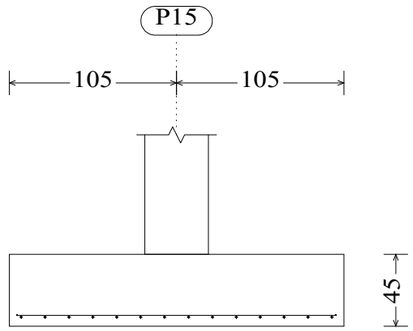


Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

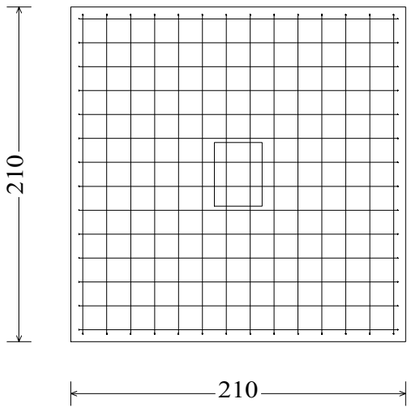
P15



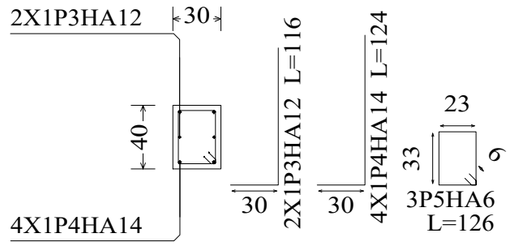
14P1HA12e=15 L=200



14P2HA12e=15 L=200



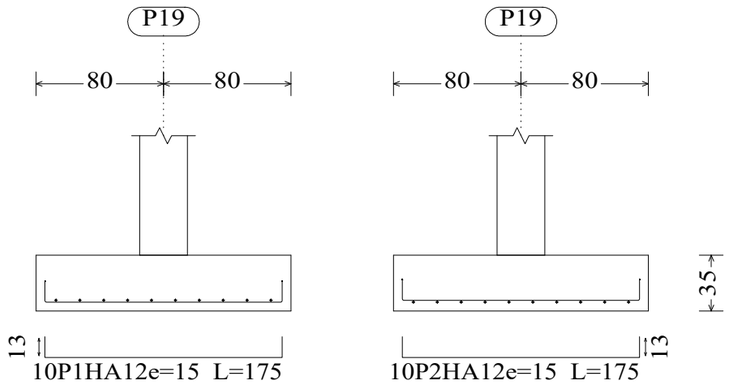
P15

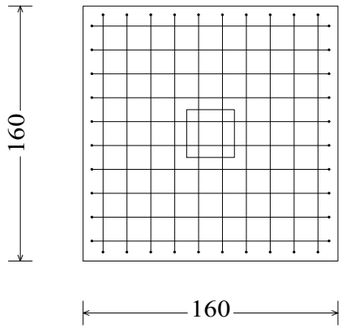


Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P15	1	HA12	14	200	2800	24.9
	2	HA12	14	200	2800	24.9
	3	HA12	2	116	232	2.1
	4	HA14	4	124	496	6.0
	5	HA6	3	126	378	0.8
					Total+10%:	64.6
					HA6:	0.9
					HA12:	57.1
					HA14:	6.6
					Total:	64.6

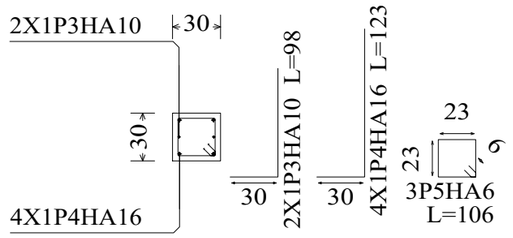
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

P19





P19

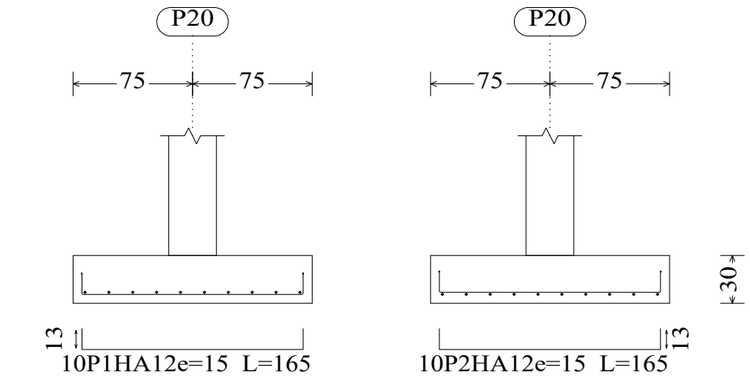


Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P19	1	HA12	10	175	1750	15.5
	2	HA12	10	175	1750	15.5
	3	HA10	2	98	196	1.2
	4	HA16	4	123	492	7.8
	5	HA6	3	106	318	0.7
					Total+10%:	44.8
					HA6:	0.8
					HA10:	1.3
					HA12:	34.1
					HA16:	8.6
					Total:	44.8

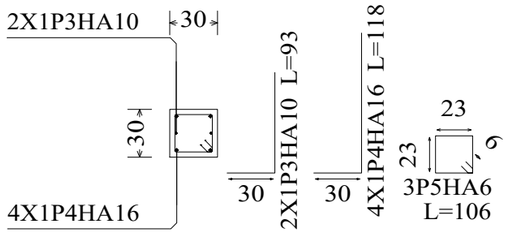
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

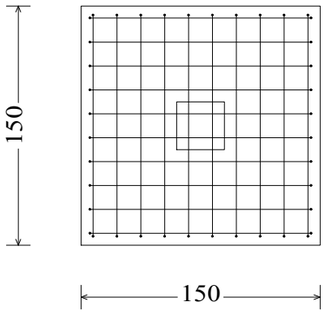
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P20	1	HA12	10	165	1650	14.6
	2	HA12	10	165	1650	14.6
	3	HA10	2	93	186	1.1
	4	HA16	4	118	472	7.4
	5	HA6	3	106	318	0.7
					Total+10%:	42.2
					HA6:	0.7
					HA10:	1.2
					HA12:	32.1
					HA16:	8.2
					Total:	42.2

P20



P20

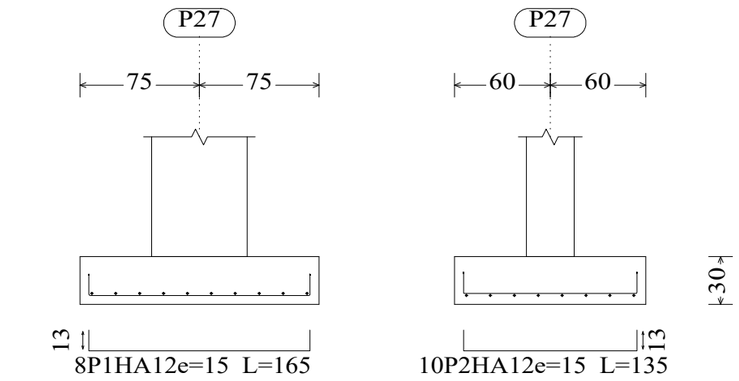


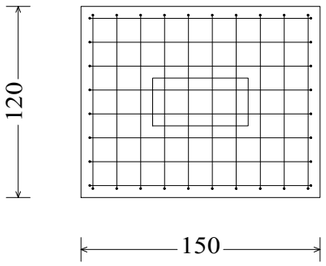


Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

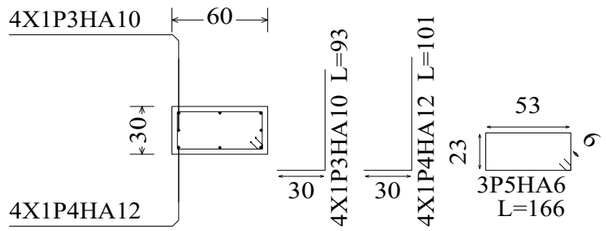
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P27	1	HA12	8	165	1320	11.7
	2	HA12	10	135	1350	12.0
	3	HA10	4	93	372	2.3
	4	HA12	4	101	404	3.6
	5	HA6	3	166	498	1.1
					Total+10%:	33.8
					HA6:	1.2
					HA10:	2.6
					HA12:	30.0
					Total:	33.8

P27



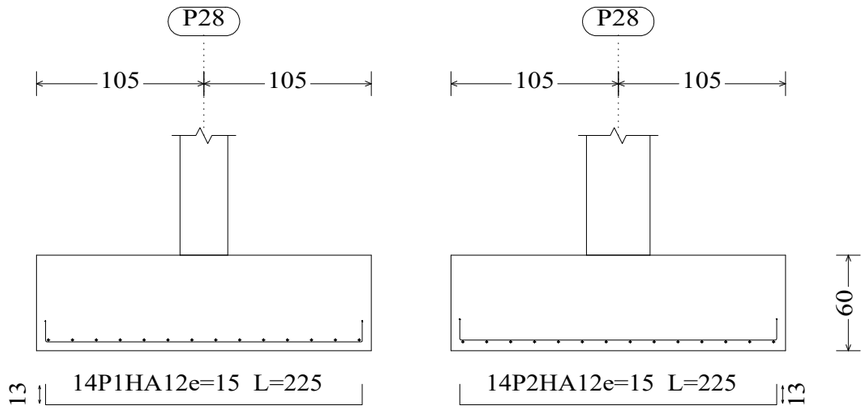


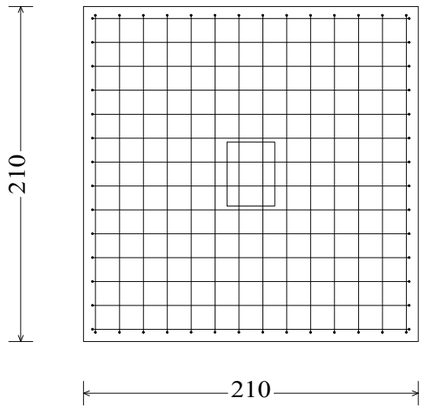
P27



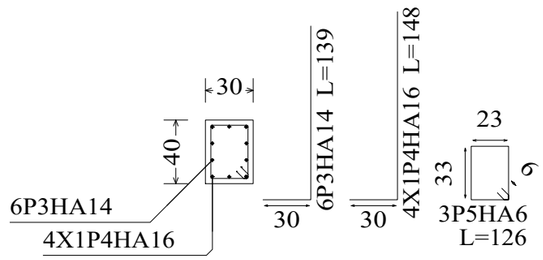
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

P28





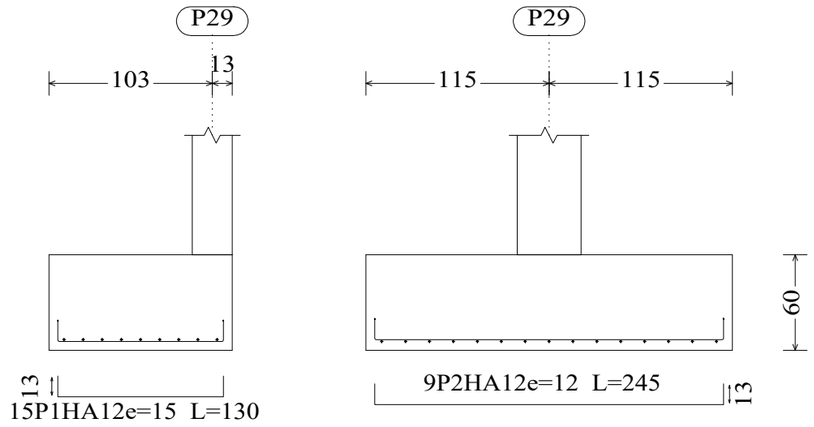
P28

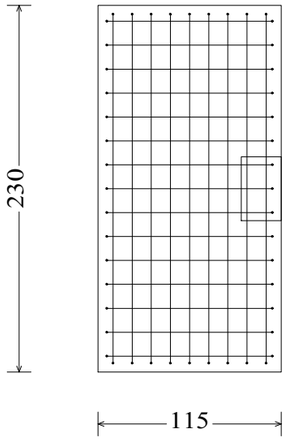


Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P28	1	HA12	14	225	3150	28.0
	2	HA12	14	225	3150	28.0
	3	HA14	6	139	834	10.1
	4	HA16	4	148	592	9.3
	5	HA6	3	126	378	0.8
					Total+10%:	83.8
					HA6:	0.9
					HA12:	61.6
					HA14:	11.1
					HA16:	10.2
					Total:	83.8

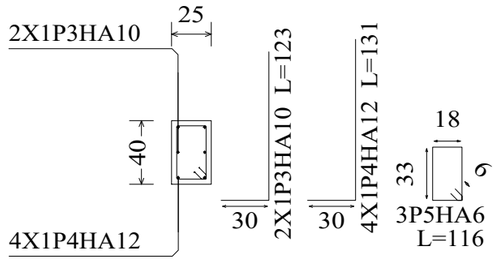
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

P29





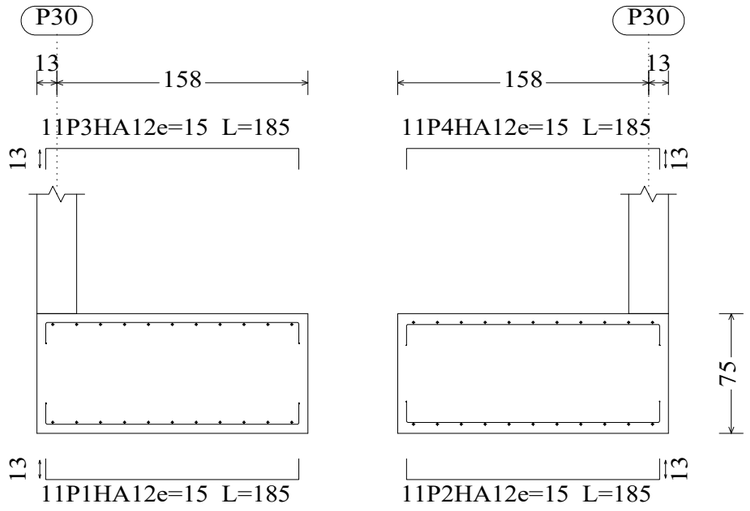
P29

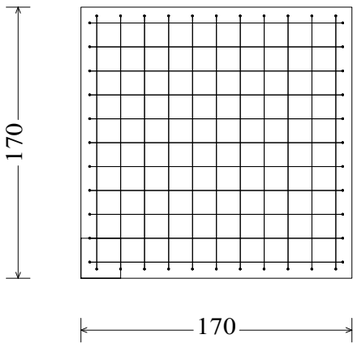


Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P29	1	HA12	15	130	1950	17.3
	2	HA12	9	245	2205	19.6
	3	HA10	2	123	246	1.5
	4	HA12	4	131	524	4.7
	5	HA6	3	116	348	0.8
					Total+10%:	48.3
					HA6:	0.9
					HA10:	1.6
					HA12:	45.8
					Total:	48.3

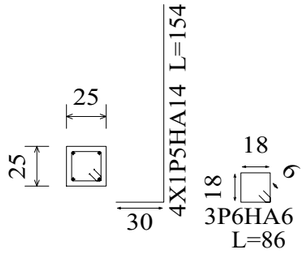
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

P30





P30

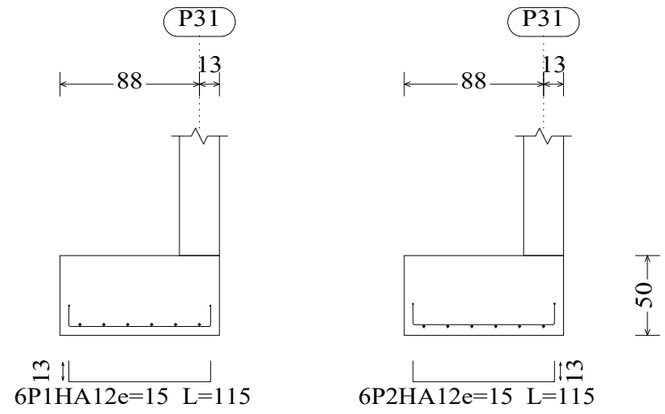


Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P30	1	HA12	11	185	2035	18.1
	2	HA12	11	185	2035	18.1
	3	HA12	11	185	2035	18.1
	4	HA12	11	185	2035	18.1
	5	HA14	4	154	616	7.4
	6	HA6	3	86	258	0.6
					Total+10%:	88.4
					HA6:	0.6
					HA12:	79.6
					HA14:	8.2
					Total:	88.4

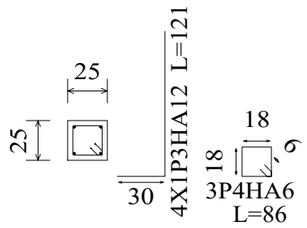
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

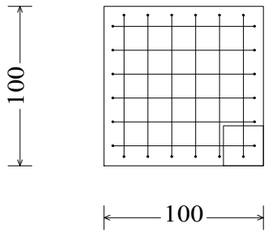
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P31	1	HA12	6	115	690	6.1
	2	HA12	6	115	690	6.1
	3	HA12	4	121	484	4.3
	4	HA6	3	86	258	0.6
					Total+10%:	18.8
					HA6:	0.6
					HA12:	18.2
					Total:	18.8

P31



P31

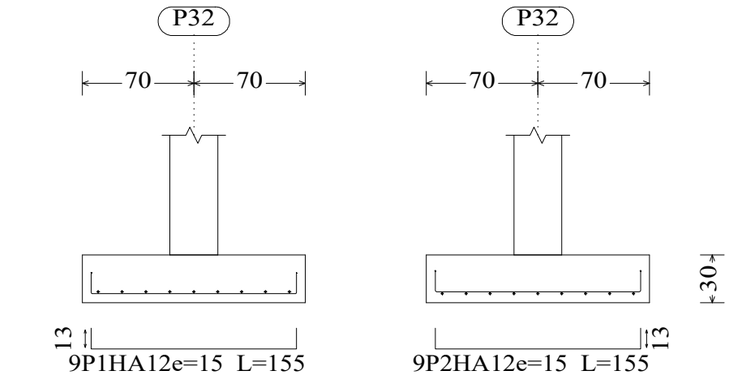




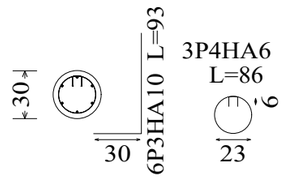
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

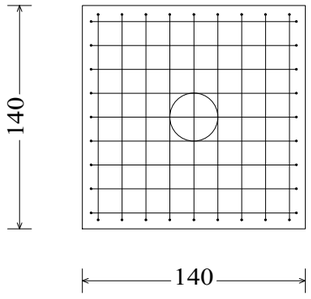
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
P32	1	HA12	9	155	1395	12.4
	2	HA12	9	155	1395	12.4
	3	HA10	6	93	558	3.4
	4	HA6	3	86	258	0.6
					Total+10%:	31.7
					HA6:	0.7
					HA10:	3.7
					HA12:	27.3
					Total:	31.7

P32



P32



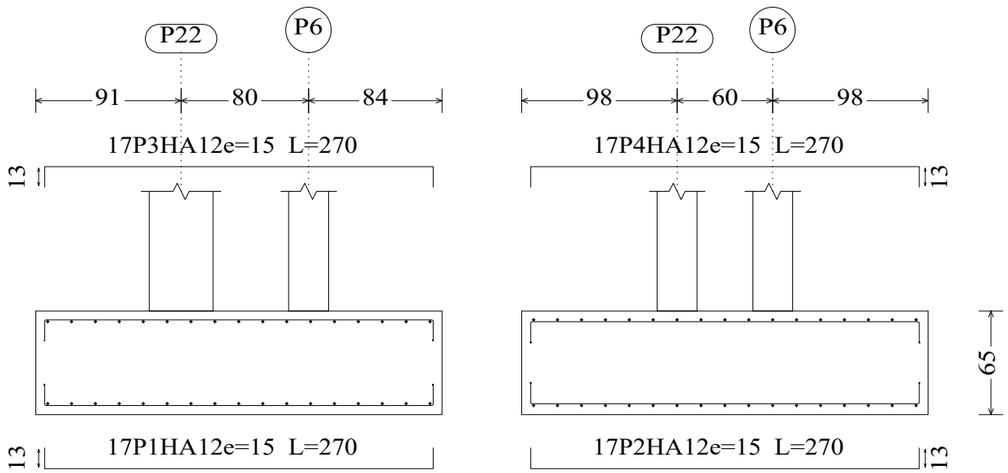


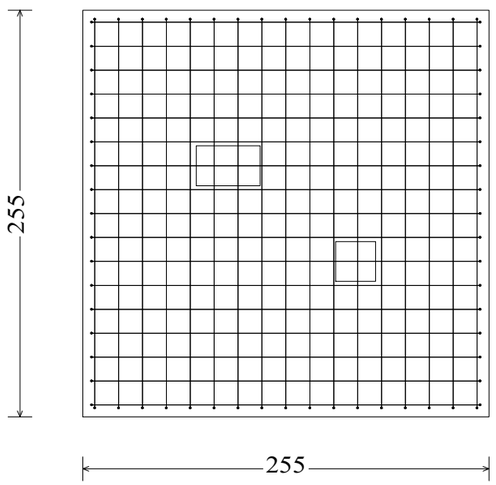
Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

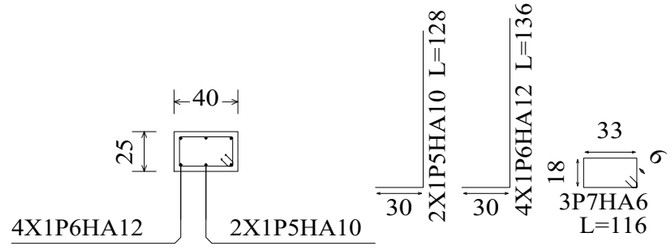
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
(P22-P6)	1	HA12	17	270	4590	40.8
	2	HA12	17	270	4590	40.8
	3	HA12	17	270	4590	40.8
	4	HA12	17	270	4590	40.8
	5	HA10	2	128	256	1.6
	6	HA12	4	136	544	4.8
	7	HA6	3	116	348	0.8
	8	HA12	4	136	544	4.8
	9	HA6	3	86	258	0.6
					Total+10%:	193.4
					HA6:	1.6
					HA10:	1.7
					HA12:	190.1
					Total:	193.4

(P22-P6)

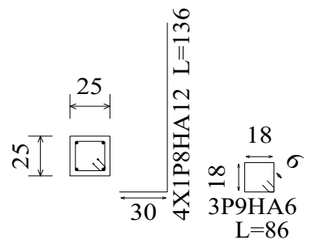




P22



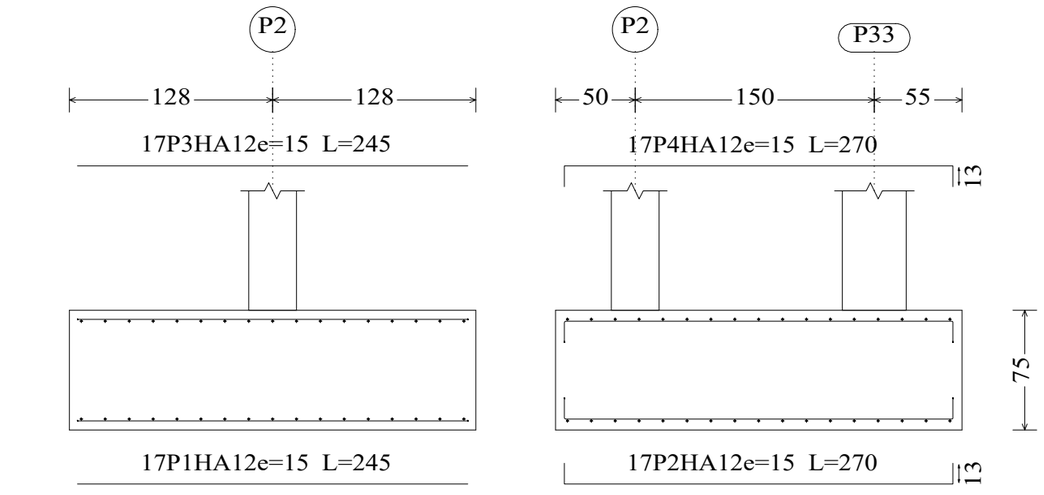
P6

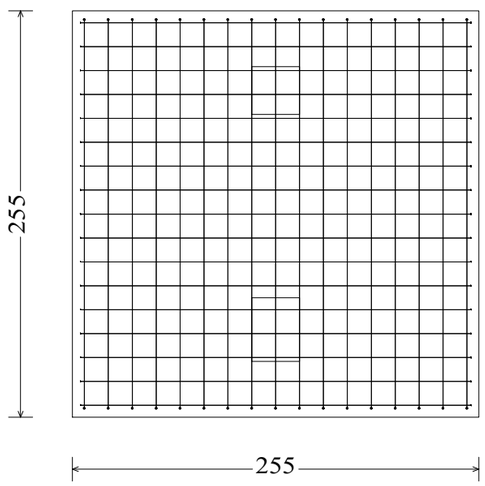


Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

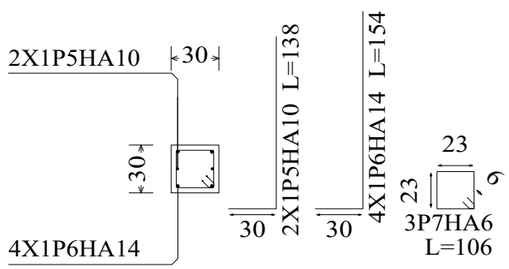
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
(P2-P33)	1	HA12	17	245	4165	37.0
	2	HA12	17	270	4590	40.8
	3	HA12	17	245	4165	37.0
	4	HA12	17	270	4590	40.8
	5	HA10	2	138	276	1.7
	6	HA14	4	154	616	7.4
	7	HA6	3	106	318	0.7
	8	HA10	4	138	552	3.4
	9	HA12	4	146	584	5.2
	10	HA6	3	126	378	0.8
					Total+10%:	192.3
					HA6:	1.7
					HA10:	5.6
					HA12:	176.9
					HA14:	8.1
					Total:	192.3

(P2-P33)

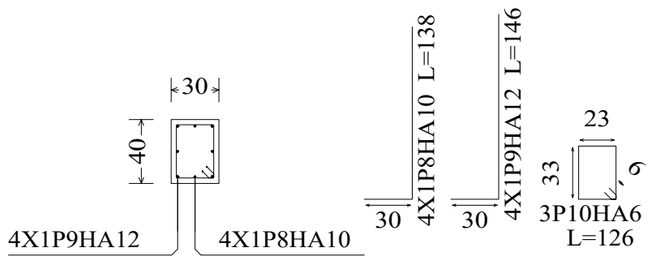




P2



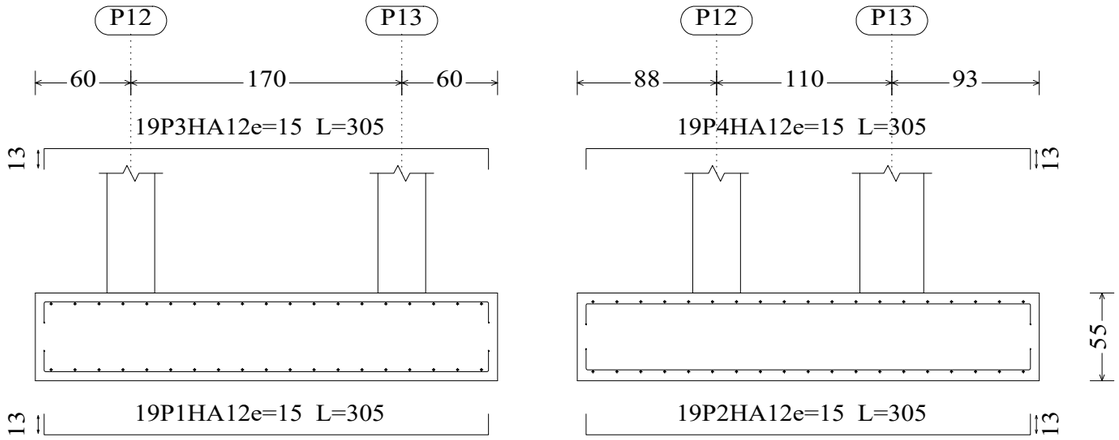
P33

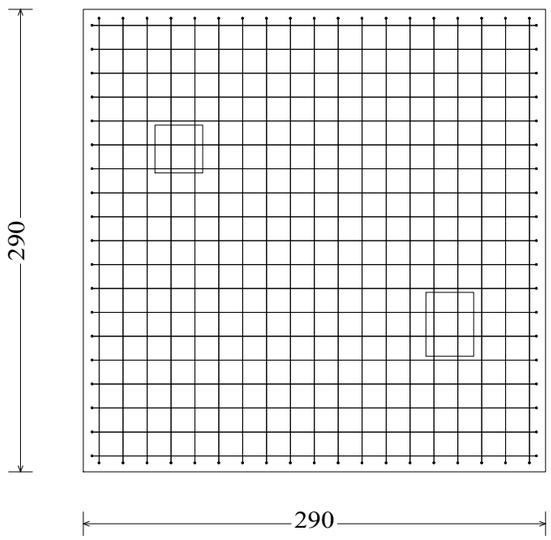


Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

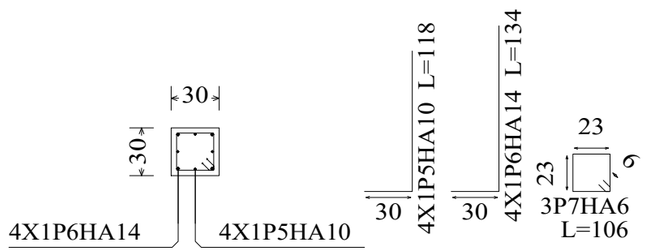
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
(P12-P13)	1	HA12	19	305	5795	51.5
	2	HA12	19	305	5795	51.5
	3	HA12	19	305	5795	51.5
	4	HA12	19	305	5795	51.5
	5	HA10	4	118	472	2.9
	6	HA14	4	134	536	6.5
	7	HA6	3	106	318	0.7
	8	HA12	2	126	252	2.2
	9	HA14	4	134	536	6.5
	10	HA6	3	126	378	0.8
					Total+10%:	248.2
					HA6:	1.7
					HA10:	3.2
					HA12:	229.0
					HA14:	14.3
					Total:	248.2

(P12-P13)

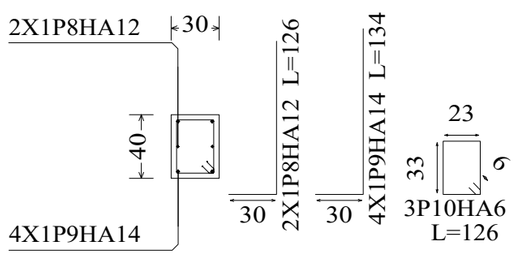




P12



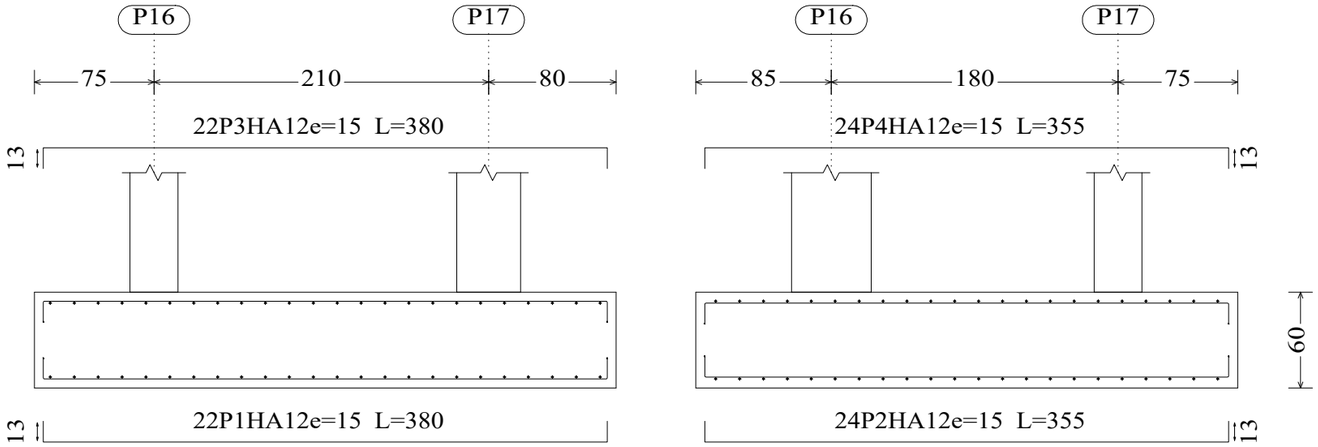
P13

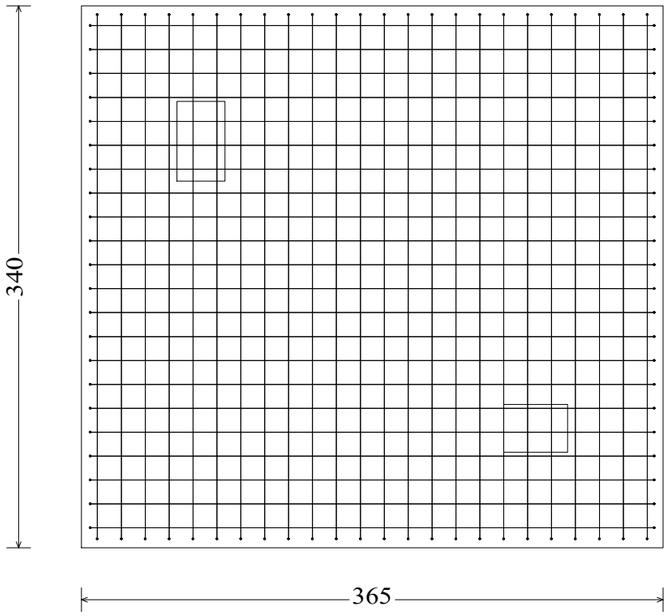


Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

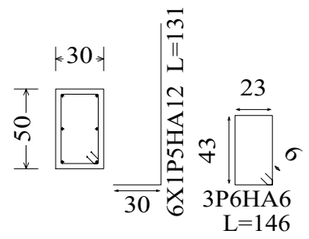
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
(P16-P17)	1	HA12	22	380	8360	74.2
	2	HA12	24	355	8520	75.6
	3	HA12	22	380	8360	74.2
	4	HA12	24	355	8520	75.6
	5	HA12	6	131	786	7.0
	6	HA6	3	146	438	1.0
	7	HA10	4	123	492	3.0
	8	HA12	4	131	524	4.7
	9	HA6	3	126	378	0.8
					Total+10%:	347.7
					HA6:	2.0
					HA10:	3.3
					HA12:	342.4
					Total:	347.7

(P16-P17)

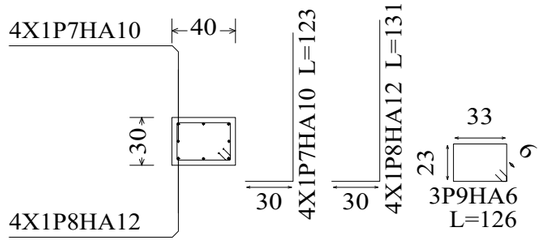




P16



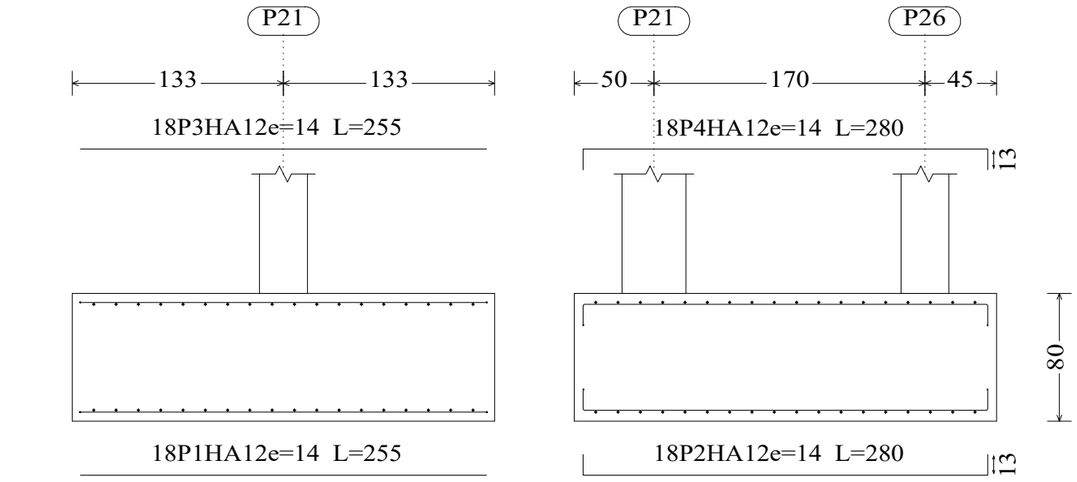
P17

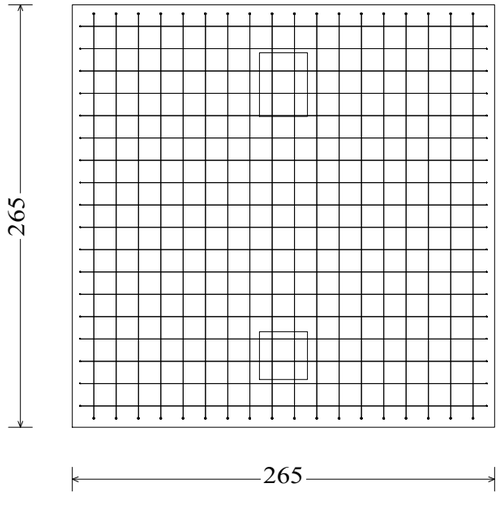


Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

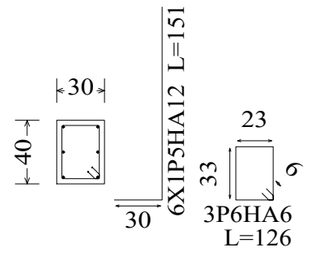
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
(P21-P26)	1	HA12	18	255	4590	40.8
	2	HA12	18	280	5040	44.7
	3	HA12	18	255	4590	40.8
	4	HA12	18	280	5040	44.7
	5	HA12	6	151	906	8.0
	6	HA6	3	126	378	0.8
	7	HA10	2	143	286	1.8
	8	HA14	4	159	636	7.7
	9	HA6	3	106	318	0.7
					Total+10%:	209.0
					HA6:	1.7
					HA10:	1.9
					HA12:	196.9
					HA14:	8.5
					Total:	209.0

(P21-P26)

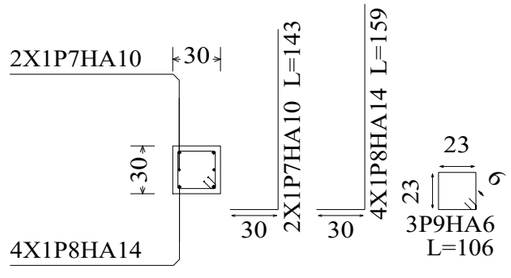




P21

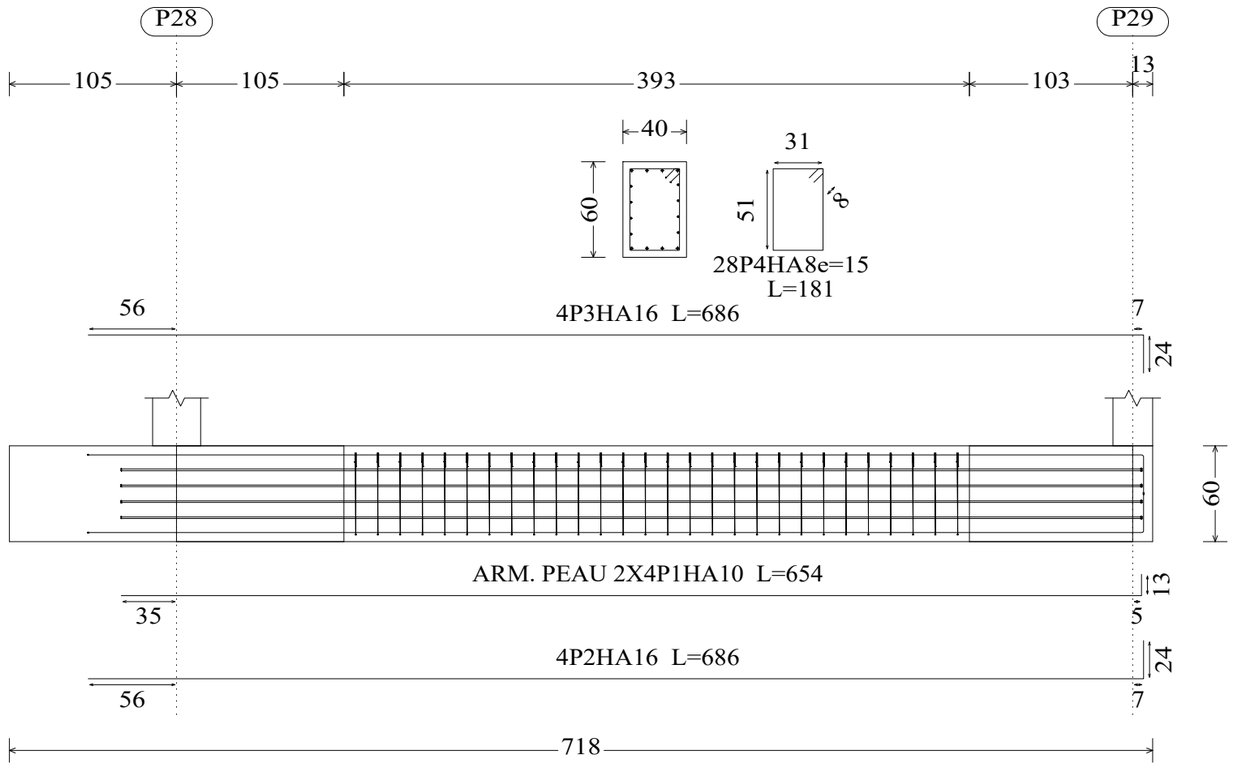


P26



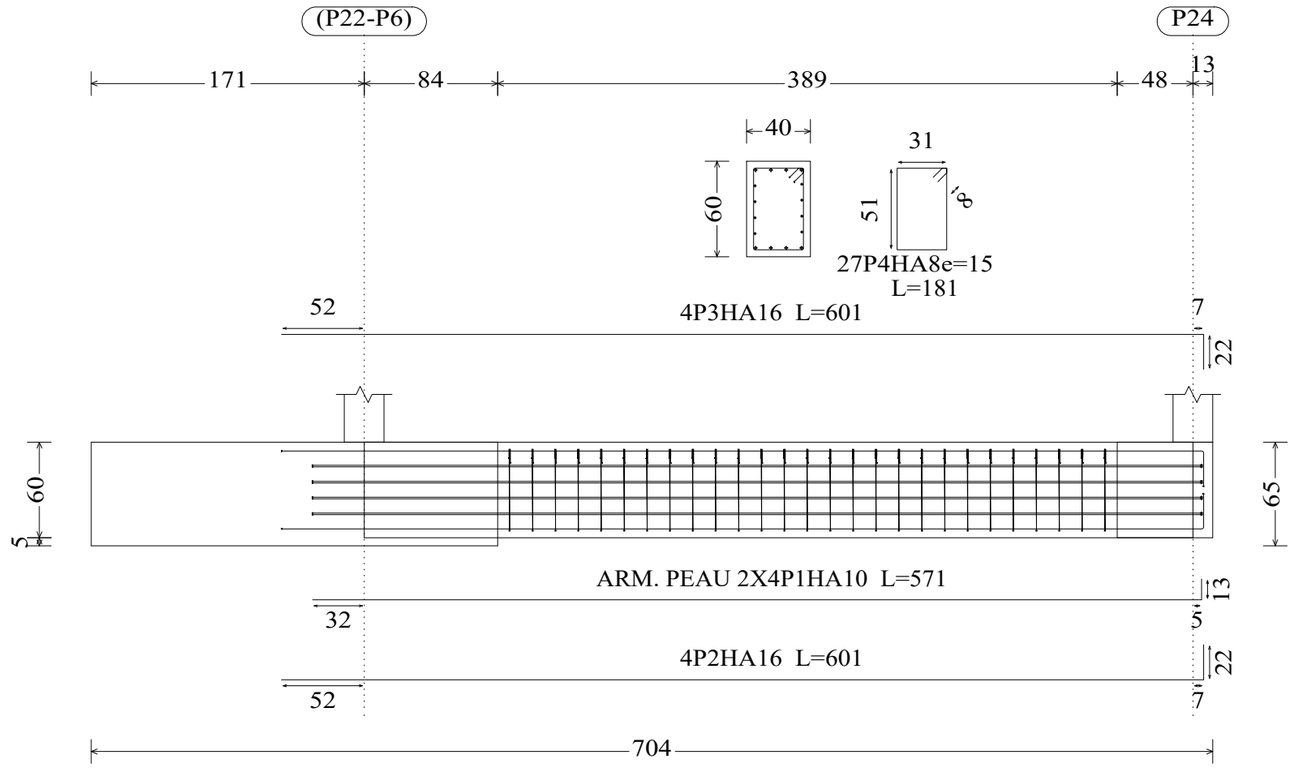
Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
R2 [P28 - P29]	1	HA10	8	654	5232	32.3
	2	HA16	4	686	2744	43.3
	3	HA16	4	686	2744	43.3
	4	HA8	28	181	5068	20.0
					Total+10%:	152.8
					HA8:	22.0
					HA10:	35.5
					HA16:	95.3
					Total:	152.8

R2 [P28 - P29]



Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

R2 [(P22-P6) - P24]

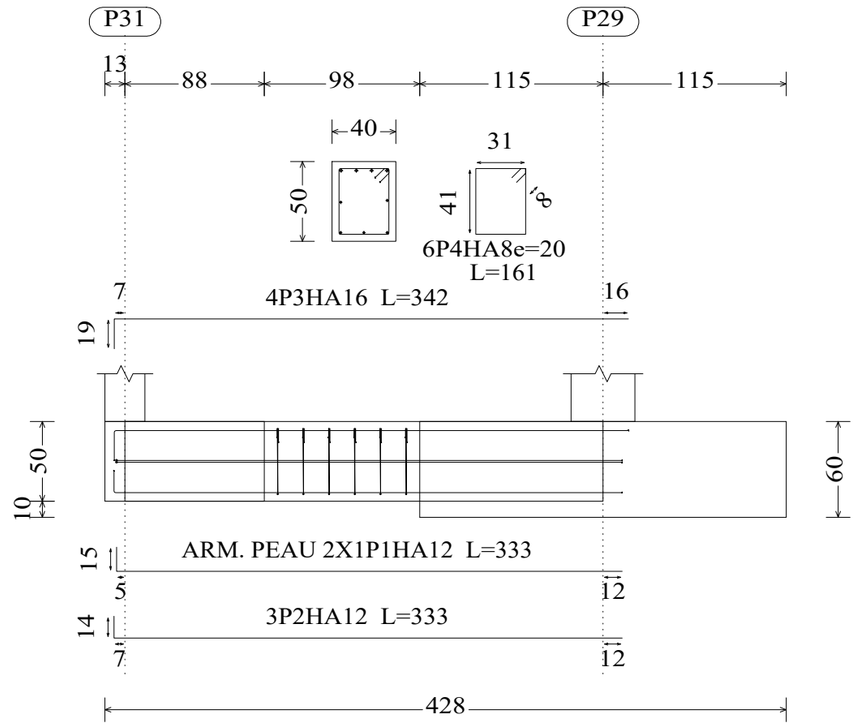


Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
R2 [(P22-P6) - P24]	1	HA10	8	571	4568	28.2
	2	HA16	4	601	2404	37.9
	3	HA16	4	601	2404	37.9
	4	HA8	27	181	4887	19.3
					Total+10%:	135.6
					HA8:	21.2
					HA10:	31.0
					HA16:	83.4
					Total:	135.6

Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
LR.T-1.1 [P31 - P29]	1	HA12	2	333	666	5.9
	2	HA12	3	333	999	8.9
	3	HA16	4	342	1368	21.6
	4	HA8	6	161	966	3.8
					Total+10%:	44.2
					HA8:	4.2
					HA12:	16.3
					HA16:	23.7
					Total:	44.2

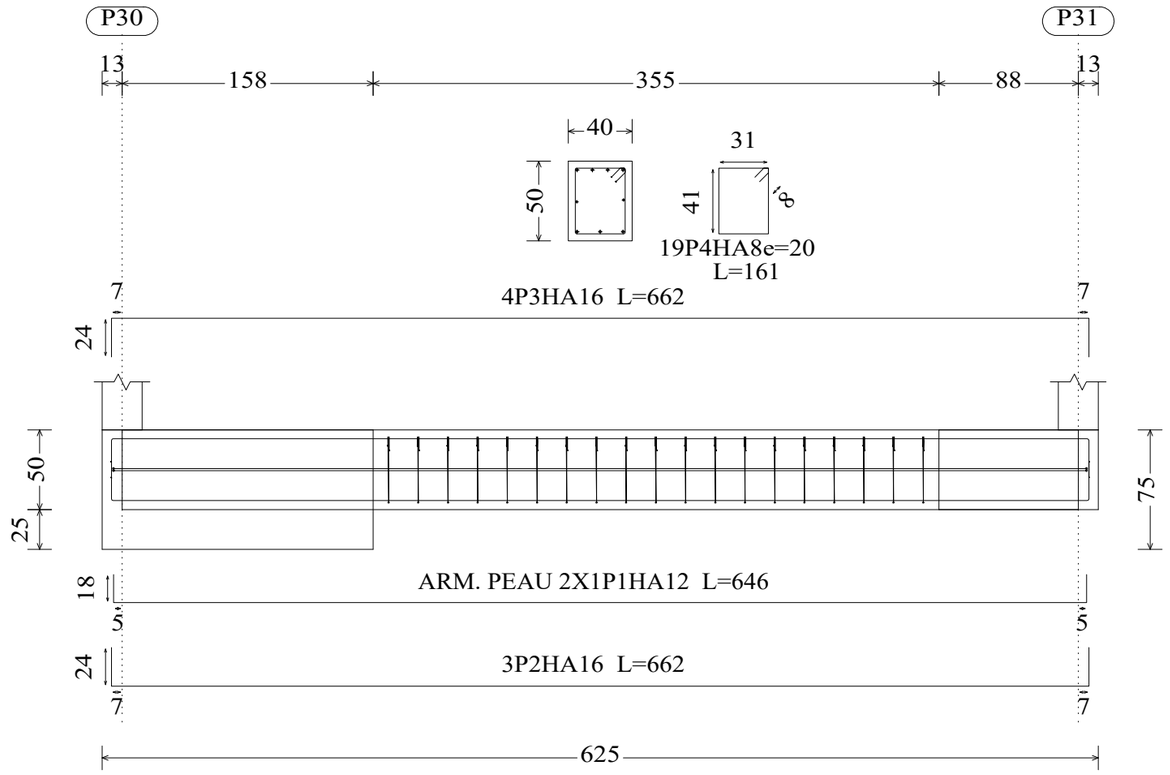
LR.T-1.1 [P31 - P29]



Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
LR.T-1.3 [P30 - P31]	1	HA12	2	646	1292	11.5
	2	HA16	3	662	1986	31.3
	3	HA16	4	662	2648	41.8
	4	HA8	19	161	3059	12.1
					Total+10%:	106.4
					HA8:	13.3
					HA12:	12.7
					HA16:	80.4
					Total:	106.4

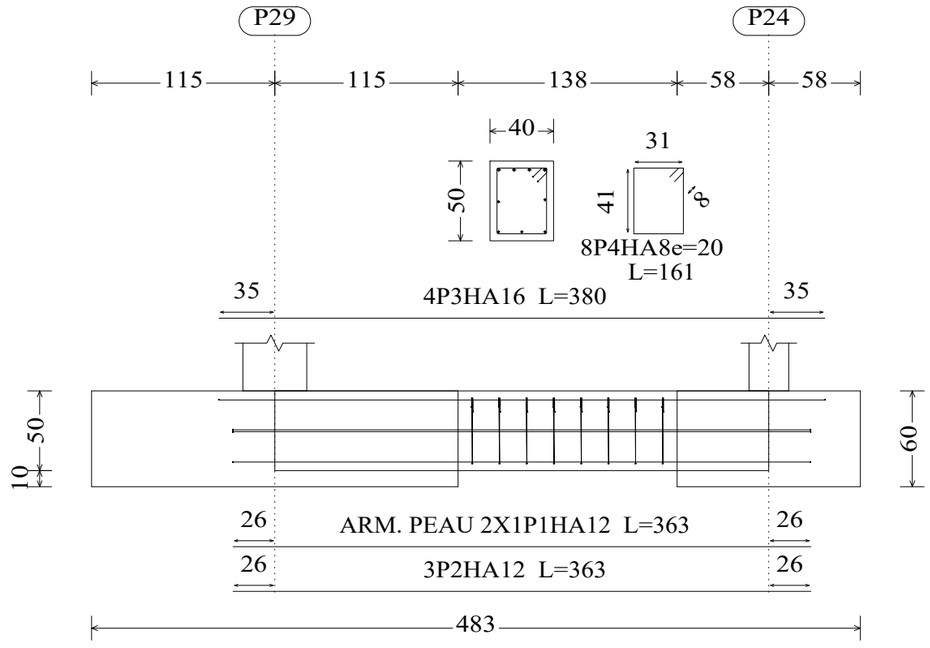
LR.T-1.3 [P30 - P31]



Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

Élément	Pos.	Diam.	No.	Long. (cm)	Total (cm)	Fe E400 (kg)
LR.T-1.1 [P29 - P24]	1	HA12	2	363	726	6.4
	2	HA12	3	363	1089	9.7
	3	HA16	4	380	1520	24.0
	4	HA8	8	161	1288	5.1
					Total+10%:	49.7
					HA8:	5.6
					HA12:	17.7
					HA16:	26.4
					Total:	49.7

LR.T-1.1 [P29 - P24]



Récapitulatif Acier Fondation Plan de ferraillage fondation		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	105.0	26	
	HA8	152.7	66	
	HA10	161.8	110	
	HA12	2143.0	2093	
	HA14	64.5	86	
	HA16	199.5	346	2727

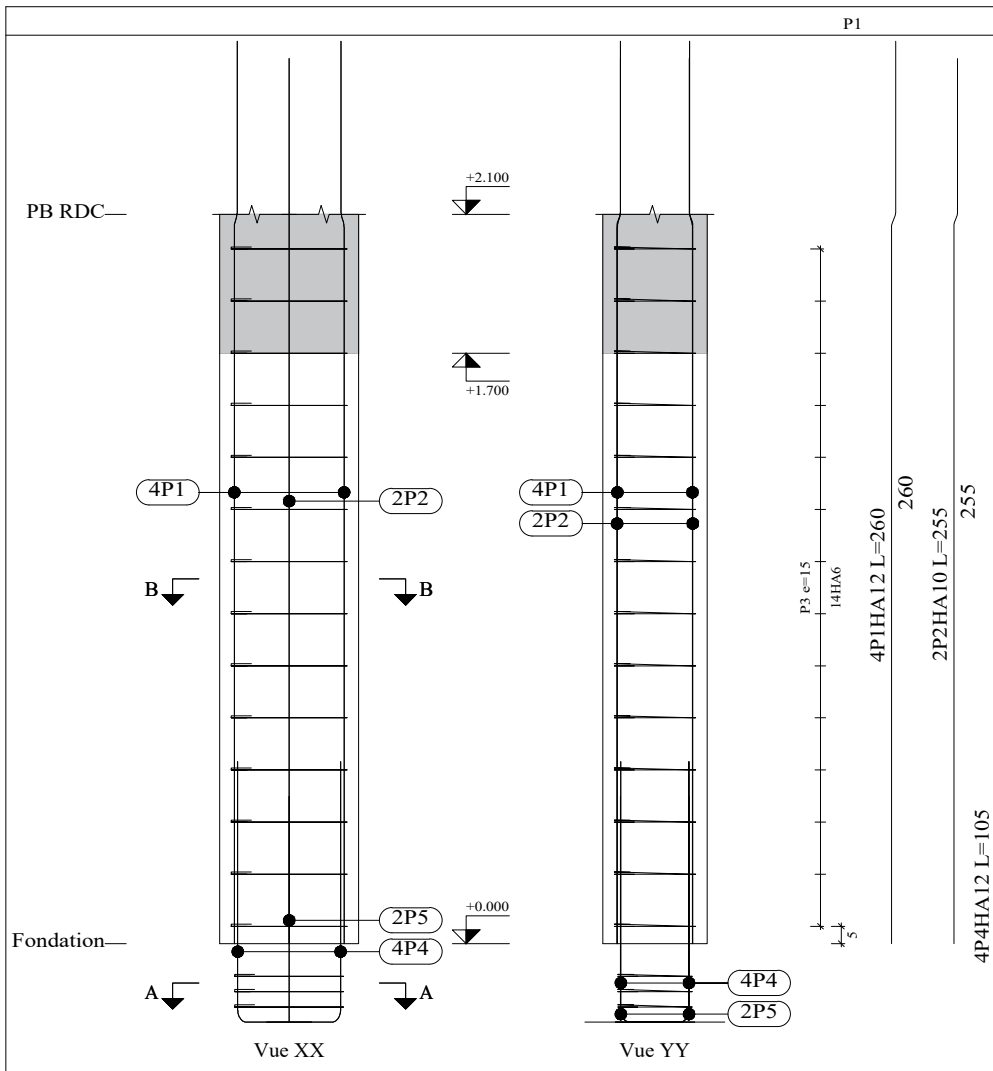
POTEAUX

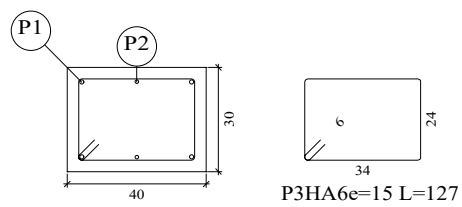
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P1	1	HA12	4	207 50	260	1040	9.2
	2	HA10	2	208 45	255	510	3.1
	3	HA6	14	24 6 34	127	1778	3.9
	4	HA12	4	30 75	105	420	3.7
	5	HA10	2	30 65	95	190	1.2
	6	HA6	3	22 6 32	119	357	0.8
						Total+10%:	24.1
						HA6:	5.2
						HA10:	4.7
						HA12:	14.2
						Total:	24.1

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P1

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA10	2	255	510
3	HA6	14	127	1778
4	HA12	4	105	420
5	HA10	2	95	190
6	HA6	3	119	357

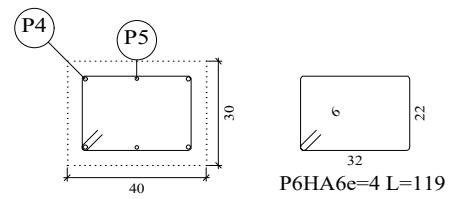




P3HA6e=15 L=127

4P1HA12
2P2HA10

Section B-B



P6HA6e=4 L=119

4P4HA12
2P5HA10

Section A-A

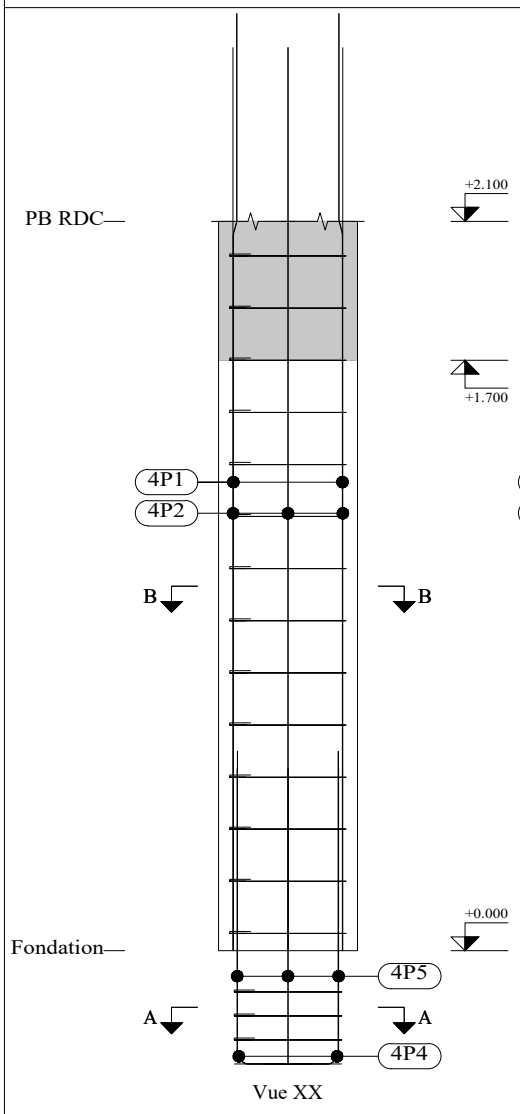
Acier: Fe E400 (18.0 kg). Ratio: 64.78 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

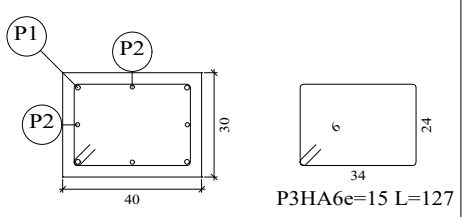
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P3	1	HA14	4	207 / 60	270	1080	13.1
	2	HA12	4	260	260	1040	9.2
	3	HA6	14	24 × 34, 6	127	1778	3.9
	4	HA14	4	30 / 90	120	480	5.8
	5	HA12	4	30 / 85	115	460	4.1
	6	HA6	3	22 × 32, 6	117	351	0.8
					Total+10%:		40.6
					HA6:		5.2
					HA12:		14.6
					HA14:		20.8
					Total:		40.6

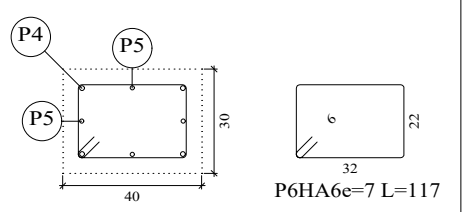
Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P3

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	270	1080
2	HA12	4	260	1040
3	HA6	14	127	1778
4	HA14	4	120	480
5	HA12	4	115	460
6	HA6	3	117	351





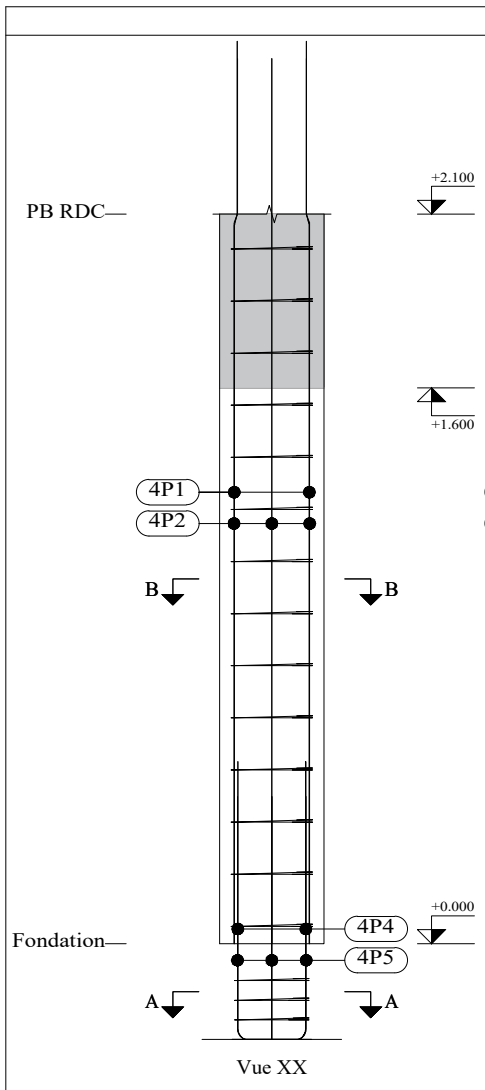


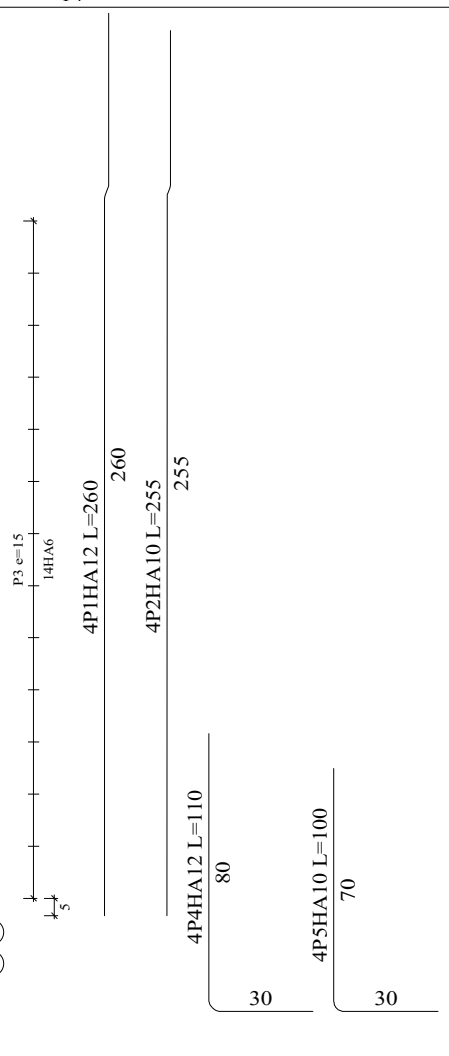
Acier: Fe E400 (28.9 kg). Ratio: 104.09 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P4	1	HA12	4	207 50	260	1040	9.2
	2	HA10	4	208 45	255	1020	6.3
	3	HA6	14	24 6 34	127	1778	3.9
	4	HA12	4	30 80	110	440	3.9
	5	HA10	4	30 70	100	400	2.5
	6	HA6	3	22 6 32	119	357	0.8
						Total+10%:	29.3
						HA6:	5.2
						HA10:	9.7
						HA12:	14.4
						Total:	29.3

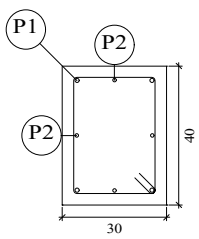
Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P4



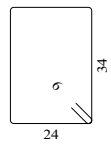


Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA10	4	255	1020
3	HA6	14	127	1778
4	HA12	4	110	440
5	HA10	4	100	400
6	HA6	3	119	357

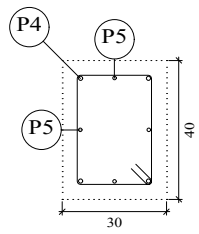


4P1HA12
4P2HA10

Section B-B

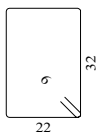


P3HA6e=15 L=127



4P4HA12
4P5HA10

Section A-A



P6HA6e=6 L=119

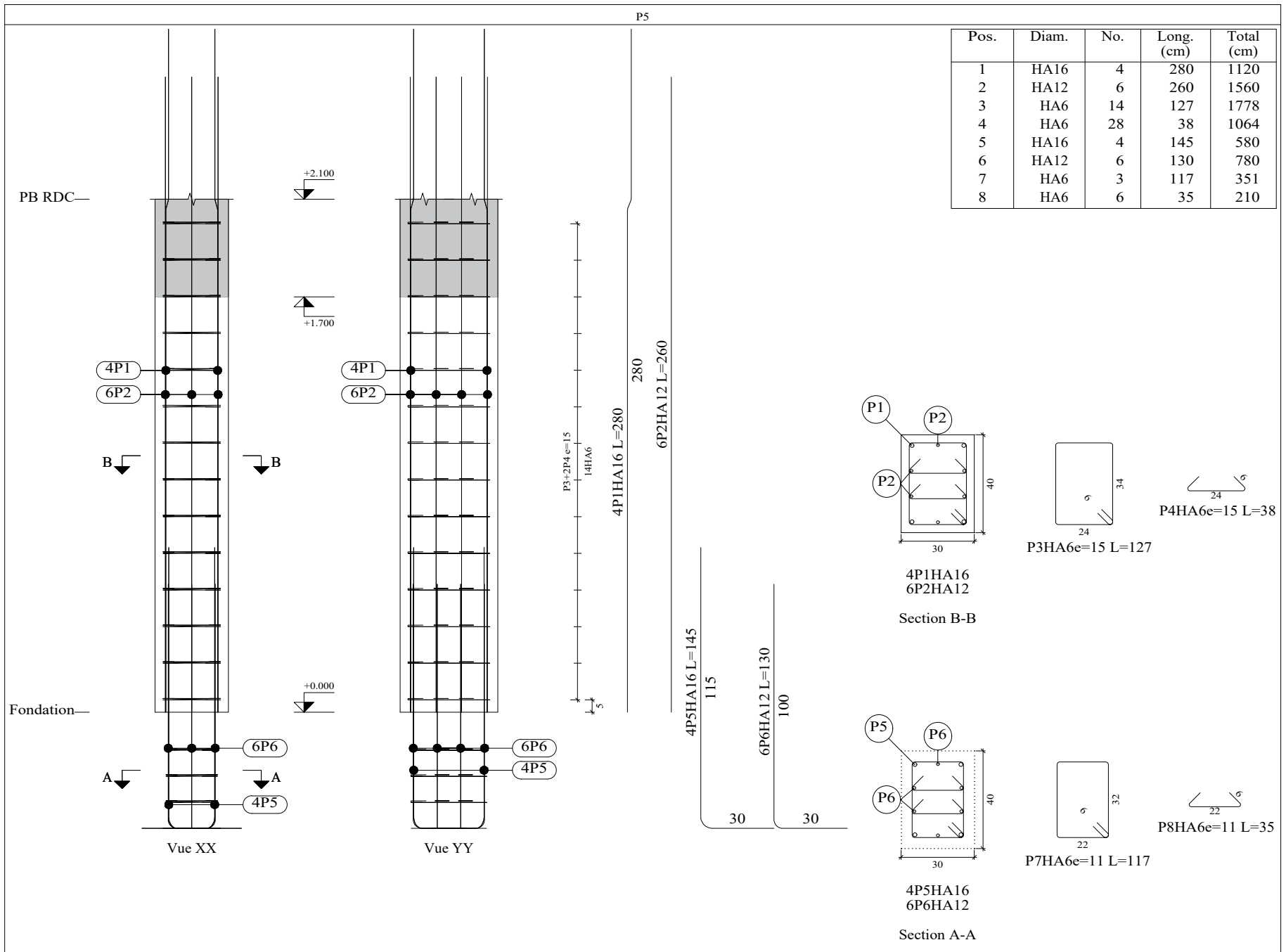
Acier: Fe E400 (21.4 kg). Ratio: 77.25 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P5	1	HA16	4	206 70	280	1120	17.7
	2	HA12	6	260	260	1560	13.9
	3	HA6	14	24 6 34	127	1778	3.9
	4	HA6	28	6 24	38	1064	2.4
	5	HA16	4	30 115	145	580	9.2
	6	HA12	6	30 100	130	780	6.9
	7	HA6	3	22 6 32	117	351	0.8
	8	HA6	6	6 22	35	210	0.5
						Total+10%:	60.8
						HA6:	8.3
						HA12:	22.9
						HA16:	29.6
						Total:	60.8

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P5

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA16	4	280	1120
2	HA12	6	260	1560
3	HA6	14	127	1778
4	HA6	28	38	1064
5	HA16	4	145	580
6	HA12	6	130	780
7	HA6	3	117	351
8	HA6	6	35	210



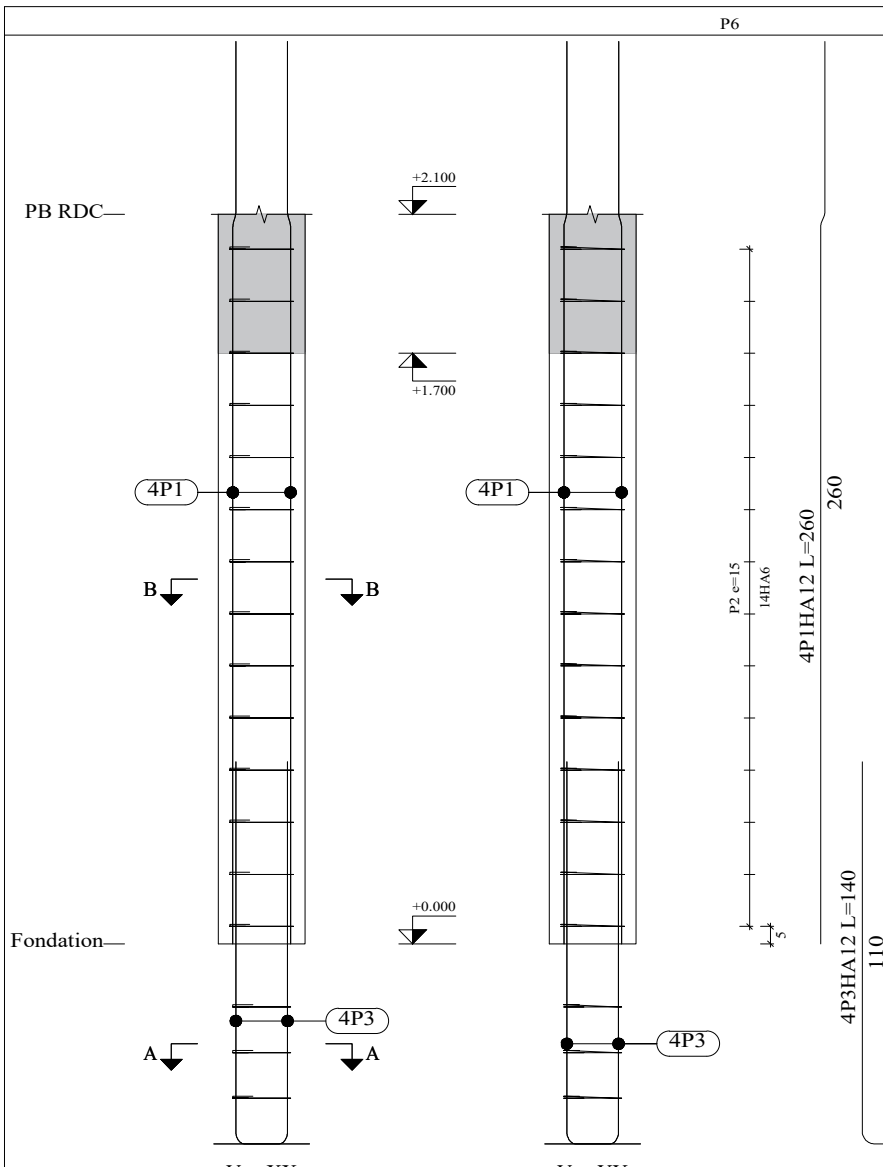
Acier: Fe E400 (41.6 kg). Ratio: 150.14 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

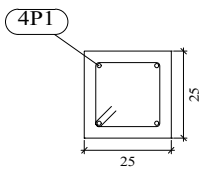
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P6	1	HA12	4	207 50	260	1040	9.2
	2	HA6	14	19 6 19	87	1218	2.7
	3	HA12	4	30 110	140	560	5.0
	4	HA6	3	17 6 17	80	240	0.5
					Total+10%:		19.1
					HA6:		3.5
					HA12:		15.6
					Total:		19.1

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P6

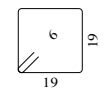
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA6	14	87	1218
3	HA12	4	140	560
4	HA6	3	80	240



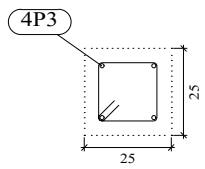


4P1HA12

Section B-B

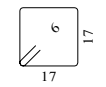


P2HA6e=15 L=87



4P3HA12

Section A-A



P4HA6e=13 L=80

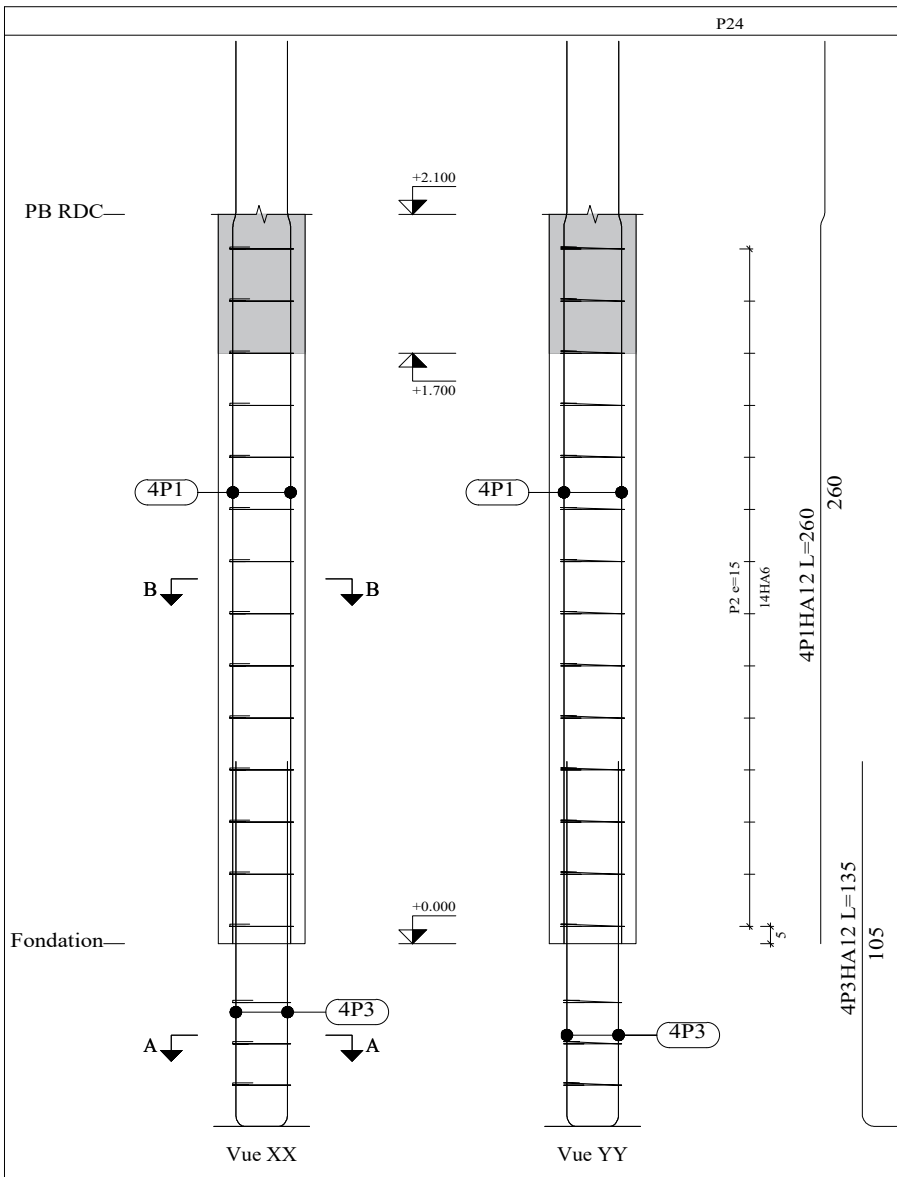
Acier: Fe E400 (13.1 kg). Ratio: 90.94 kg/m3		Niveau: PB RDC
Béton: B20 (0.13 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.10 m2	Enrobage géométrique: 3 cm	

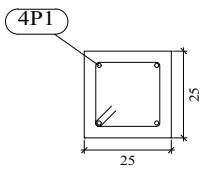
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P24	1	HA12	4	207 50	260	1040	9.2
	2	HA6	14	19 6 19	87	1218	2.7
	3	HA12	4	30 105	135	540	4.8
	4	HA6	3	17 6 17	80	240	0.5
						Total+10%:	18.9
						HA6:	3.5
						HA12:	15.4
						Total:	18.9

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P24

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA6	14	87	1218
3	HA12	4	135	540
4	HA6	3	80	240



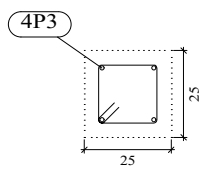


4P1HA12

Section B-B



P2HA6e=15 L=87



4P3HA12

Section A-A



P4HA6e=12 L=80

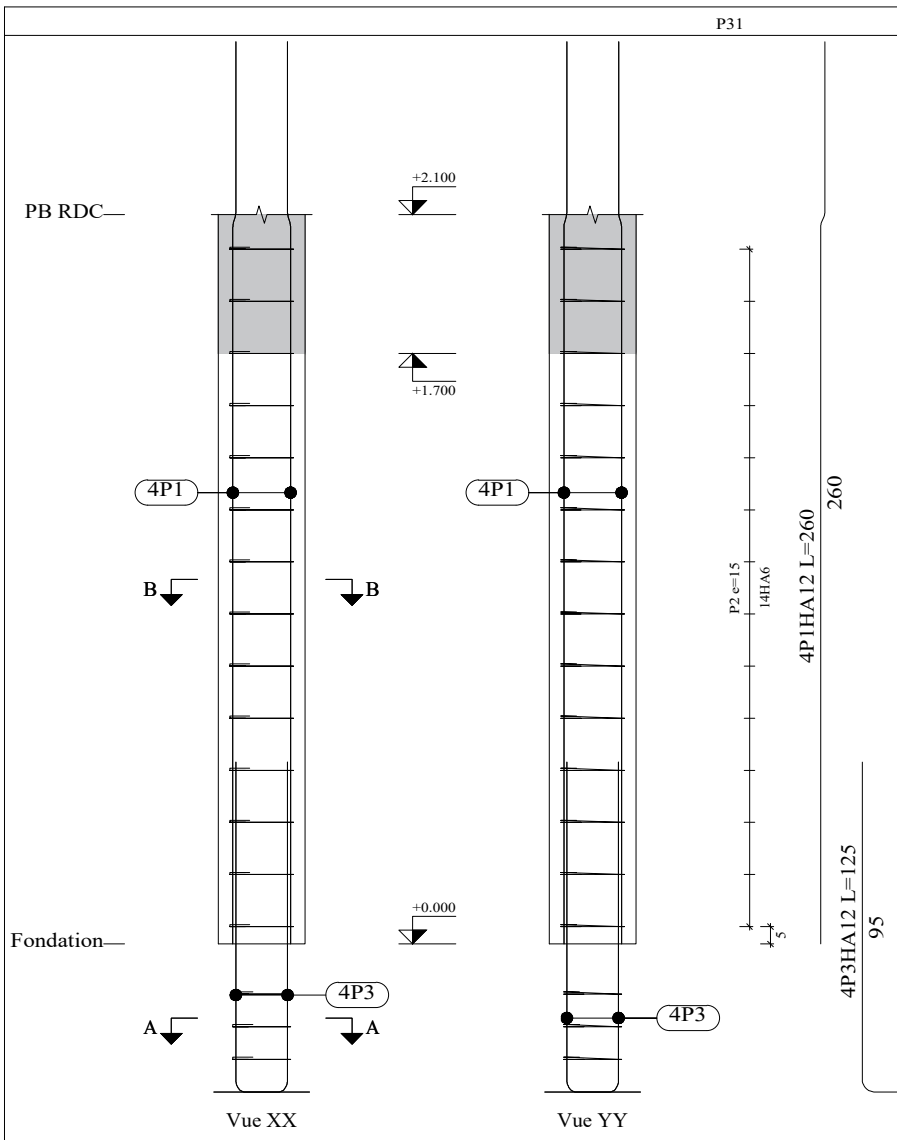
Acier: Fe E400 (13.1 kg). Ratio: 90.94 kg/m3		Niveau: PB RDC
Béton: B20 (0.13 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.10 m2	Enrobage géométrique: 3 cm	

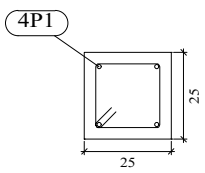
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P31	1	HA12	4	207 50	260	1040	9.2
	2	HA6	14	19 6 19	87	1218	2.7
	3	HA12	4	30 95	125	500	4.4
	4	HA6	3	17 6 17	80	240	0.5
					Total+10%:		18.5
						HA6:	3.5
						HA12:	15.0
						Total:	18.5

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P31

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA6	14	87	1218
3	HA12	4	125	500
4	HA6	3	80	240



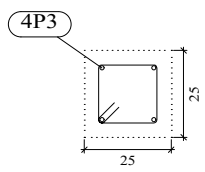


4P1HA12

Section B-B



P2HA6e=15 L=87



4P3HA12

Section A-A



P4HA6e=9 L=80

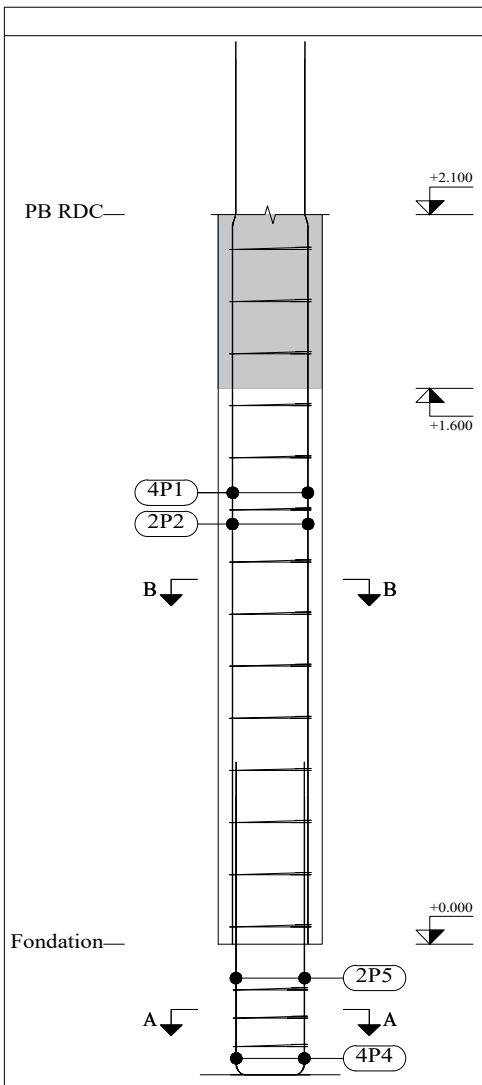
Acier: Fe E400 (13.1 kg). Ratio: 90.94 kg/m3		Niveau: PB RDC
Béton: B20 (0.13 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.10 m2	Enrobage géométrique: 3 cm	

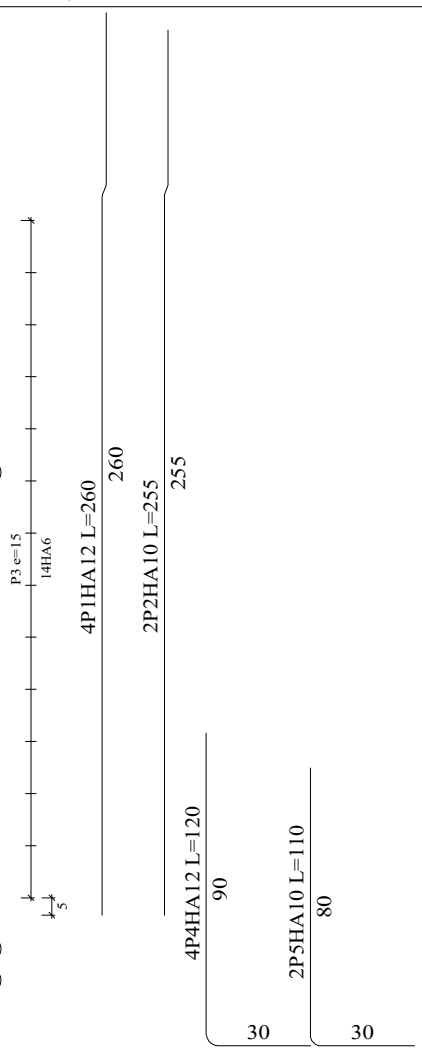
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P10	1	HA12	4	207 50	260	1040	9.2
	2	HA10	2	208 45	255	510	3.1
	3	HA6	14	24 6 34	127	1778	3.9
	4	HA12	4	30 90	120	480	4.3
	5	HA10	2	30 80	110	220	1.4
	6	HA6	3	22 6 32	119	357	0.8
						Total+10%:	25.0
						HA6:	5.2
						HA10:	4.9
						HA12:	14.9
						Total:	25.0

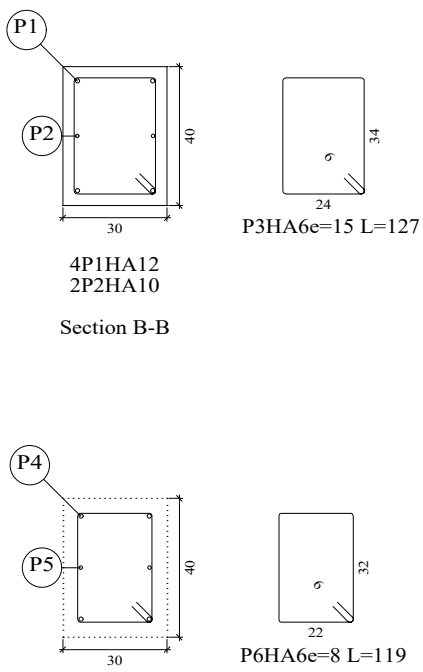
Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P10

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA10	2	255	510
3	HA6	14	127	1778
4	HA12	4	120	480
5	HA10	2	110	220
6	HA6	3	119	357







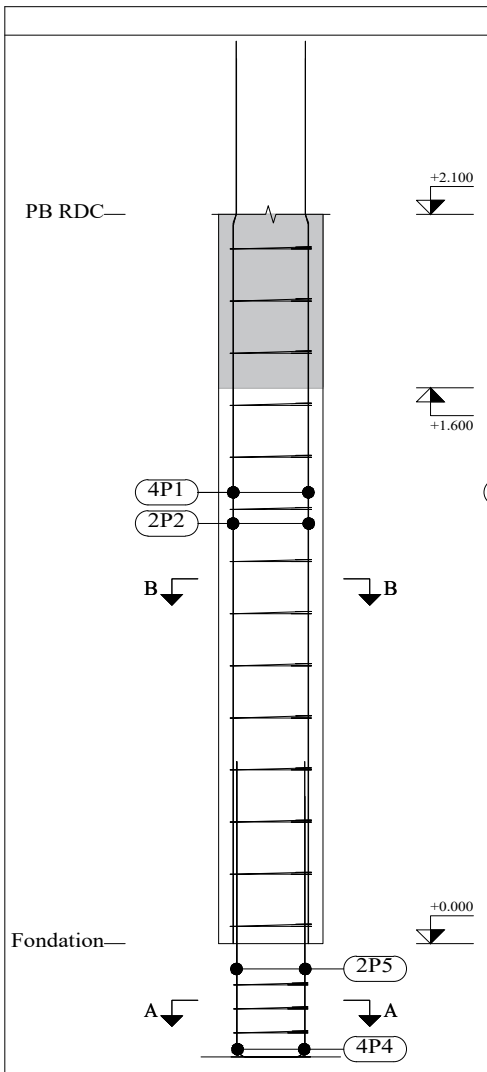
Acier: Fe E400 (18.0 kg). Ratio: 64.78 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

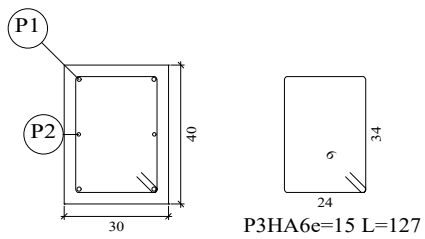
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P14	1	HA12	4	207 50	260	1040	9.2
	2	HA10	2	208 45	255	510	3.1
	3	HA6	14	24 6 34	127	1778	3.9
	4	HA12	4	30 85	115	460	4.1
	5	HA10	2	30 75	105	210	1.3
	6	HA6	3	22 6 32	119	357	0.8
						Total+10%:	24.6
						HA6:	5.1
						HA10:	4.9
						HA12:	14.6
						Total:	24.6

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P14

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA10	2	255	510
3	HA6	14	127	1778
4	HA12	4	115	460
5	HA10	2	105	210
6	HA6	3	119	357

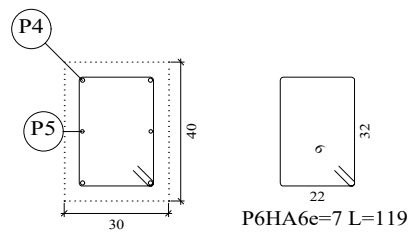




4P1HA12
2P2HA10

Section B-B

P3HA6e=15 L=127



4P4HA12
2P5HA10

Section A-A

P6HA6e=7 L=119

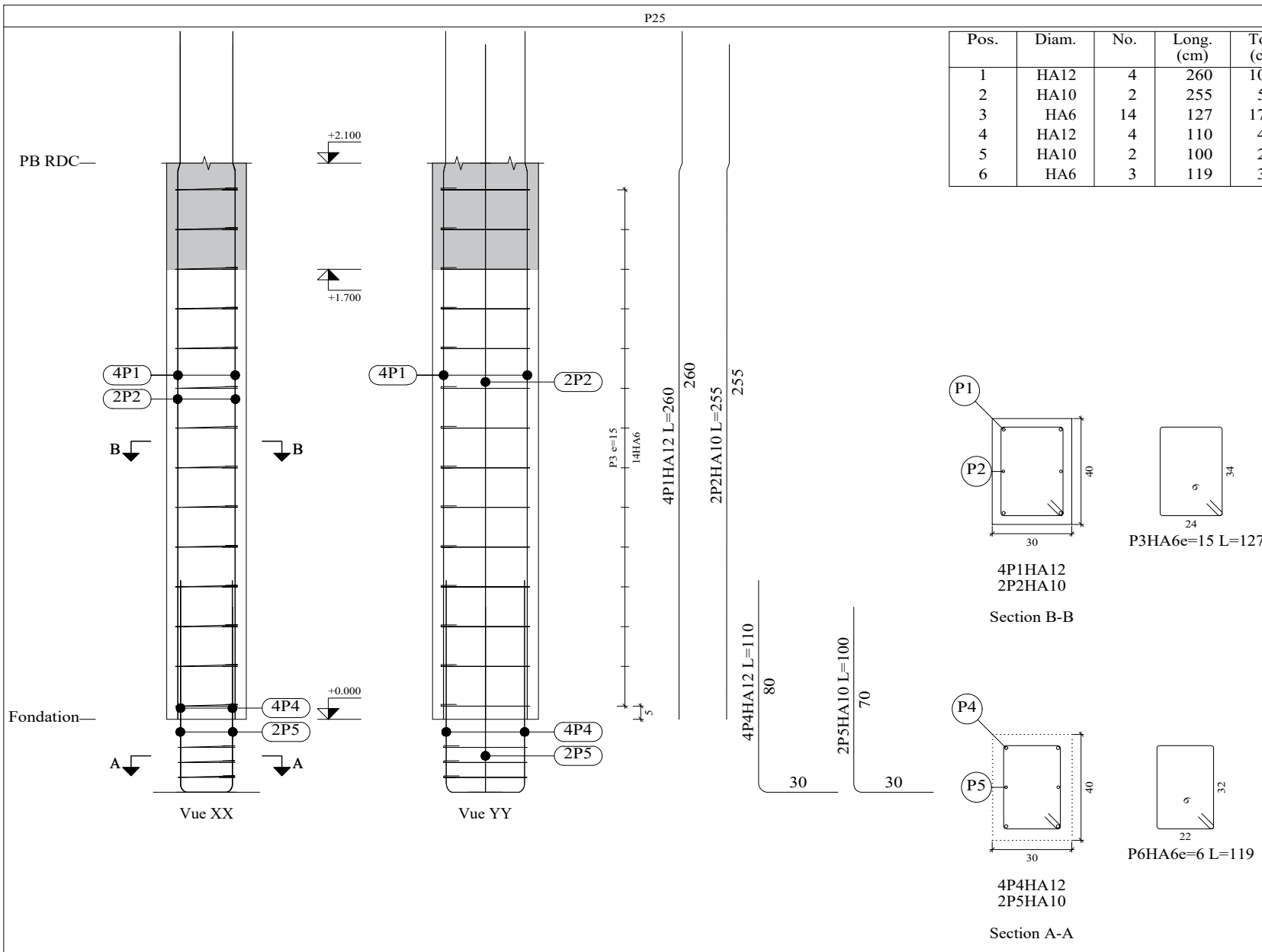
Acier: Fe E400 (18.0 kg). Ratio: 64.78 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P25	1	HA12	4	207 50	260	1040	9.2
	2	HA10	2	208 45	255	510	3.1
	3	HA6	14	24 6 34	127	1778	3.9
	4	HA12	4	30 80	110	440	3.9
	5	HA10	2	30 70	100	200	1.2
	6	HA6	3	22 6 32	119	357	0.8
						Total+10%:	24.3
						HA6:	5.2
						HA10:	4.7
						HA12:	14.4
						Total:	24.3

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P25

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA10	2	255	510
3	HA6	14	127	1778
4	HA12	4	110	440
5	HA10	2	100	200
6	HA6	3	119	357



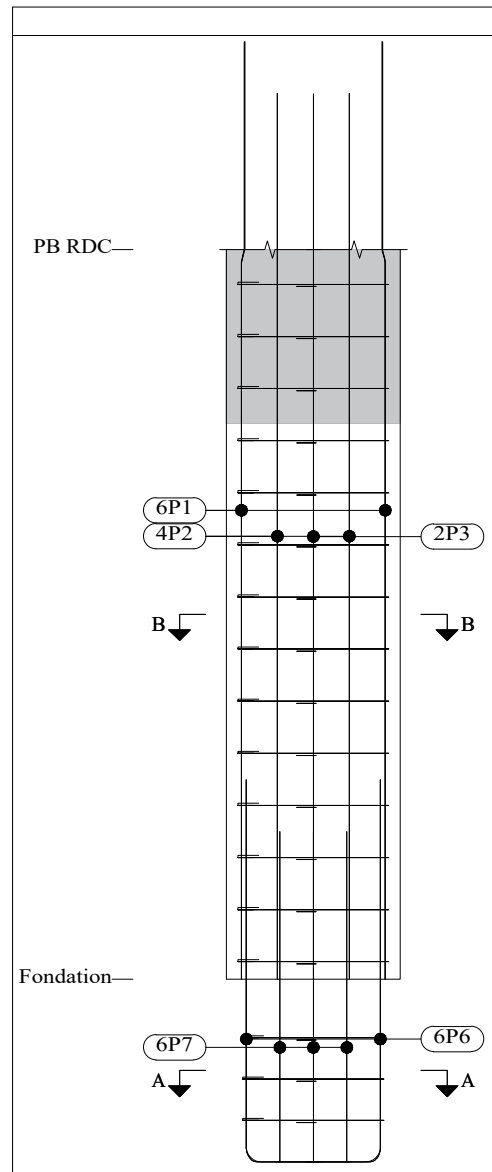
Acier: Fe E400 (18.0 kg). Ratio: 64.78 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P11	1	HA14	6	207 60	270	1620	19.6
	2	HA10	4	255	255	1020	6.3
	3	HA10	2	208 45	255	510	3.1
	4	HA6	14	24 6 44	147	2058	4.6
	5	HA6	14	6 24	38	532	1.2
	6	HA14	6	30 110	140	840	10.2
	7	HA10	6	30 95	125	750	4.6
	8	HA6	3	21 6 41	136	408	0.9
	9	HA6	3	6 21	35	105	0.2
						Total+10%:	55.8
						HA6:	7.6
						HA10:	15.4
						HA14:	32.8
						Total:	55.8

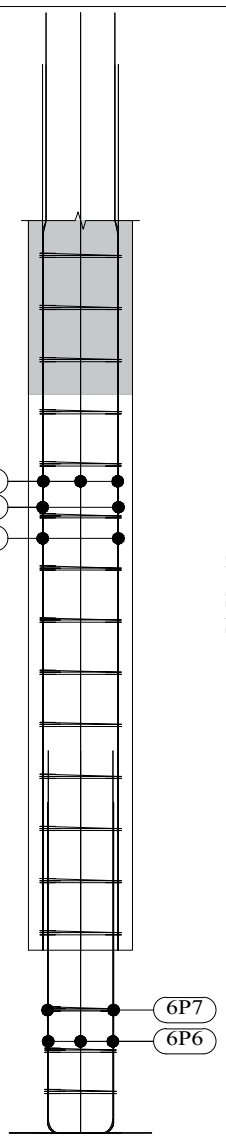
Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P11

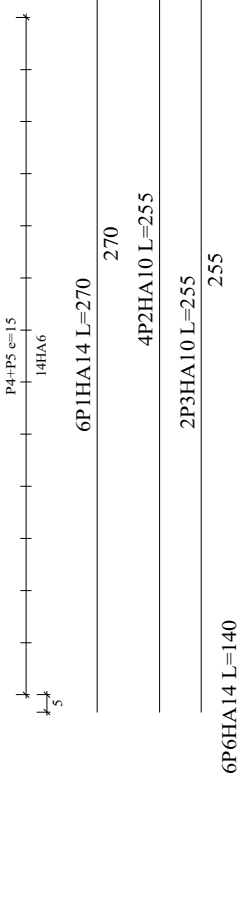
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	6	270	1620
2	HA10	4	255	1020
3	HA10	2	255	510
4	HA6	14	147	2058
5	HA6	14	38	532
6	HA14	6	140	840
7	HA10	6	125	750
8	HA6	3	136	408
9	HA6	3	35	105

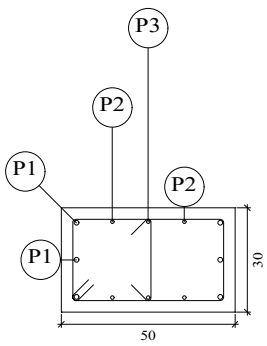


Vue XX



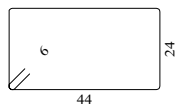
Vue YY



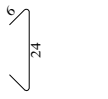


6P1HA14
4P2HA10
2P3HA10

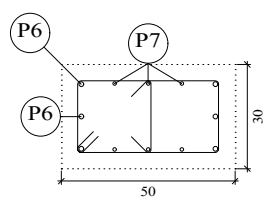
Section B-B



P4HA6e=15 L=147

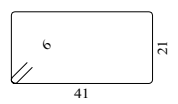


P5HA6e=15 L=38

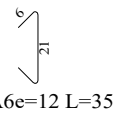


6P6HA14
6P7HA10

Section A-A



P8HA6e=12 L=136



P9HA6e=12 L=35

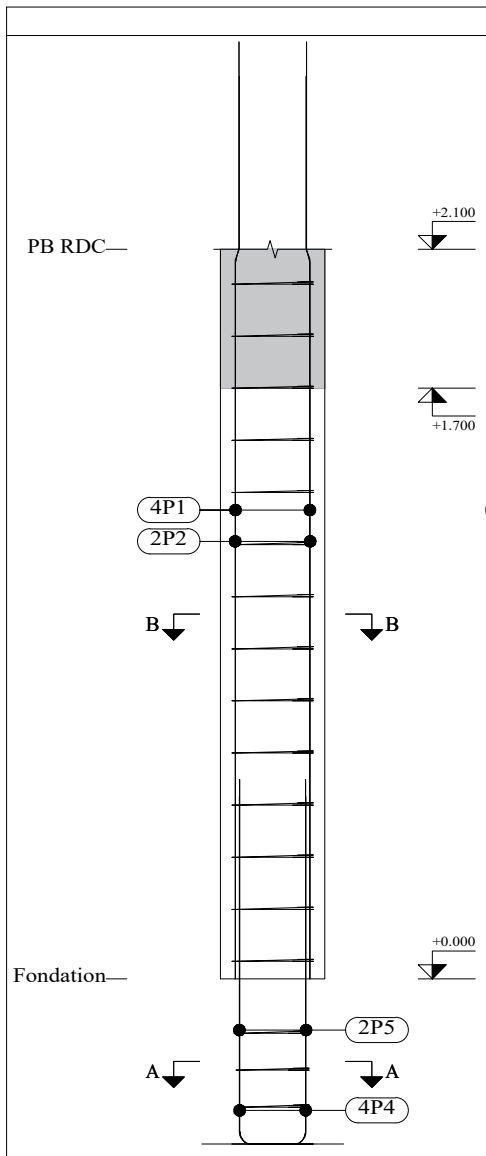
Acier: Fe E400 (38.2 kg). Ratio: 110.34 kg/m3		Niveau: PB RDC
Béton: B20 (0.32 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 3.36 m2	Enrobage géométrique: 3 cm	

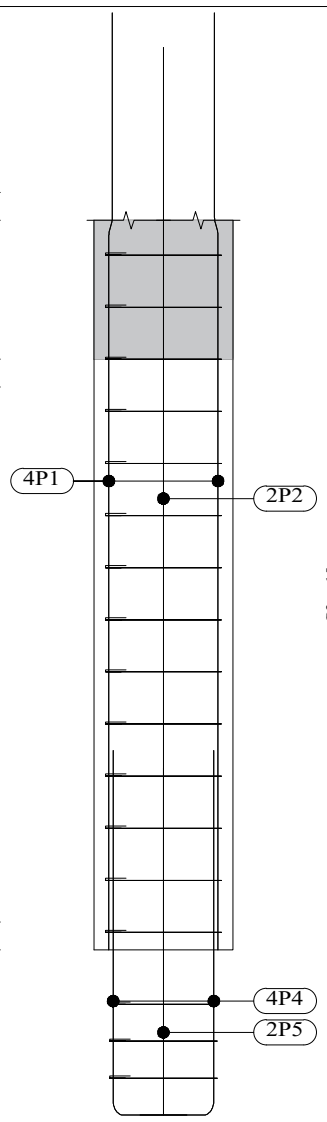
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P13	1	HA14	4	207 60	270	1080	13.1
	2	HA12	2	207 50	260	520	4.6
	3	HA6	14	24 6 34	127	1778	3.9
	4	HA14	4	30 105	135	540	6.5
	5	HA12	2	30 100	130	260	2.3
	6	HA6	3	22 6 32	117	351	0.8
						Total+10%:	34.3
						HA6:	5.1
						HA12:	7.6
						HA14:	21.6
						Total:	34.3

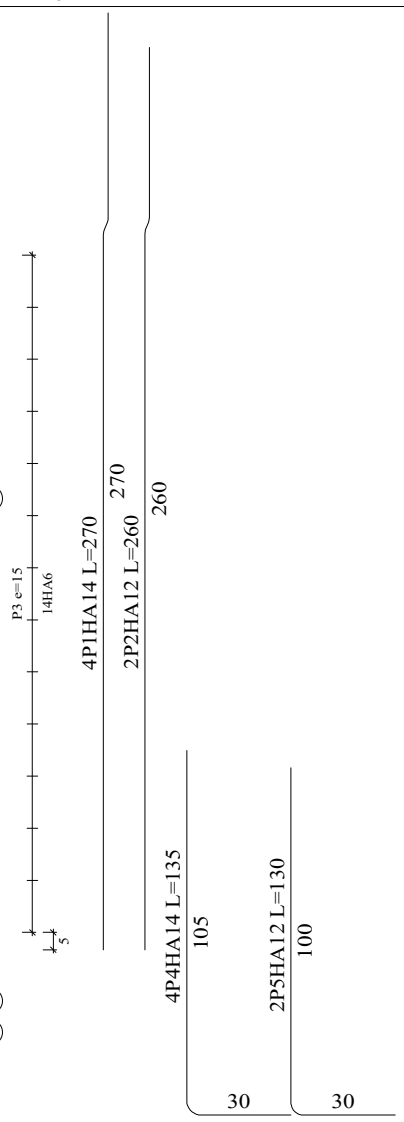
Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

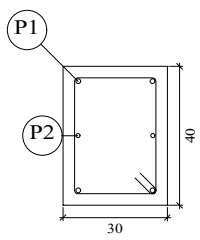
P13

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	270	1080
2	HA12	2	260	520
3	HA6	14	127	1778
4	HA14	4	135	540
5	HA12	2	130	260
6	HA6	3	117	351



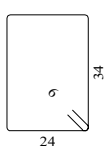




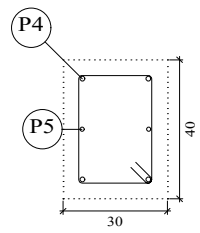


4P1HA14
2P2HA12

Section B-B

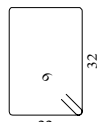


P3HA6e=15 L=127



4P4HA14
2P5HA12

Section A-A



P6HA6e=11 L=117

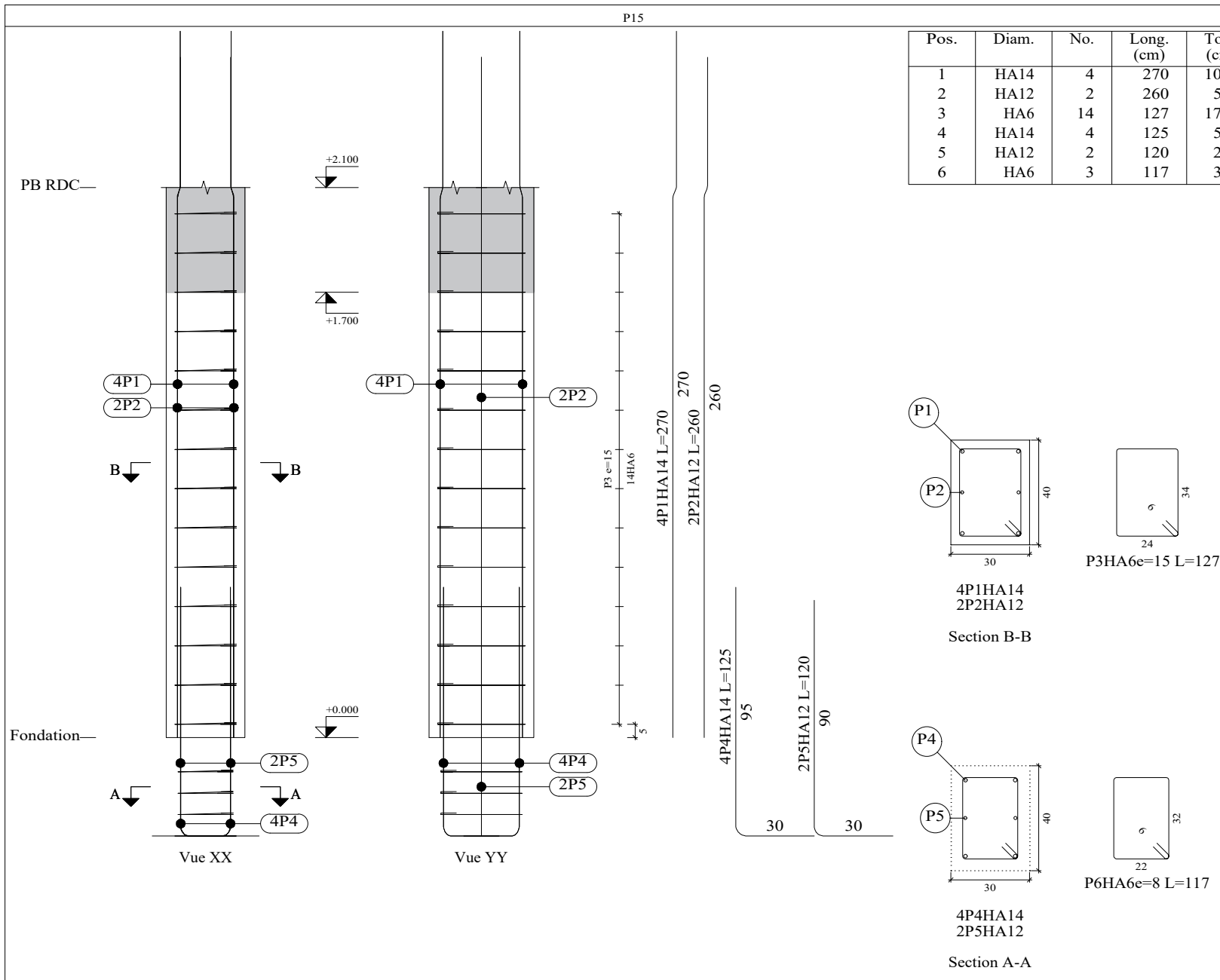
Acier: Fe E400 (23.8 kg). Ratio: 85.77 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P15	1	HA14	4	207 60	270	1080	13.1
	2	HA12	2	207 50	260	520	4.6
	3	HA6	14	24 6 34	127	1778	3.9
	4	HA14	4	30 95	125	500	6.0
	5	HA12	2	30 90	120	240	2.1
	6	HA6	3	22 6 32	117	351	0.8
						Total+10%:	33.6
						HA6:	5.2
						HA12:	7.4
						HA14:	21.0
						Total:	33.6

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P15

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	270	1080
2	HA12	2	260	520
3	HA6	14	127	1778
4	HA14	4	125	500
5	HA12	2	120	240
6	HA6	3	117	351



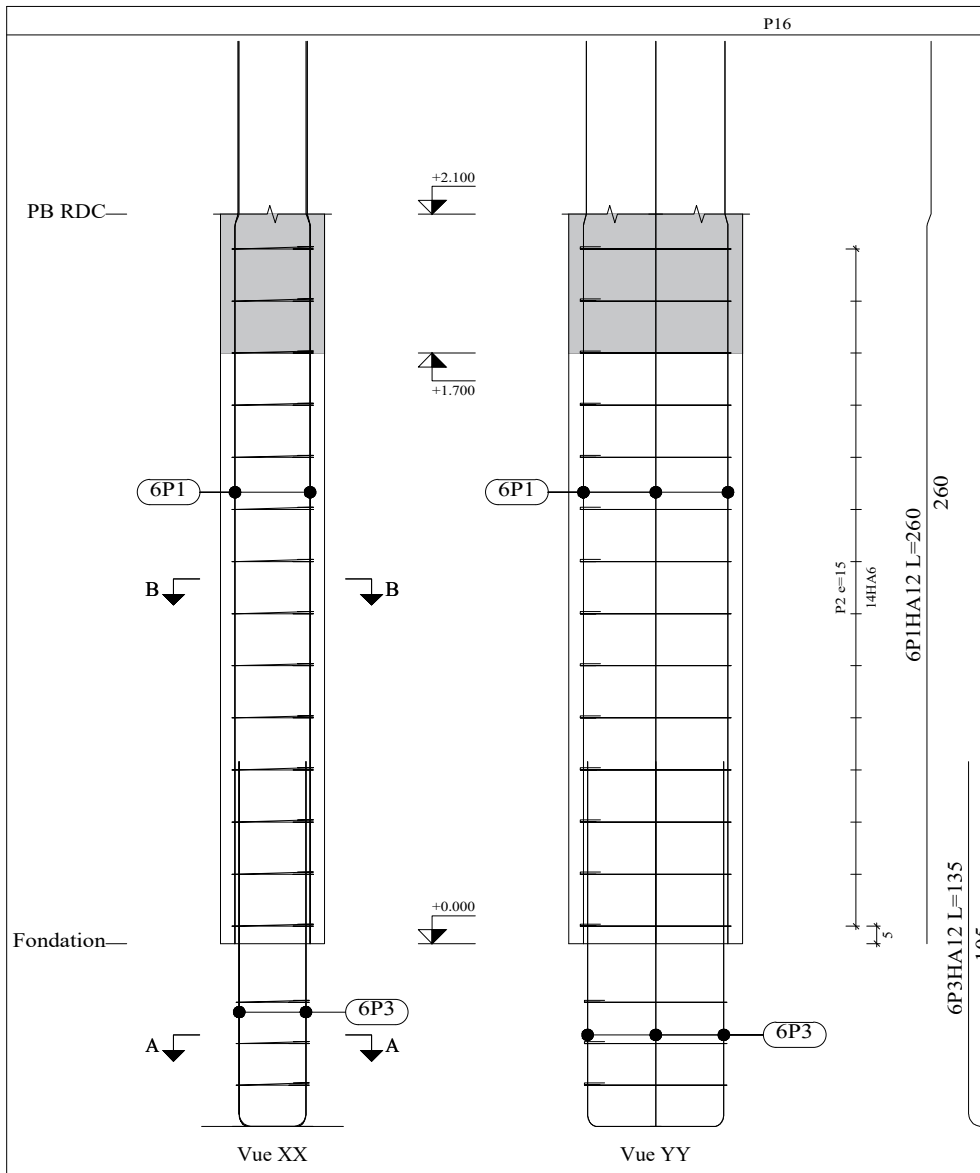
Acier: Fe E400 (23.8 kg). Ratio: 85.77 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

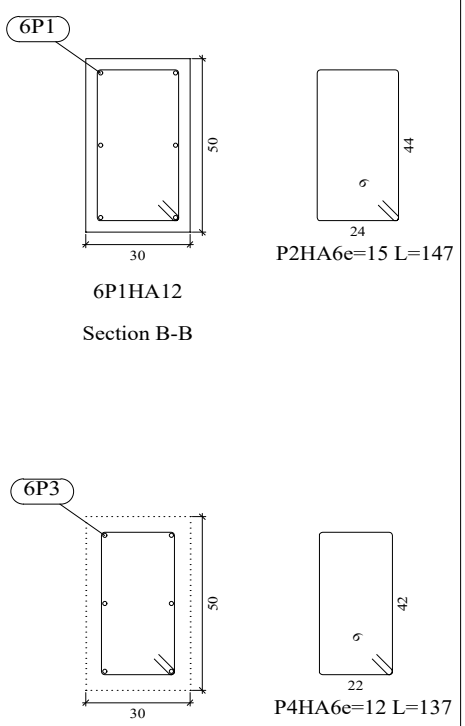
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P16	1	HA12	6	207 50	260	1560	13.9
	2	HA6	14	24 6 44	147	2058	4.6
	3	HA12	6	30 105	135	810	7.2
	4	HA6	3	22 6 42	137	411	0.9
						Total+10%:	29.3
						HA6:	6.1
						HA12:	23.2
						Total:	29.3

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P16

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	6	260	1560
2	HA6	14	147	2058
3	HA12	6	135	810
4	HA6	3	137	411





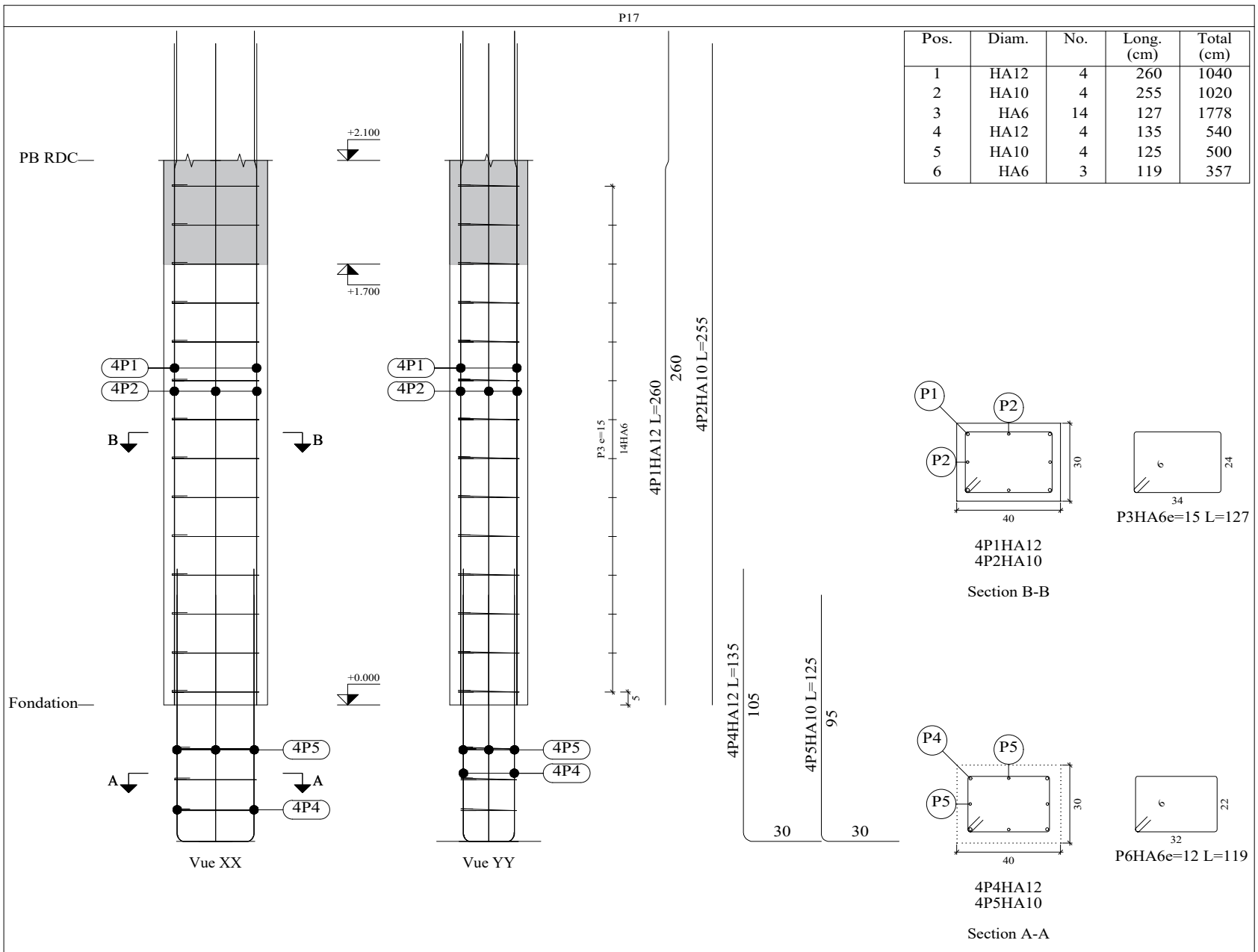
Acier: Fe E400 (20.3 kg). Ratio: 58.47 kg/m3		Niveau: PB RDC
Béton: B20 (0.32 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 3.36 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P17	1	HA12	4	207 50	260	1040	9.2
	2	HA10	4	255	255	1020	6.3
	3	HA6	14	24 6 34	127	1778	3.9
	4	HA12	4	30 105	135	540	4.8
	5	HA10	4	30 95	125	500	3.1
	6	HA6	3	22 6 32	119	357	0.8
						Total+10%:	30.9
						HA6:	5.2
						HA10:	10.3
						HA12:	15.4
						Total:	30.9

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P17

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA10	4	255	1020
3	HA6	14	127	1778
4	HA12	4	135	540
5	HA10	4	125	500
6	HA6	3	119	357



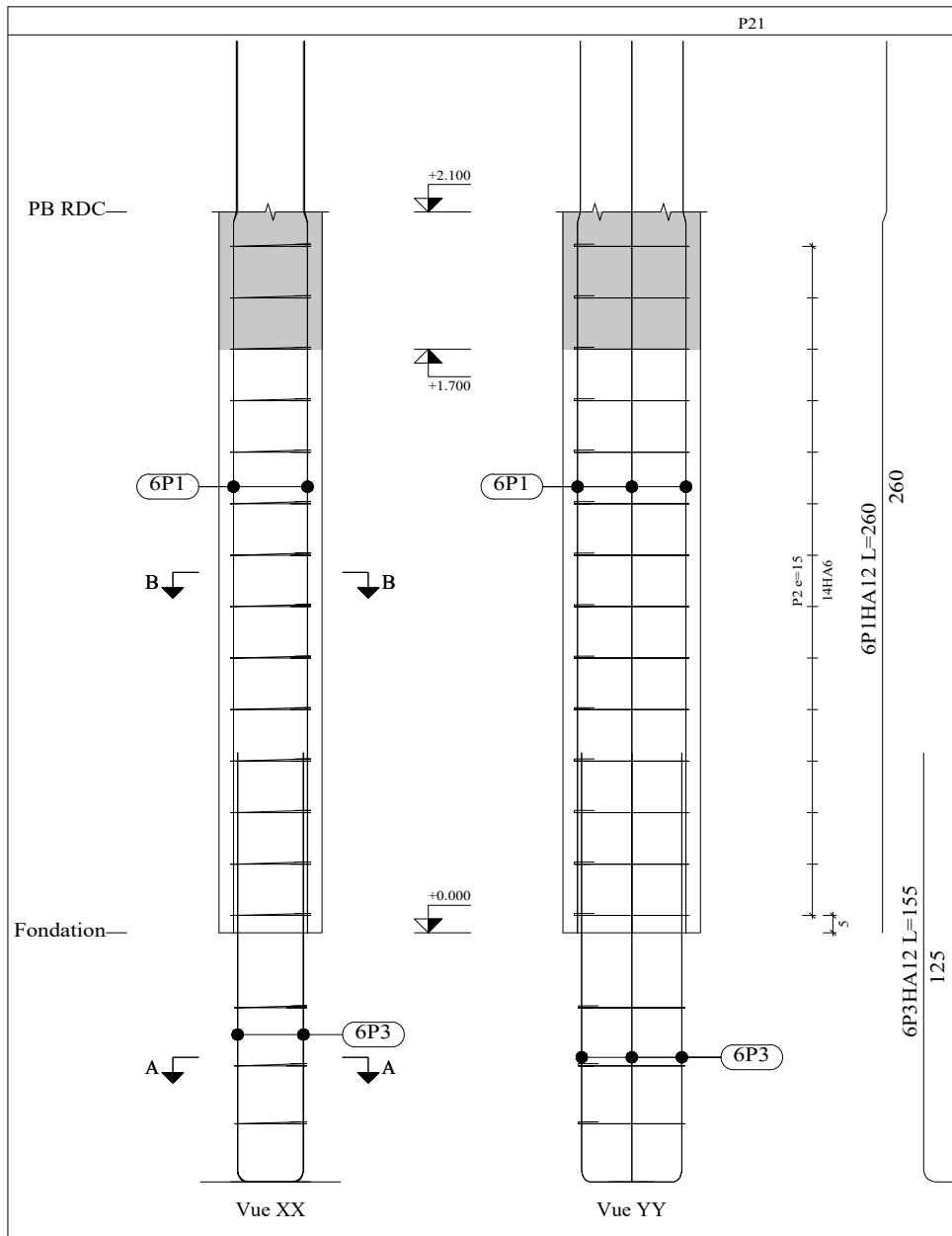
Acier: Fe E400 (21.4 kg). Ratio: 77.25 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P21	1	HA12	6	207 50	260	1560	13.9
	2	HA6	14	24 6 34	127	1778	3.9
	3	HA12	6	30 125	155	930	8.3
	4	HA6	3	22 6 32	117	351	0.8
						Total+10%:	29.6
						HA6:	5.2
						HA12:	24.4
						Total:	29.6

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

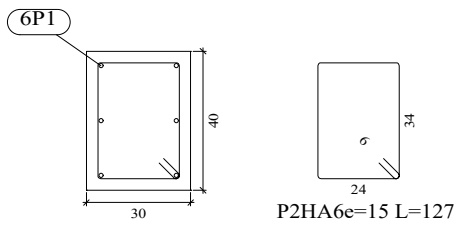
P21

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	6	260	1560
2	HA6	14	127	1778
3	HA12	6	155	930
4	HA6	3	117	351



Vue XX

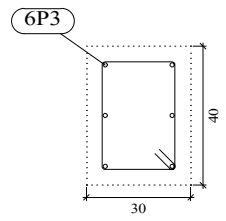
Vue YY



6P1HA12

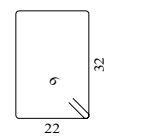
Section B-B

P2HA6e=15 L=127



6P3HA12

Section A-A



P4HA6e=17 L=117

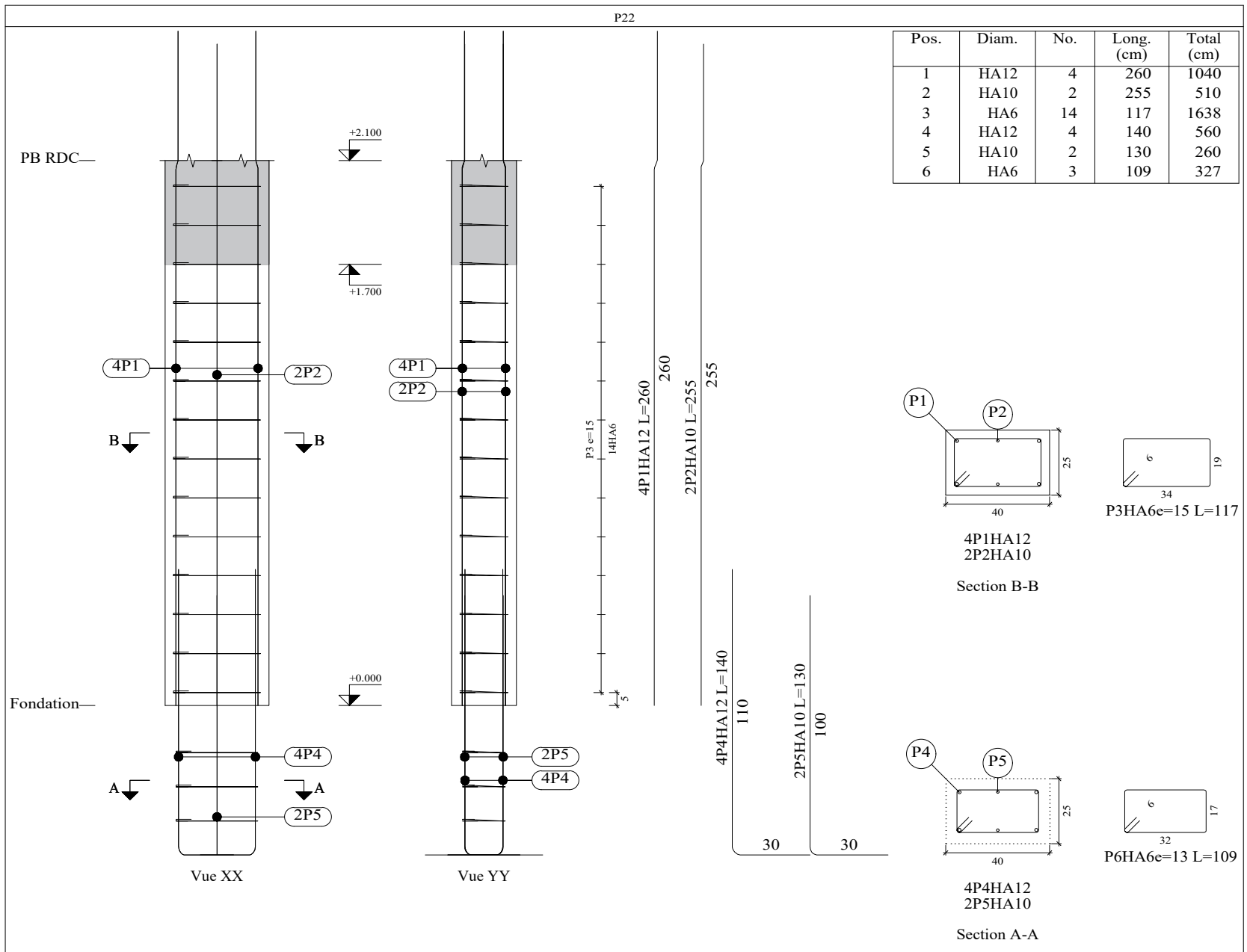
Acier: Fe E400 (19.6 kg). Ratio: 70.62 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P22	1	HA12	4	207 50	260	1040	9.2
	2	HA10	2	208 45	255	510	3.1
	3	HA6	14	19 6 34	117	1638	3.6
	4	HA12	4	30 110	140	560	5.0
	5	HA10	2	30 100	130	260	1.6
	6	HA6	3	17 6 32	109	327	0.7
						Total+10%:	25.5
						HA6:	4.7
						HA10:	5.2
						HA12:	15.6
						Total:	25.5

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P22

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA10	2	255	510
3	HA6	14	117	1638
4	HA12	4	140	560
5	HA10	2	130	260
6	HA6	3	109	327



4P1HA12
2P2HA10

Section B-B

P3HA6e=15 L=117

4P4HA12
2P5HA10

Section A-A

P6HA6e=13 L=109

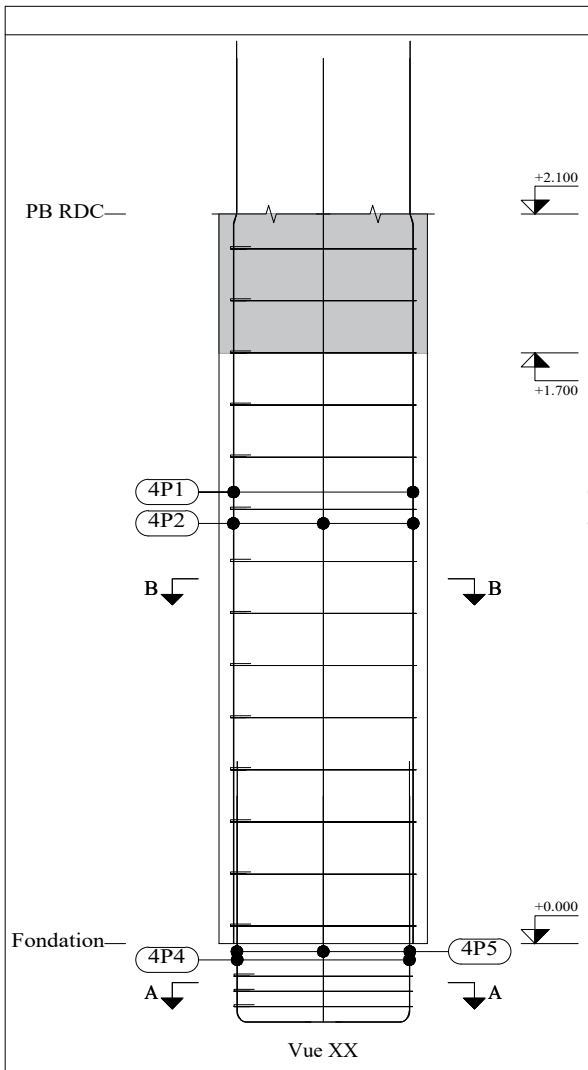
Acier: Fe E400 (17.6 kg). Ratio: 76.25 kg/m3		Niveau: PB RDC
Béton: B20 (0.21 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.73 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P27	1	HA12	4	207 50	260	1040	9.2
	2	HA10	4	208 45	255	1020	6.3
	3	HA6	14	24 6 54	167	2338	5.2
	4	HA12	4	30 75	105	420	3.7
	5	HA10	4	30 65	95	380	2.3
	6	HA6	3	22 6 52	159	477	1.1
						Total+10%:	30.6
						HA6:	6.9
						HA10:	9.5
						HA12:	14.2
						Total:	30.6

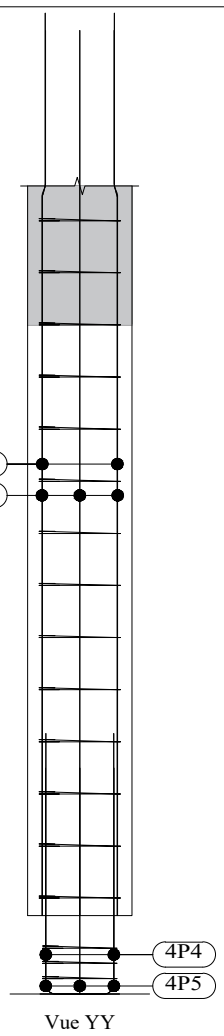
Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P27

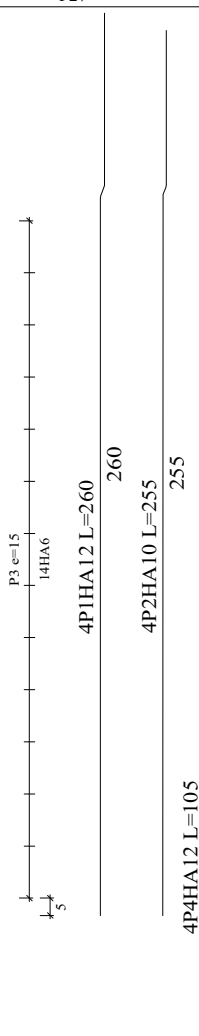
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA10	4	255	1020
3	HA6	14	167	2338
4	HA12	4	105	420
5	HA10	4	95	380
6	HA6	3	159	477

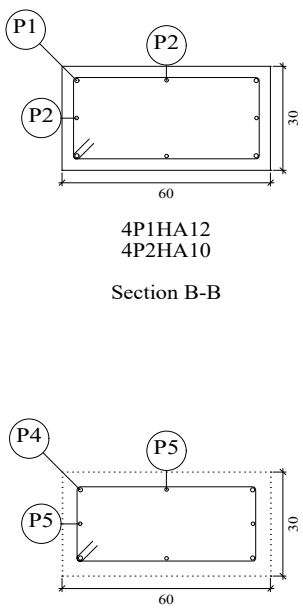


Vue XX



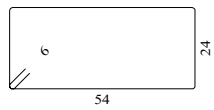
Vue YY





4P1HA12
4P2HA10

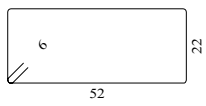
Section B-B



P3HA6e=15 L=167

4P4HA12
4P5HA10

Section A-A



P6HA6e=4 L=159

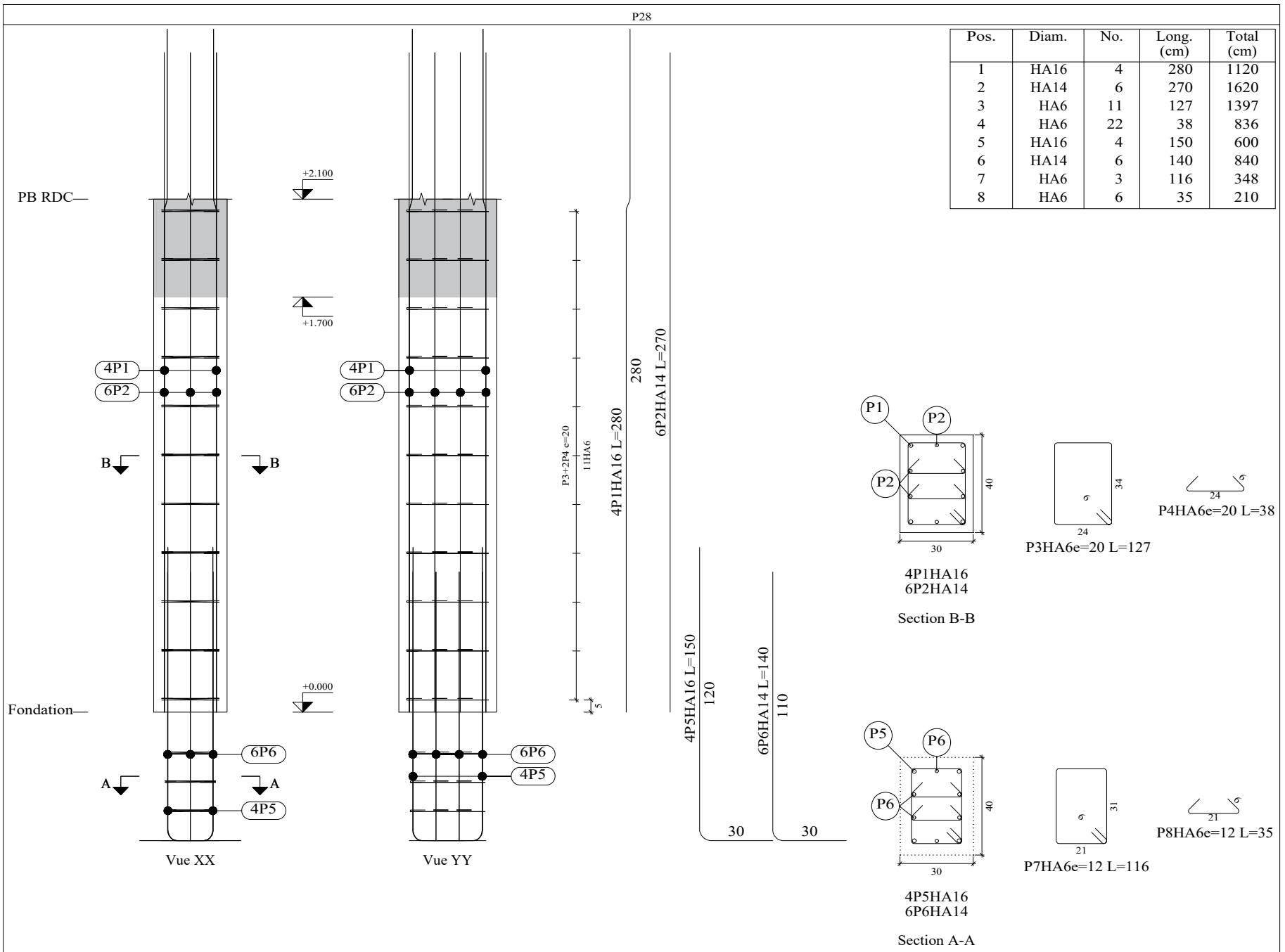
Acier: Fe E400 (22.8 kg). Ratio: 54.79 kg/m3		Niveau: PB RDC
Béton: B20 (0.38 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 3.78 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P28	1	HA16	4	206 70	280	1120	17.7
	2	HA14	6	270	270	1620	19.6
	3	HA6	11	24 6 34	127	1397	3.1
	4	HA6	22	6 24	38	836	1.9
	5	HA16	4	30 120	150	600	9.5
	6	HA14	6	30 110	140	840	10.2
	7	HA6	3	21 6 31	116	348	0.8
	8	HA6	6	6 21	35	210	0.5
					Total+10%:		69.6
					HA6:		6.9
					HA14:		32.8
					HA16:		29.9
					Total:		69.6

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P28

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA16	4	280	1120
2	HA14	6	270	1620
3	HA6	11	127	1397
4	HA6	22	38	836
5	HA16	4	150	600
6	HA14	6	140	840
7	HA6	3	116	348
8	HA6	6	35	210



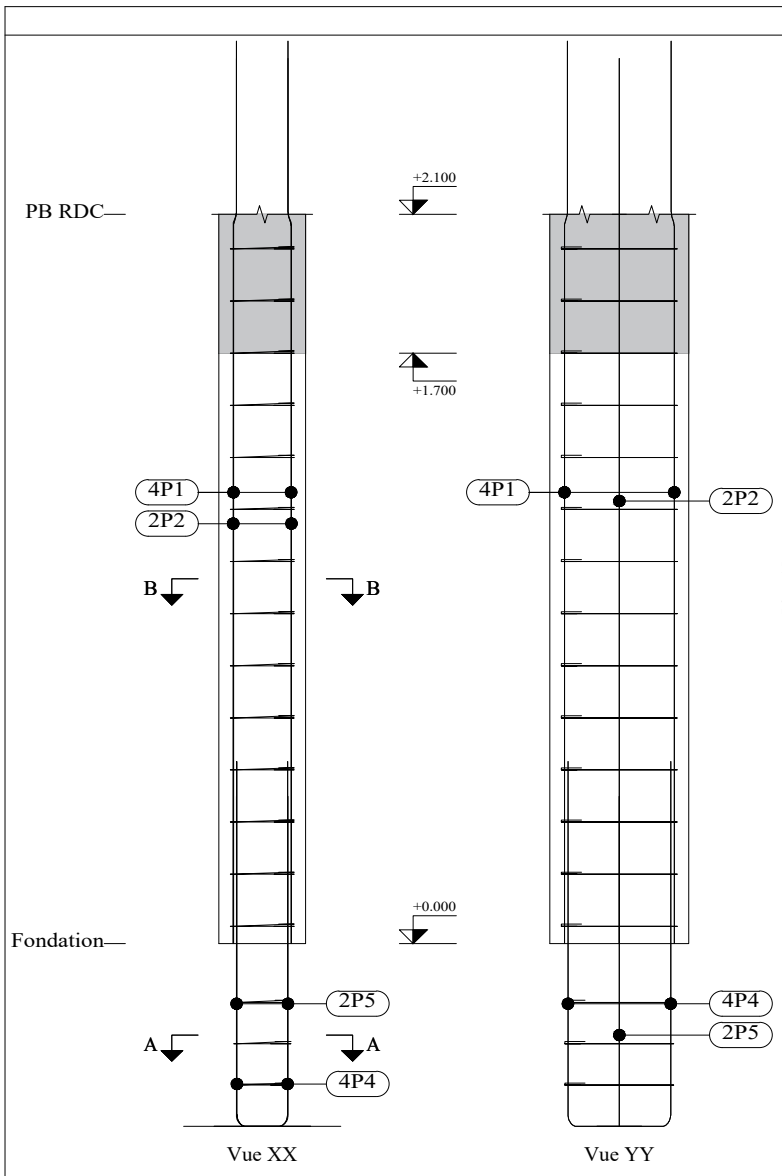
Acier: Fe E400 (46.4 kg). Ratio: 167.50 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P29	1	HA12	4	207 50	260	1040	9.2
	2	HA10	2	208 45	255	510	3.1
	3	HA6	14	19 6 34	117	1638	3.6
	4	HA12	4	30 105	135	540	4.8
	5	HA10	2	30 95	125	250	1.5
	6	HA6	3	17 6 32	109	327	0.7
						Total+10%:	25.2
						HA6:	4.7
						HA10:	5.1
						HA12:	15.4
						Total:	25.2

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

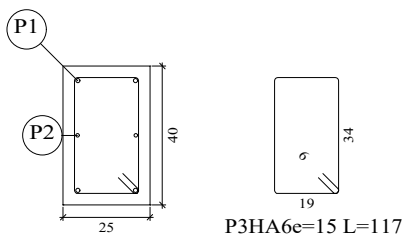
P29

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA10	2	255	510
3	HA6	14	117	1638
4	HA12	4	135	540
5	HA10	2	125	250
6	HA6	3	109	327



Vue XX

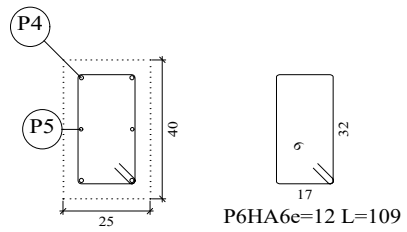
Vue YY



P3HA6e=15 L=117

4P1HA12
2P2HA10

Section B-B



P6HA6e=12 L=109

4P4HA12
2P5HA10

Section A-A

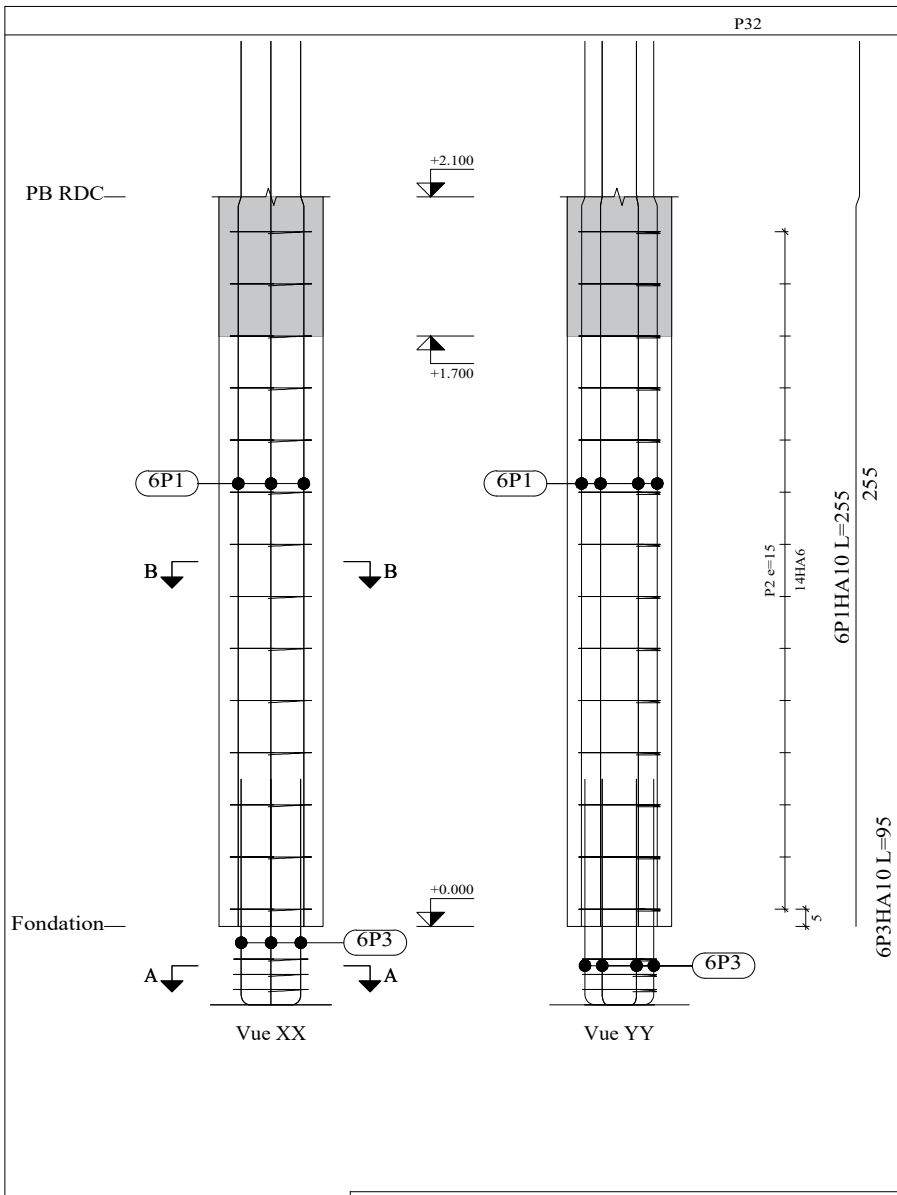
Acier: Fe E400 (17.6 kg). Ratio: 76.25 kg/m3		Niveau: PB RDC
Béton: B20 (0.21 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.73 m2	Enrobage géométrique: 3 cm	

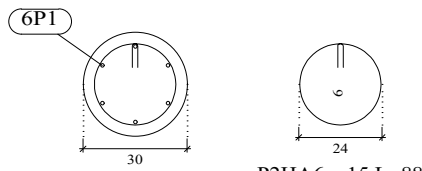
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P32	1	HA10	6	208 45	255	1530	9.4
	2	HA6	14	24 6	88	1232	2.7
	3	HA10	6	30 65	95	570	3.5
	4	HA6	3	22 6	82	246	0.5
					Total+10%:		17.7
					HA6:		3.5
					HA10:		14.2
					Total:		17.7

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P32

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA10	6	255	1530
2	HA6	14	88	1232
3	HA10	6	95	570
4	HA6	3	82	246

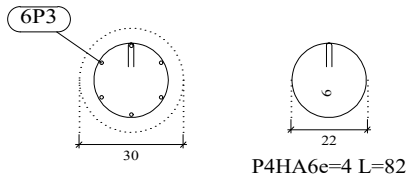




6P1HA10

Section B-B

P2HA6e=15 L=88



6P3HA10

Section A-A

P4HA6e=4 L=82

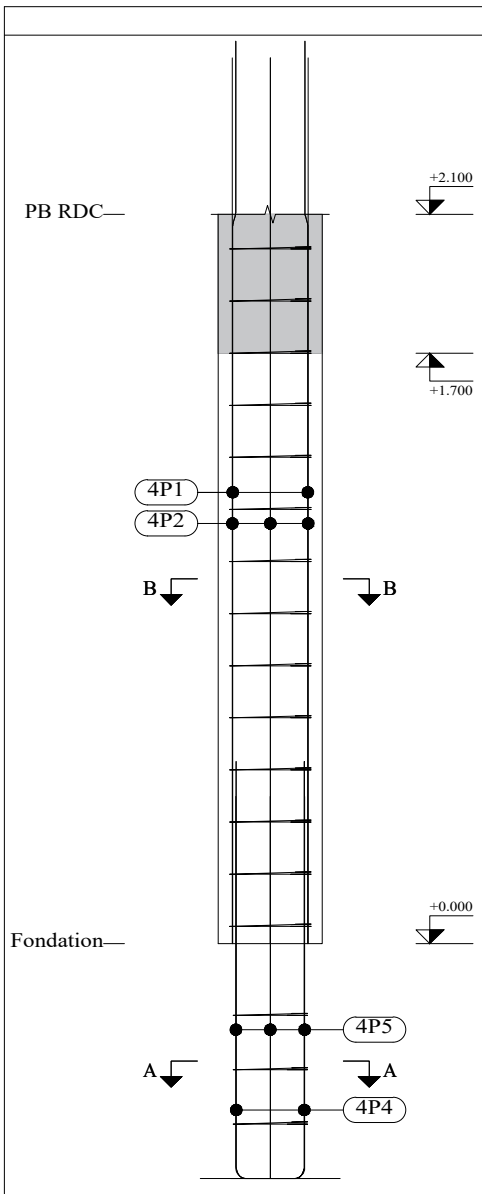
Acier: Fe E400 (13.4 kg). Ratio: 81.97 kg/m3		Niveau: PB RDC
Béton: B20 (0.15 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 1.98 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P33	1	HA12	4	207 50	260	1040	9.2
	2	HA10	4	255	255	1020	6.3
	3	HA6	14	24 6 34	127	1778	3.9
	4	HA12	4	30 120	150	600	5.3
	5	HA10	4	30 110	140	560	3.5
	6	HA6	3	22 6 32	119	357	0.8
						Total+10%:	31.9
						HA6:	5.2
						HA10:	10.7
						HA12:	16.0
						Total:	31.9

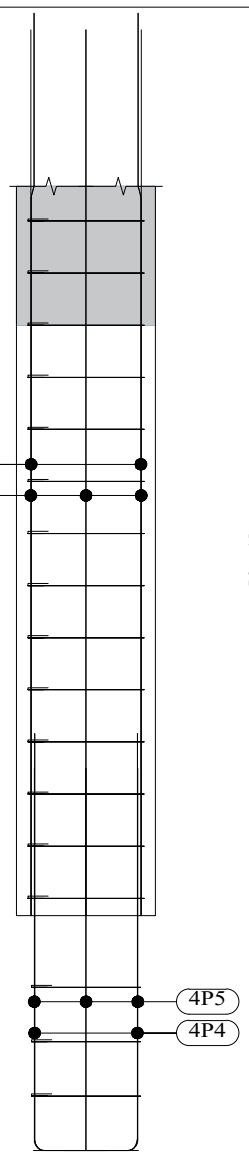
Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P33

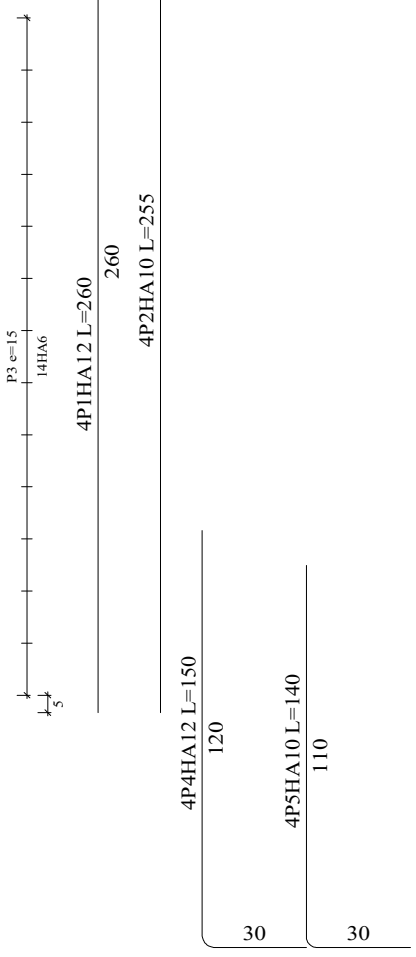
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	260	1040
2	HA10	4	255	1020
3	HA6	14	127	1778
4	HA12	4	150	600
5	HA10	4	140	560
6	HA6	3	119	357

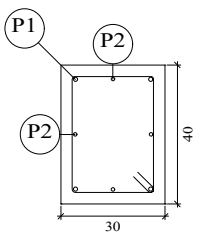


Vue XX



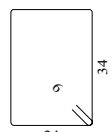
Vue YY



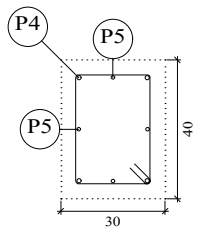


4P1HA12
4P2HA10

Section B-B

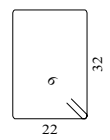


P3HA6e=15 L=127



4P4HA12
4P5HA10

Section A-A



P6HA6e=16 L=119

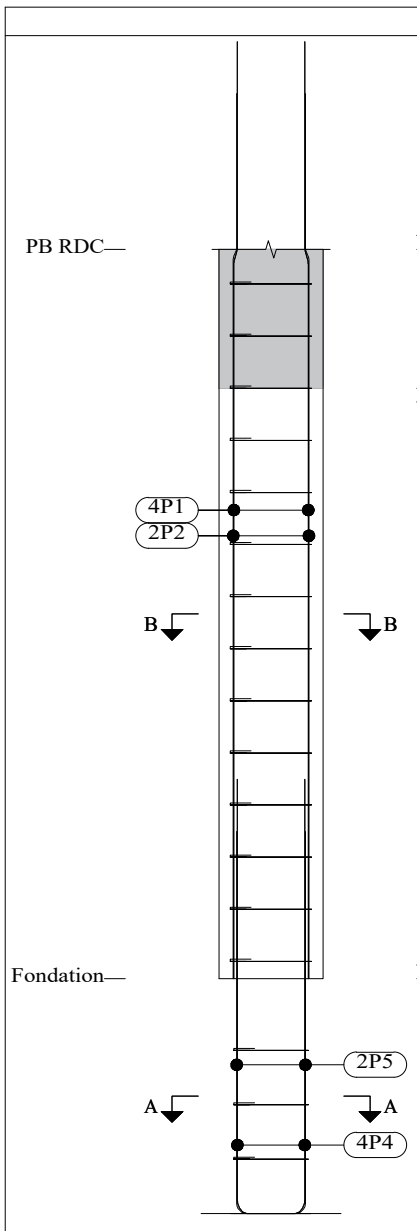
Acier: Fe E400 (21.4 kg). Ratio: 77.25 kg/m3		Niveau: PB RDC
Béton: B20 (0.25 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.94 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P2	1	HA14	4	207 60	270	1080	13.1
	2	HA10	2	208 45	255	510	3.1
	3	HA6	14	24 6 24	107	1498	3.3
	4	HA14	4	30 125	155	620	7.5
	5	HA10	2	30 110	140	280	1.7
	6	HA6	3	22 6 22	99	297	0.7
					Total+10%:		32.3
					HA6:		4.4
					HA10:		5.2
					HA14:		22.7
					Total:		32.3

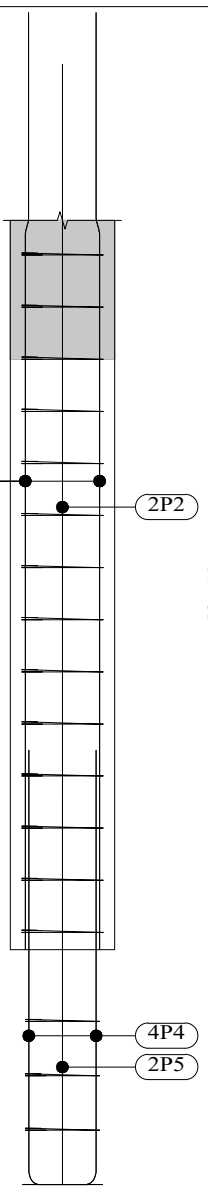
Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P2

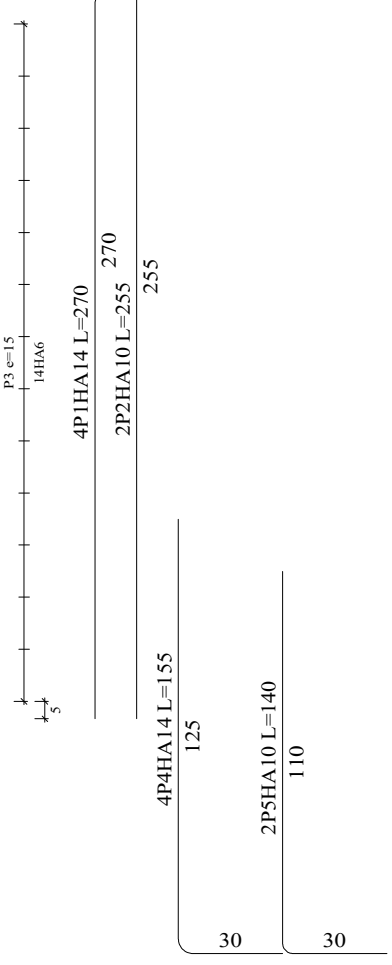
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	270	1080
2	HA10	2	255	510
3	HA6	14	107	1498
4	HA14	4	155	620
5	HA10	2	140	280
6	HA6	3	99	297

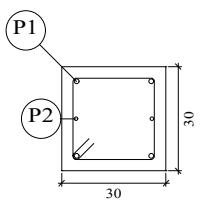


Vue XX



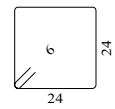
Vue YY



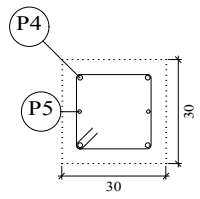


4P1HA14
2P2HA10

Section B-B

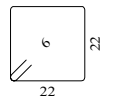


P3HA6e=15 L=107



4P4HA14
2P5HA10

Section A-A



P6HA6e=16 L=99

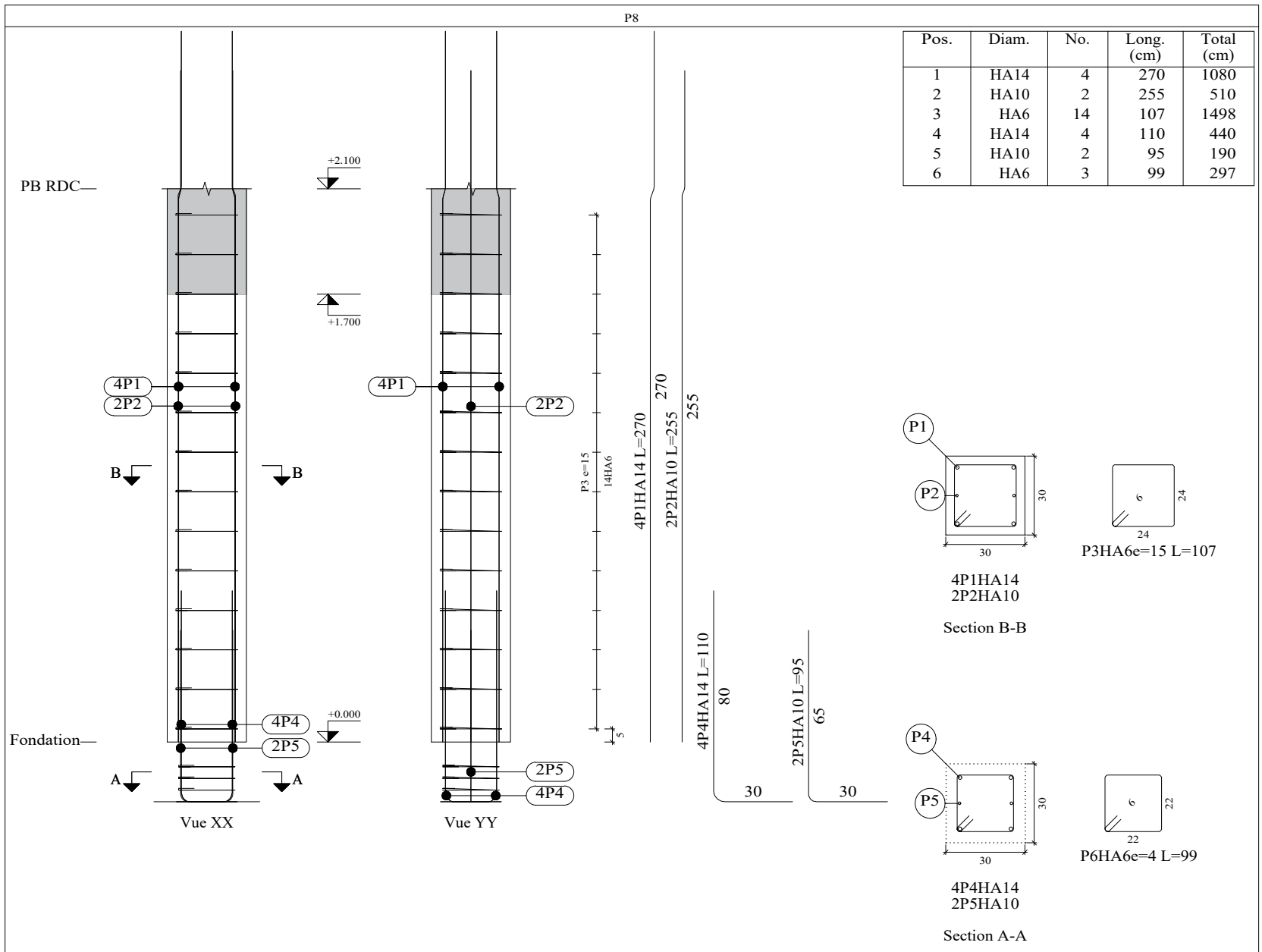
Acier: Fe E400 (21.5 kg). Ratio: 103.28 kg/m3	Niveau: PB RDC
Béton: B20 (0.19 m3) / Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.52 m2 / Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P8	1	HA14	4	207 60	270	1080	13.1
	2	HA10	2	208 45	255	510	3.1
	3	HA6	14	24 6 24	107	1498	3.3
	4	HA14	4	30 80	110	440	5.3
	5	HA10	2	30 65	95	190	1.2
	6	HA6	3	22 6 22	99	297	0.7
						Total+10%:	29.4
						HA6:	4.4
						HA10:	4.8
						HA14:	20.2
						Total:	29.4

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P8

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	270	1080
2	HA10	2	255	510
3	HA6	14	107	1498
4	HA14	4	110	440
5	HA10	2	95	190
6	HA6	3	99	297



4P1HA14
2P2HA10

Section B-B

P3HA6e=15 L=107

4P4HA14
2P5HA10

Section A-A

P6HA6e=4 L=99

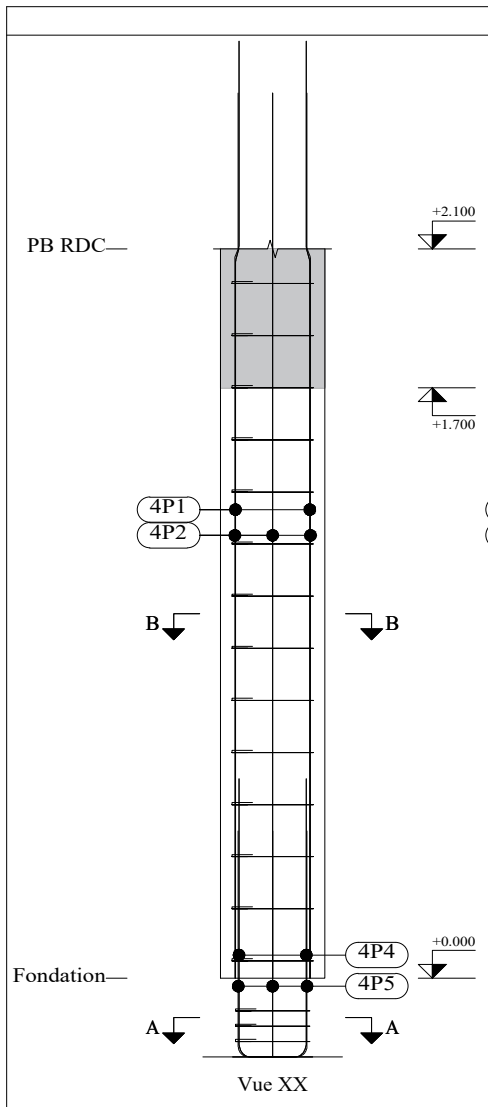
Acier: Fe E400 (21.5 kg). Ratio: 103.28 kg/m3	Niveau: PB RDC
Béton: B20 (0.19 m3) — Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.52 m2 — Enrobage géométrique: 3 cm	

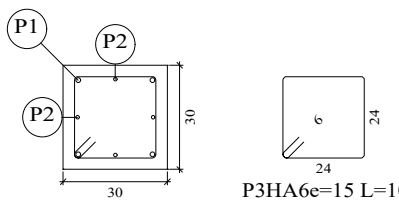
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P7	1	HA14	4	207 / 60	270	1080	13.1
	2	HA10	4	208 / 45	255	1020	6.3
	3	HA6	14	24 × 24, 6	107	1498	3.3
	4	HA14	4	30 / 80	110	440	5.3
	5	HA10	4	30 / 65	95	380	2.3
	6	HA6	3	22 × 22, 6	99	297	0.7
						Total+10%:	34.1
						HA6:	4.4
						HA10:	9.5
						HA14:	20.2
						Total:	34.1

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P7

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	270	1080
2	HA10	4	255	1020
3	HA6	14	107	1498
4	HA14	4	110	440
5	HA10	4	95	380
6	HA6	3	99	297

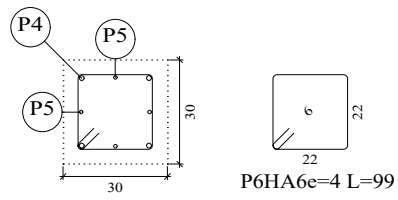




4P1HA14
4P2HA10

Section B-B

P3HA6e=15 L=107



4P4HA14
4P5HA10

Section A-A

P6HA6e=4 L=99

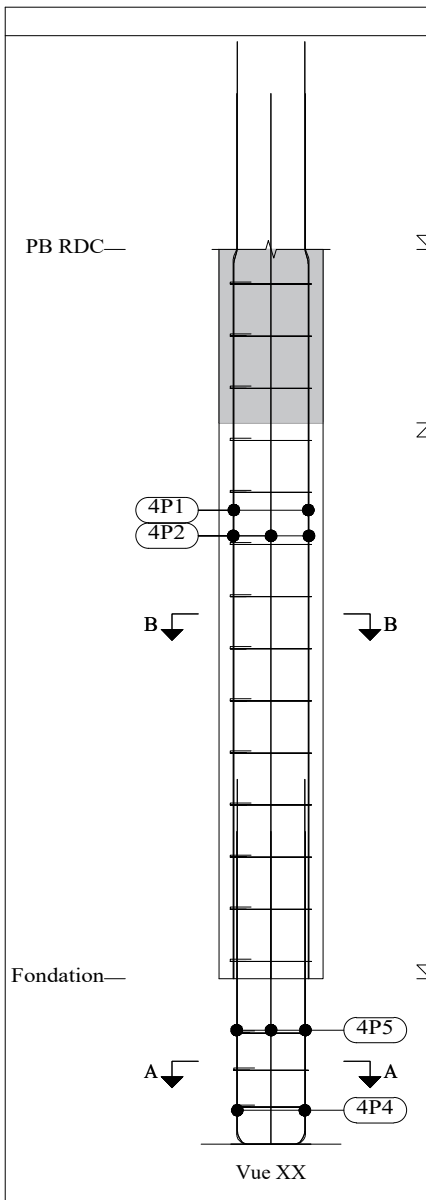
Acier: Fe E400 (24.9 kg). Ratio: 119.92 kg/m3	Niveau: PB RDC
Béton: B20 (0.19 m3) / Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.52 m2 / Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P12	1	HA14	4	207 60	270	1080	13.1
	2	HA10	4	208 45	255	1020	6.3
	3	HA6	14	24 6 24	107	1498	3.3
	4	HA14	4	30 105	135	540	6.5
	5	HA10	4	30 90	120	480	3.0
	6	HA6	3	22 6 22	99	297	0.7
					Total+10%:		36.2
					HA6:		4.4
					HA10:		10.2
					HA14:		21.6
					Total:		36.2

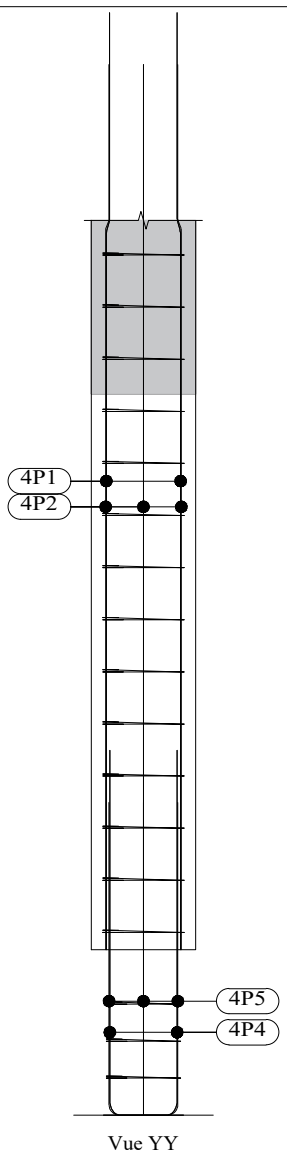
Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P12

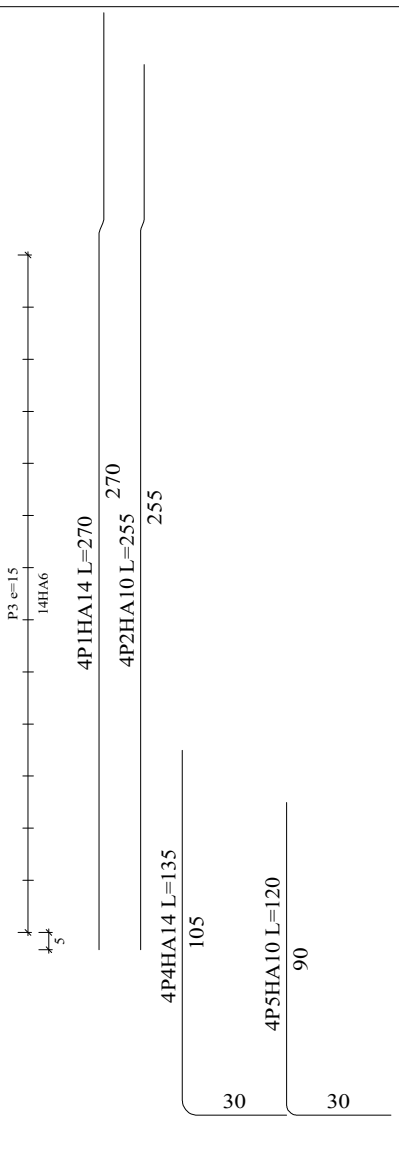
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	270	1080
2	HA10	4	255	1020
3	HA6	14	107	1498
4	HA14	4	135	540
5	HA10	4	120	480
6	HA6	3	99	297

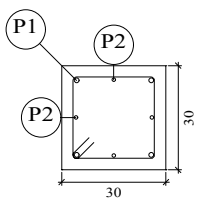


Vue XX



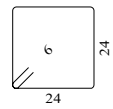
Vue YY



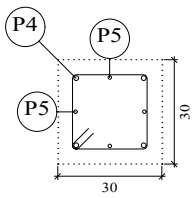


4P1HA14
4P2HA10

Section B-B

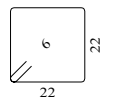


P3HA6e=15 L=107



4P4HA14
4P5HA10

Section A-A



P6HA6e=11 L=99

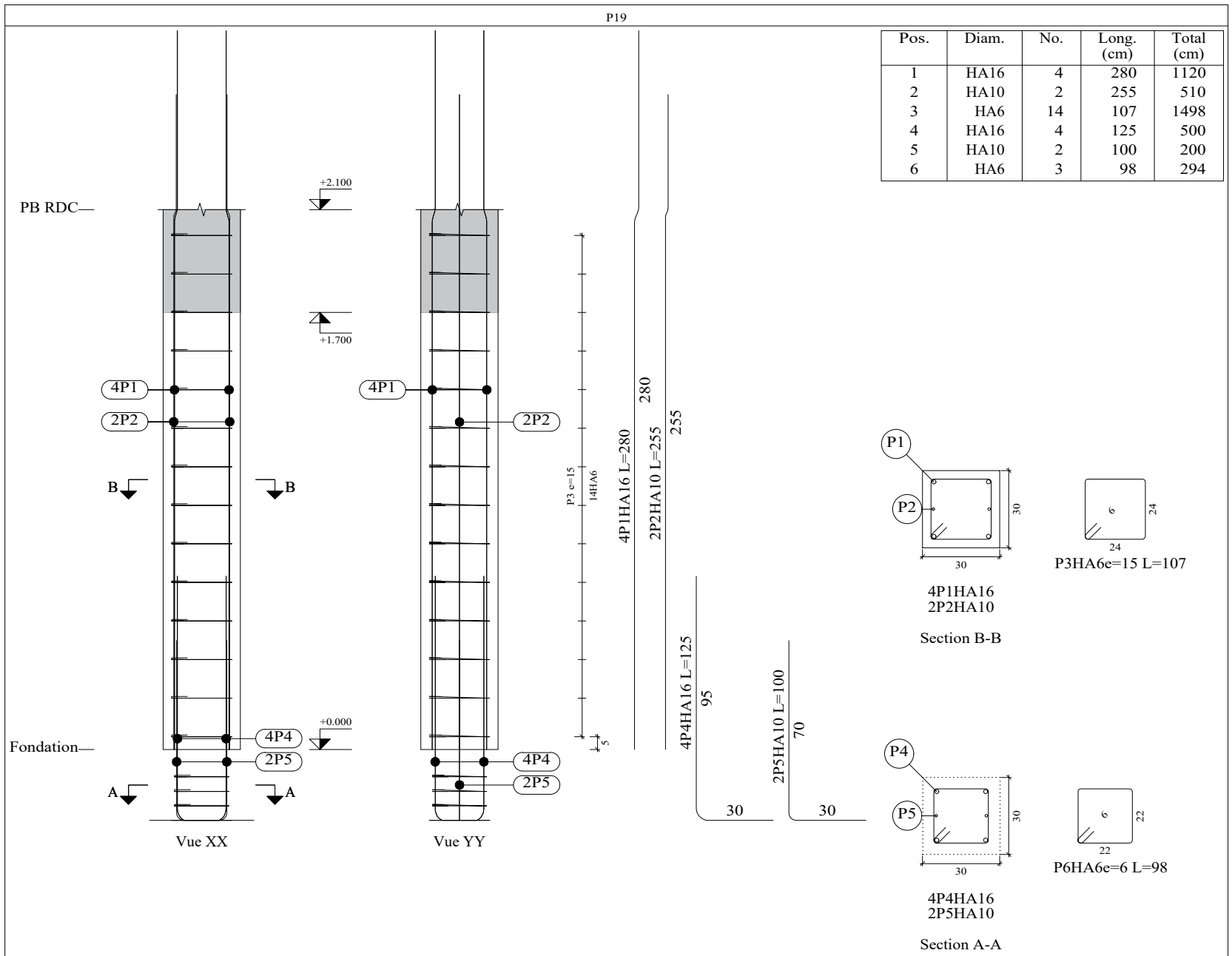
Acier: Fe E400 (24.9 kg). Ratio: 119.92 kg/m3		Niveau: PB RDC
Béton: B20 (0.19 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.52 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P19	1	HA16	4	206 / 70	280	1120	17.7
	2	HA10	2	208 / 45	255	510	3.1
	3	HA6	14	24 / 6 / 24	107	1498	3.3
	4	HA16	4	30 / 95	125	500	7.9
	5	HA10	2	30 / 70	100	200	1.2
	6	HA6	3	22 / 6 / 22	98	294	0.7
						Total+10%:	37.3
						HA6:	4.4
						HA10:	4.7
						HA16:	28.2
						Total:	37.3

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P19

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA16	4	280	1120
2	HA10	2	255	510
3	HA6	14	107	1498
4	HA16	4	125	500
5	HA10	2	100	200
6	HA6	3	98	294



4P1HA16
2P2HA10

Section B-B

P3HA6e=15 L=107

4P4HA16
2P5HA10

Section A-A

P6HA6e=6 L=98

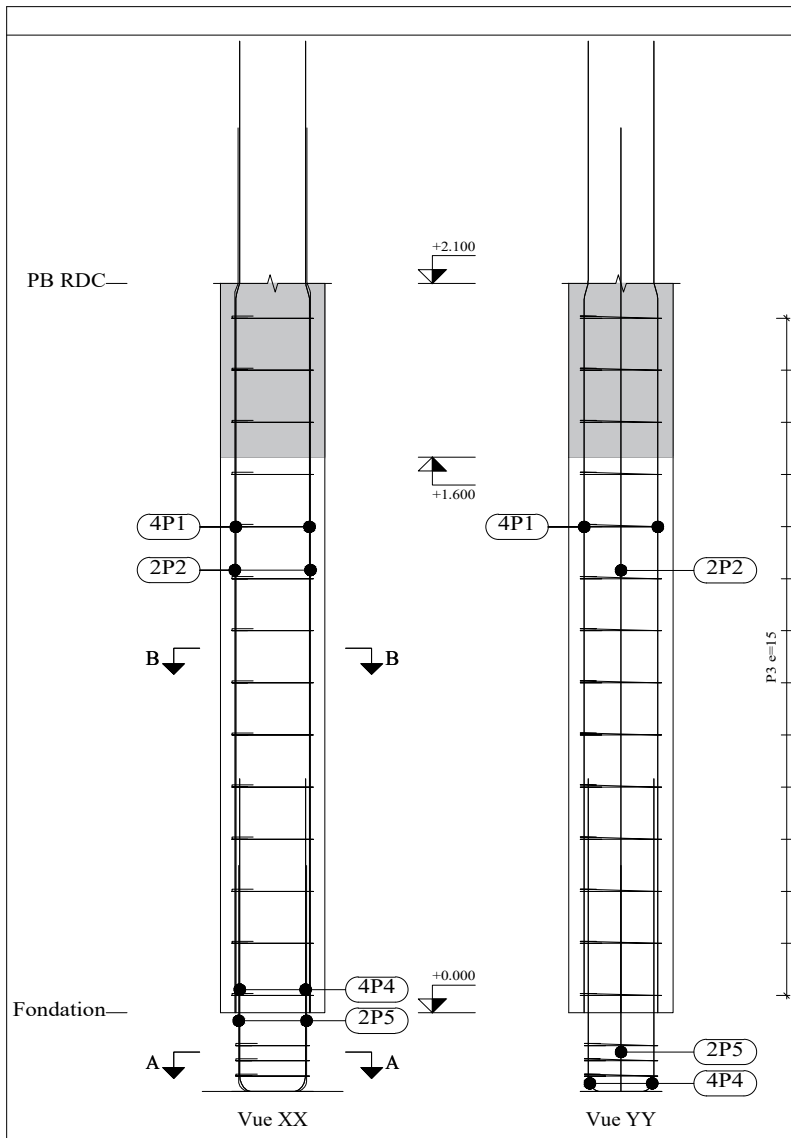
Acier: Fe E400 (26.6 kg). Ratio: 127.76 kg/m3		Niveau: PB RDC
Béton: B20 (0.19 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.52 m2	Enrobage géométrique: 3 cm	

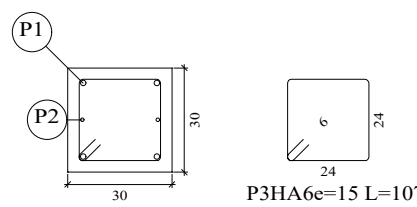
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P20	1	HA16	4	206 / 70	280	1120	17.7
	2	HA10	2	208 / 45	255	510	3.1
	3	HA6	14	24 × 24, 6	107	1498	3.3
	4	HA16	4	30 / 90	120	480	7.6
	5	HA10	2	30 / 65	95	190	1.2
	6	HA6	3	22 × 22, 6	98	294	0.7
						Total+10%:	37.0
						HA6:	4.4
						HA10:	4.8
						HA16:	27.8
						Total:	37.0

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P20

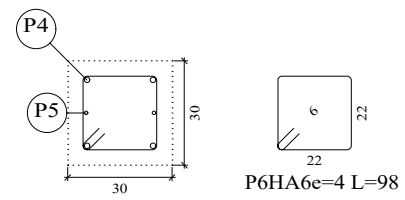
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA16	4	280	1120
2	HA10	2	255	510
3	HA6	14	107	1498
4	HA16	4	120	480
5	HA10	2	95	190
6	HA6	3	98	294





Section B-B

P3HA6e=15 L=107



Section A-A

P6HA6e=4 L=98

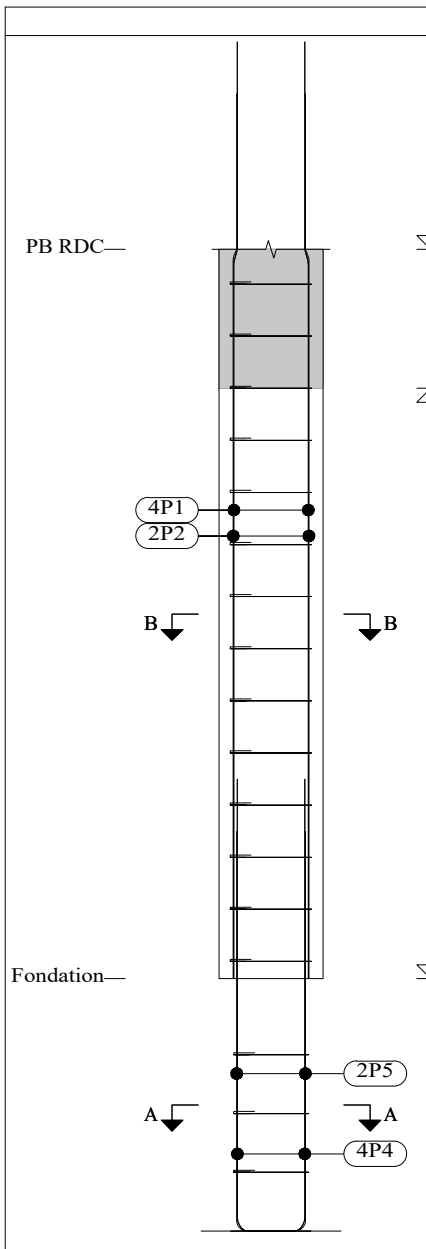
Acier: Fe E400 (26.6 kg). Ratio: 127.76 kg/m3		Niveau: PB RDC
Béton: B20 (0.19 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.52 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P26	1	HA14	4	207 / 60	270	1080	13.1
	2	HA10	2	208 / 45	255	510	3.1
	3	HA6	14	24 / 6 / 24	107	1498	3.3
	4	HA14	4	30 / 130	160	640	7.7
	5	HA10	2	30 / 115	145	290	1.8
	6	HA6	3	22 / 6 / 22	99	297	0.7
					Total+10%:		32.7
					HA6:		4.4
					HA10:		5.4
					HA14:		22.9
					Total:		32.7

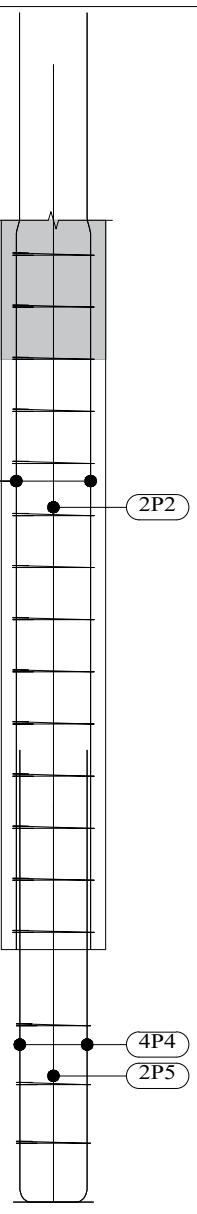
Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P26

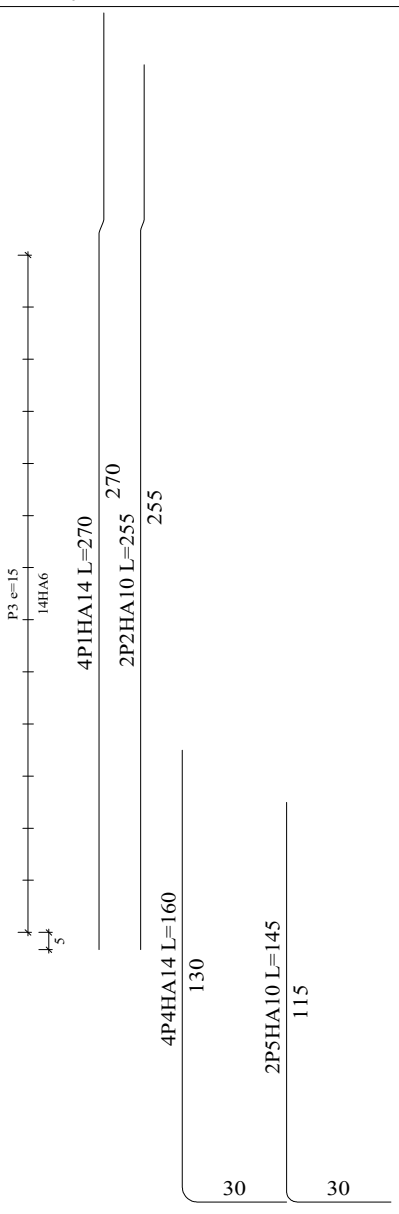
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	270	1080
2	HA10	2	255	510
3	HA6	14	107	1498
4	HA14	4	160	640
5	HA10	2	145	290
6	HA6	3	99	297

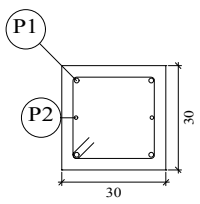


Vue XX



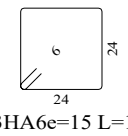
Vue YY



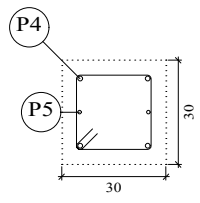


4P1HA14
2P2HA10

Section B-B

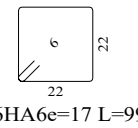


P3HA6e=15 L=107



4P4HA14
2P5HA10

Section A-A



P6HA6e=17 L=99

Acier: Fe E400 (21.5 kg). Ratio: 103.28 kg/m3		Niveau: PB RDC
Béton: B20 (0.19 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.52 m2	Enrobage géométrique: 3 cm	

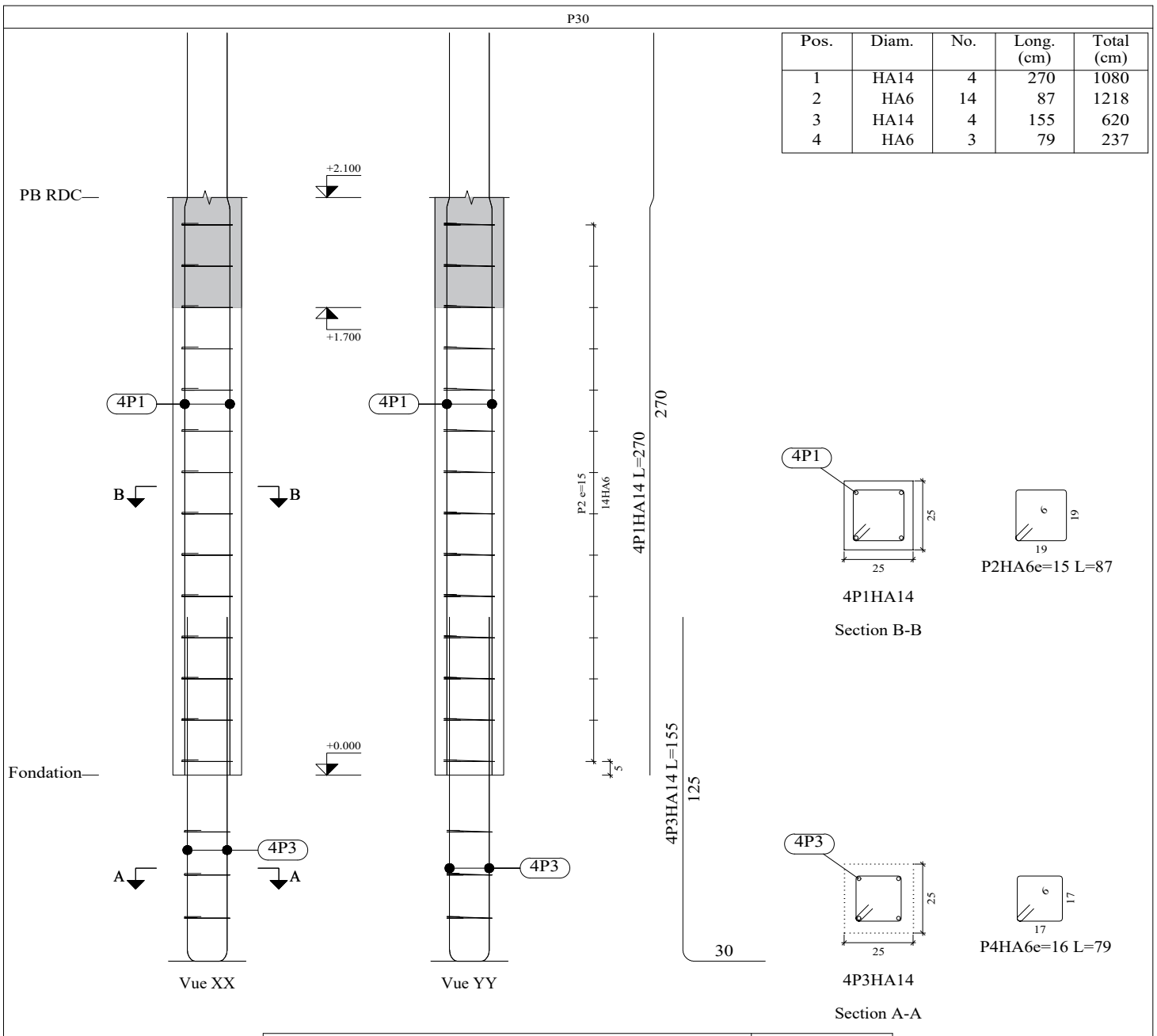
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P30	1	HA14	4	207 60	270	1080	13.1
	2	HA6	14	19 19 6	87	1218	2.7
	3	HA14	4	30 125	155	620	7.5
	4	HA6	3	17 17 6	79	237	0.5
					Total+10%:		26.2
					HA6:		3.5
					HA14:		22.7
					Total:		26.2

Niveau: PB RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Récapitulatif Acier Poteaux		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	618.4	151	
	HA10	212.9	144	
	HA12	302.6	296	
	HA14	194.6	259	
	HA16	66.4	115	965

P30

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	270	1080
2	HA6	14	87	1218
3	HA14	4	155	620
4	HA6	3	79	237



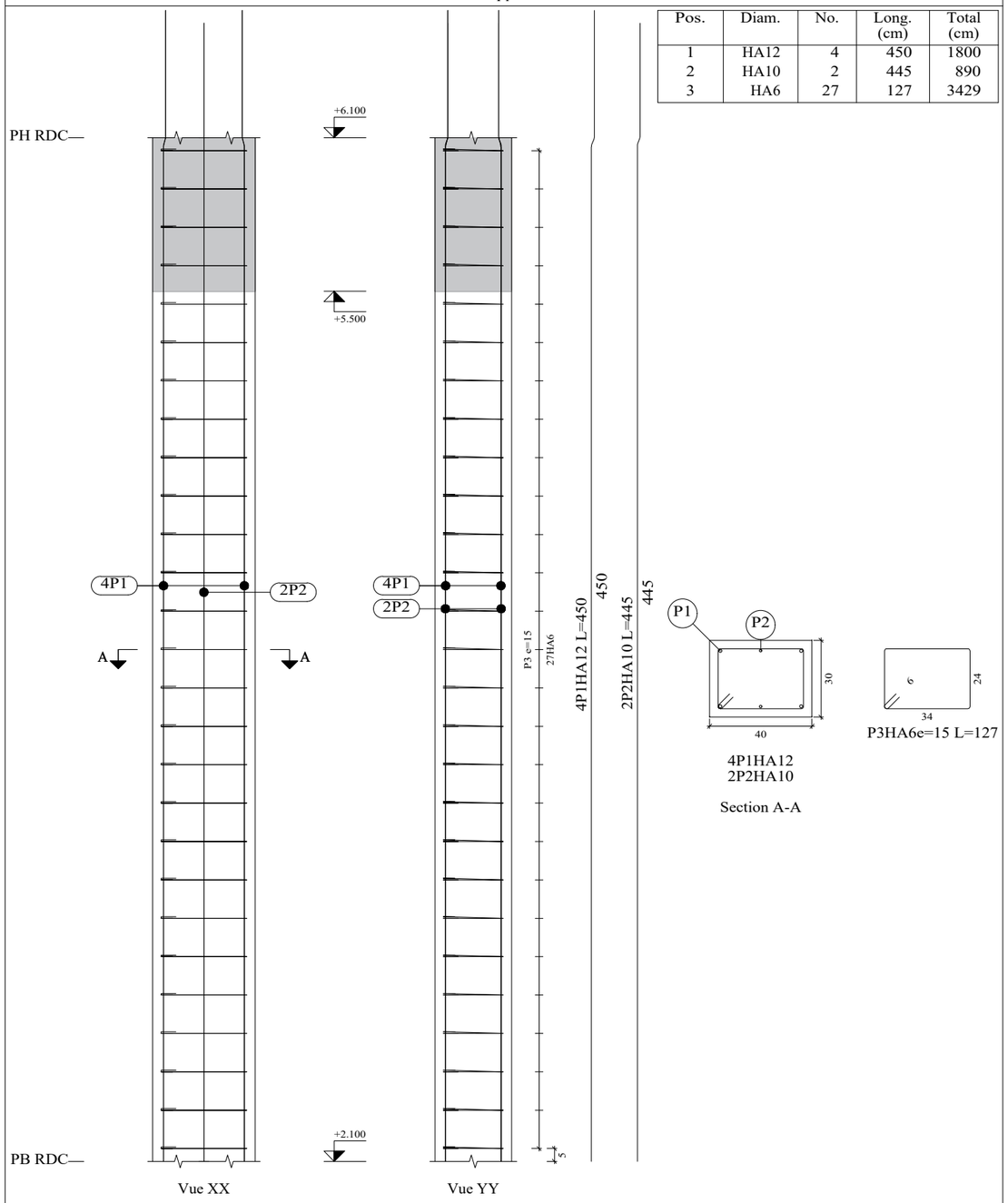
Acier: Fe E400 (17.3 kg). Ratio: 120.03 kg/m3		Niveau: PB RDC
Béton: B20 (0.13 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.10 m2	Enrobage géométrique: 3 cm	

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P1	1	HA12	4	397 50	450	1800	16.0
	2	HA10	2	398 45	445	890	5.5
	3	HA6	27	24 6 34	127	3429	7.6
						Total+10%:	32.0
P31	1	HA12	4	277	277	1108	9.8
	2	HA6	19	19 6 19	87	1653	3.7
						Total+10%:	14.9
						HA6:	12.4
						HA10:	6.1
						HA12:	28.4
						Total:	46.9

P1

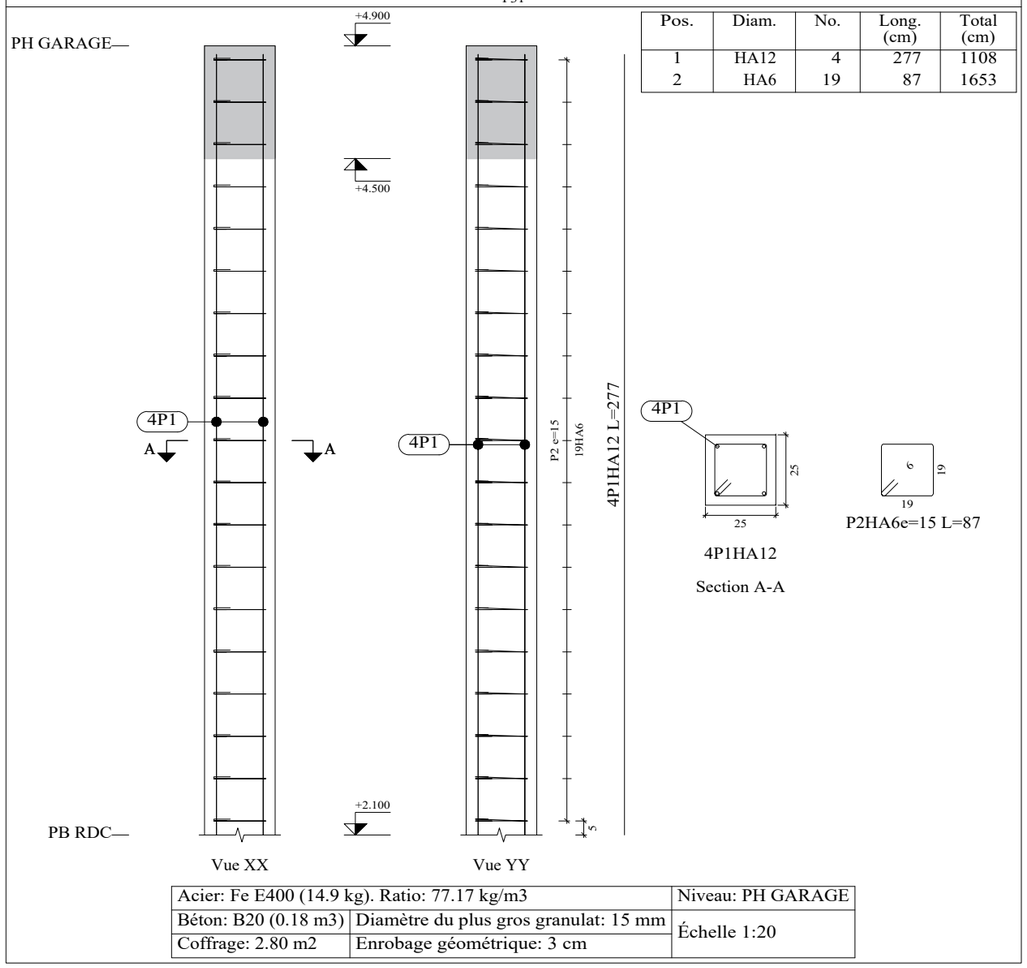
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	450	1800
2	HA10	2	445	890
3	HA6	27	127	3429



Acier: Fe E400 (32.0 kg). Ratio: 60.58 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.48 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

P31

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	277	1108
2	HA6	19	87	1653



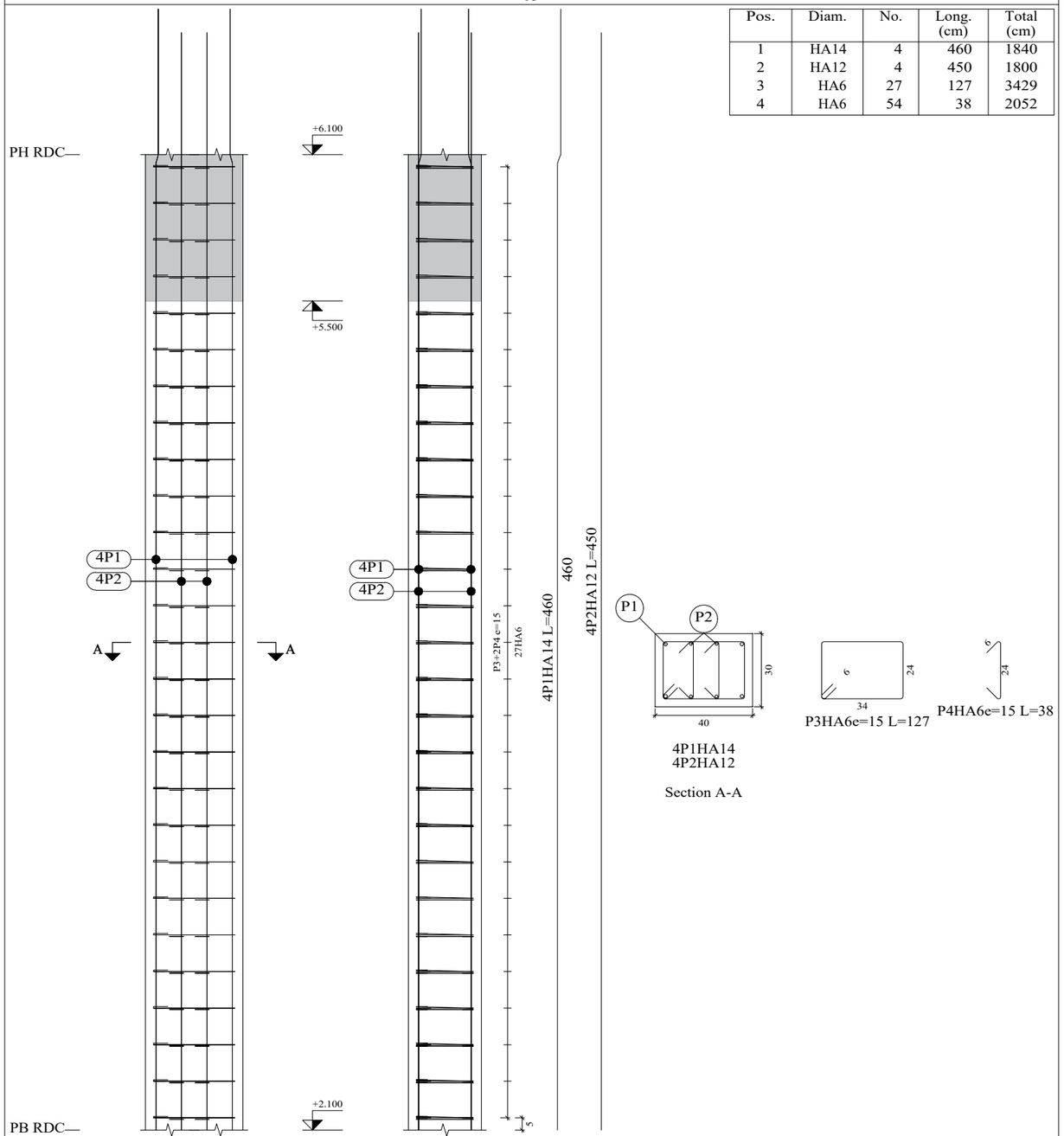
Acier: Fe E400 (14.9 kg). Ratio: 77.17 kg/m3		Niveau: PH GARAGE
Béton: B20 (0.18 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.80 m2	Enrobage géométrique: 3 cm	

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P3	1	HA14	4	397 60	460	1840	22.2
	2	HA12	4	450	450	1800	16.0
	3	HA6	27	24 6 34	127	3429	7.6
	4	HA6	54	6 24	38	2052	4.6
					Total+10%:		55.4
P24	1	HA12	4	277	277	1108	9.8
	2	HA6	19	19 6 19	87	1653	3.7
					Total+10%:		14.9
					HA6:		17.5
					HA12:		28.4
					HA14:		24.4
					Total:		70.3

P3

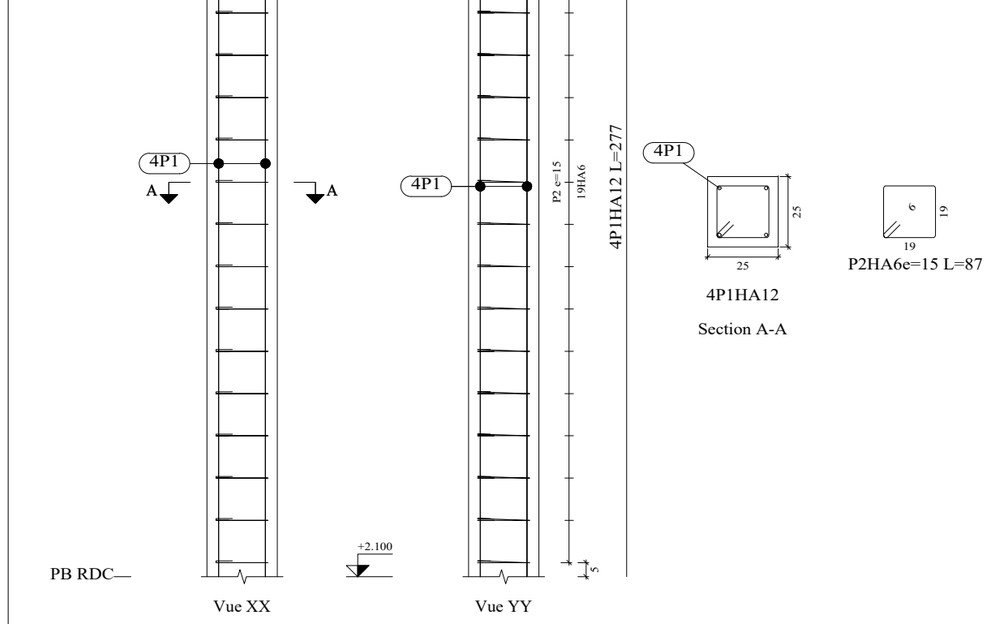
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	460	1840
2	HA12	4	450	1800
3	HA6	27	127	3429
4	HA6	54	38	2052



Acier: Fe E400 (55.4 kg). Ratio: 104.96 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.48 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

P24

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	277	1108
2	HA6	19	87	1653



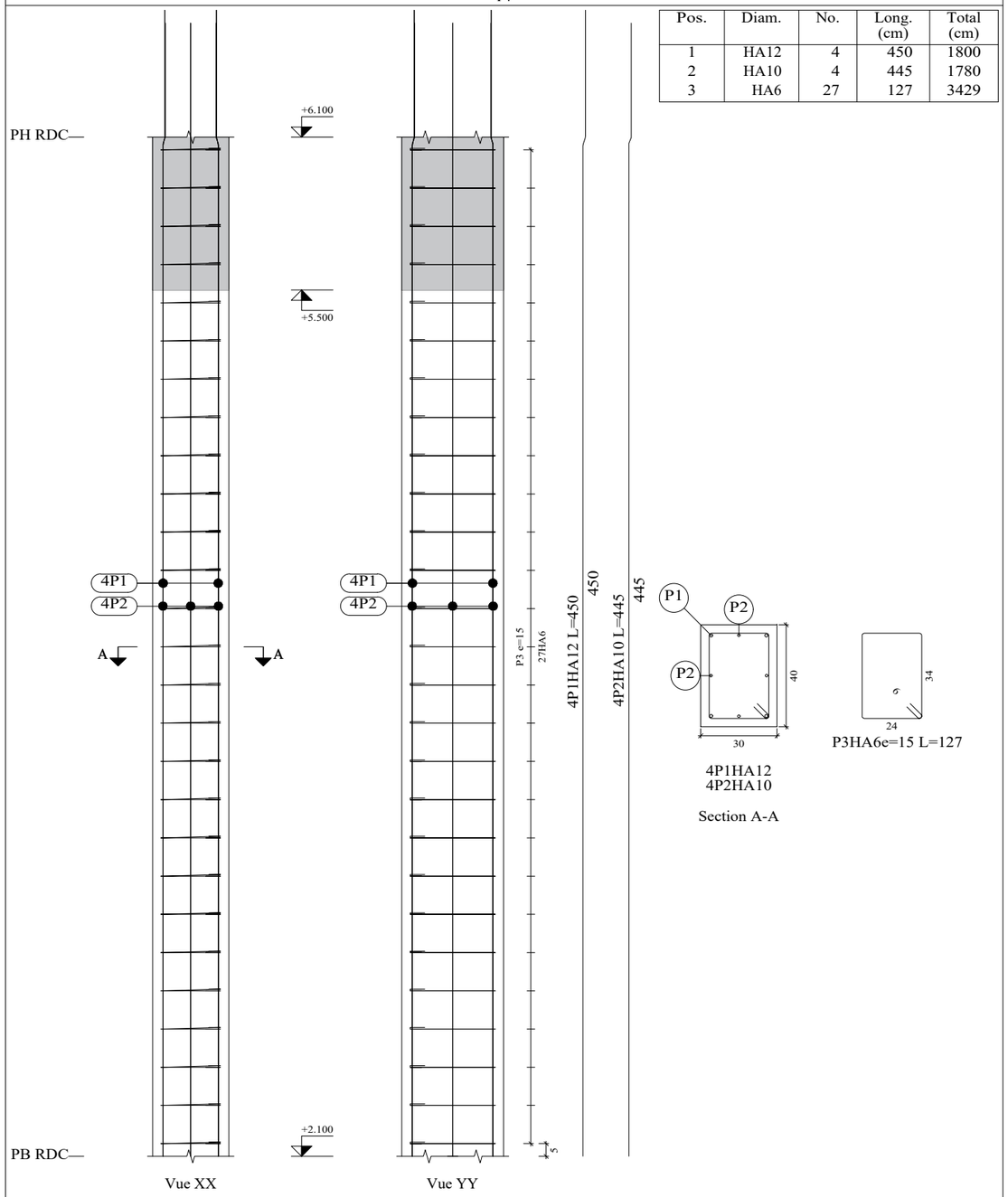
Acier: Fe E400 (14.9 kg). Ratio: 77.17 kg/m3		Niveau: PH GARAGE
Béton: B20 (0.18 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.80 m2	Enrobage géométrique: 3 cm	

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P4	1	HA12	4	397 50	450	1800	16.0
	2	HA10	4	398 45	445	1780	11.0
	3	HA6	27	24 6 34	127	3429	7.6
						Total+10%:	38.1
P6	1	HA12	4	277	277	1108	9.8
	2	HA6	19	19 6 19	87	1653	3.7
						Total+10%:	14.9
						HA6:	12.5
						HA10:	12.1
						HA12:	28.4
						Total:	53.0

P4

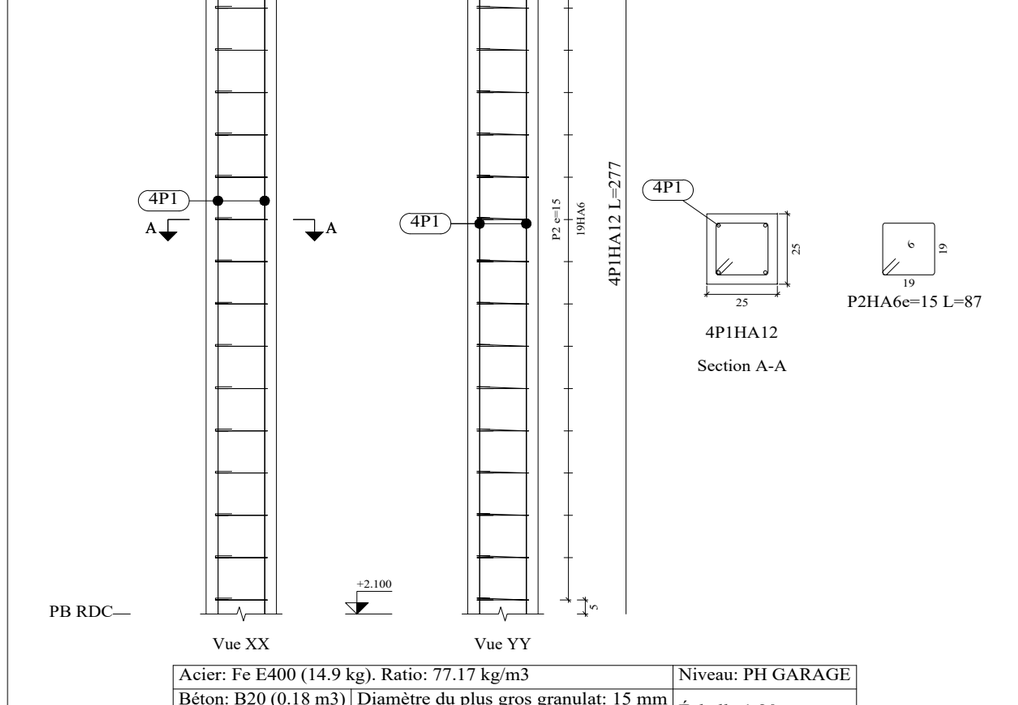
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	450	1800
2	HA10	4	445	1780
3	HA6	27	127	3429



Acier: Fe E400 (38.0 kg). Ratio: 72.01 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.48 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

P6

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	277	1108
2	HA6	19	87	1653



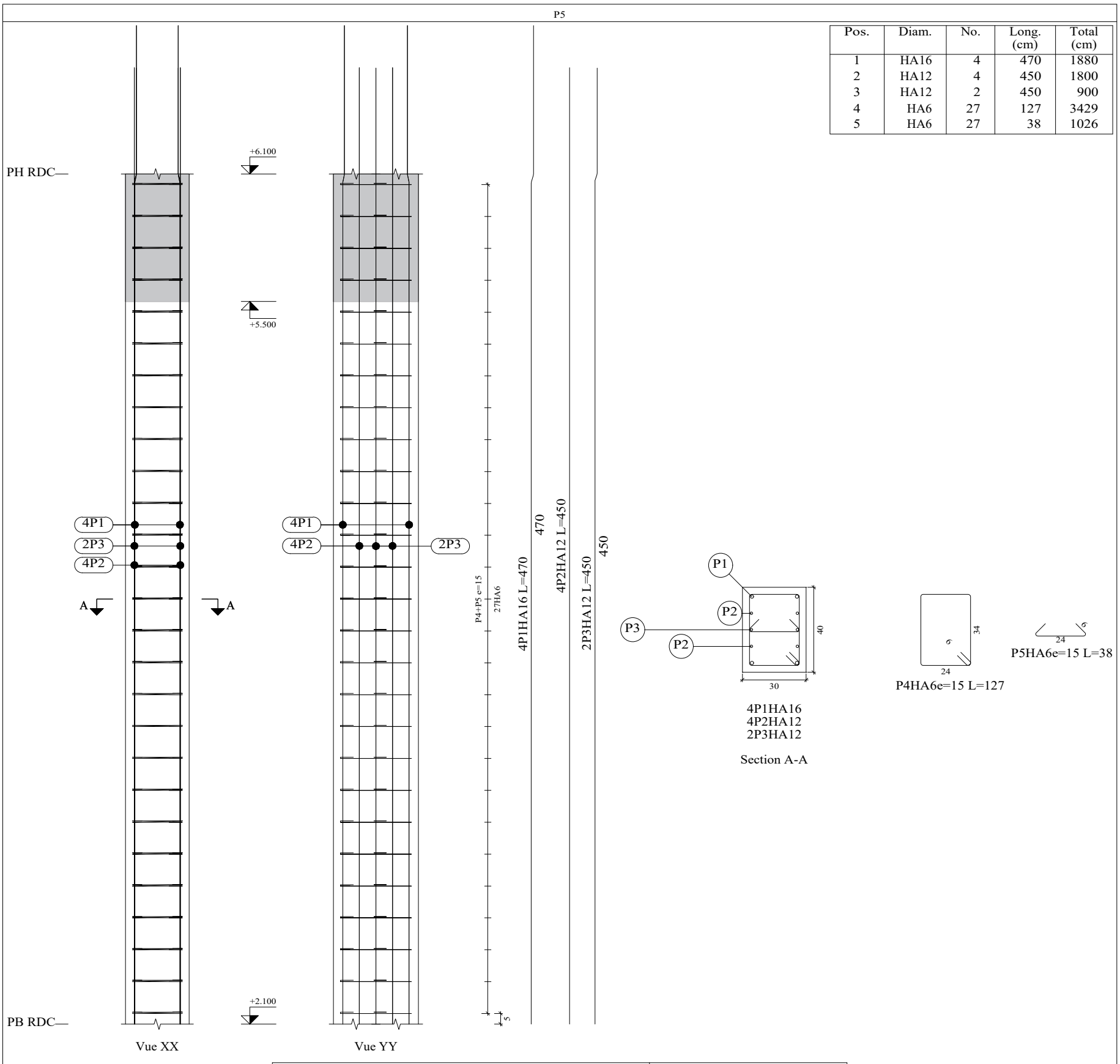
Acier: Fe E400 (14.9 kg). Ratio: 77.17 kg/m3		Niveau: PH GARAGE
Béton: B20 (0.18 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.80 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P5	1	HA16	4	396 70	470	1880	29.7
	2	HA12	4	450	450	1800	16.0
	3	HA12	2	397 50	450	900	8.0
	4	HA6	27	24 6 34	127	3429	7.6
	5	HA6	27	6 24	38	1026	2.3
					Total+10%:		70.0
					HA6:		10.9
					HA12:		26.4
					HA16:		32.7
					Total:		70.0

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P5

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA16	4	470	1880
2	HA12	4	450	1800
3	HA12	2	450	900
4	HA6	27	127	3429
5	HA6	27	38	1026



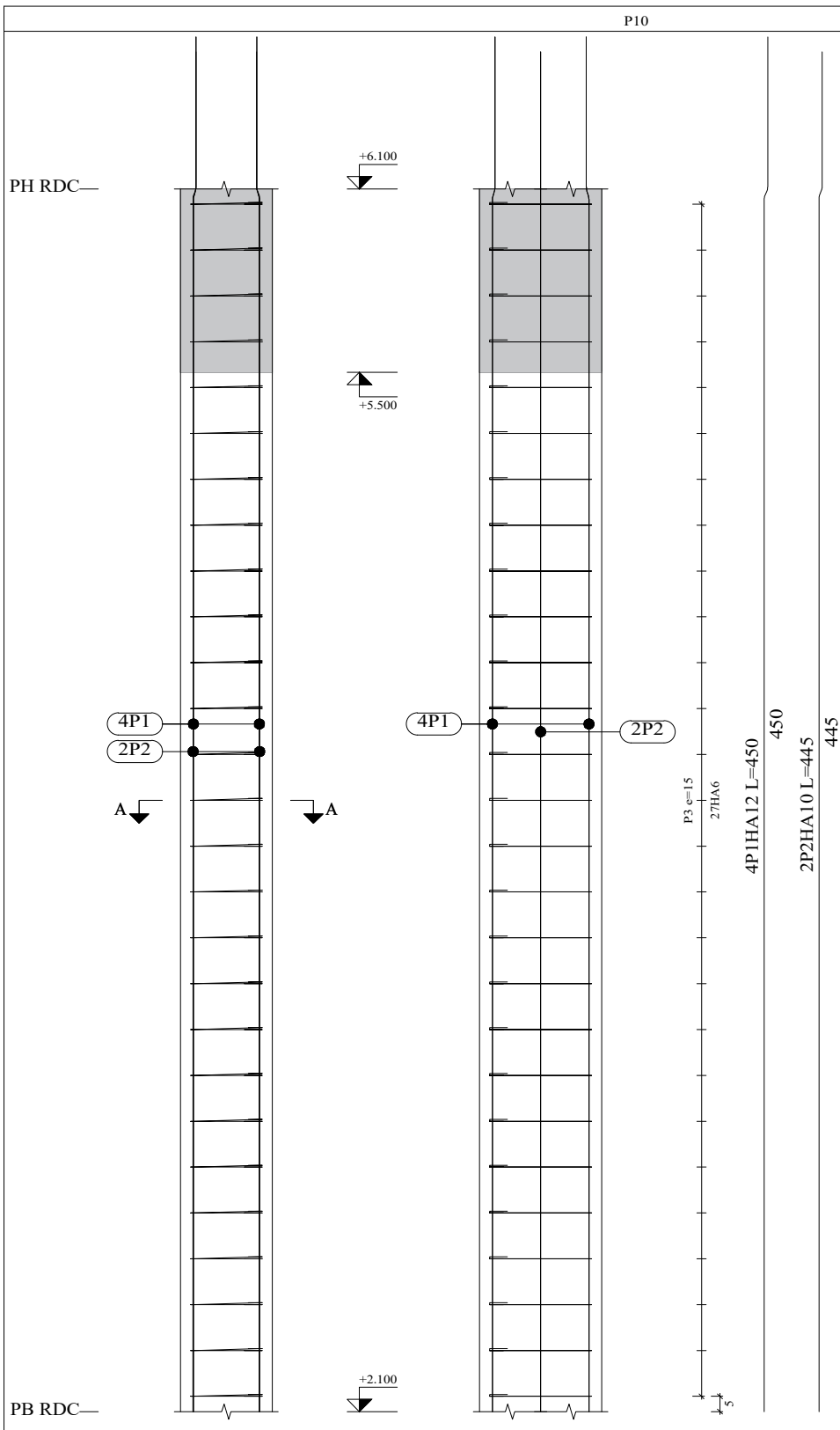
Acier: Fe E400 (69.9 kg). Ratio: 132.36 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.48 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P10	1	HA12	4	397 50	450	1800	16.0
	2	HA10	2	398 45	445	890	5.5
	3	HA6	27	24 6 34	127	3429	7.6
						Total+10%:	32.0
						HA6:	8.3
						HA10:	6.1
						HA12:	17.6
						Total:	32.0

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

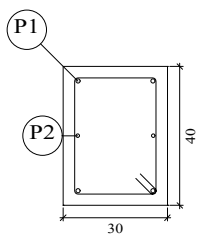
P10

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	450	1800
2	HA10	2	445	890
3	HA6	27	127	3429



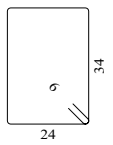
Vue XX

Vue YY



4P1HA12
2P2HA10

Section A-A



P3HA6e=15 L=127

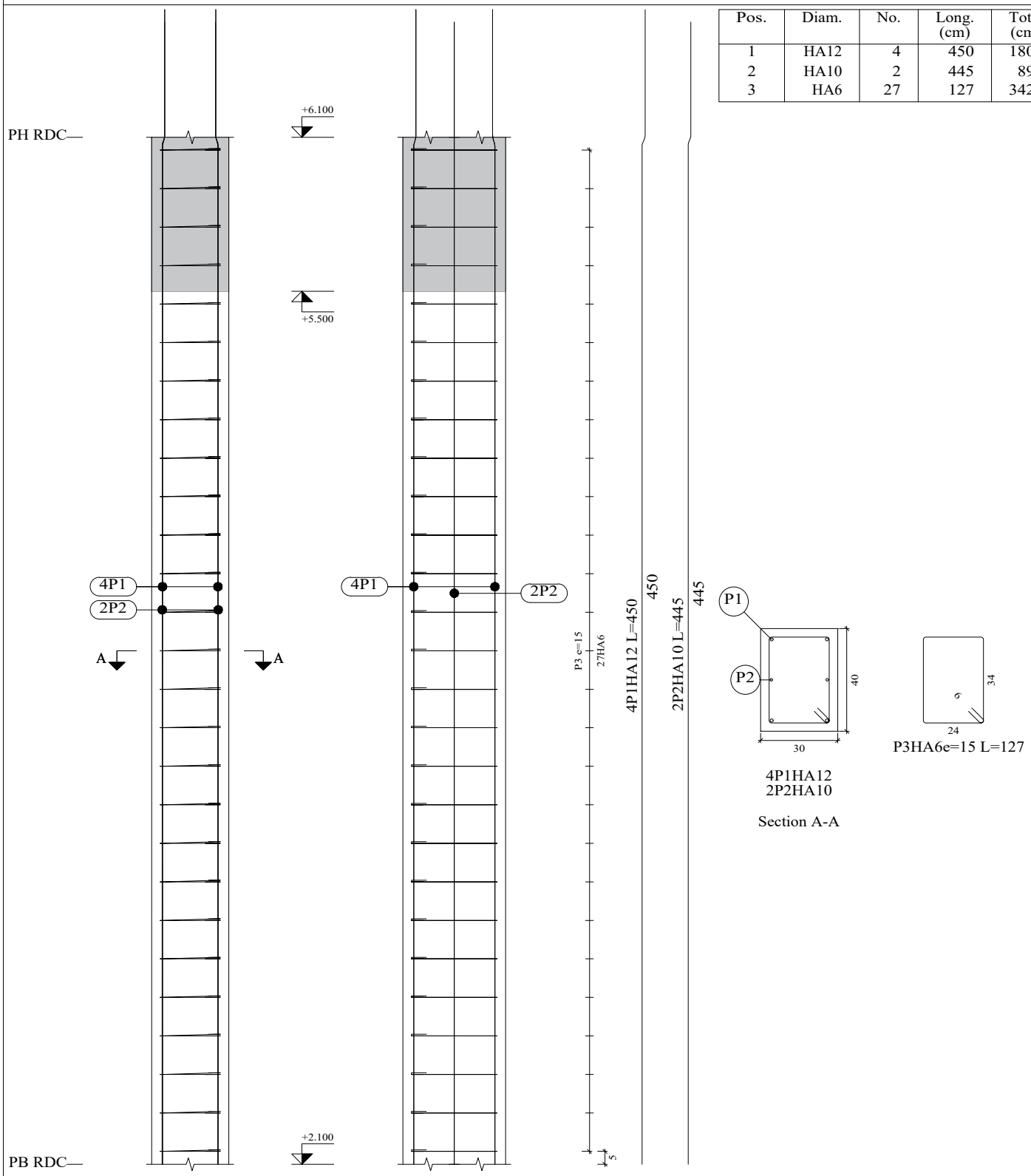
Acier: Fe E400 (32.0 kg). Ratio: 60.58 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.48 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P14	1	HA12	4	397 50	450	1800	16.0
	2	HA10	2	398 45	445	890	5.5
	3	HA6	27	6 / 24 / 34	127	3429	7.6
						Total+10%:	32.0
						HA6:	8.3
						HA10:	6.1
						HA12:	17.6
						Total:	32.0

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P14

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	450	1800
2	HA10	2	445	890
3	HA6	27	127	3429



4P1HA12
2P2HA10

Section A-A

P3HA6e=15 L=127

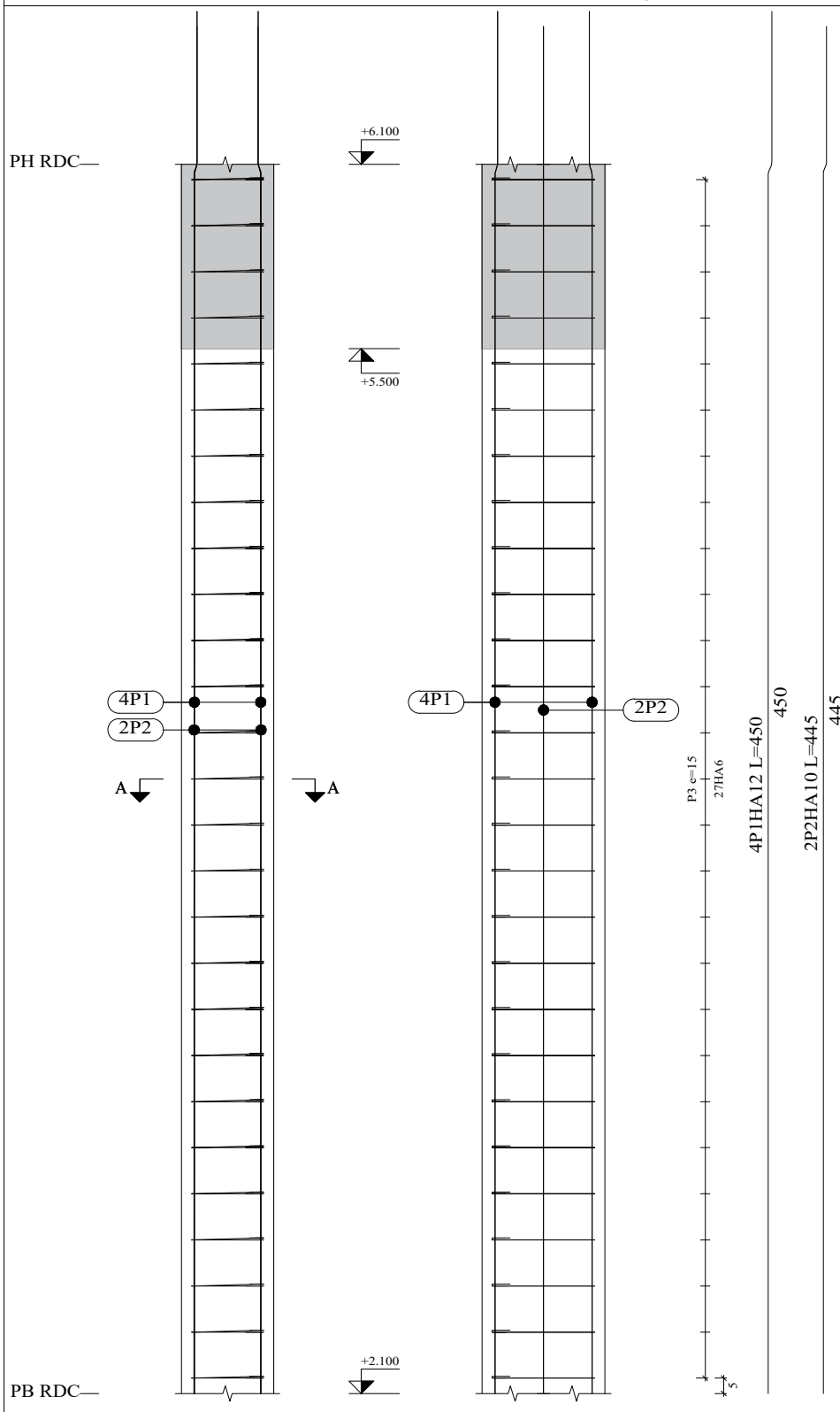
Acier: Fe E400 (32.0 kg). Ratio: 60.58 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.48 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

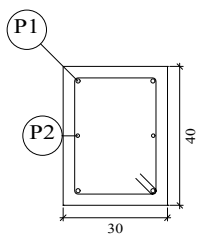
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P25	1	HA12	4	397 50	450	1800	16.0
	2	HA10	2	398 45	445	890	5.5
	3	HA6	27	24 6 34	127	3429	7.6
						Total+10%:	32.0
						HA6:	8.3
						HA10:	6.1
						HA12:	17.6
						Total:	32.0

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P25

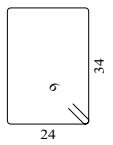
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	450	1800
2	HA10	2	445	890
3	HA6	27	127	3429





4P1HA12
2P2HA10

Section A-A



P3HA6e=15 L=127

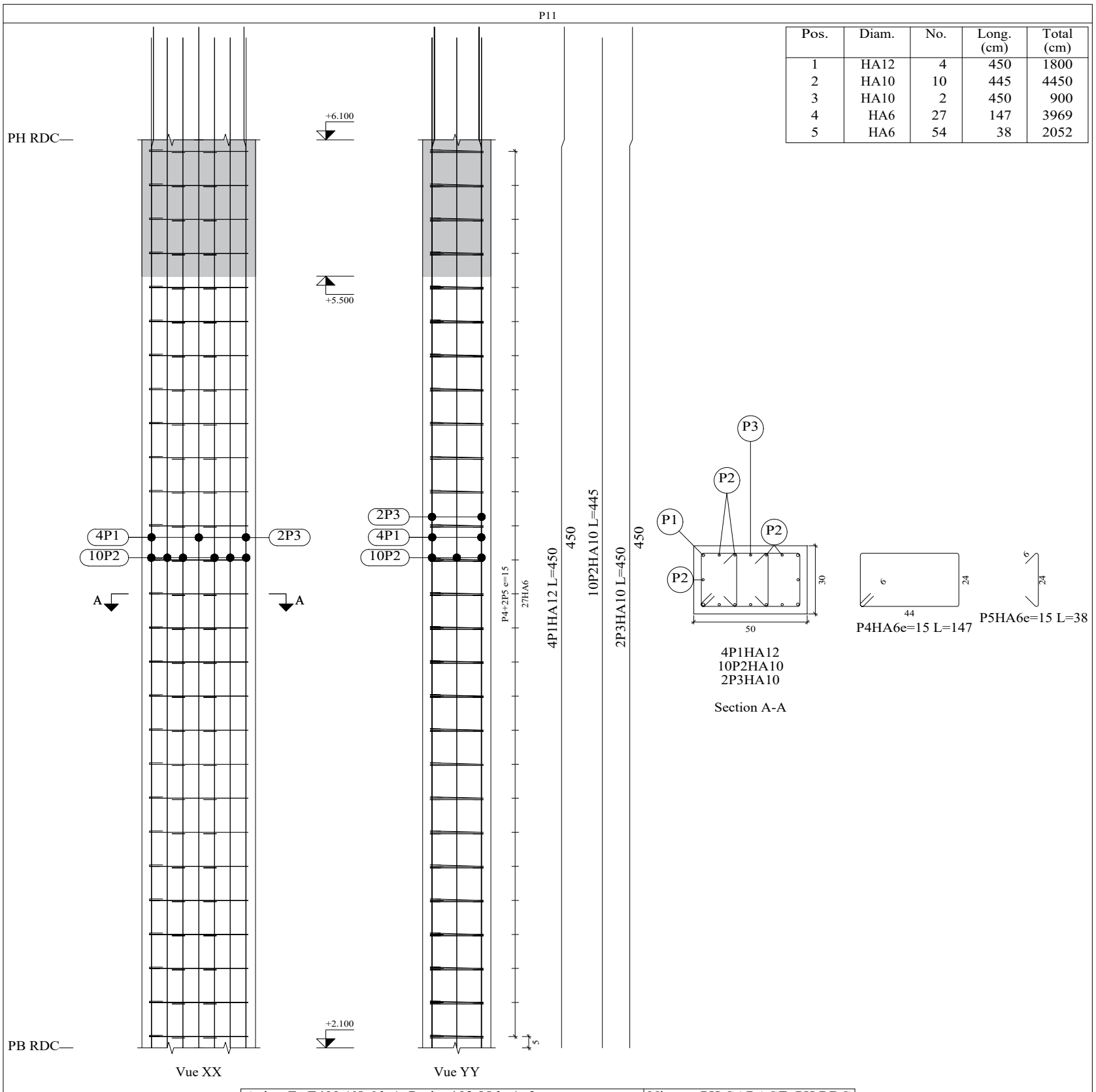
Acier: Fe E400 (32.0 kg). Ratio: 60.58 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.48 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P11	1	HA12	4	397 50	450	1800	16.0
	2	HA10	10	445	445	4450	27.4
	3	HA10	2	398 50	450	900	5.5
	4	HA6	27	24 6 44	147	3969	8.8
	5	HA6	54	6 24	38	2052	4.6
					Total+10%:		68.5
					HA6:		14.7
					HA10:		36.2
					HA12:		17.6
					Total:		68.5

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P11

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	450	1800
2	HA10	10	445	4450
3	HA10	2	450	900
4	HA6	27	147	3969
5	HA6	54	38	2052



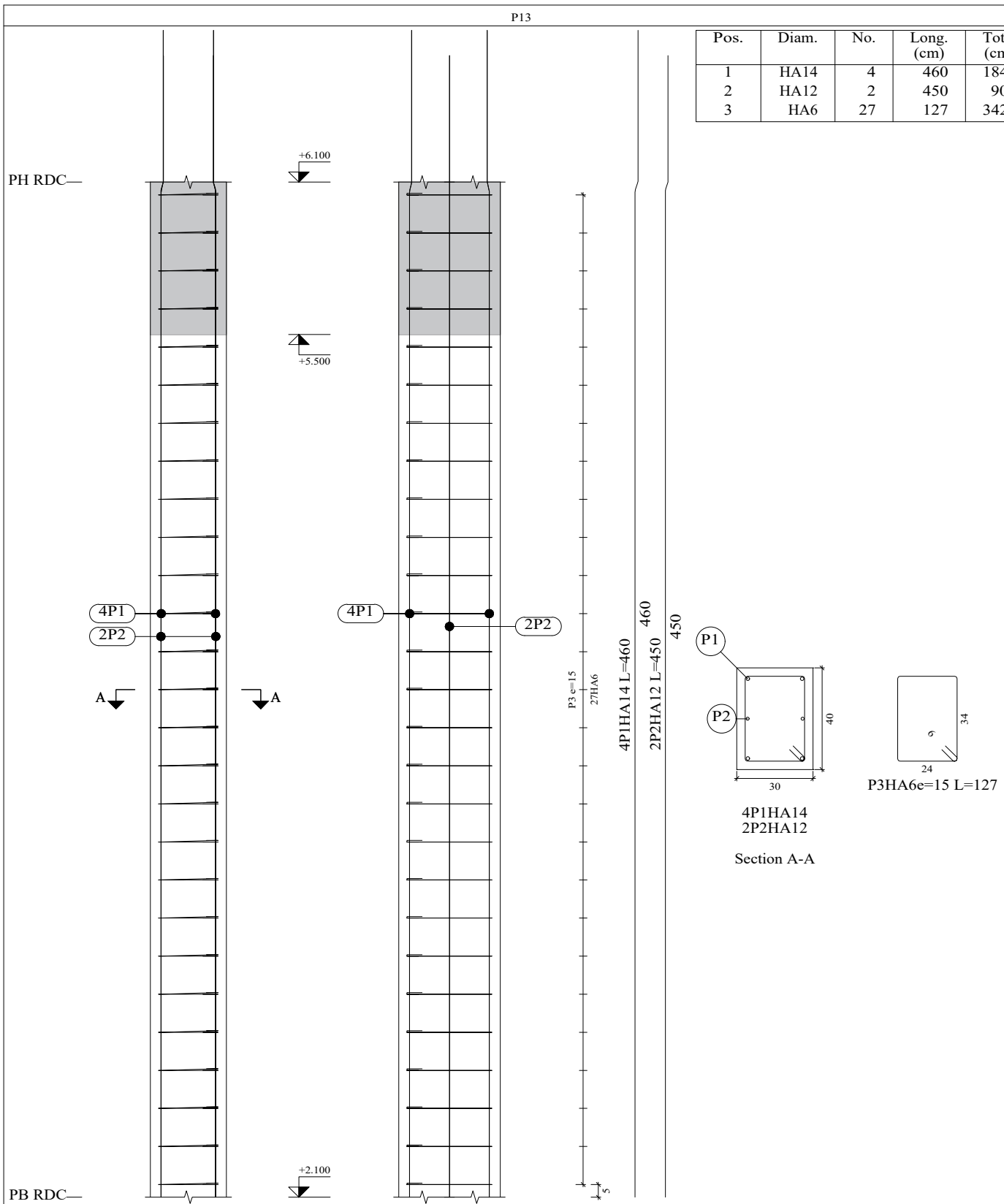
Acier: Fe E400 (68.6 kg). Ratio: 103.88 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.60 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 6.40 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P13	1	HA14	4	397 60	460	1840	22.2
	2	HA12	2	397 50	450	900	8.0
	3	HA6	27	24 / 6 / 34	127	3429	7.6
					Total+10%:		41.6
					HA6:		8.4
					HA12:		8.8
					HA14:		24.4
					Total:		41.6

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P13

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	460	1840
2	HA12	2	450	900
3	HA6	27	127	3429



4P1HA14
2P2HA12

Section A-A

P3HA6e=15 L=127

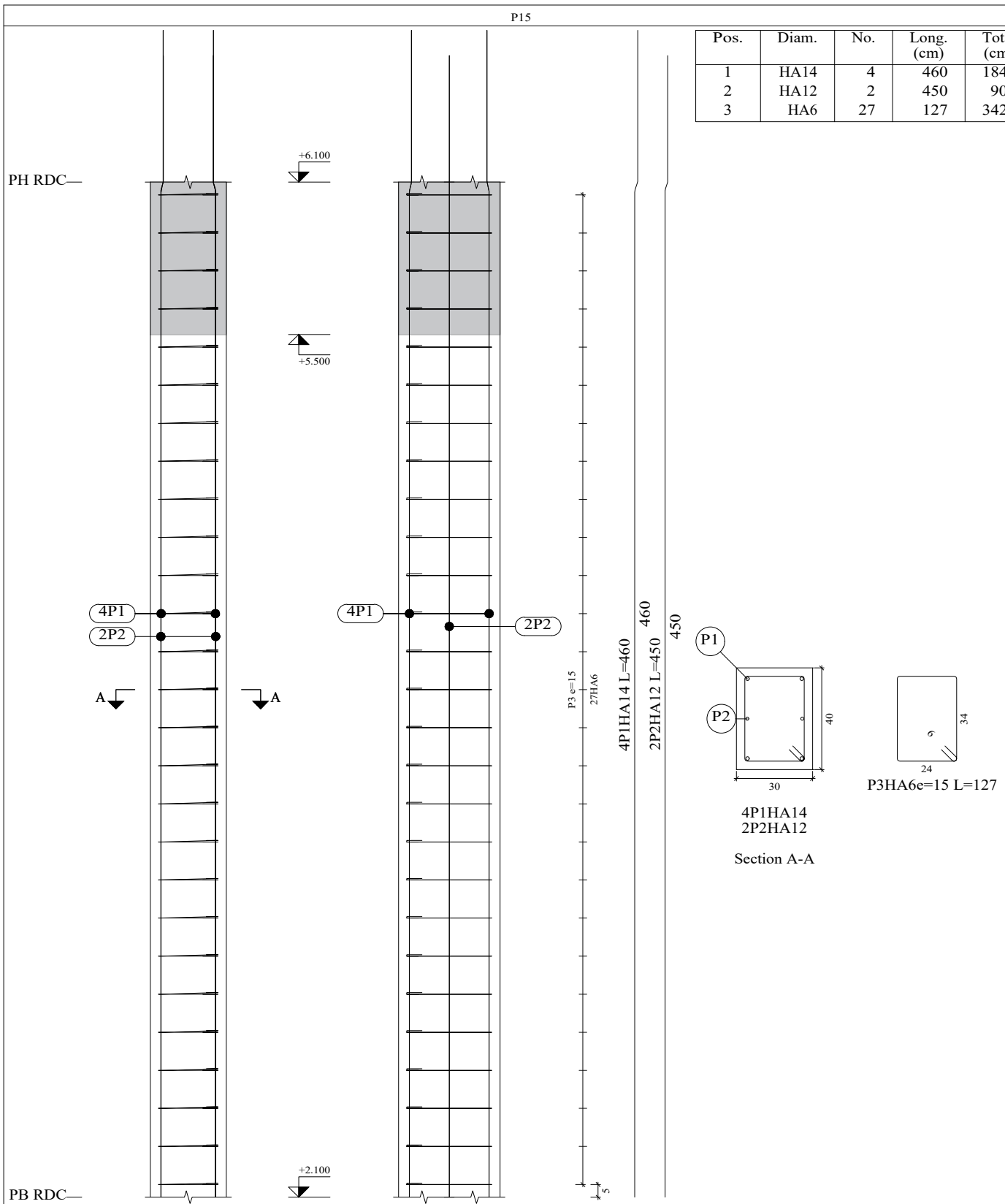
Acier: Fe E400 (41.6 kg). Ratio: 78.82 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.48 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P15	1	HA14	4	397 60	460	1840	22.2
	2	HA12	2	397 50	450	900	8.0
	3	HA6	27	24 6 34	127	3429	7.6
					Total+10%:		41.6
					HA6:		8.4
					HA12:		8.8
					HA14:		24.4
					Total:		41.6

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P15

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	460	1840
2	HA12	2	450	900
3	HA6	27	127	3429



4P1HA14
2P2HA12

Section A-A

P3HA6e=15 L=127

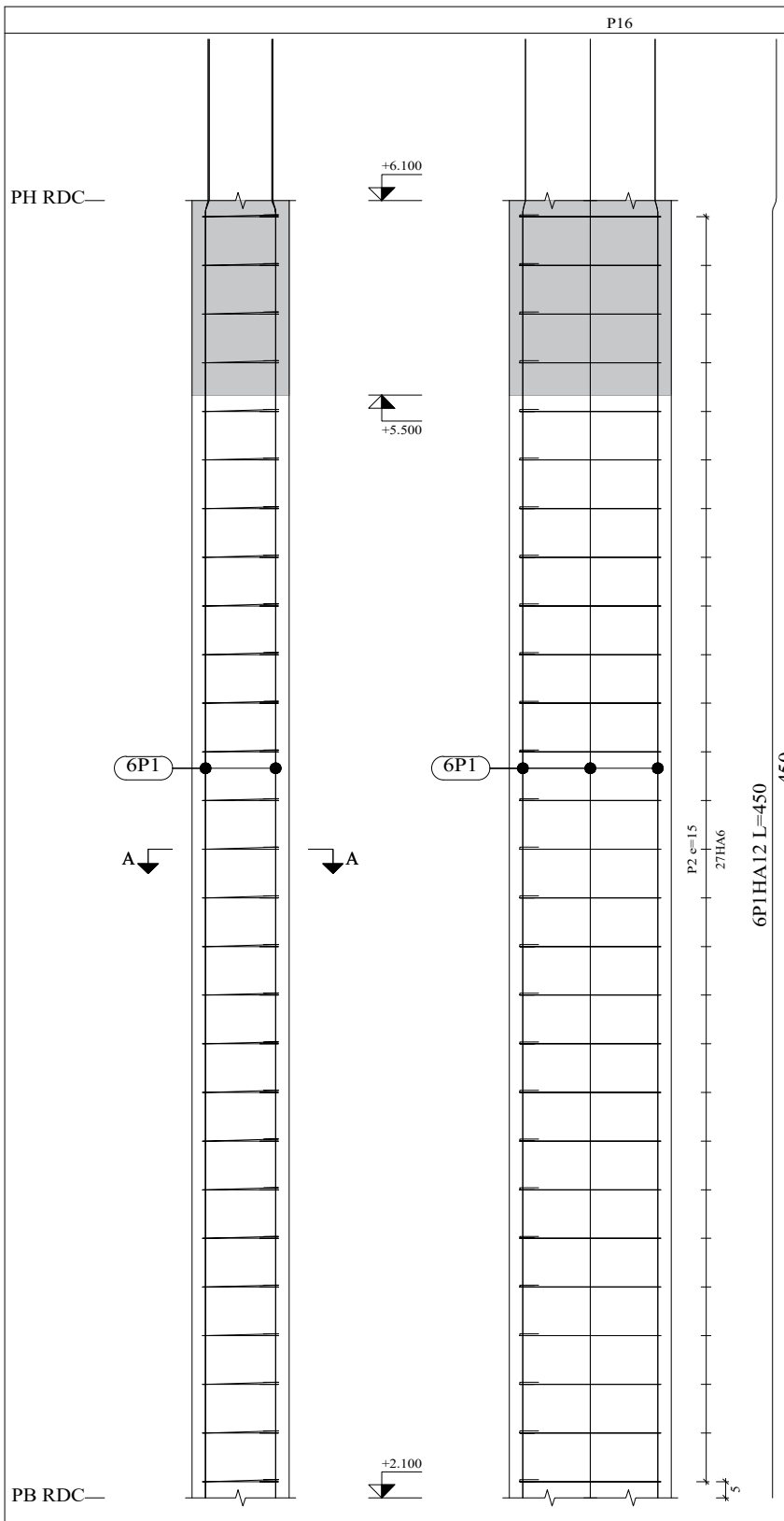
Acier: Fe E400 (41.6 kg). Ratio: 78.82 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.48 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

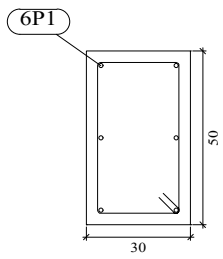
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P16	1	HA12	6	397 50	450	2700	24.0
	2	HA6	27	24 6 44	147	3969	8.8
					Total+10%:		36.1
					HA6:		9.7
					HA12:		26.4
					Total:		36.1

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P16

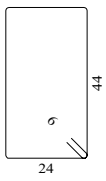
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	6	450	2700
2	HA6	27	147	3969





6P1HA12

Section A-A



P2HA6e=15 L=147

Vue XX

Vue YY

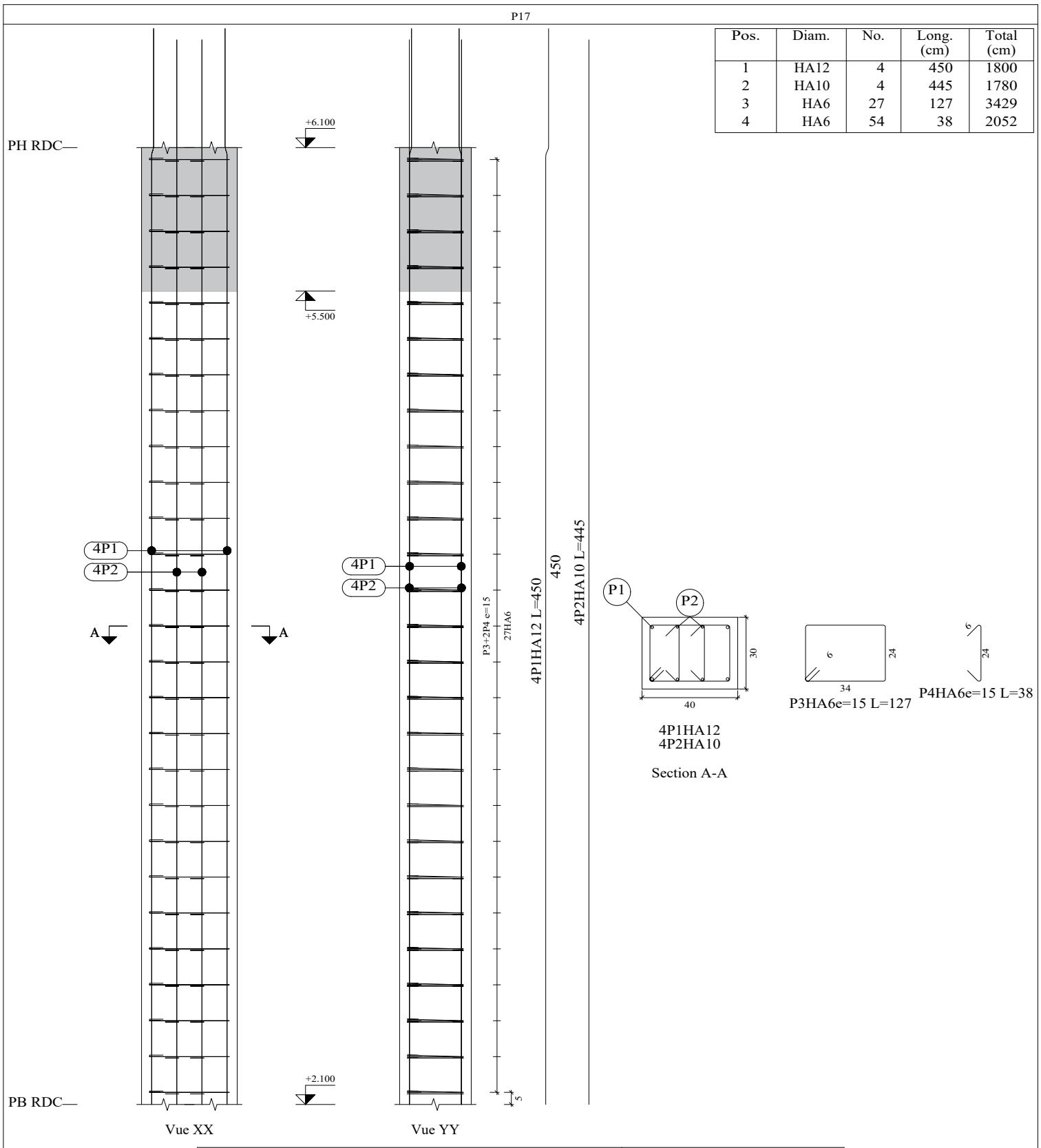
Acier: Fe E400 (36.1 kg). Ratio: 54.63 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.60 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 6.40 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P17	1	HA12	4	397 50	450	1800	16.0
	2	HA10	4	445	445	1780	11.0
	3	HA6	27	24 6 34	127	3429	7.6
	4	HA6	54	6 24	38	2052	4.6
					Total+10%:		43.1
					HA6:		13.4
					HA10:		12.1
					HA12:		17.6
					Total:		43.1

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P17

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	450	1800
2	HA10	4	445	1780
3	HA6	27	127	3429
4	HA6	54	38	2052



Vue XX

Vue YY

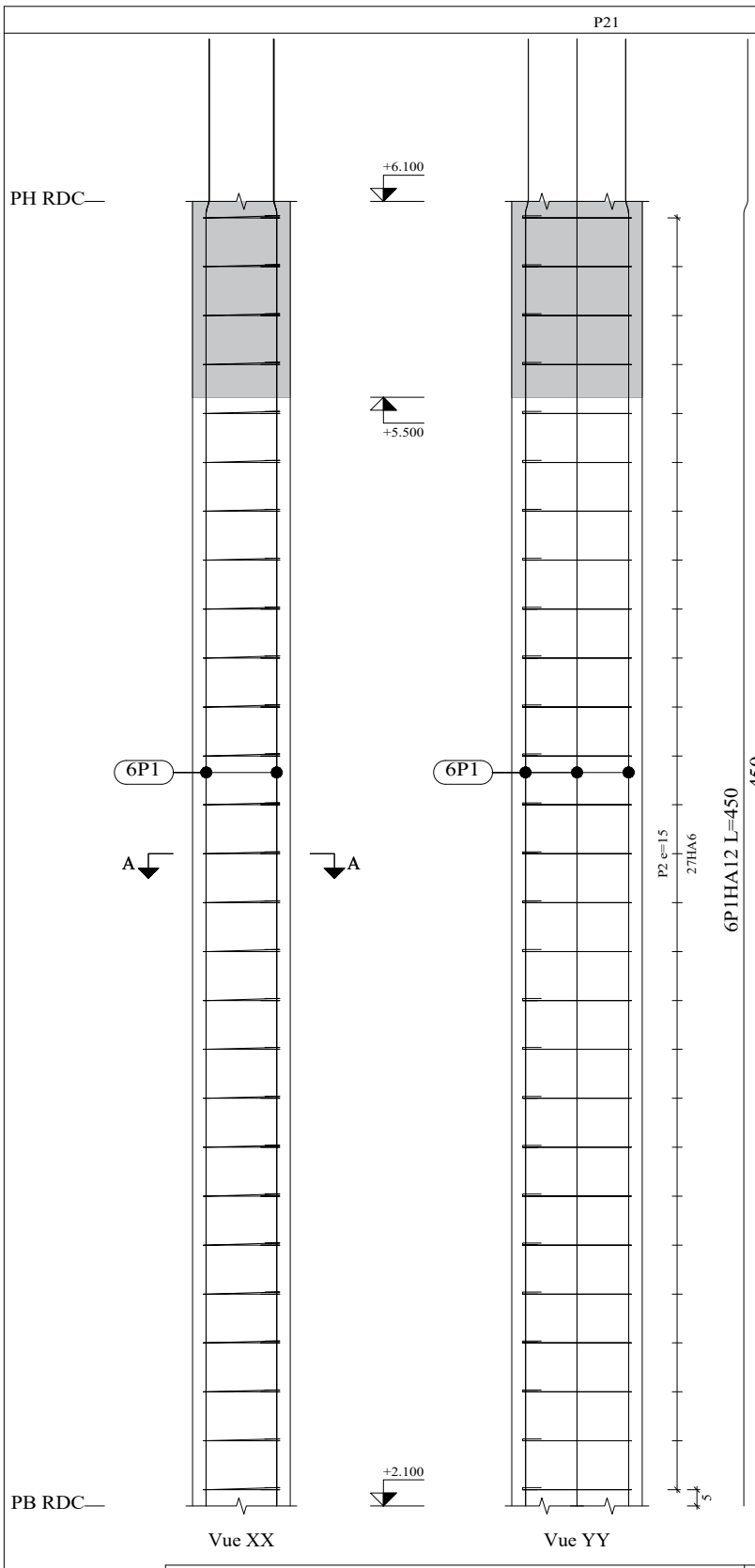
Acier: Fe E400 (43.0 kg). Ratio: 81.50 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.48 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

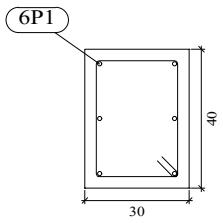
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P21	1	HA12	6	397 50	450	2700	24.0
	2	HA6	27	24 6 34	127	3429	7.6
					Total+10%:		34.8
					HA6:		8.4
					HA12:		26.4
					Total:		34.8

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P21

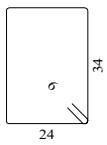
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	6	450	2700
2	HA6	27	127	3429





6P1HA12

Section A-A



P2HA6e=15 L=127

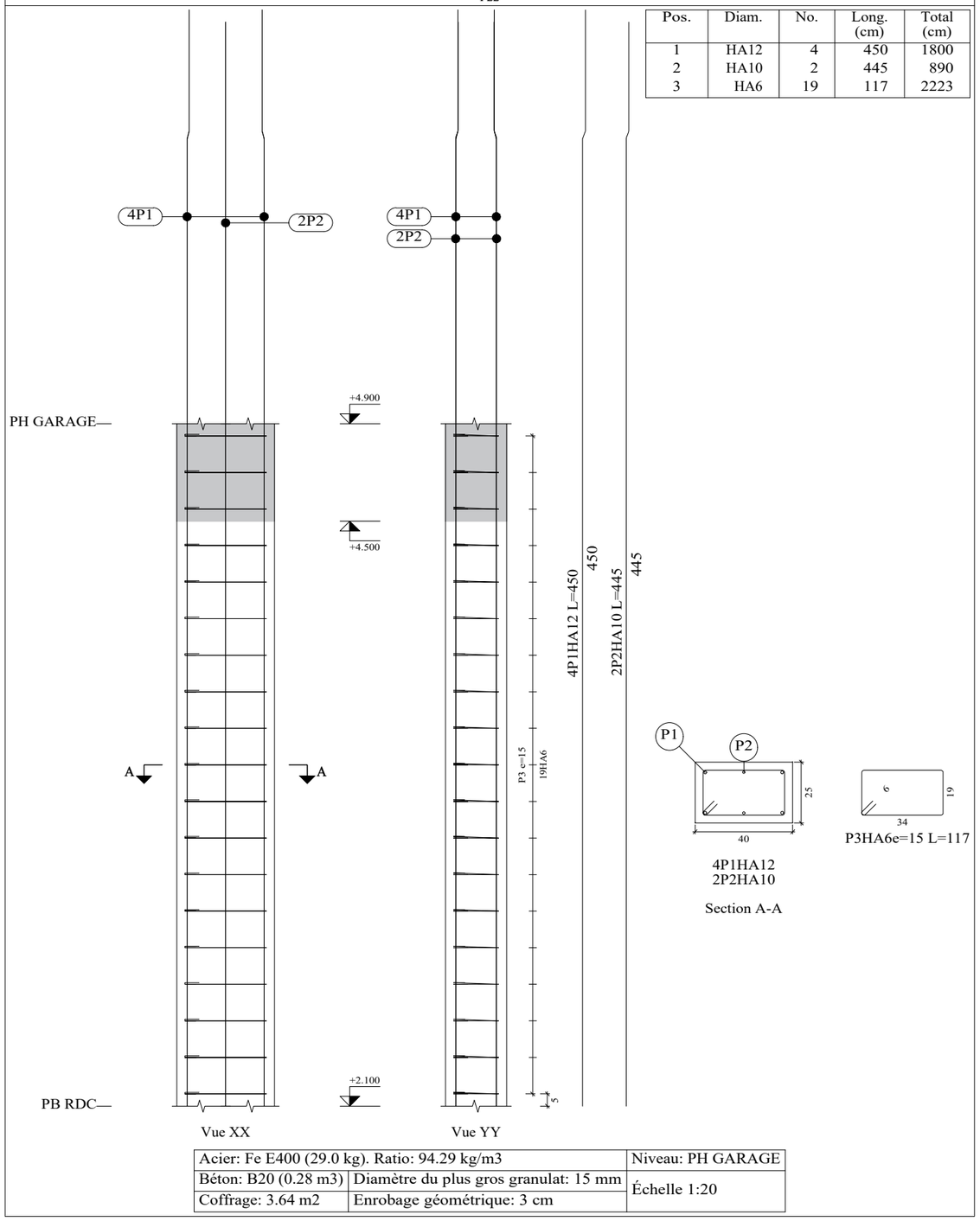
Acier: Fe E400 (34.7 kg). Ratio: 65.79 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.48 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P22	1	HA12	4	397 50	450	1800	16.0
	2	HA10	2	398 45	445	890	5.5
	3	HA6	19	19 6 34	117	2223	4.9
					Total+10%:		29.0
P29	1	HA12	4	277	277	1108	9.8
	2	HA10	2	277	277	554	3.4
	3	HA6	19	19 6 34	117	2223	4.9
					Total+10%:		19.9
					HA6:		10.7
					HA10:		9.8
					HA12:		28.4
					Total:		48.9

P22

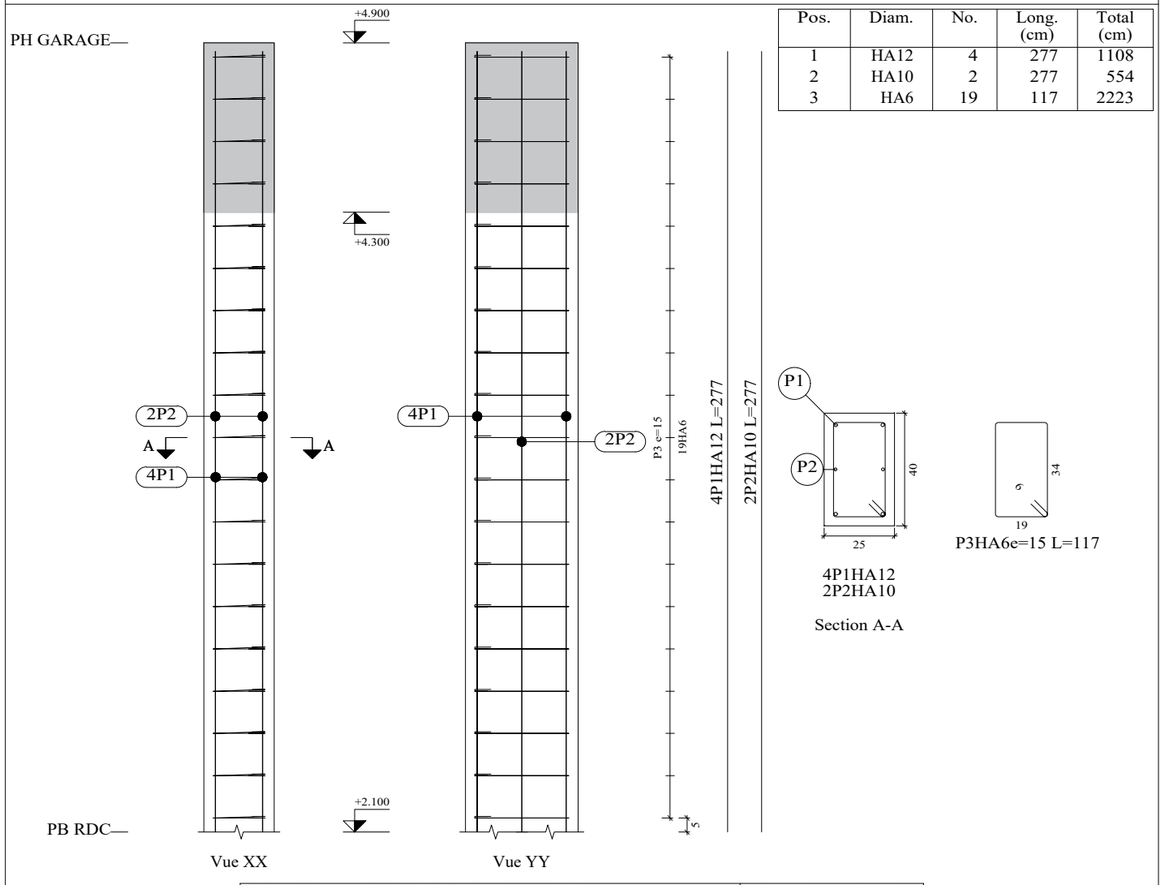
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	450	1800
2	HA10	2	445	890
3	HA6	19	117	2223



Acier: Fe E400 (29.0 kg). Ratio: 94.29 kg/m3		Niveau: PH GARAGE
Béton: B20 (0.28 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 3.64 m2	Enrobage géométrique: 3 cm	

P29

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	277	1108
2	HA10	2	277	554
3	HA6	19	117	2223



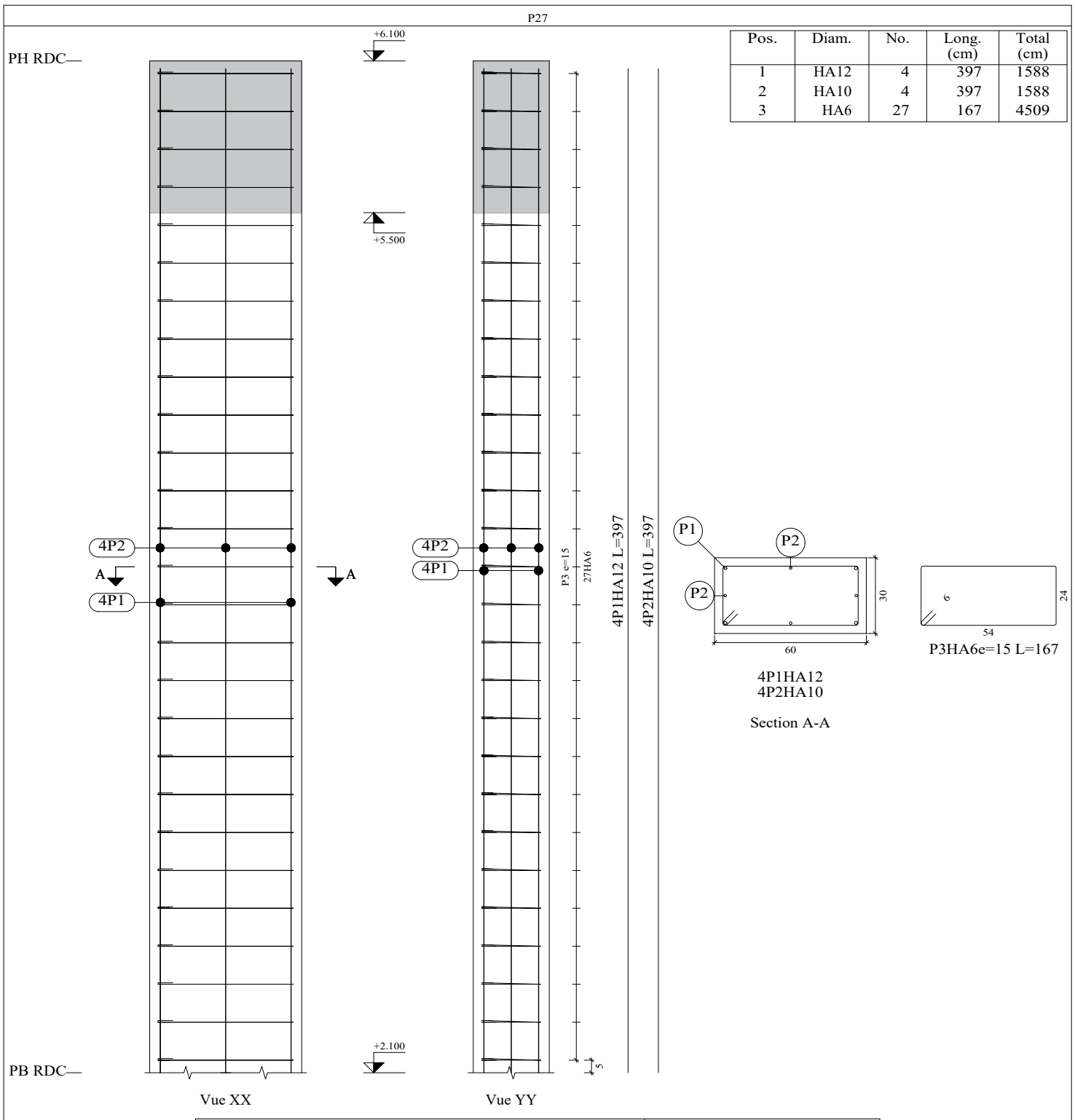
Acier: Fe E400 (20.0 kg). Ratio: 64.95 kg/m3		Niveau: PH GARAGE
Béton: B20 (0.28 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 3.64 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P27	1	HA12	4	397	397	1588	14.1
	2	HA10	4	397	397	1588	9.8
	3	HA6	27	24 / 6 / 54	167	4509	10.0
					Total+10%:		37.3
						HA6:	11.0
						HA10:	10.8
						HA12:	15.5
						Total:	37.3

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P27

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	397	1588
2	HA10	4	397	1588
3	HA6	27	167	4509



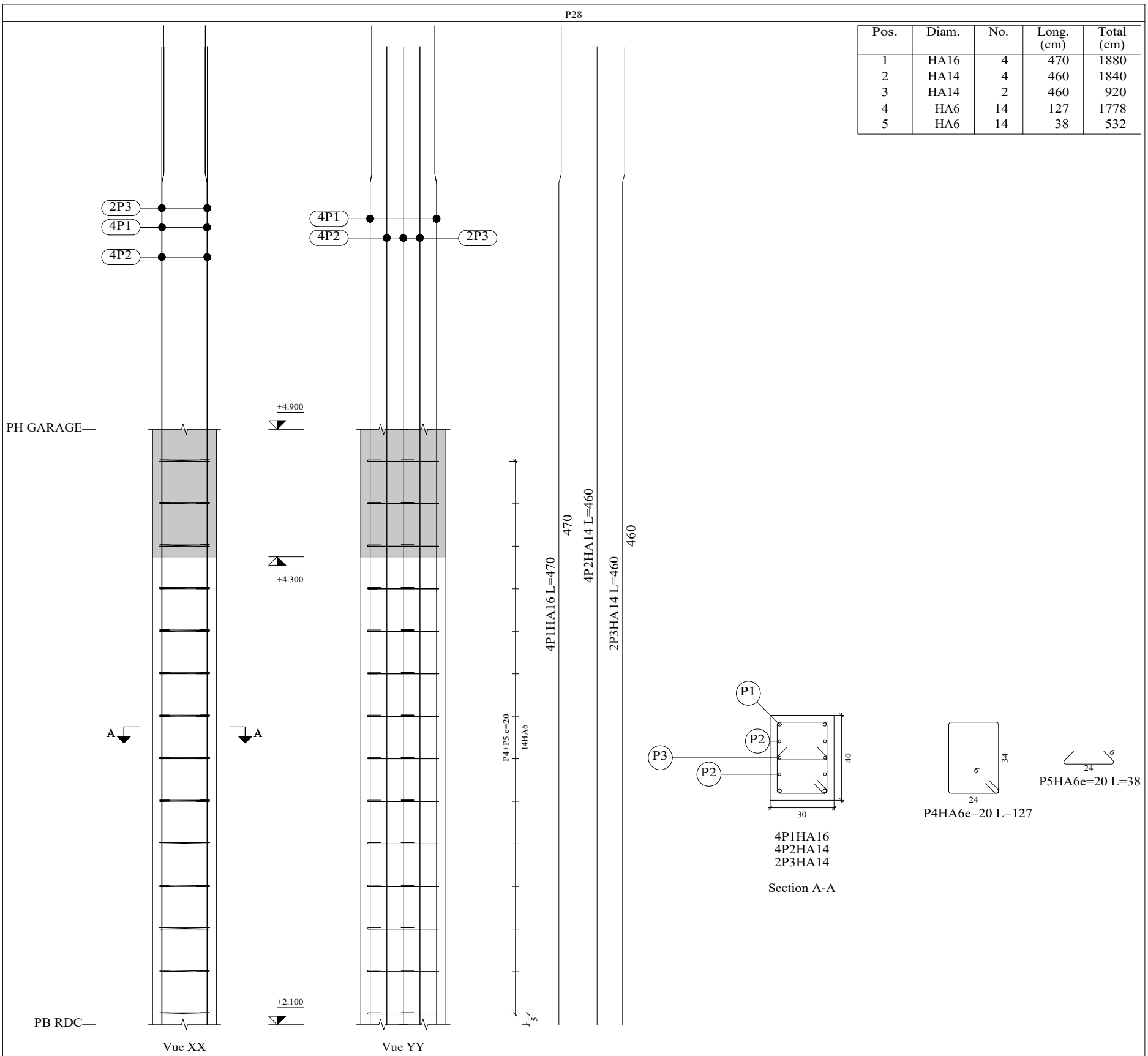
Acier: Fe E400 (37.3 kg). Ratio: 47.08 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.72 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 7.20 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P28	1	HA16	4	396 70	470	1880	29.7
	2	HA14	4	460	460	1840	22.2
	3	HA14	2	397 60	460	920	11.1
	4	HA6	14	24 6 34	127	1778	3.9
	5	HA6	14	6 24	38	532	1.2
					Total+10%:		74.9
					HA6:		5.6
					HA14:		36.6
					HA16:		32.7
					Total:		74.9

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P28

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA16	4	470	1880
2	HA14	4	460	1840
3	HA14	2	460	920
4	HA6	14	127	1778
5	HA6	14	38	532



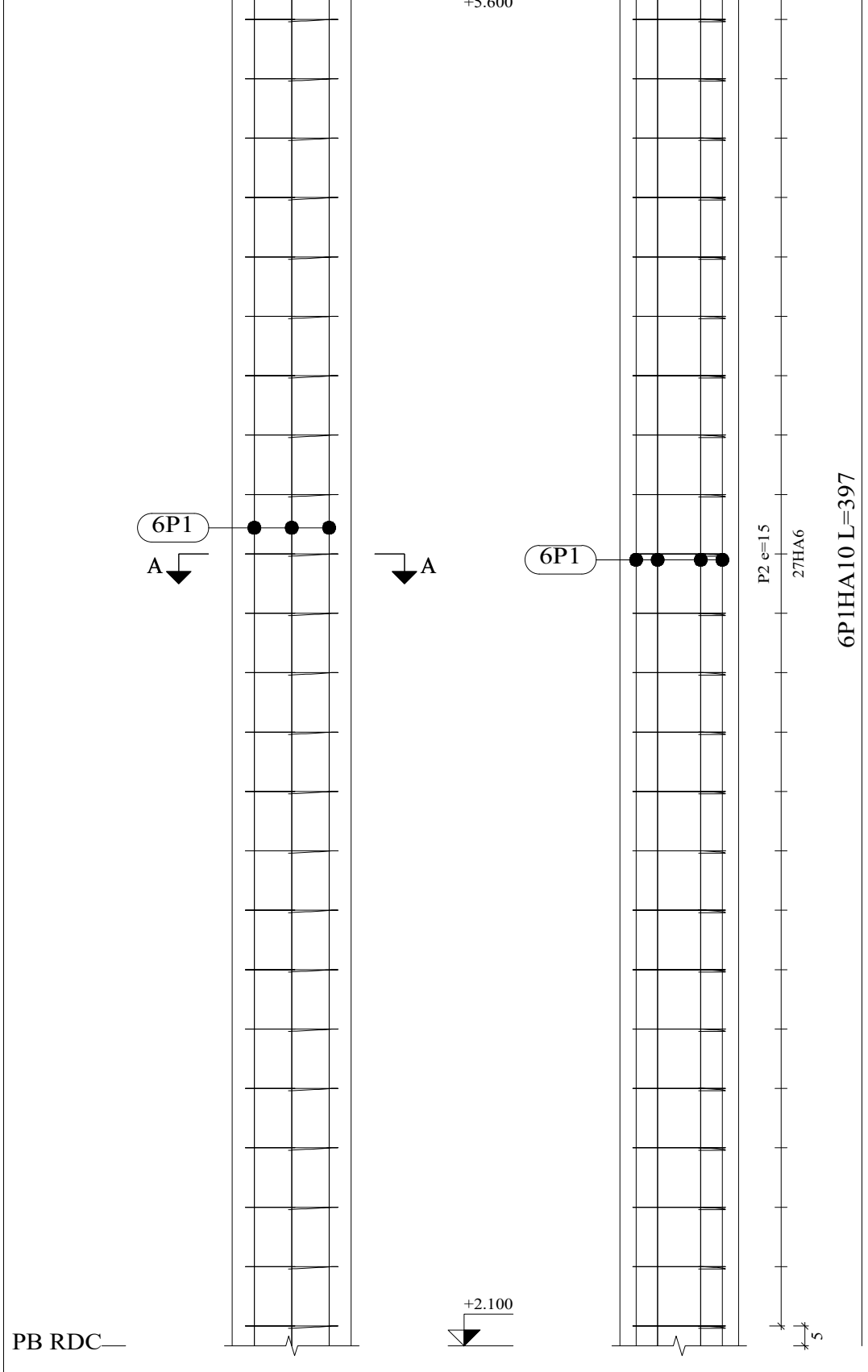
Acier: Fe E400 (75.0 kg). Ratio: 202.83 kg/m3		Niveau: PH GARAGE
Béton: B20 (0.34 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 3.92 m2	Enrobage géométrique: 3 cm	

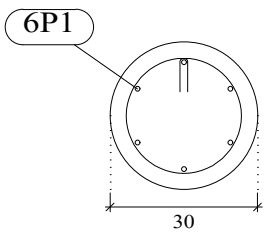
Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P32	1	HA10	6	397	397	2382	14.7
	2	HA6	27	24 6	88	2376	5.3
					Total+10%:		22.0
					HA6:		5.8
					HA10:		16.2
					Total:		22.0

P32

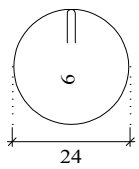
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA10	6	397	2382
2	HA6	27	88	2376





6P1HA10

Section A-A



P2HA6e=15 L=88

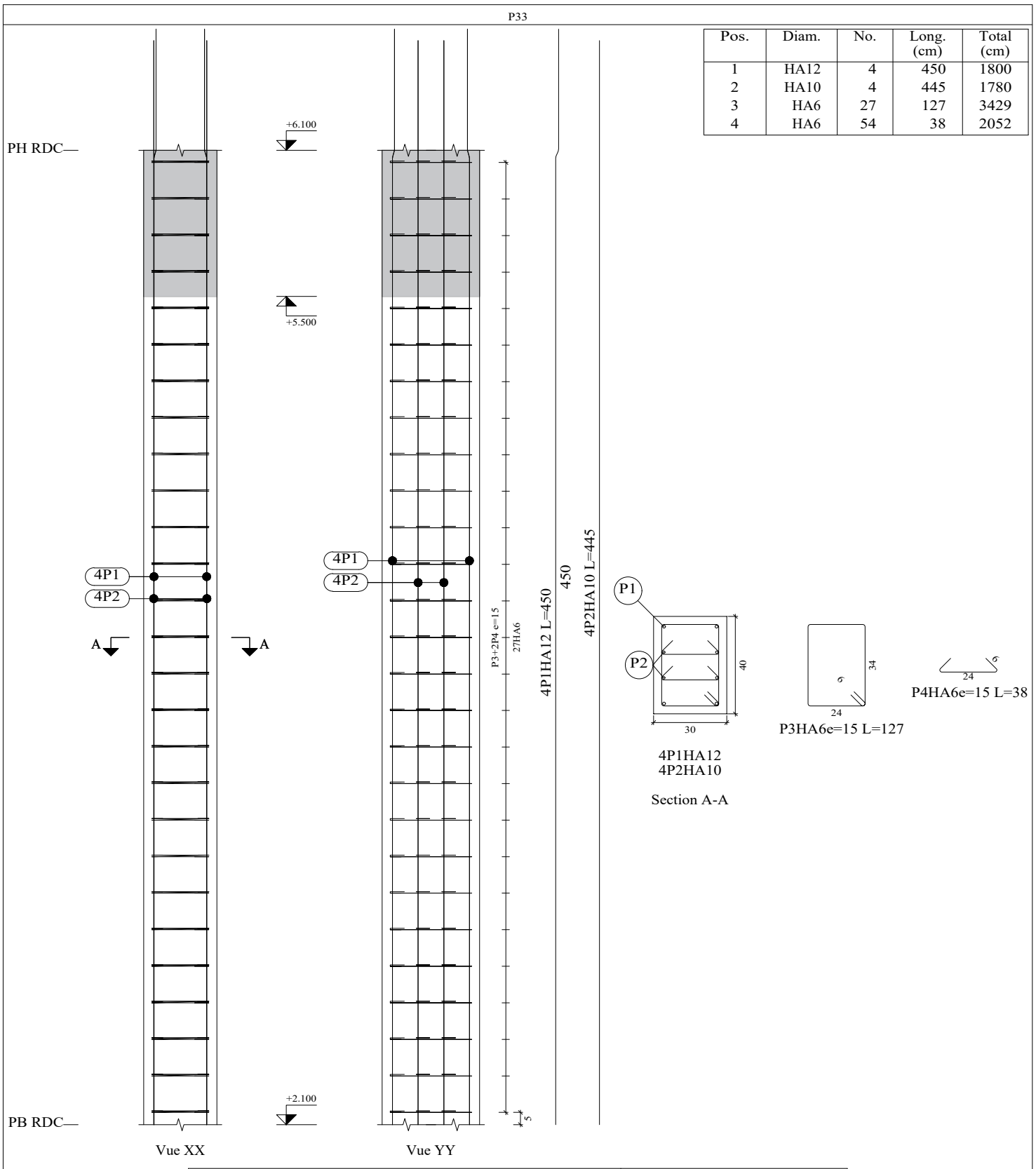
Acier: Fe E400 (22.0 kg). Ratio: 70.59 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.28 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 3.77 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P33	1	HA12	4	397 50	450	1800	16.0
	2	HA10	4	445	445	1780	11.0
	3	HA6	27	24 6 34	127	3429	7.6
	4	HA6	54	6 24	38	2052	4.6
					Total+10%:		43.1
					HA6:		13.4
					HA10:		12.1
					HA12:		17.6
					Total:		43.1

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P33

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	450	1800
2	HA10	4	445	1780
3	HA6	27	127	3429
4	HA6	54	38	2052



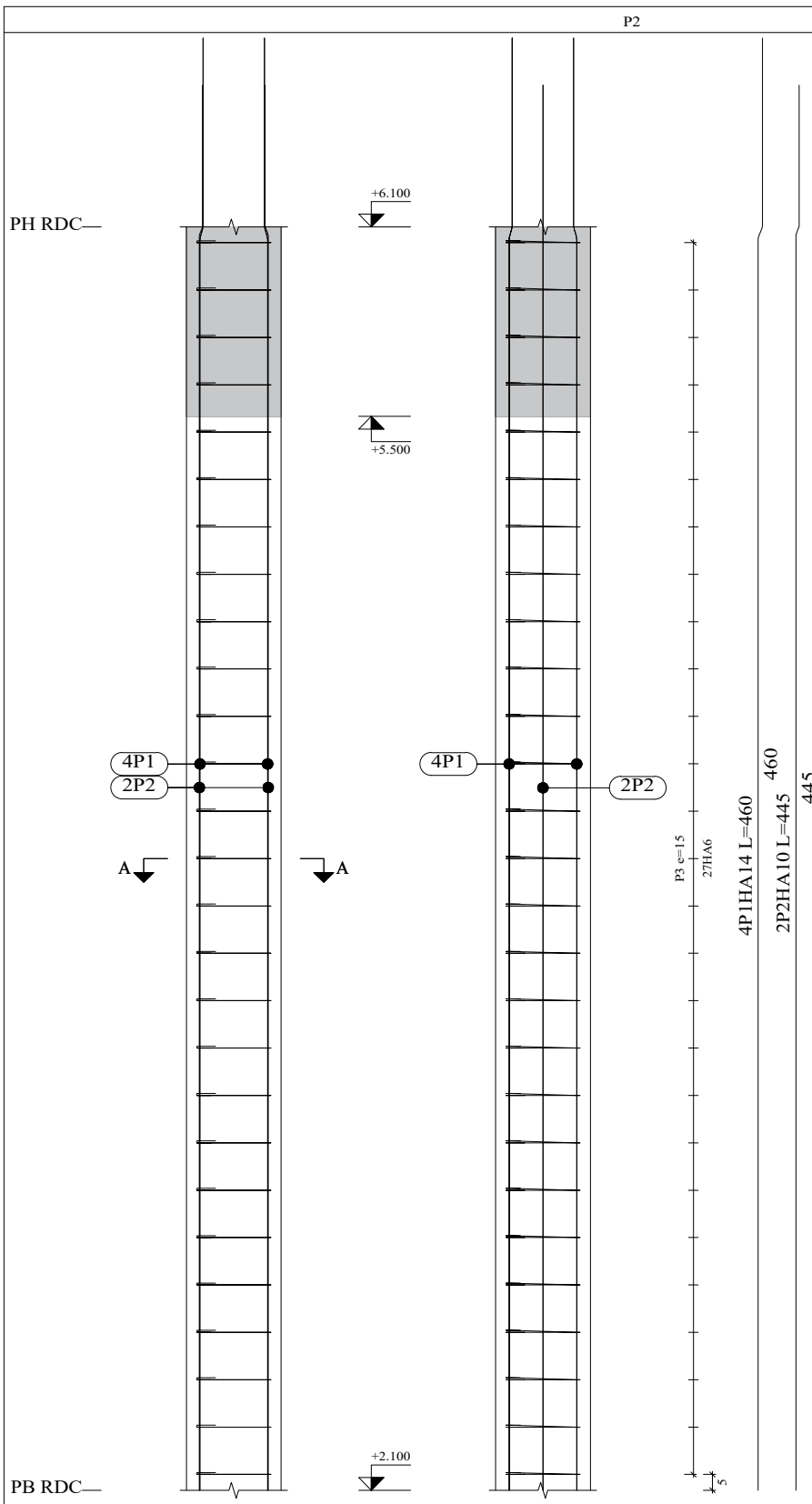
Acier: Fe E400 (43.0 kg). Ratio: 81.50 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.48 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P2	1	HA14	4	397 60	460	1840	22.2
	2	HA10	2	398 45	445	890	5.5
	3	HA6	27	24 24 6	107	2889	6.4
					Total+10%:		37.5
					HA6:		7.0
					HA10:		6.1
					HA14:		24.4
					Total:		37.5

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

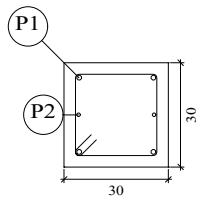
P2

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	460	1840
2	HA10	2	445	890
3	HA6	27	107	2889



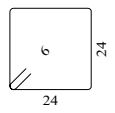
Vue XX

Vue YY



4P1HA14
2P2HA10

Section A-A



P3HA6e=15 L=107

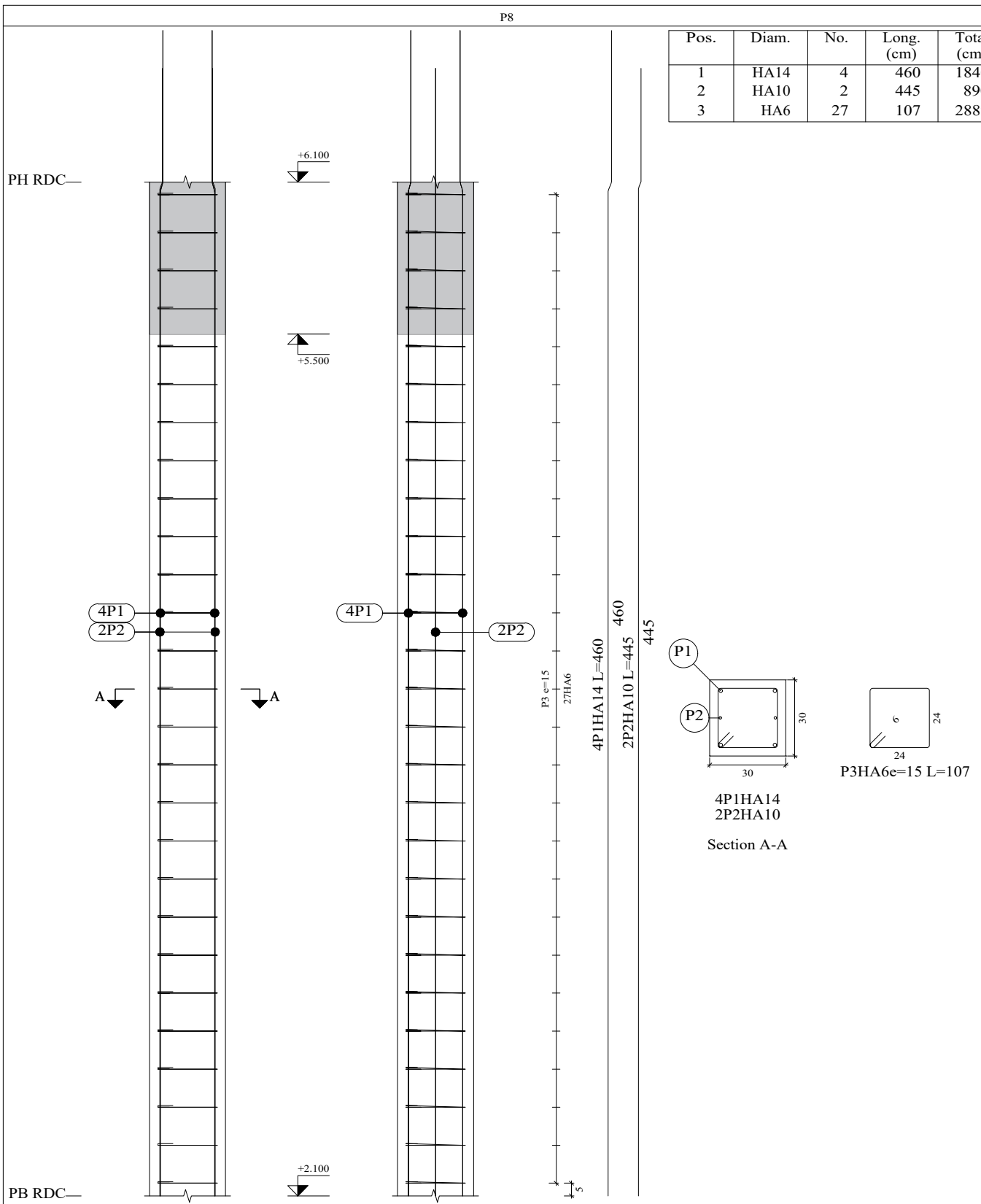
Acier: Fe E400 (37.5 kg). Ratio: 94.82 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.36 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.80 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P8	1	HA14	4	397 60	460	1840	22.2
	2	HA10	2	398 45	445	890	5.5
	3	HA6	27	24 6 24	107	2889	6.4
					Total+10%:		37.5
					HA6:		7.0
					HA10:		6.1
					HA14:		24.4
					Total:		37.5

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P8

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	460	1840
2	HA10	2	445	890
3	HA6	27	107	2889



4P1HA14
2P2HA10

Section A-A

P3HA6e=15 L=107

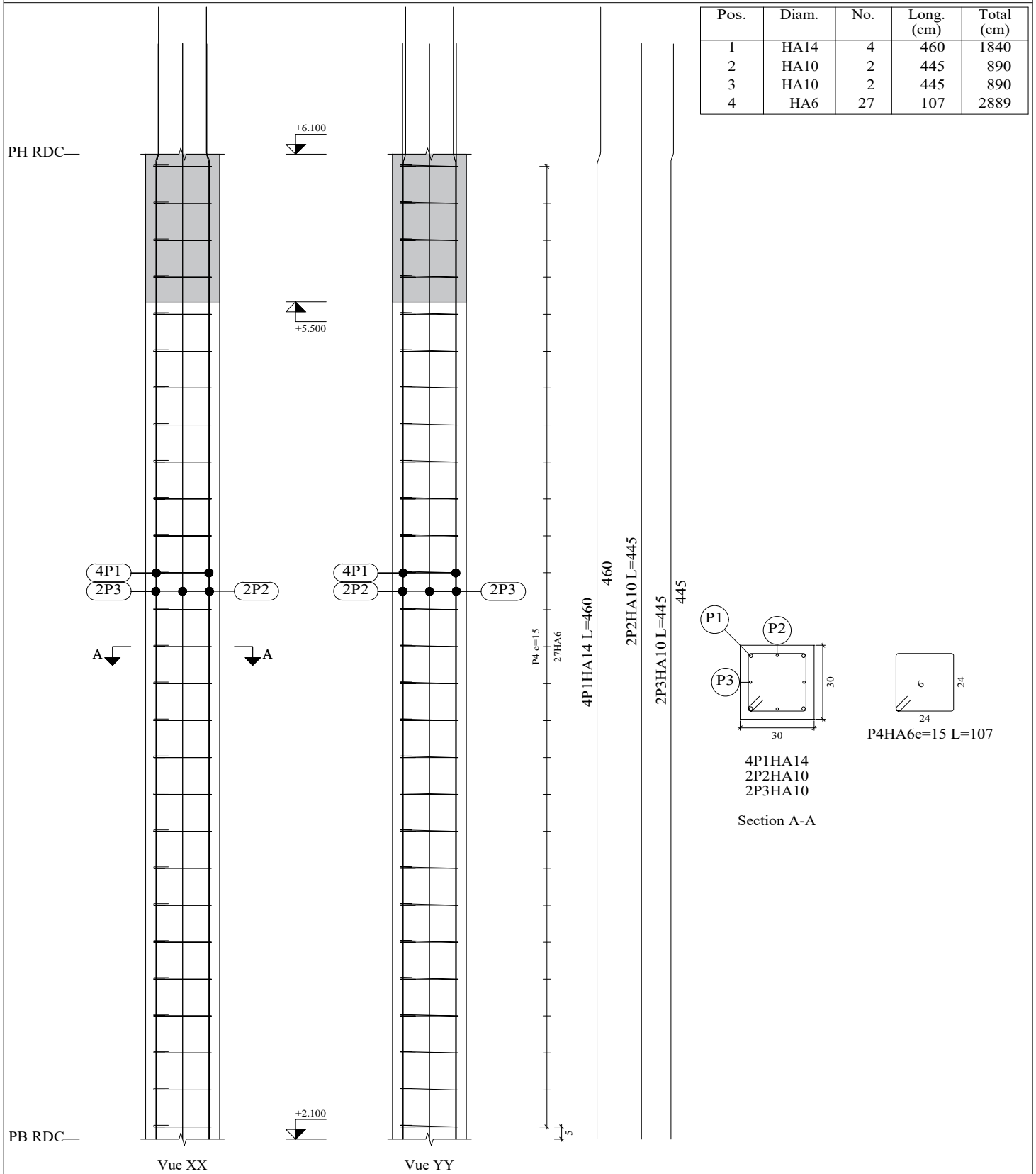
Acier: Fe E400 (37.5 kg). Ratio: 94.82 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.36 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.80 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P7	1	HA14	4	397 60	460	1840	22.2
	2	HA10	2	445	445	890	5.5
	3	HA10	2	398 45	445	890	5.5
	4	HA6	27	24 6 24	107	2889	6.4
					Total+10%:		43.6
					HA6:		7.1
					HA10:		12.1
					HA14:		24.4
					Total:		43.6

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P7

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	460	1840
2	HA10	2	445	890
3	HA10	2	445	890
4	HA6	27	107	2889



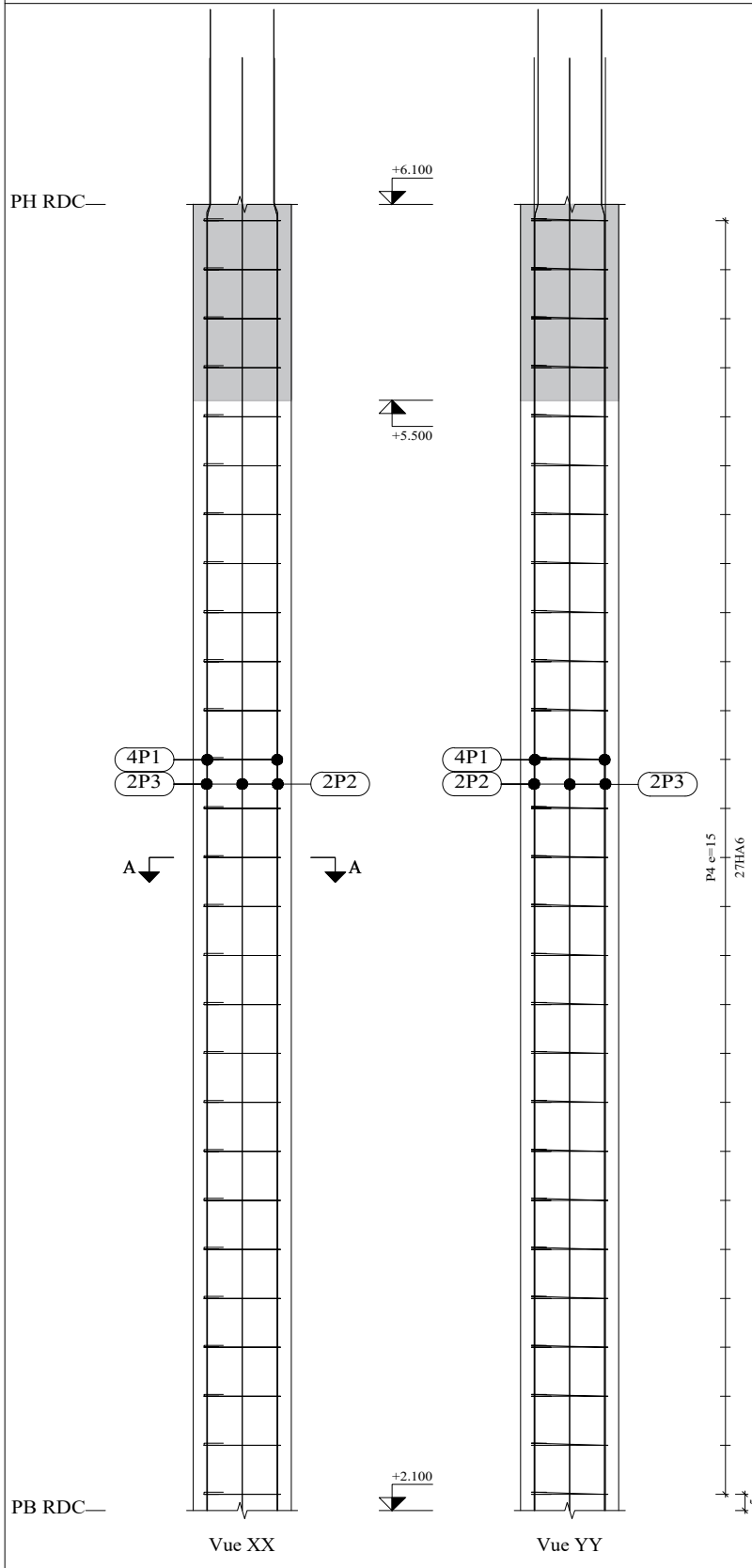
Acier: Fe E400 (43.6 kg). Ratio: 110.06 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.36 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.80 m2	Enrobage géométrique: 3 cm	

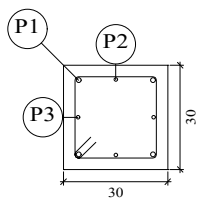
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P12	1	HA14	4	397 60	460	1840	22.2
	2	HA10	2	445	445	890	5.5
	3	HA10	2	398 45	445	890	5.5
	4	HA6	27	24 6 24	107	2889	6.4
					Total+10%:		43.6
					HA6:		7.1
					HA10:		12.1
					HA14:		24.4
					Total:		43.6

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P12

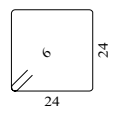
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	460	1840
2	HA10	2	445	890
3	HA10	2	445	890
4	HA6	27	107	2889





4P1HA14
2P2HA10
2P3HA10

Section A-A



P4HA6e=15 L=107

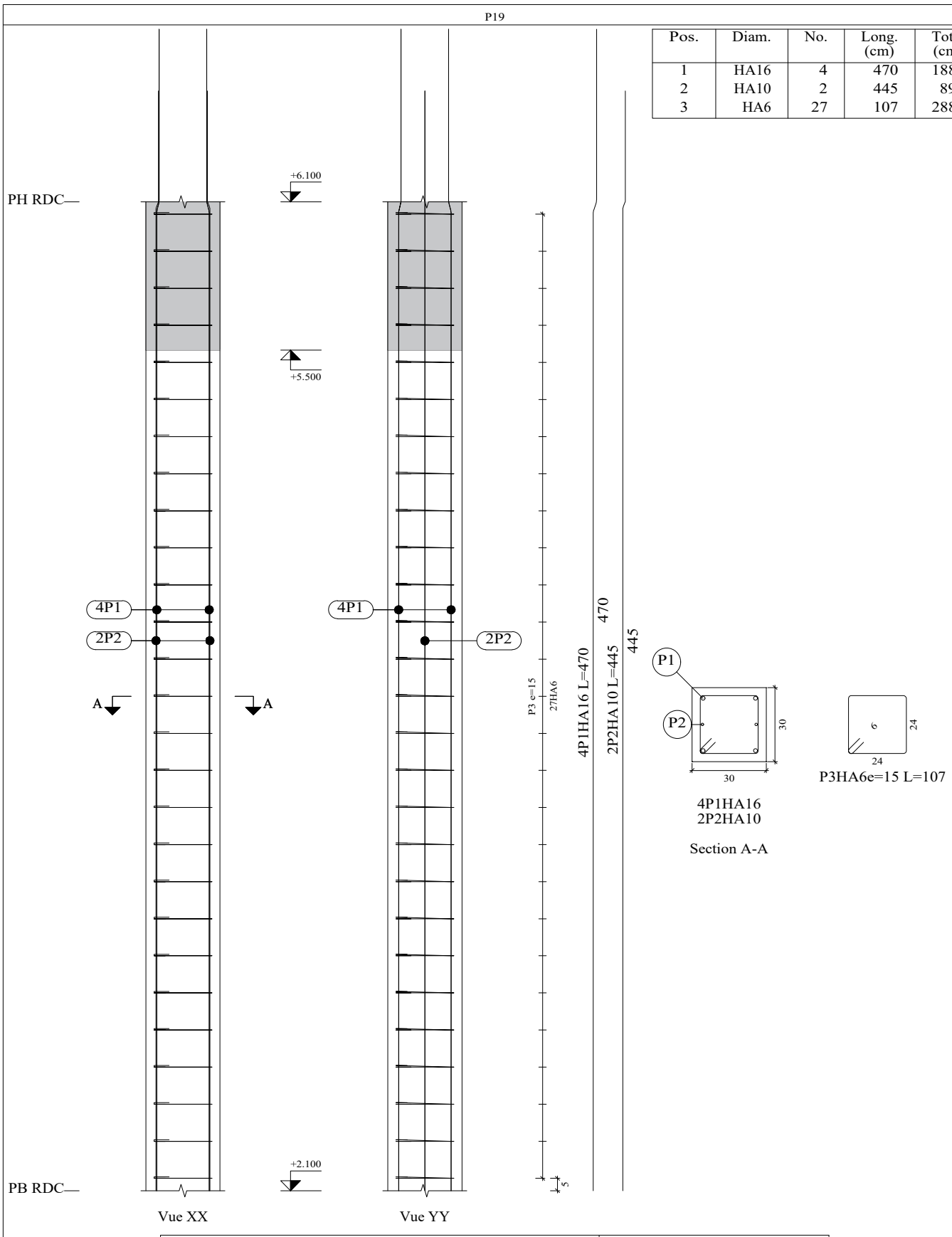
Acier: Fe E400 (43.6 kg). Ratio: 110.06 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.36 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.80 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P19	1	HA16	4	396 / 70	470	1880	29.7
	2	HA10	2	398 / 45	445	890	5.5
	3	HA6	27	24 × 24, 6	107	2889	6.4
					Total+10%:		45.8
					HA6:		7.1
					HA10:		6.0
					HA16:		32.7
					Total:		45.8

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P19

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA16	4	470	1880
2	HA10	2	445	890
3	HA6	27	107	2889



4P1HA16
2P2HA10

Section A-A

P3HA6e=15 L=107

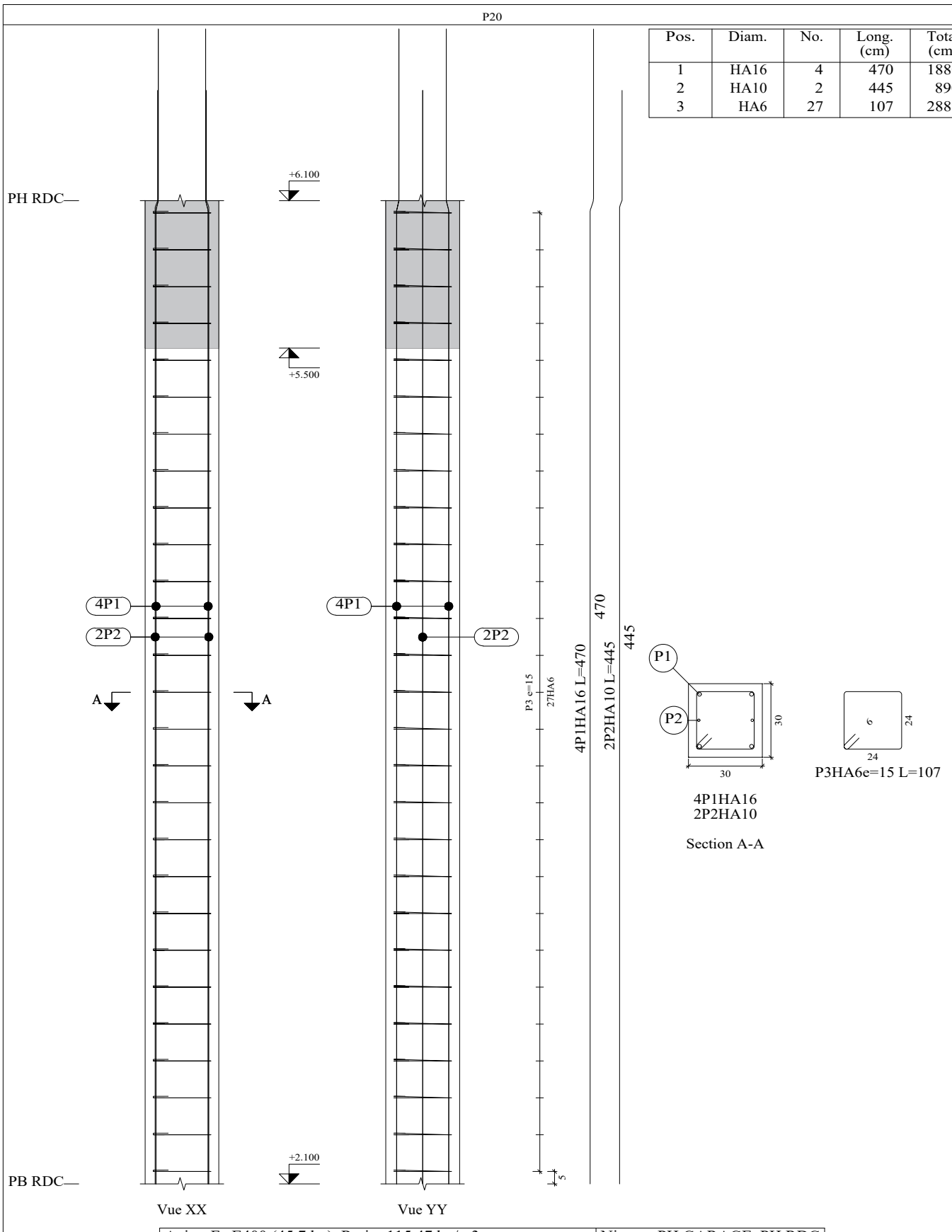
Acier: Fe E400 (45.7 kg). Ratio: 115.47 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.36 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.80 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P20	1	HA16	4	396 / 70	470	1880	29.7
	2	HA10	2	398 / 45	445	890	5.5
	3	HA6	27	24 × 24, 6	107	2889	6.4
					Total+10%:		45.8
					HA6:		7.1
					HA10:		6.0
					HA16:		32.7
					Total:		45.8

Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P20

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA16	4	470	1880
2	HA10	2	445	890
3	HA6	27	107	2889



4P1HA16
2P2HA10

Section A-A

P3HA6e=15 L=107

Acier: Fe E400 (45.7 kg). Ratio: 115.47 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.36 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.80 m2	Enrobage géométrique: 3 cm	

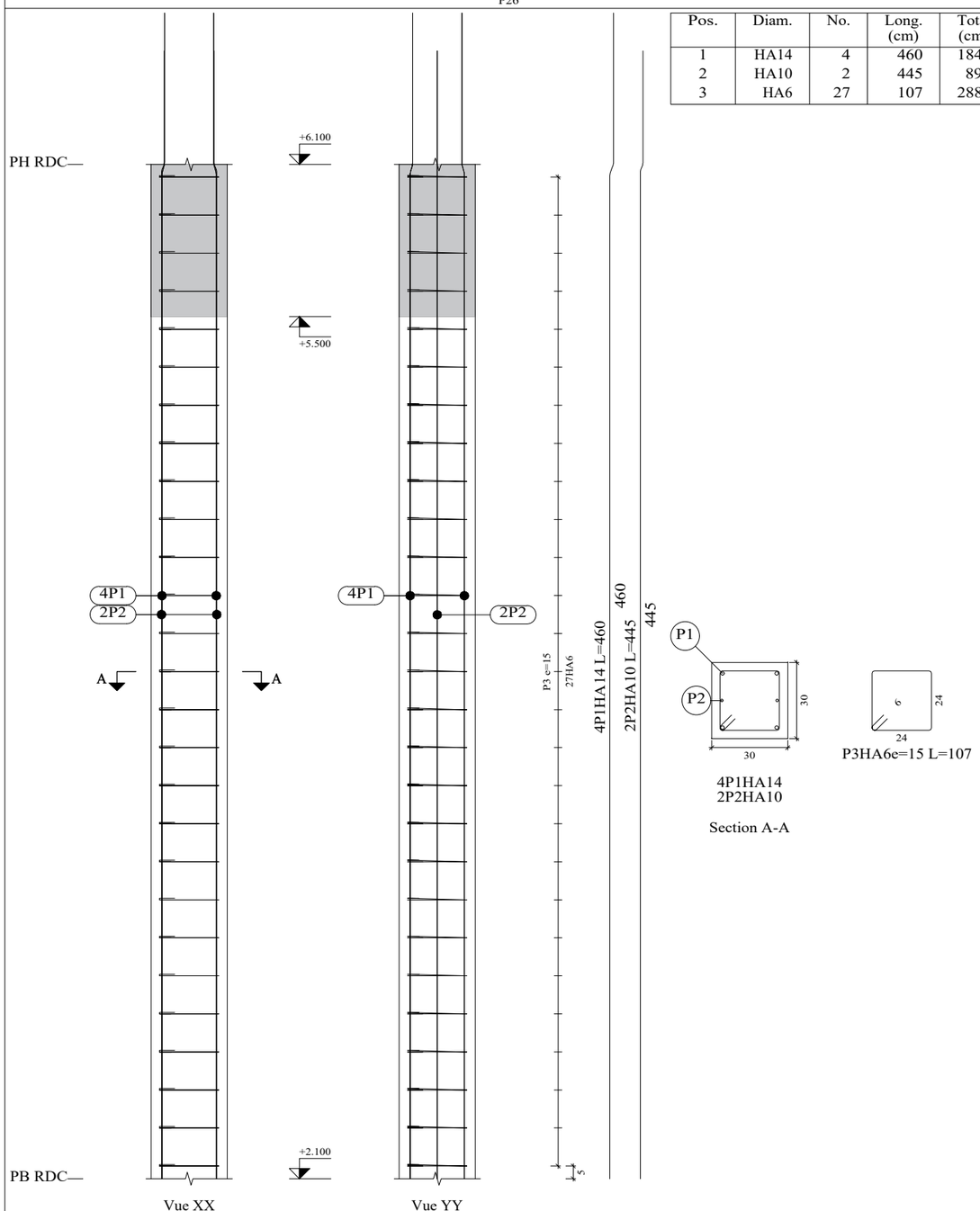
Niveau: PH GARAGE
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Récapitulatif Acier Poteaux		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	988.0	241	
	HA10	282.5	192	
	HA12	339.2	331	
	HA14	185.9	247	
	HA16	75.2	131	1142

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P26	1	HA14	4	397 60	460	1840	22.2
	2	HA10	2	398 45	445	890	5.5
	3	HA6	27	24 6 24	107	2889	6.4
					Total+10%:		37.5
P30	1	HA14	4	277	277	1108	13.4
	2	HA10	2	275 15	290	580	3.6
	3	HA6	19	19 6 19	87	1653	3.7
					Total+10%:		22.8
					HA6:		11.1
					HA10:		10.1
					HA14:		39.1
					Total:		60.3

P26

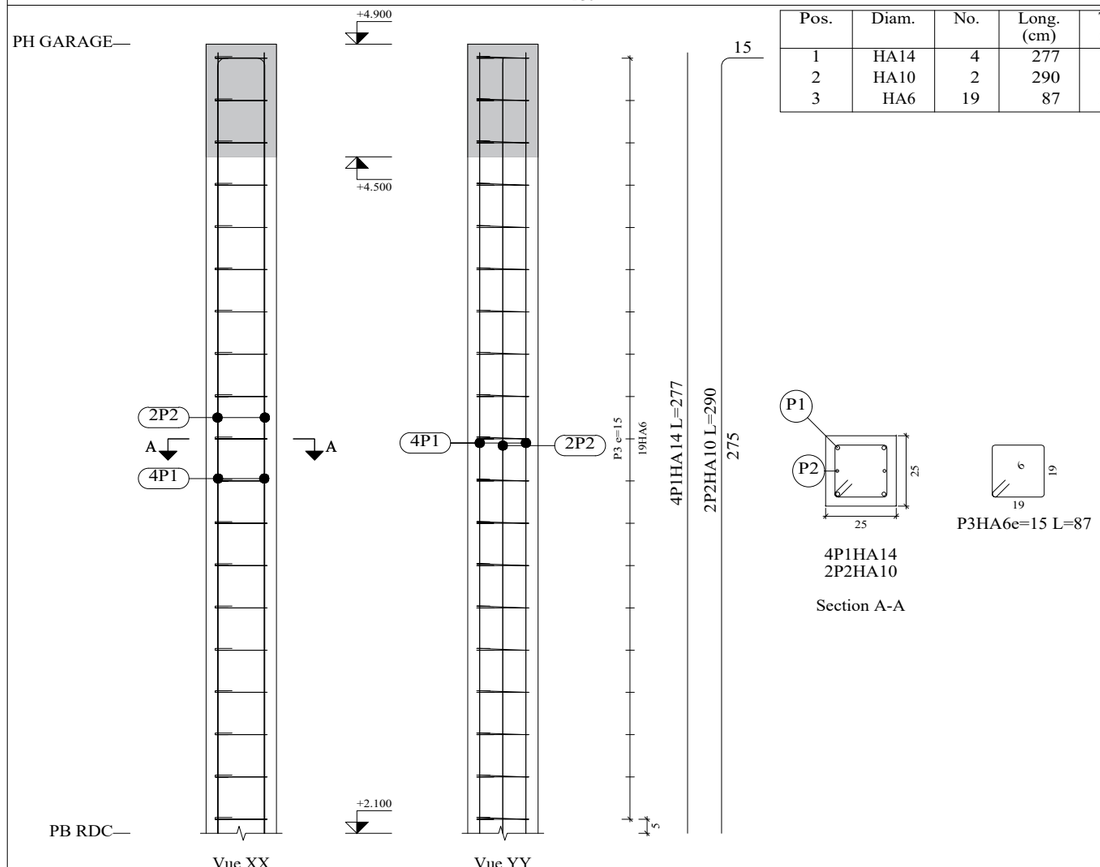
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	460	1840
2	HA10	2	445	890
3	HA6	27	107	2889



Acier: Fe E400 (37.5 kg). Ratio: 94.82 kg/m3		Niveau: PH GARAGE, PH RDC
Béton: B20 (0.36 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.80 m2	Enrobage géométrique: 3 cm	

P30

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	277	1108
2	HA10	2	290	580
3	HA6	19	87	1653



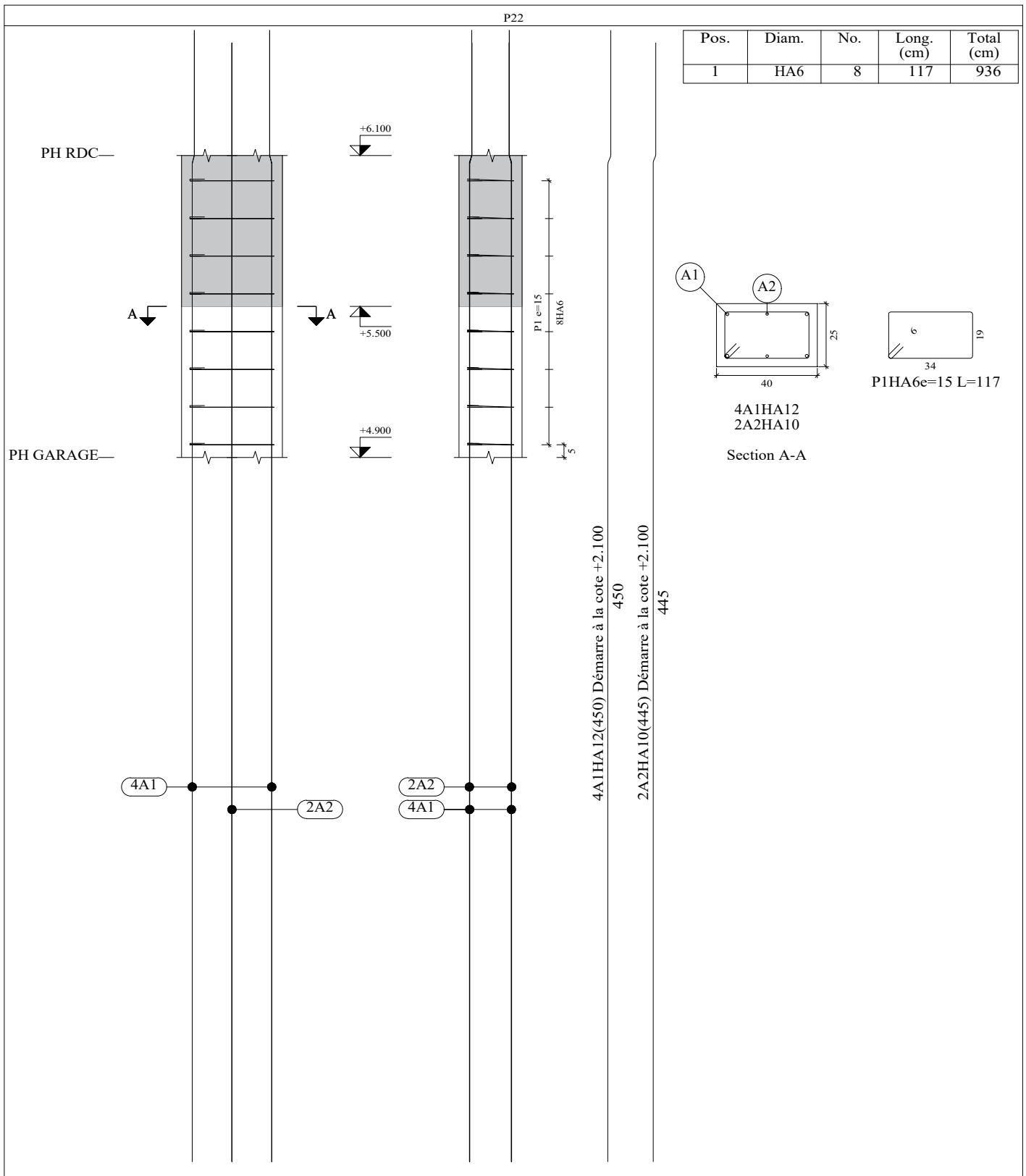
Acier: Fe E400 (22.7 kg). Ratio: 117.91 kg/m3		Niveau: PH GARAGE
Béton: B20 (0.18 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.80 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P22	1	HA6	8	19, 6, 34	117	936	2.1
					Total+10%:		2.3
						HA6:	2.3
						Total:	2.3

Niveau: PH RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P22

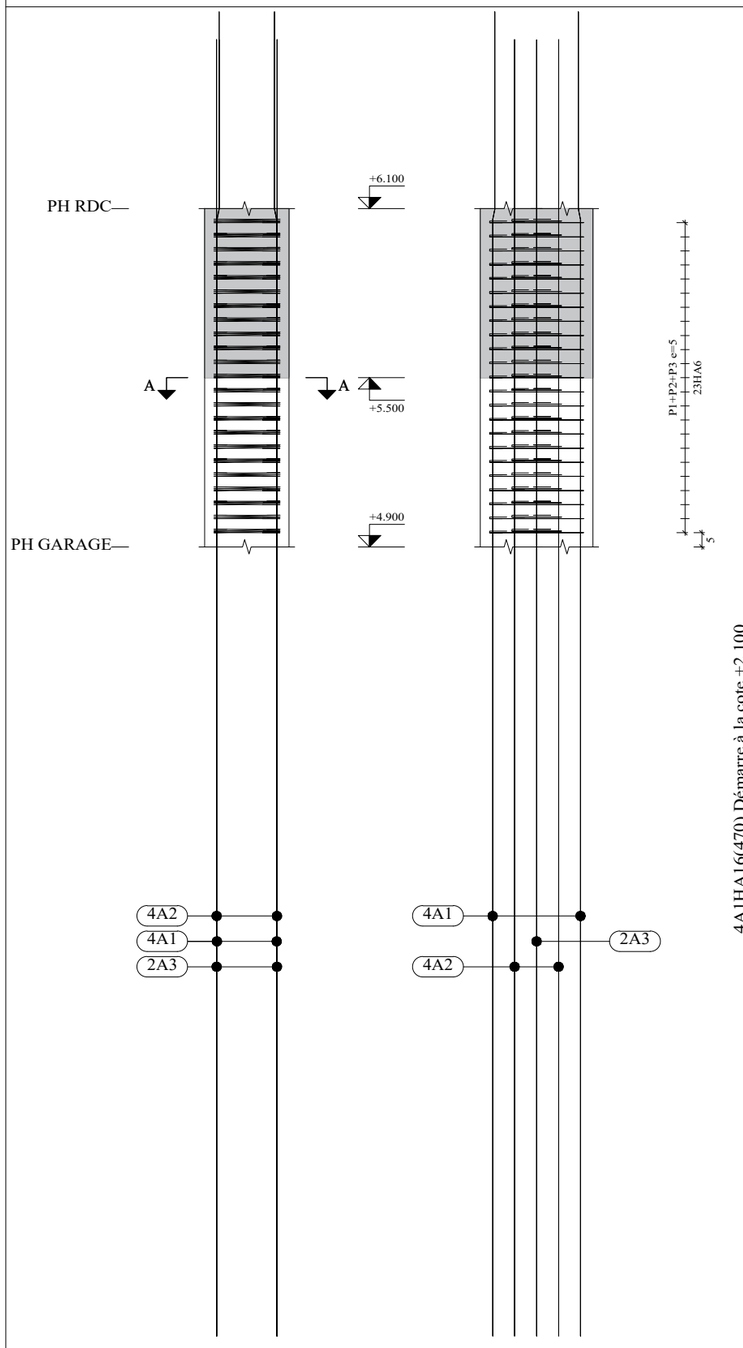
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA6	8	117	936



Acier: Fe E400 (2.3 kg). Ratio: 17.31 kg/m3		Niveau: PH RDC
Béton: B20 (0.12 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 1.56 m2	Enrobage géométrique: 3 cm	

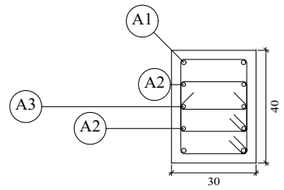
P28

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA6	23	127	2921
2	HA6	23	95	2185
3	HA6	23	38	874



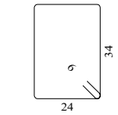
Vue XX

Vue YY



4A1HA16
4A2HA14
2A3HA14

Section A-A



P1HA6e=5 L=127



P2HA6e=5 L=95



P3HA6e=5 L=38

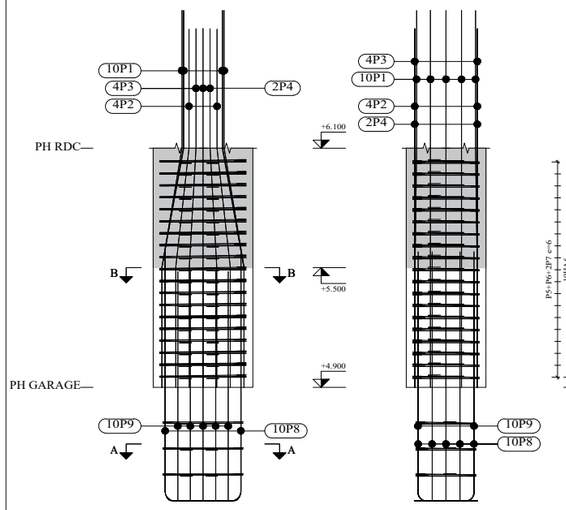
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P28	1	HA6	23		127	2921	6.5
	2	HA6	23		95	2185	4.8
	3	HA6	23		38	874	1.9
				Total+10%:			14.5
				HA6:			14.5
				Total:			14.5

Acier: Fe E400 (14.6 kg). Ratio: 92.16 kg/m3	Niveau: PH RDC
Béton: B20 (0.14 m3) — Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 1.68 m2 — Enrobage géométrique: 3 cm	

Niveau: PH RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

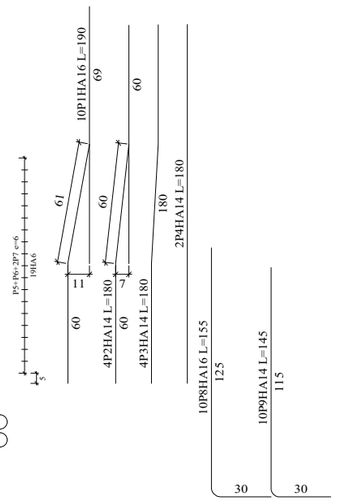
P35

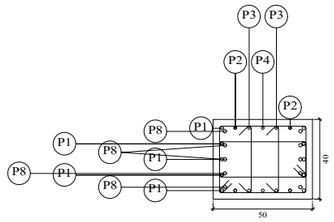
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA16	10	190	1900
2	HA14	4	180	720
3	HA14	4	180	720
4	HA14	2	180	360
5	HA6	19	167	3173
6	HA6	19	136	2584
7	HA6	38	48	1824
8	HA16	10	155	1550
9	HA14	10	145	1450
10	HA6	3	154	462
11	HA6	6	45	270
12	HA6	3	55	165



Vue XX

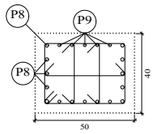
Vue YY





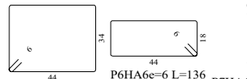
10P8HA16
14P1HA16
8P2HA14
4P3HA14
2P4HA14

Section B-B



10P8HA16
10P9HA14

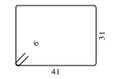
Section A-A



P5HA6e=6 L=167

P6HA6e=6 L=136

P7HA6e=6 L=48



P10HA6e=13 L=154

P11HA6e=13 L=45

P12HA6e=13 L=55

Acier: Fe E400 (75.4 kg). Ratio: 285.68 kg/m3		Niveau: PH RDC
Béton: B20 (0.24 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.16 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P35	1	HA16	10	60 61 69	190	1900	30.0
	2	HA14	4	60 60 60	180	720	8.7
	3	HA14	4	60 60 60	180	720	8.7
	4	HA14	2	180	180	360	4.4
	5	HA6	19	34 / 6 / 44	167	3173	7.0
	6	HA6	19	44 / 6 / 18	136	2584	5.7
	7	HA6	38	6 / 34	48	1824	4.0
	8	HA16	10	30 / 125	155	1550	24.5
	9	HA14	10	30 / 115	145	1450	17.5
	10	HA6	3	31 / 6 / 41	154	462	1.0
	11	HA6	6	6 / 31	45	270	0.6
	12	HA6	3	6 / 41	55	165	0.4
				Total+10%:			123.8
					HA6:		20.6
					HA14:		43.2
					HA16:		60.0
					Total:		123.8

Niveau: PH RDC
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

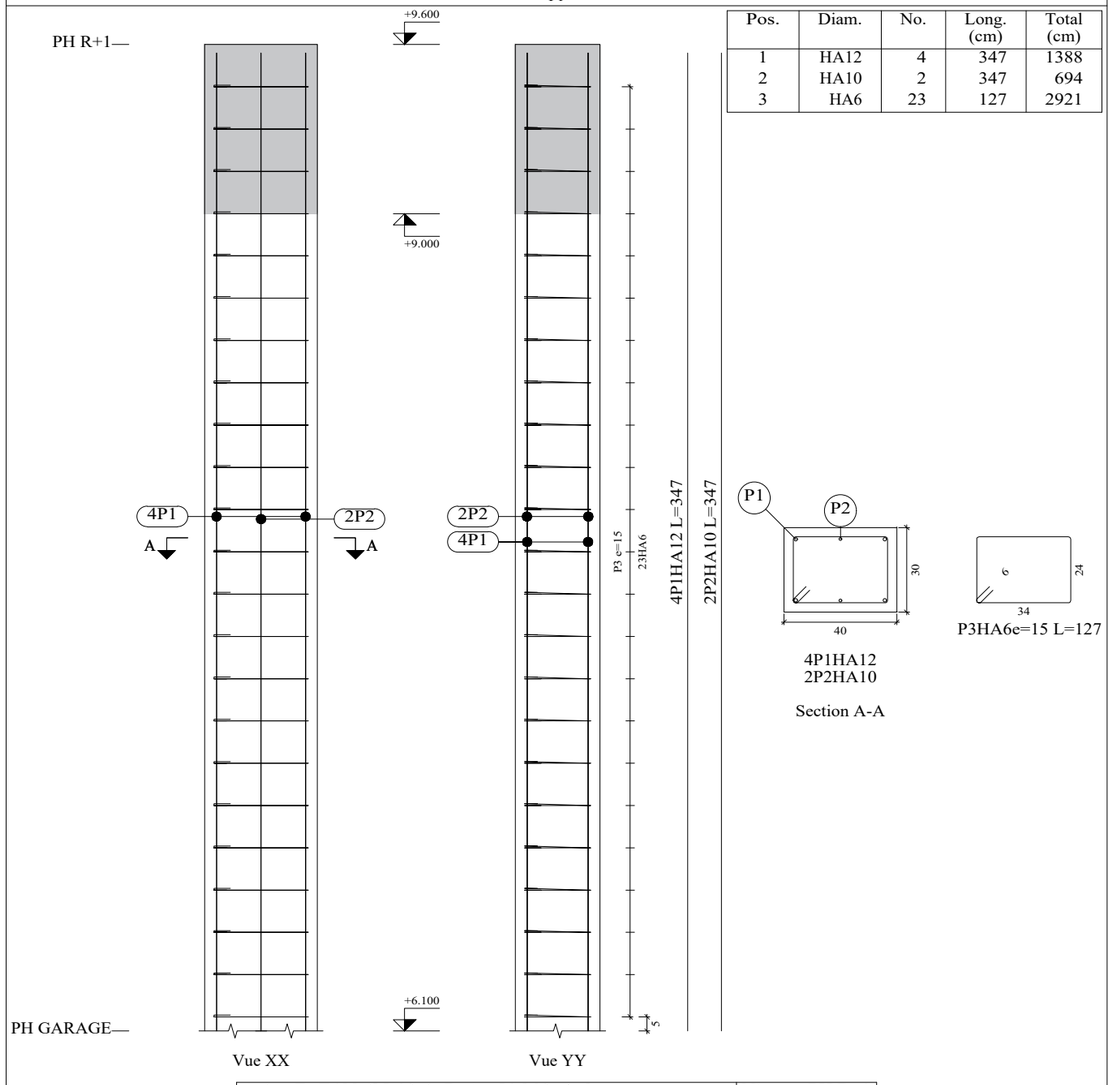
Récapitulatif Acier Poteaux		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	153.9	38	
	HA14	32.5	43	
	HA16	34.5	60	141

Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P1	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24, 6, 34	127	2921	6.5
						Total+10%:	25.4
P3	1	HA14	4	347	347	1388	16.8
	2	HA12	2	347	347	694	6.2
	3	HA6	23	24, 6, 34	127	2921	6.5
						Total+10%:	32.5
						HA6:	14.3
						HA10:	4.8
						HA12:	20.3
						HA14:	18.5
						Total:	57.9

P1

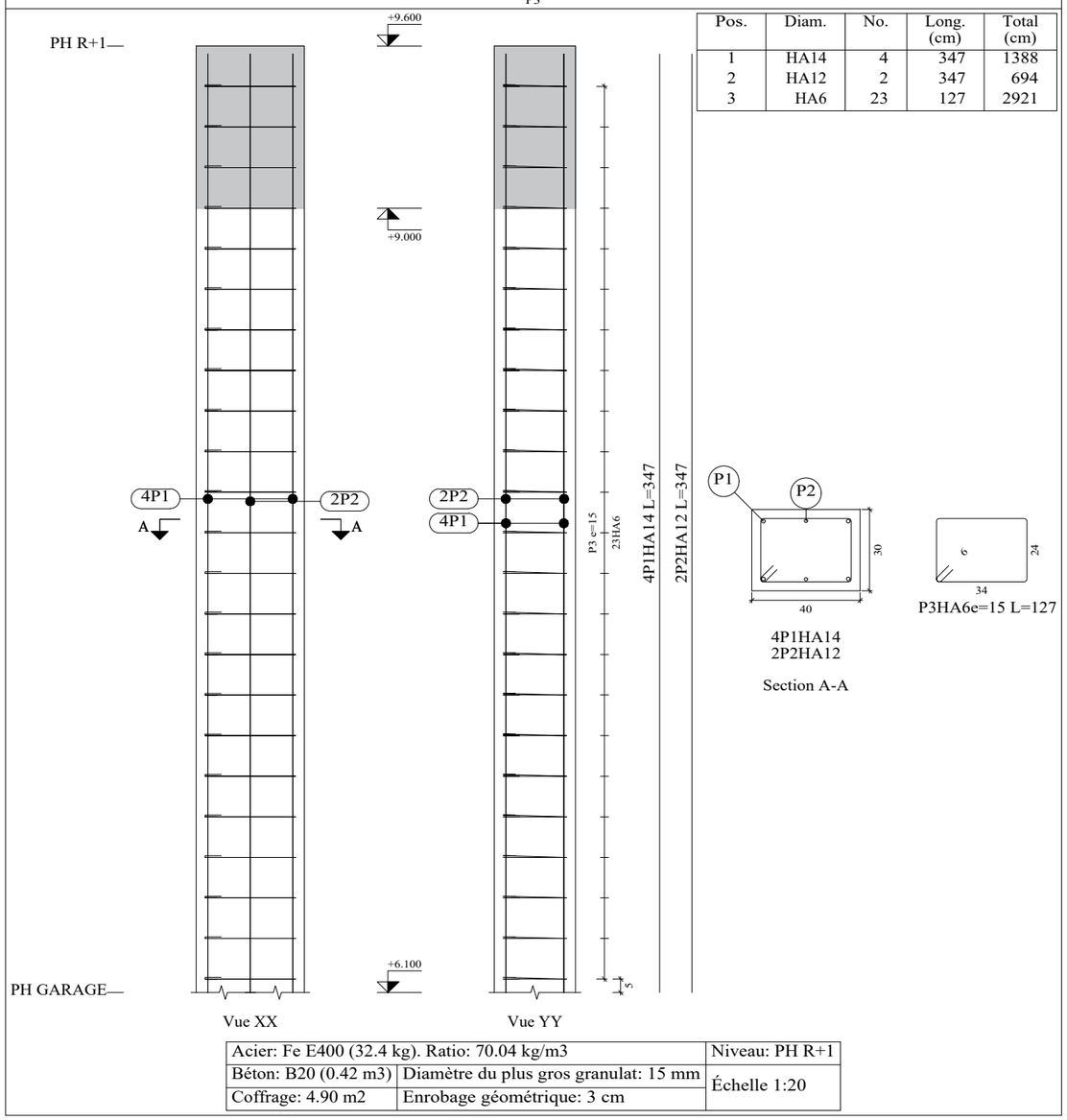
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	127	2921



Acier: Fe E400 (25.4 kg). Ratio: 54.96 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

P3

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	347	1388
2	HA12	2	347	694
3	HA6	23	127	2921



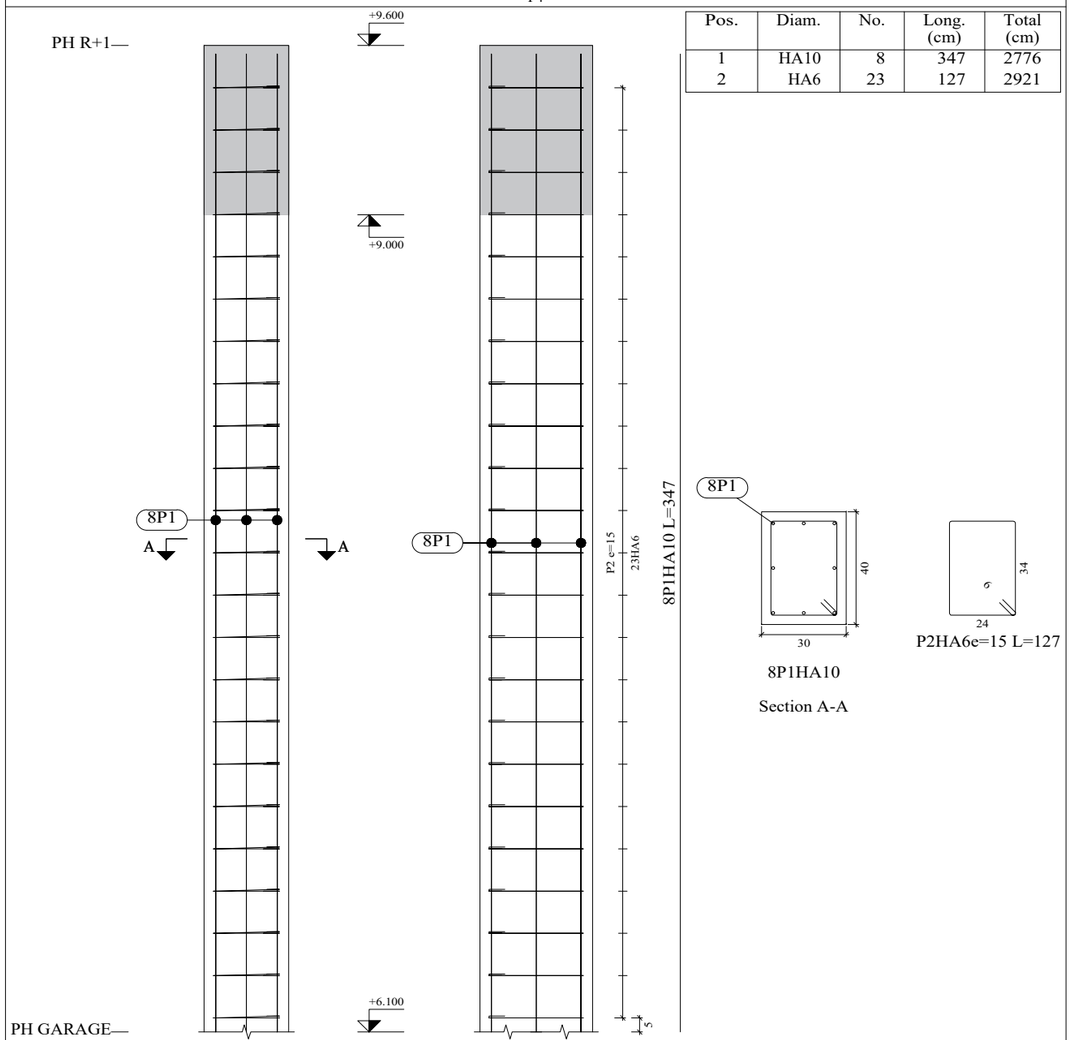
Acier: Fe E400 (32.4 kg). Ratio: 70.04 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

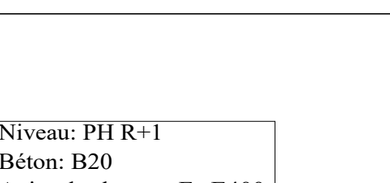
Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P4	1	HA10	8	347	347	2776	17.1
	2	HA6	23	24, 6, 34	127	2921	6.5
						Total+10%:	26.0
P5	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24, 6, 34	127	2921	6.5
						Total+10%:	25.4
						HA6:	14.3
						HA10:	23.6
						HA12:	13.5
						Total:	51.4

P4

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA10	8	347	2776
2	HA6	23	127	2921

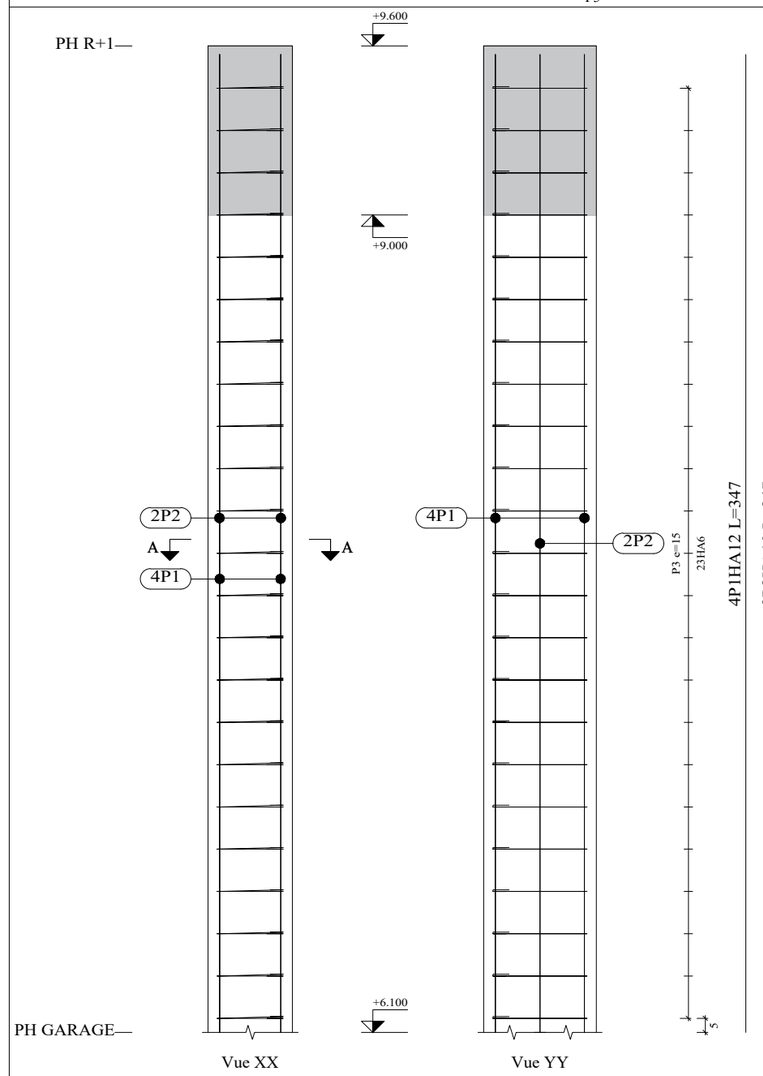


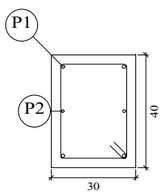


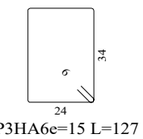
Acier: Fe E400 (26.0 kg). Ratio: 56.18 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

P5

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	127	2921







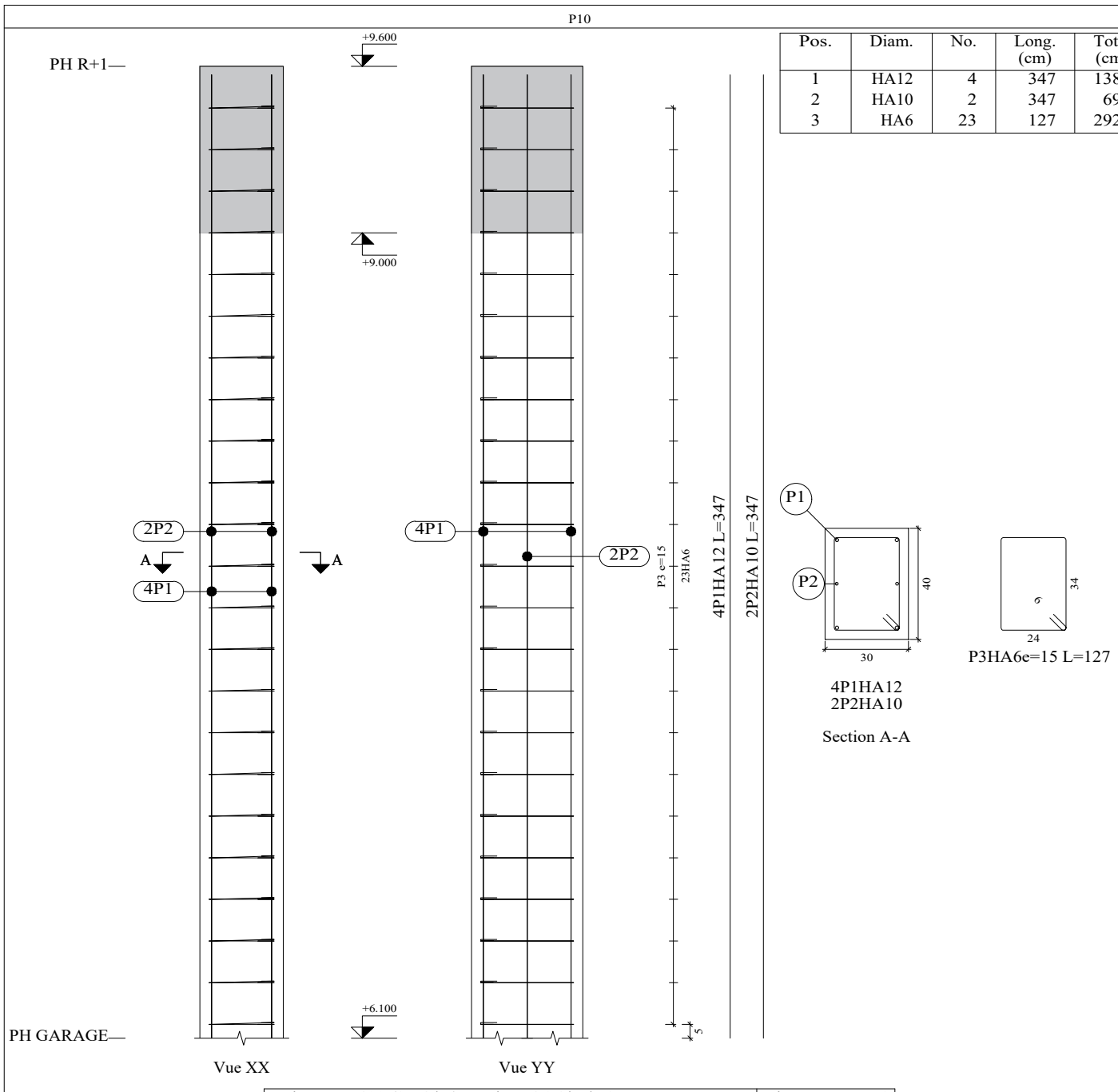
Acier: Fe E400 (25.4 kg). Ratio: 54.96 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P10	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24, 6, 34	127	2921	6.5
						Total+10%:	25.4
						HA6:	7.1
						HA10:	4.8
						HA12:	13.5
						Total:	25.4

Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P10

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	127	2921



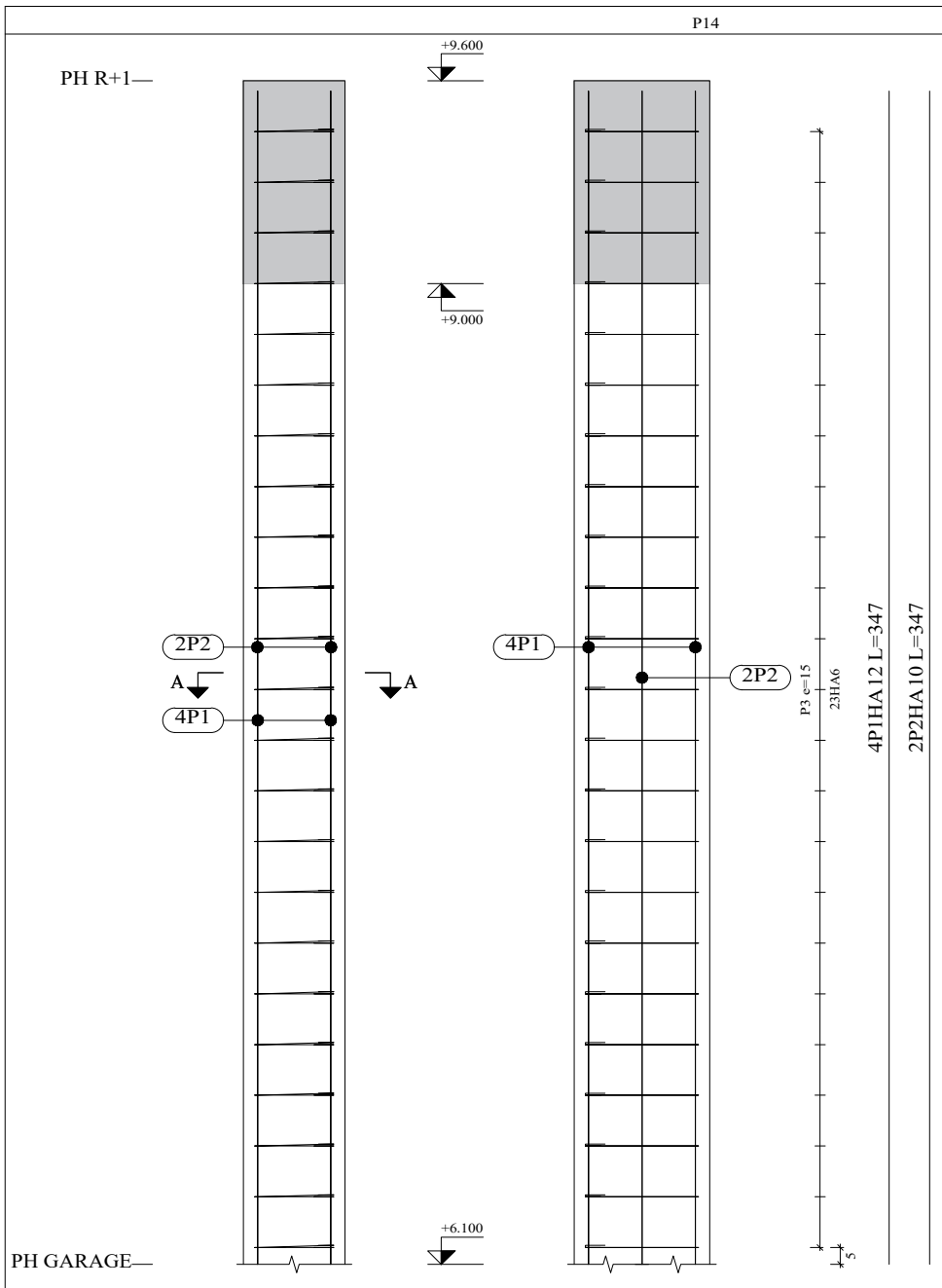
Acier: Fe E400 (25.4 kg). Ratio: 54.96 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

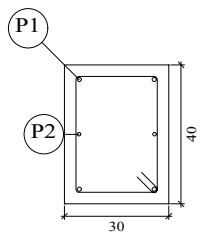
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P14	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24 × 34, 6	127	2921	6.5
						Total+10%:	25.4
						HA6:	7.1
						HA10:	4.8
						HA12:	13.5
						Total:	25.4

Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P14

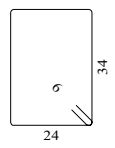
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	127	2921





4P1HA12
2P2HA10

Section A-A



P3HA6e=15 L=127

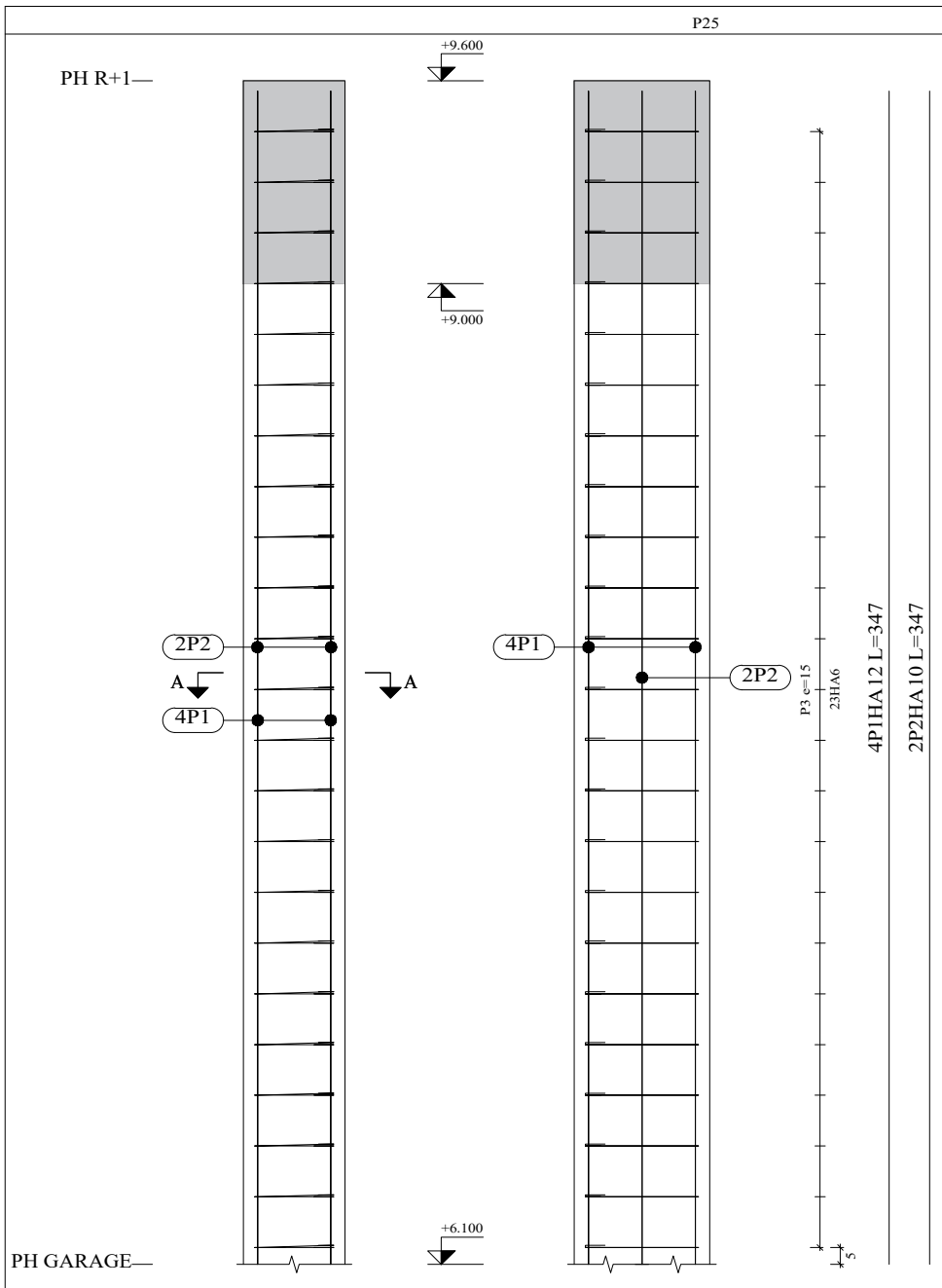
Acier: Fe E400 (25.4 kg). Ratio: 54.96 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P25	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	6 / 24 / 34	127	2921	6.5
						Total+10%:	25.4
						HA6:	7.1
						HA10:	4.8
						HA12:	13.5
						Total:	25.4

Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P25

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	127	2921



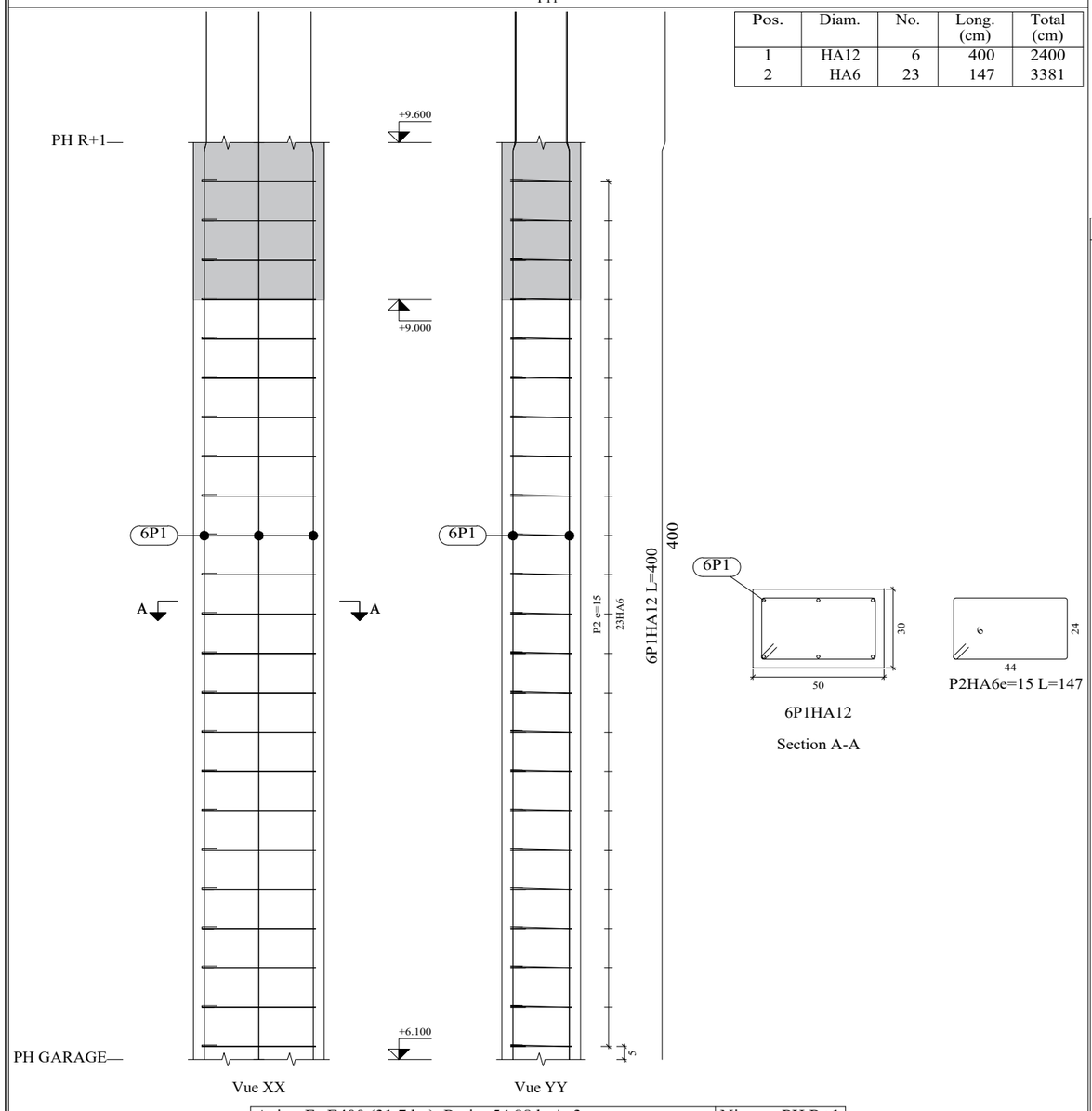
Acier: Fe E400 (25.4 kg). Ratio: 54.96 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P11	1	HA12	6	347 50	400	2400	21.3
	2	HA6	23	24 6 44	147	3381	7.5
						Total+10%:	31.7
P8	1	HA14	4	347	347	1388	16.8
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24 6 24	107	2461	5.5
						Total+10%:	29.3
						HA6:	14.4
						HA10:	4.7
						HA12:	23.4
						HA14:	18.5
						Total:	61.0

P11

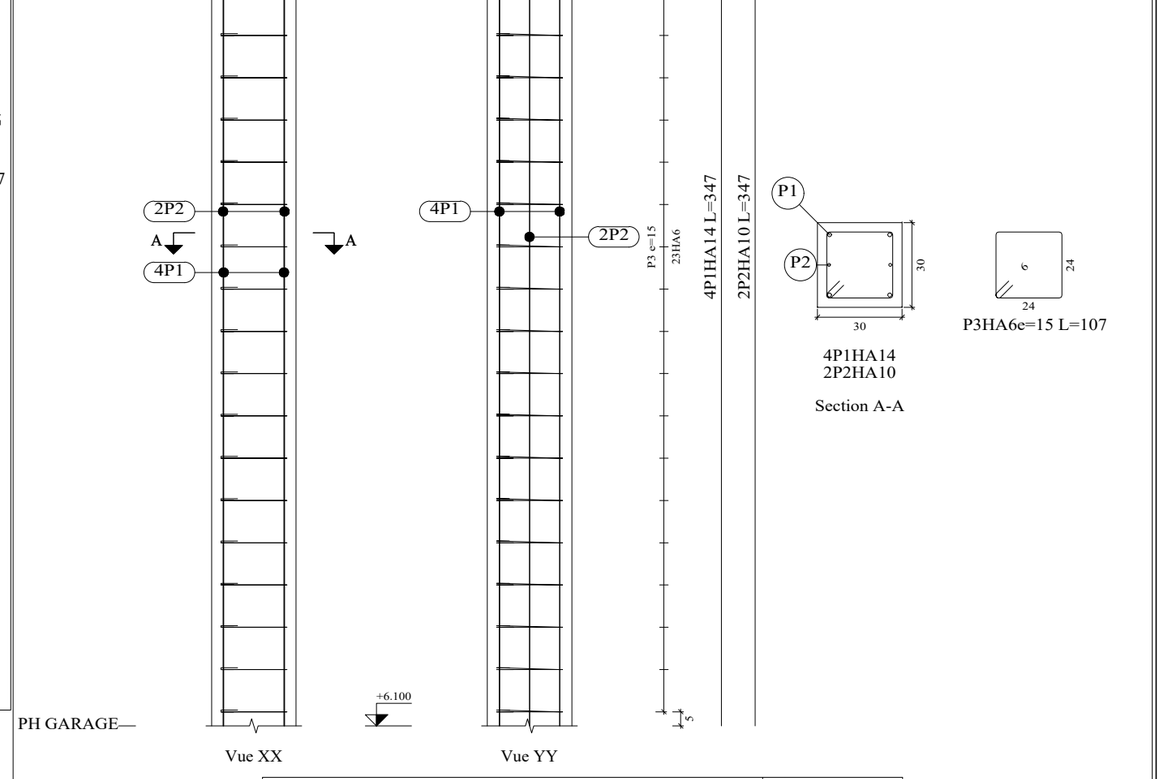
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	6	400	2400
2	HA6	23	147	3381



Acier: Fe E400 (31.7 kg). Ratio: 54.88 kg/m3		Niveau: PH R+1
Béton: B20 (0.53 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

P8

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	347	1388
2	HA10	2	347	694
3	HA6	23	107	2461



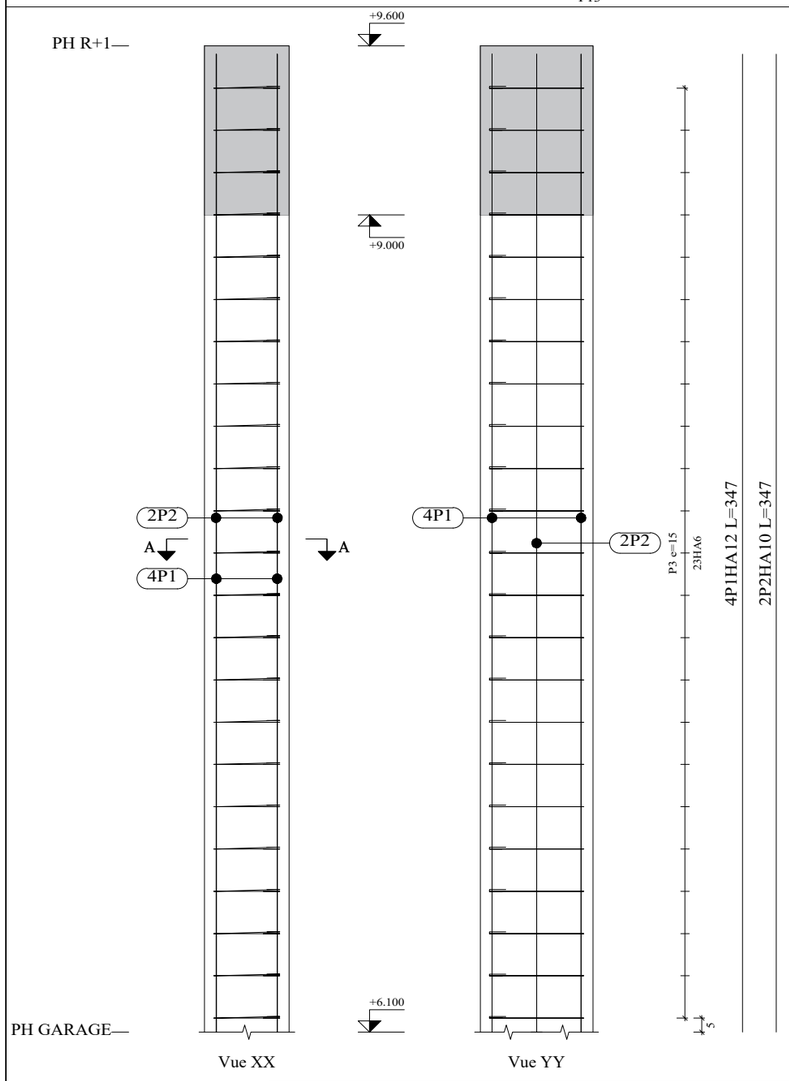
Acier: Fe E400 (29.2 kg). Ratio: 84.17 kg/m3		Niveau: PH R+1
Béton: B20 (0.32 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.20 m2	Enrobage géométrique: 3 cm	

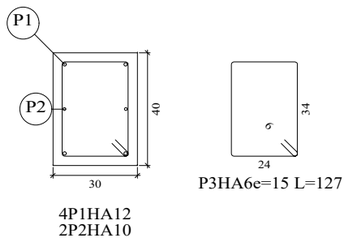
Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P13	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24, 6, 34	127	2921	6.5
					Total+10%:		25.4
P2	1	HA14	4	347	347	1388	16.8
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24, 6, 24	107	2461	5.5
					Total+10%:		29.3
					HA6:		13.2
					HA10:		9.5
					HA12:		13.5
					HA14:		18.5
					Total:		54.7

P13

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	127	2921

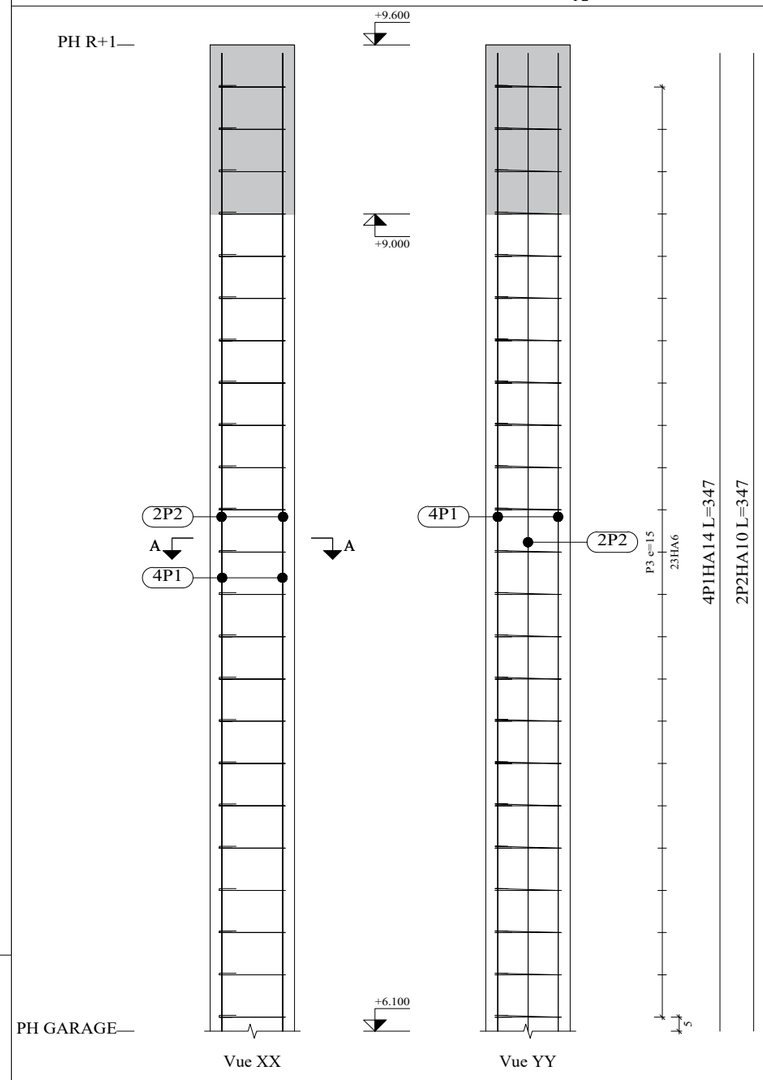


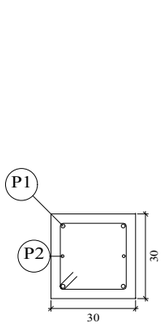


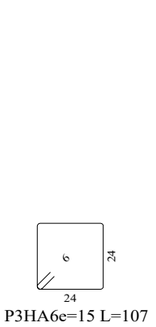
Acier: Fe E400 (25.4 kg). Ratio: 54.96 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

P2

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	347	1388
2	HA10	2	347	694
3	HA6	23	107	2461





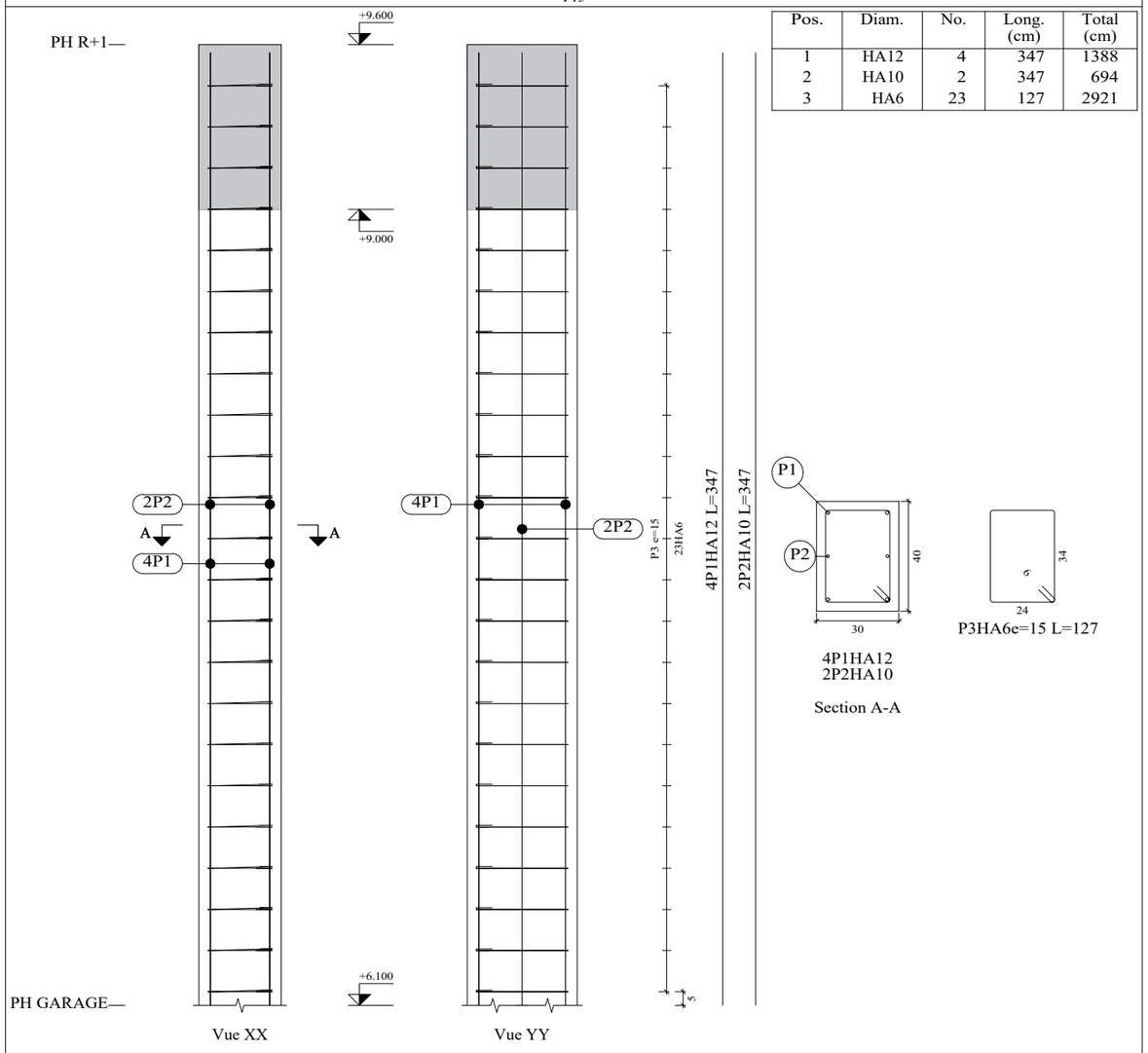


Acier: Fe E400 (29.2 kg). Ratio: 84.17 kg/m3		Niveau: PH R+1
Béton: B20 (0.32 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.20 m2	Enrobage géométrique: 3 cm	

Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P15	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24, 6, 34	127	2921	6.5
					Total+10%:		25.4
P16	1	HA12	6	347	347	2082	18.5
	2	HA6	23	24, 6, 44	147	3381	7.5
					Total+10%:		28.6
					HA6:		15.3
					HA10:		4.8
					HA12:		33.9
					Total:		54.0

P15



Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	127	2921

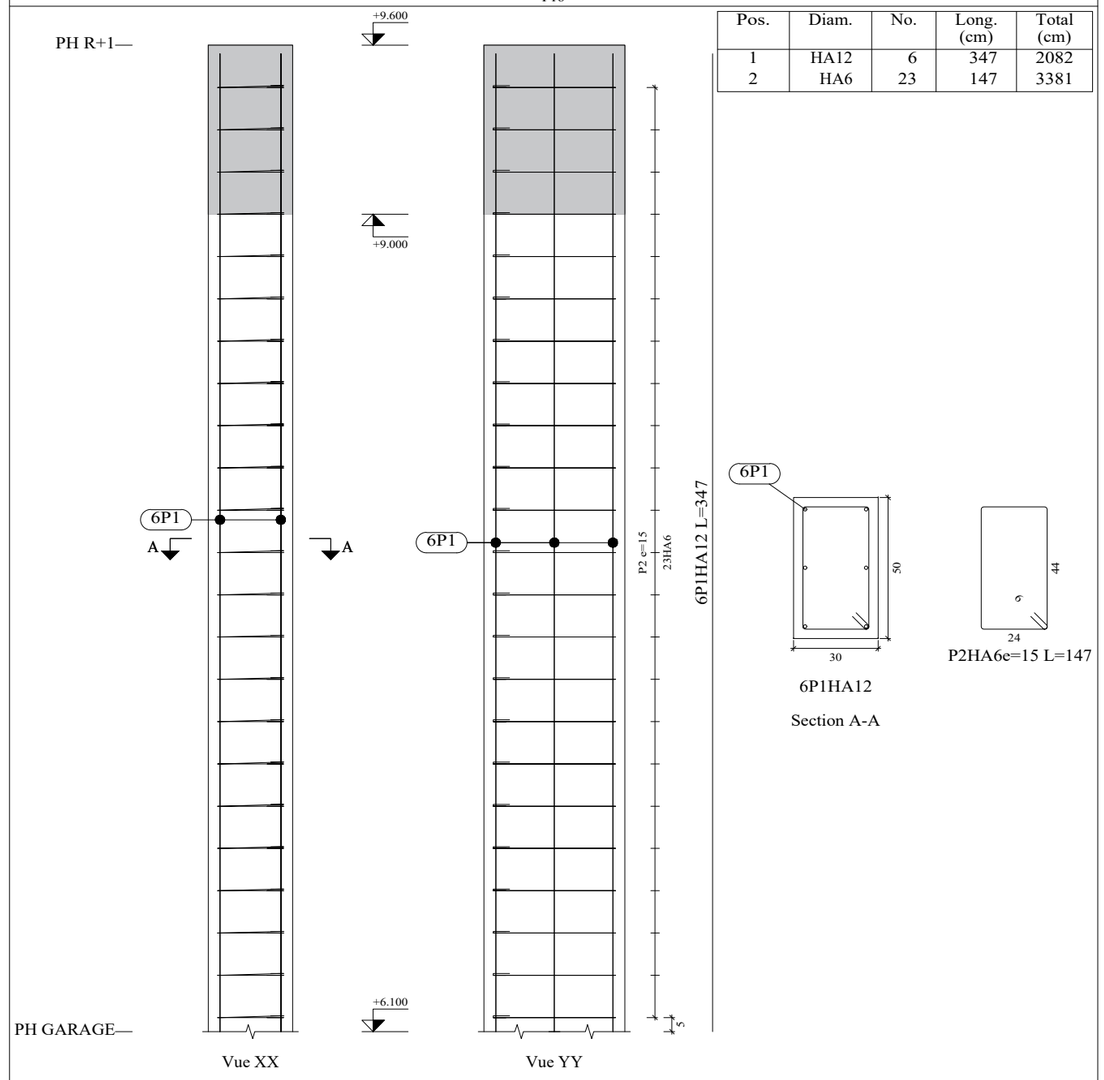
4P1HA12
2P2HA10

Section A-A

P3HA6e=15 L=127

Acier: Fe E400 (25.4 kg). Ratio: 54.96 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

P16



Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	6	347	2082
2	HA6	23	147	3381

6P1HA12

Section A-A

P2HA6e=15 L=147

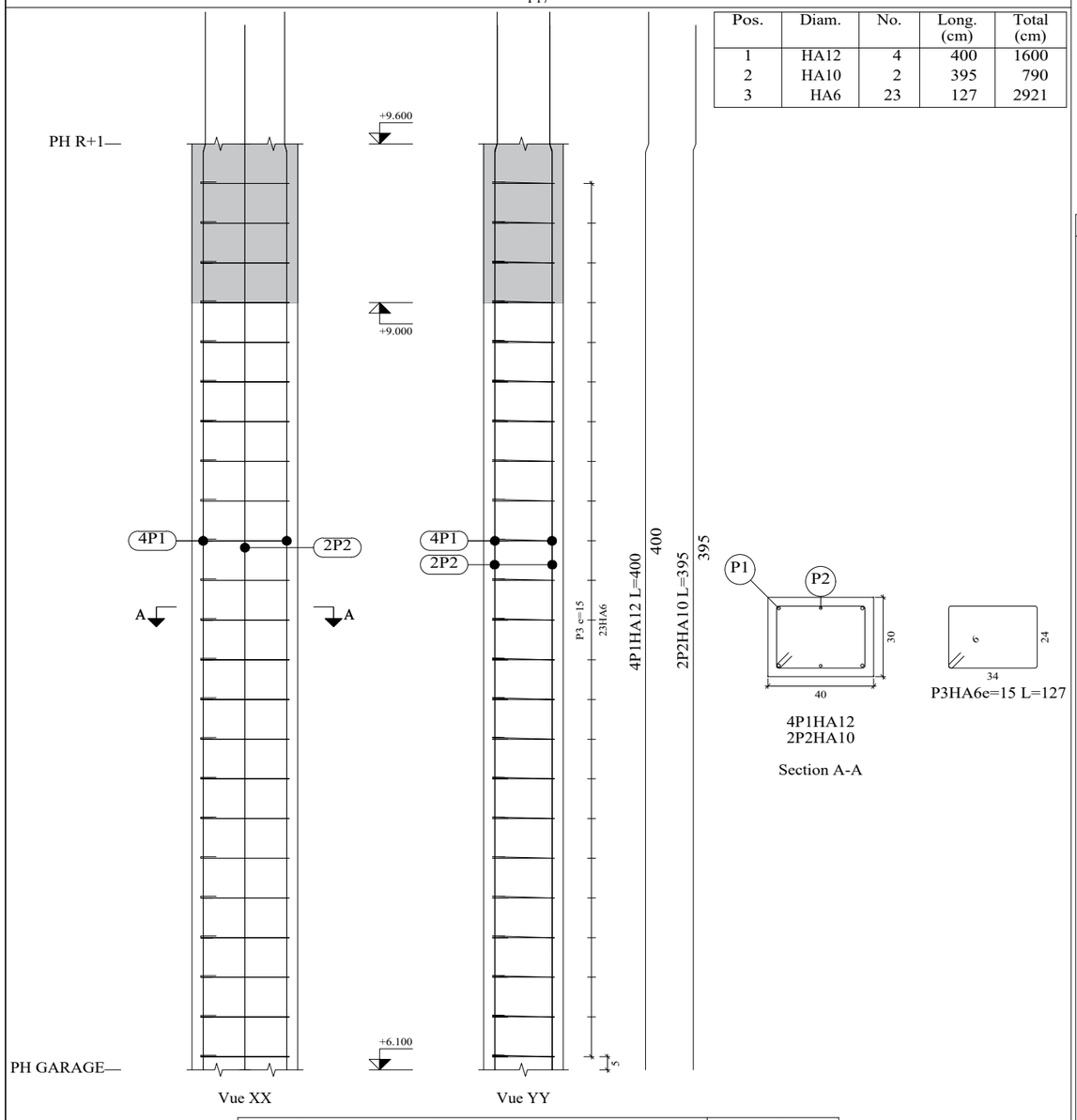
Acier: Fe E400 (28.6 kg). Ratio: 49.50 kg/m3		Niveau: PH R+1
Béton: B20 (0.53 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P17	1	HA12	4	347 50	400	1600	14.2
	2	HA10	2	348 45	395	790	4.9
	3	HA6	23	24 6 34	127	2921	6.5
					Total+10%:		28.2
P21	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24 6 34	127	2921	6.5
					Total+10%:		25.4
					HA6:		14.3
					HA10:		10.2
					HA12:		29.1
					Total:		53.6

P17

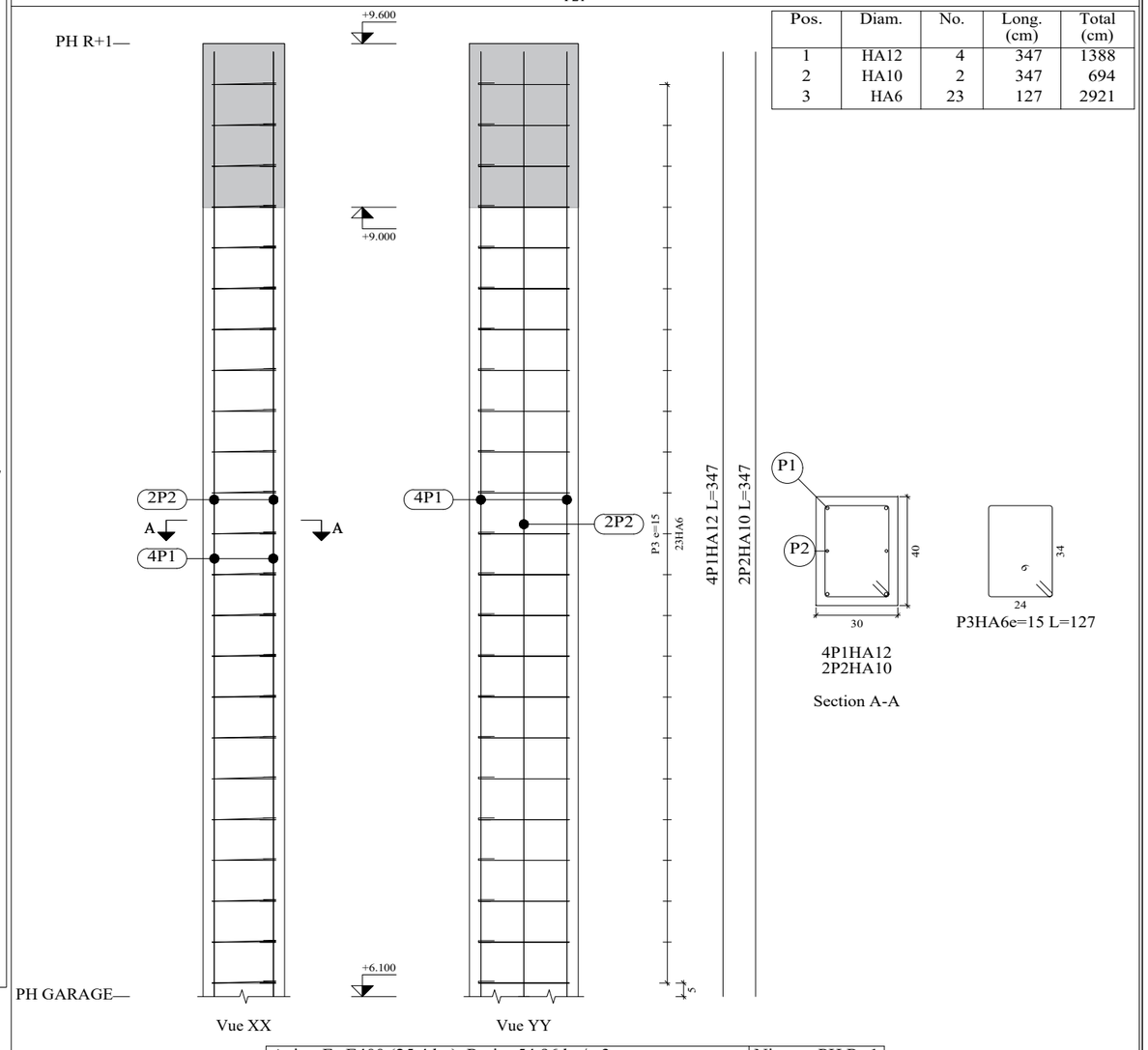
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	400	1600
2	HA10	2	395	790
3	HA6	23	127	2921



Acier: Fe E400 (28.1 kg). Ratio: 60.85 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

P21

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	127	2921



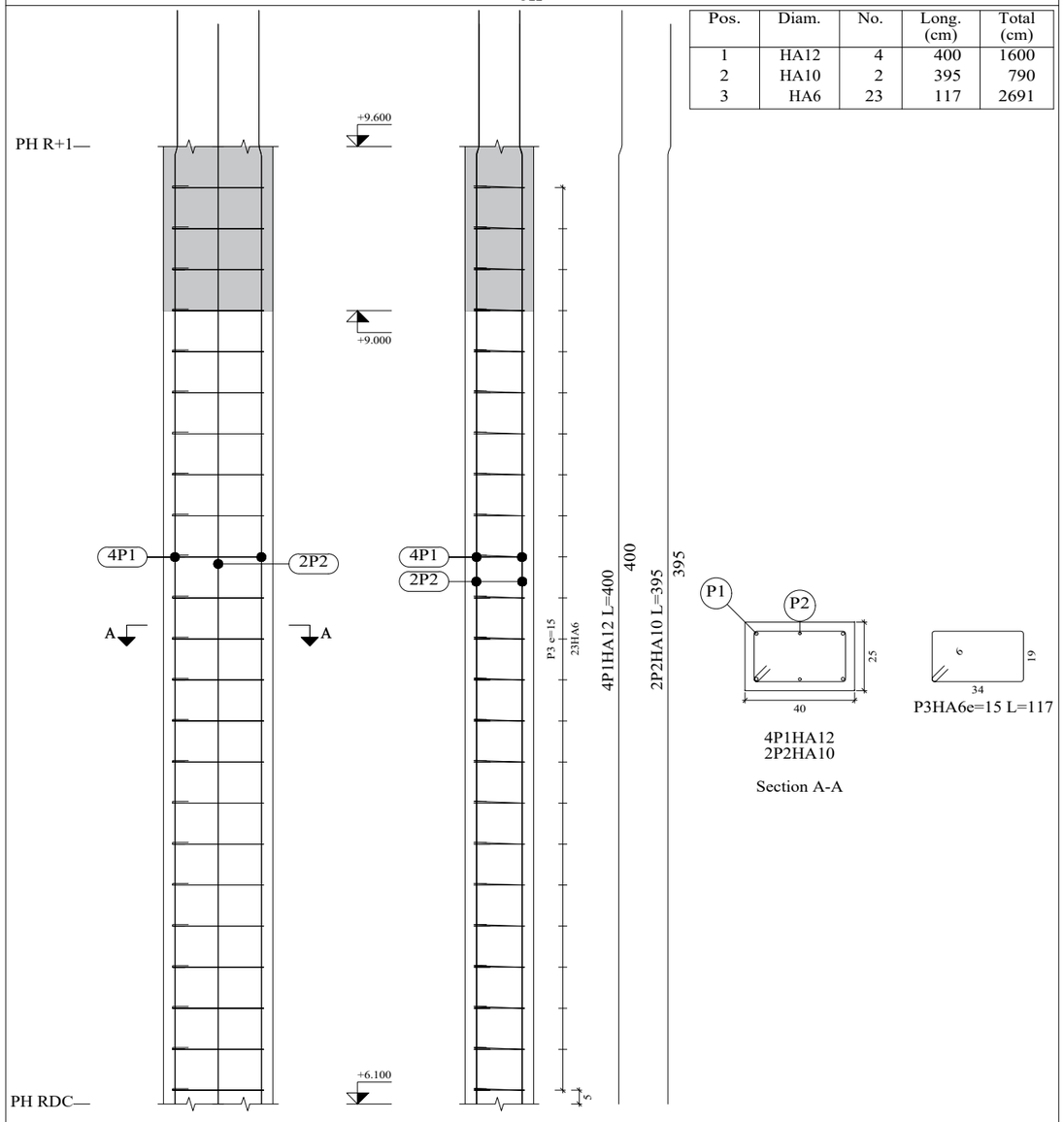
Acier: Fe E400 (25.4 kg). Ratio: 54.96 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P22	1	HA12	4	347 50	400	1600	14.2
	2	HA10	2	348 45	395	790	4.9
	3	HA6	23	19 6 34	117	2691	6.0
					Total+10%:		27.6
P28	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24 6 34	127	2921	6.5
					Total+10%:		25.4
					HA6:		13.7
					HA10:		10.2
					HA12:		29.1
					Total:		53.0

P22

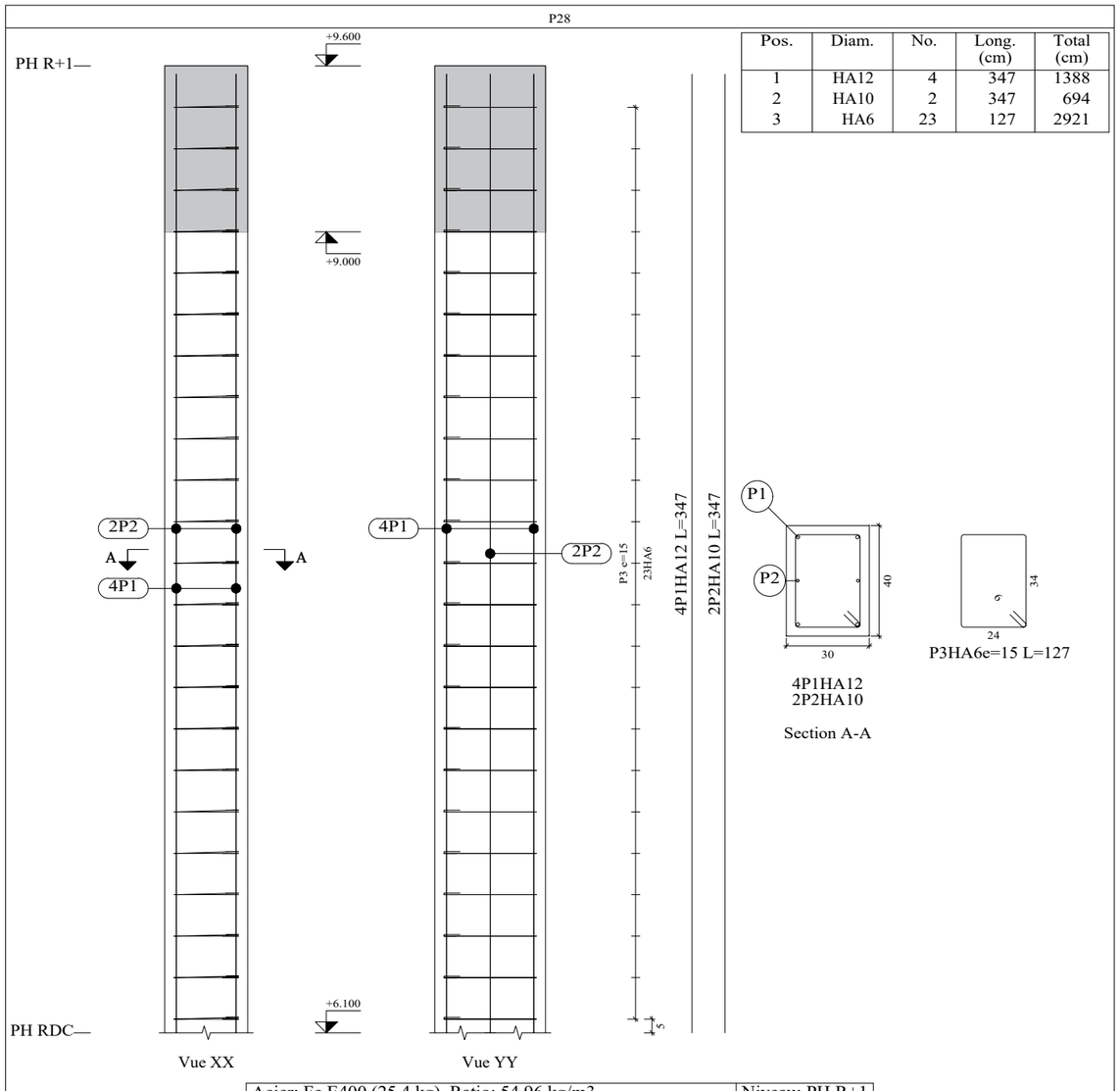
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	400	1600
2	HA10	2	395	790
3	HA6	23	117	2691



Acier: Fe E400 (27.6 kg). Ratio: 71.57 kg/m3		Niveau: PH R+1
Béton: B20 (0.35 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.55 m2	Enrobage géométrique: 3 cm	

P28

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	127	2921



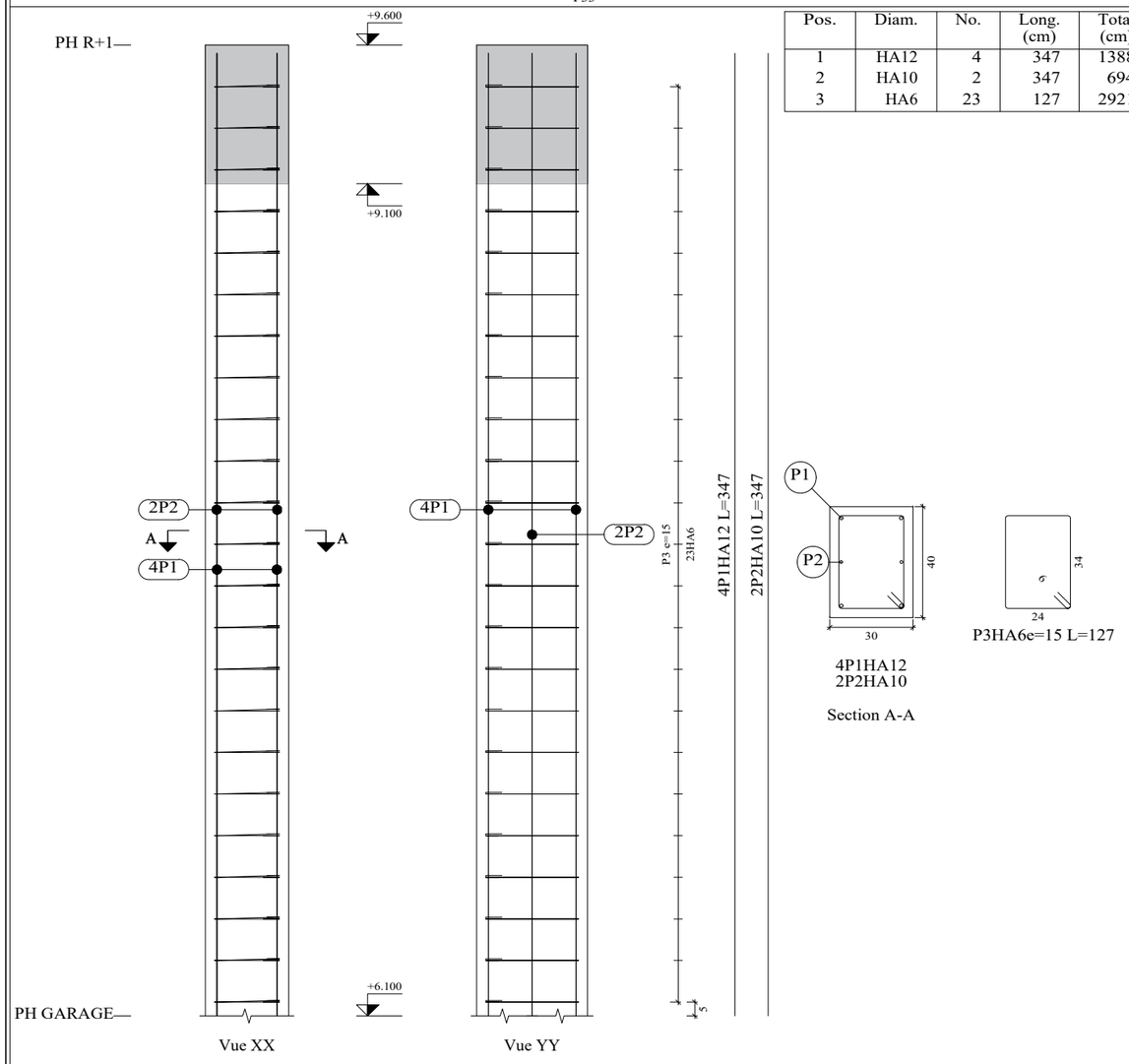
Acier: Fe E400 (25.4 kg). Ratio: 54.96 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P33	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24, 6, 34	127	2921	6.5
					Total+10%:		25.4
P35	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24, 6, 34	127	2921	6.5
					Total+10%:		25.4
					HA6:		14.2
					HA10:		9.6
					HA12:		27.0
					Total:		50.8

P33

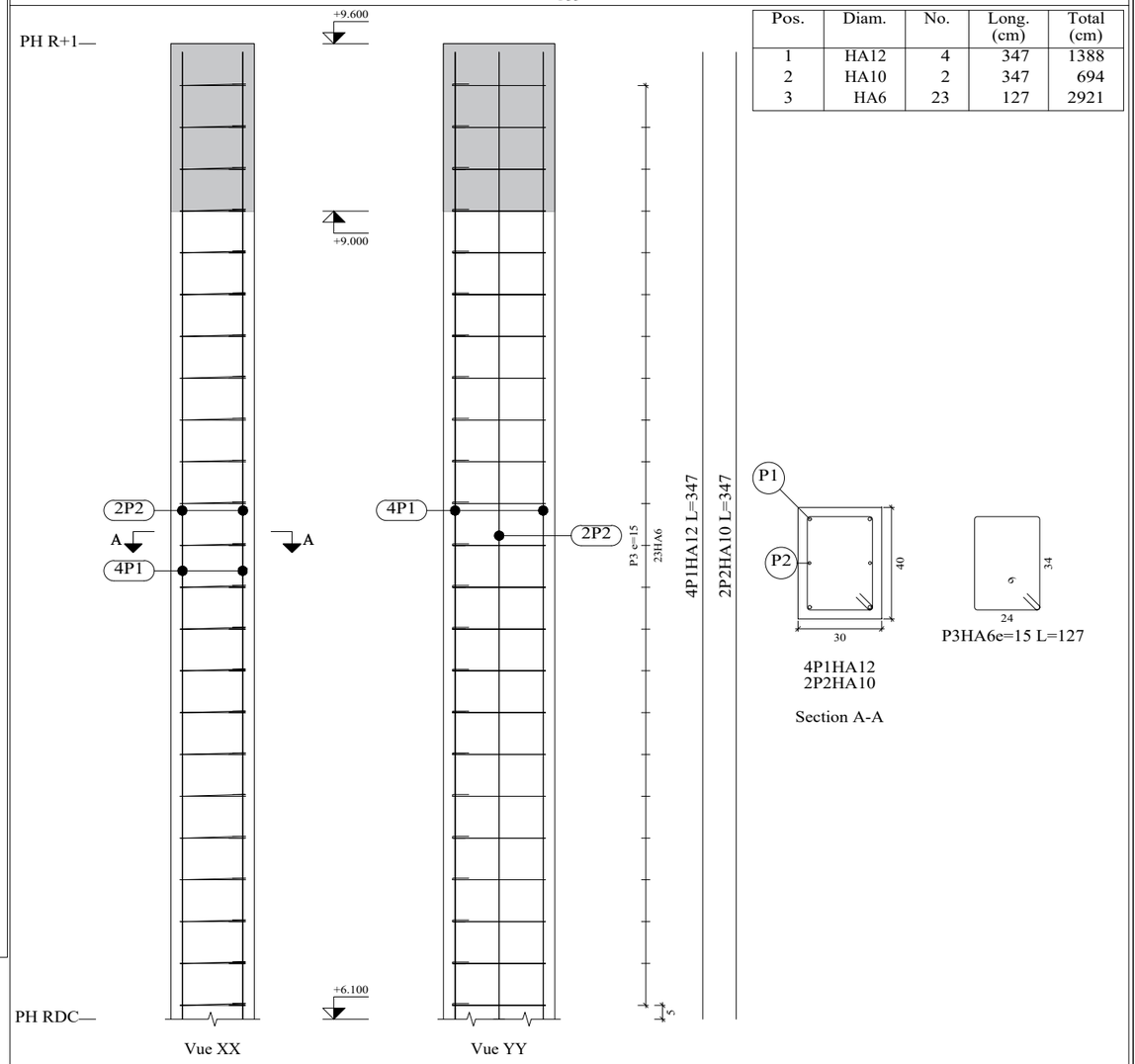
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	127	2921

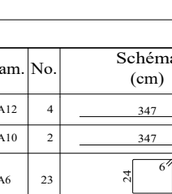


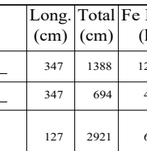
Acier: Fe E400 (25.4 kg). Ratio: 54.96 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

P35

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	127	2921







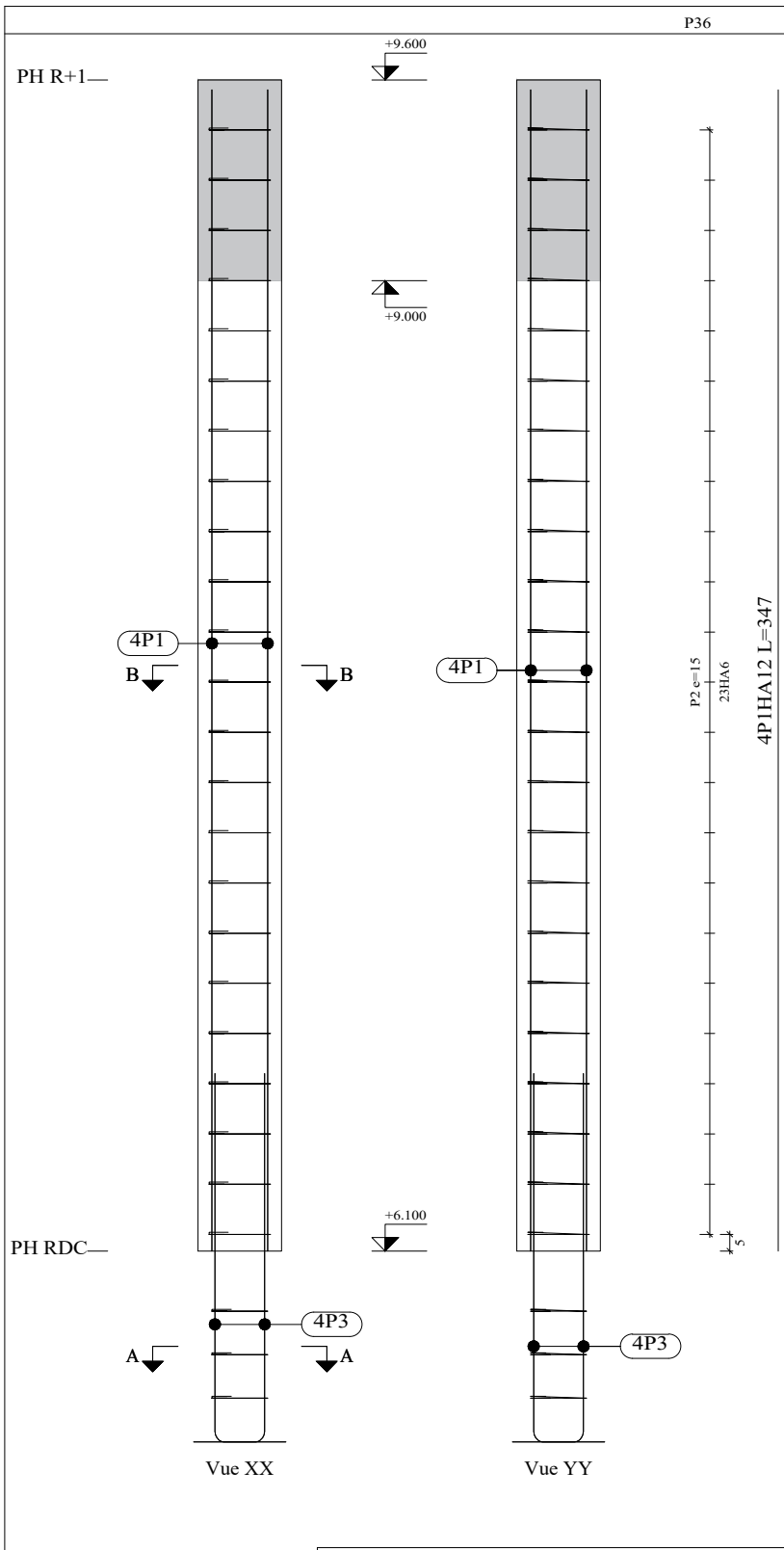
Acier: Fe E400 (25.4 kg). Ratio: 54.96 kg/m3		Niveau: PH R+1
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

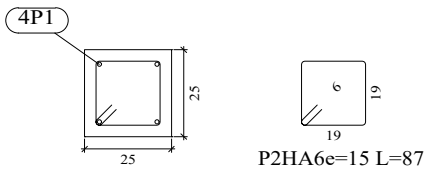
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P36	1	HA12	4	347	347	1388	12.3
	2	HA6	23	19 / 19 / 6	87	2001	4.4
	3	HA12	4	30 / 110	140	560	5.0
	4	HA6	3	17 / 17 / 6	80	240	0.5
				Total+10%:			24.4
						HA6:	5.4
						HA12:	19.0
						Total:	24.4

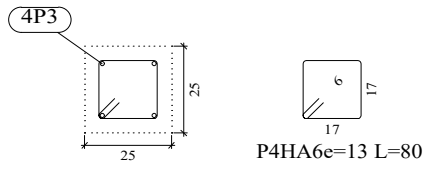
Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P36

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA6	23	87	2001
3	HA12	4	140	560
4	HA6	3	80	240







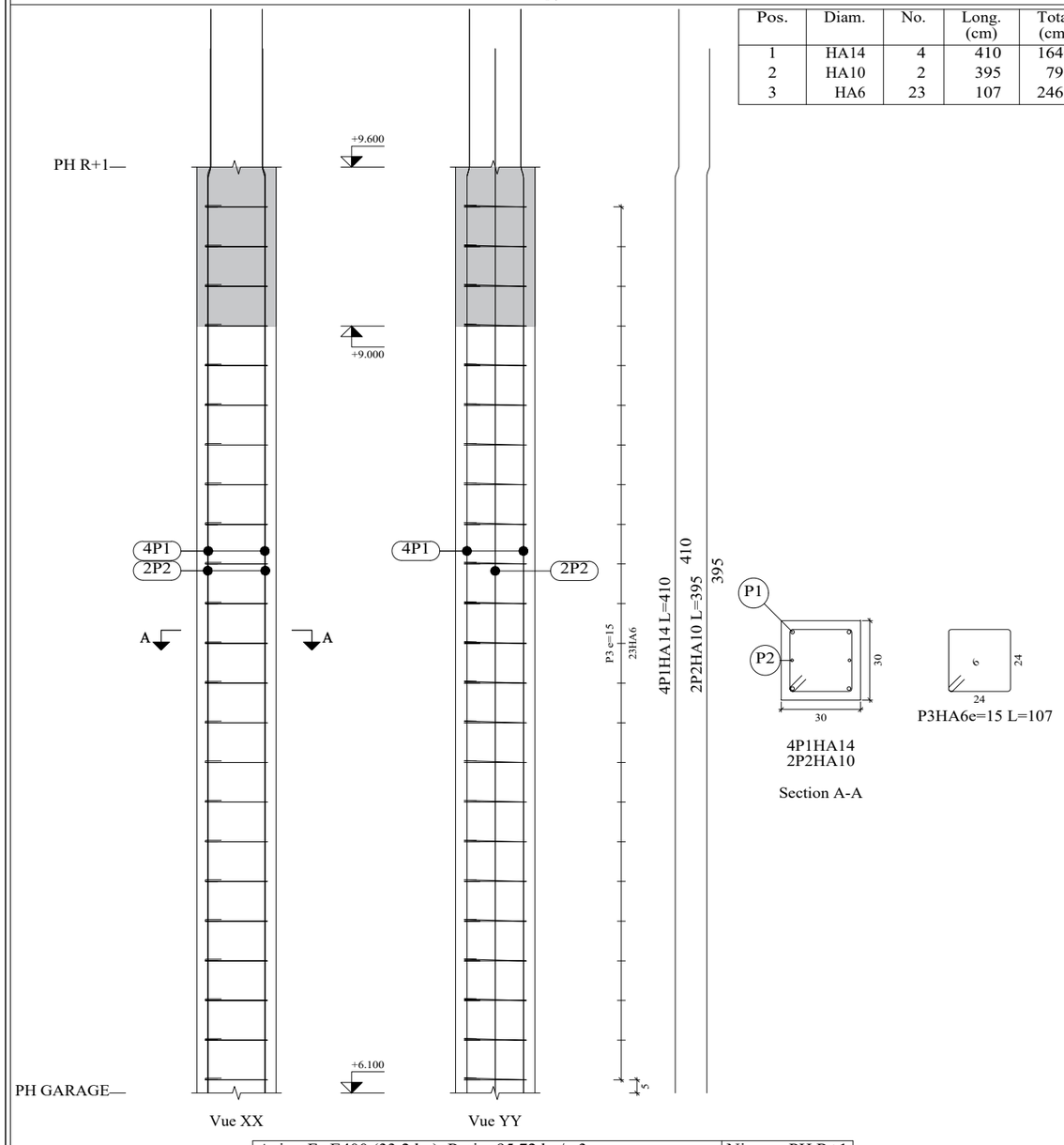
Acier: Fe E400 (18.4 kg). Ratio: 76.63 kg/m3		Niveau: PH R+1
Béton: B20 (0.22 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 3.50 m2	Enrobage géométrique: 3 cm	

Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P7	1	HA14	4	347 60	410	1640	19.8
	2	HA10	2	348 45	395	790	4.9
	3	HA6	23	24 6 24	107	2461	5.5
						Total+10%:	33.2
P20	1	HA14	4	347	347	1388	16.8
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24 6 24	107	2461	5.5
						Total+10%:	29.3
						HA6:	12.1
						HA10:	10.1
						HA14:	40.3
						Total:	62.5

P7

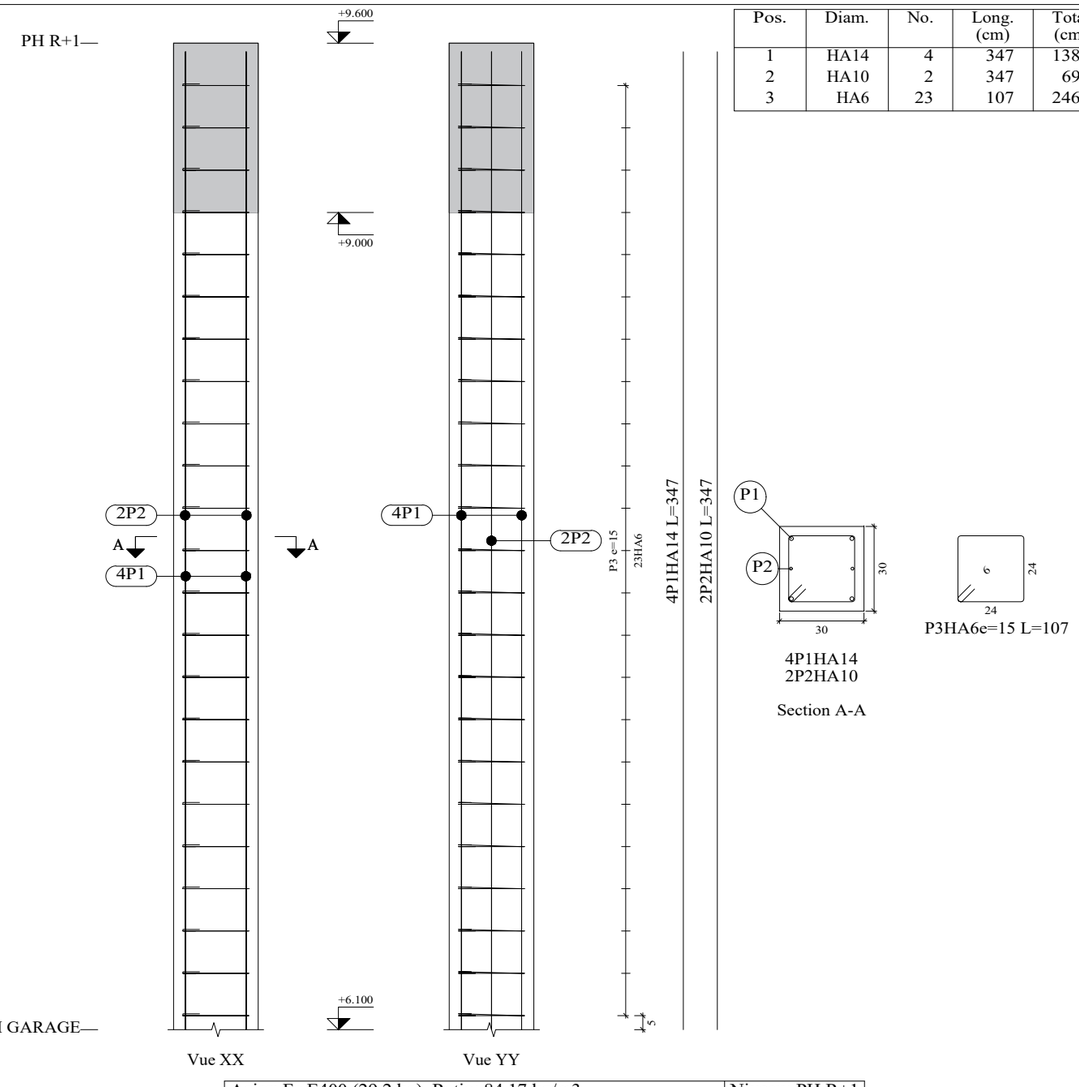
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	410	1640
2	HA10	2	395	790
3	HA6	23	107	2461



Acier: Fe E400 (33.2 kg). Ratio: 95.72 kg/m3		Niveau: PH R+1
Béton: B20 (0.32 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.20 m2	Enrobage géométrique: 3 cm	

P20

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	347	1388
2	HA10	2	347	694
3	HA6	23	107	2461



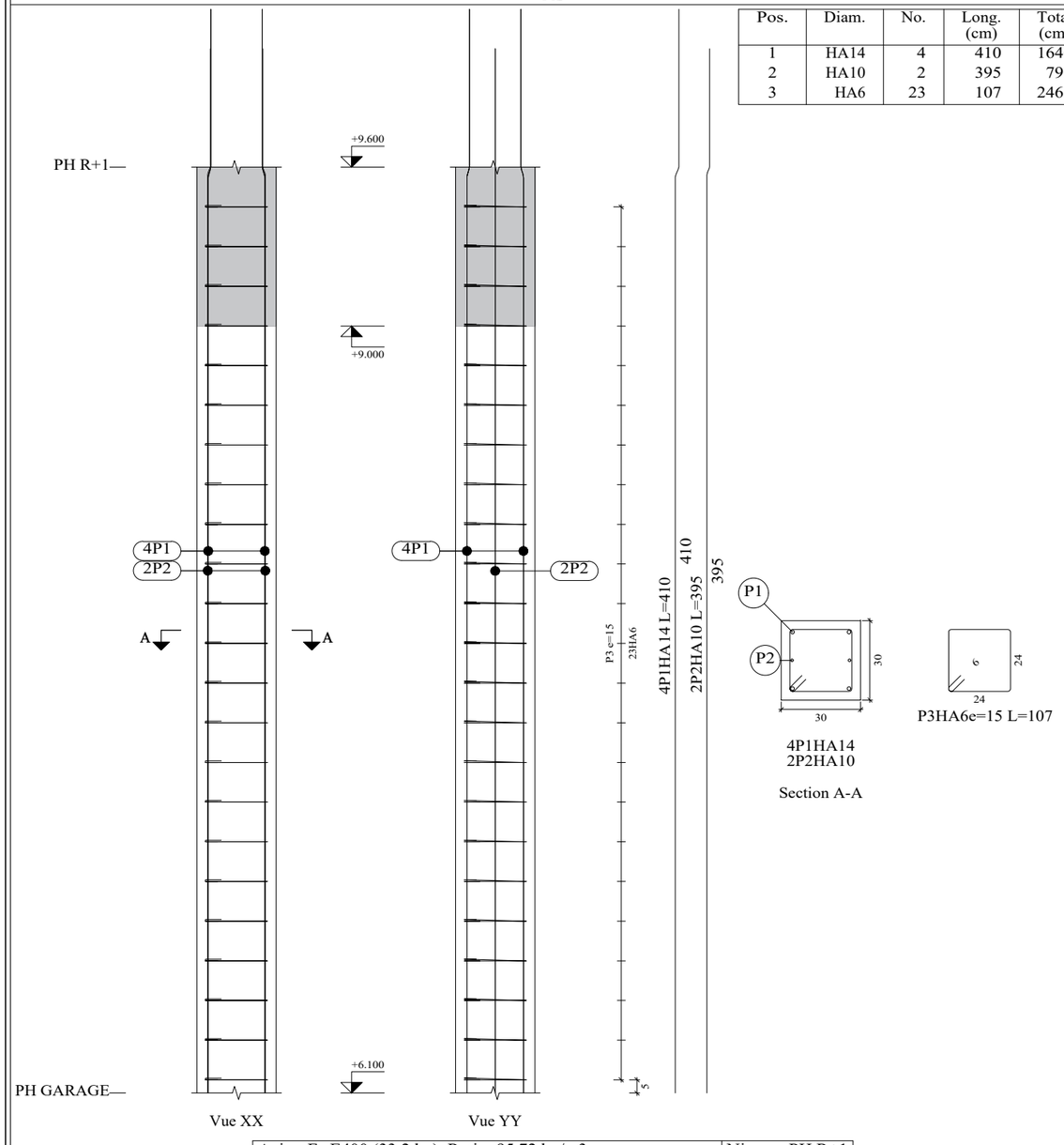
Acier: Fe E400 (29.2 kg). Ratio: 84.17 kg/m3		Niveau: PH R+1
Béton: B20 (0.32 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.20 m2	Enrobage géométrique: 3 cm	

Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P12	1	HA14	4	347 60	410	1640	19.8
	2	HA10	2	348 45	395	790	4.9
	3	HA6	23	24 6 24	107	2461	5.5
						Total+10%:	33.2
P19	1	HA14	4	347	347	1388	16.8
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24 6 24	107	2461	5.5
						Total+10%:	29.3
						HA6:	12.1
						HA10:	10.1
						HA14:	40.3
						Total:	62.5

P12

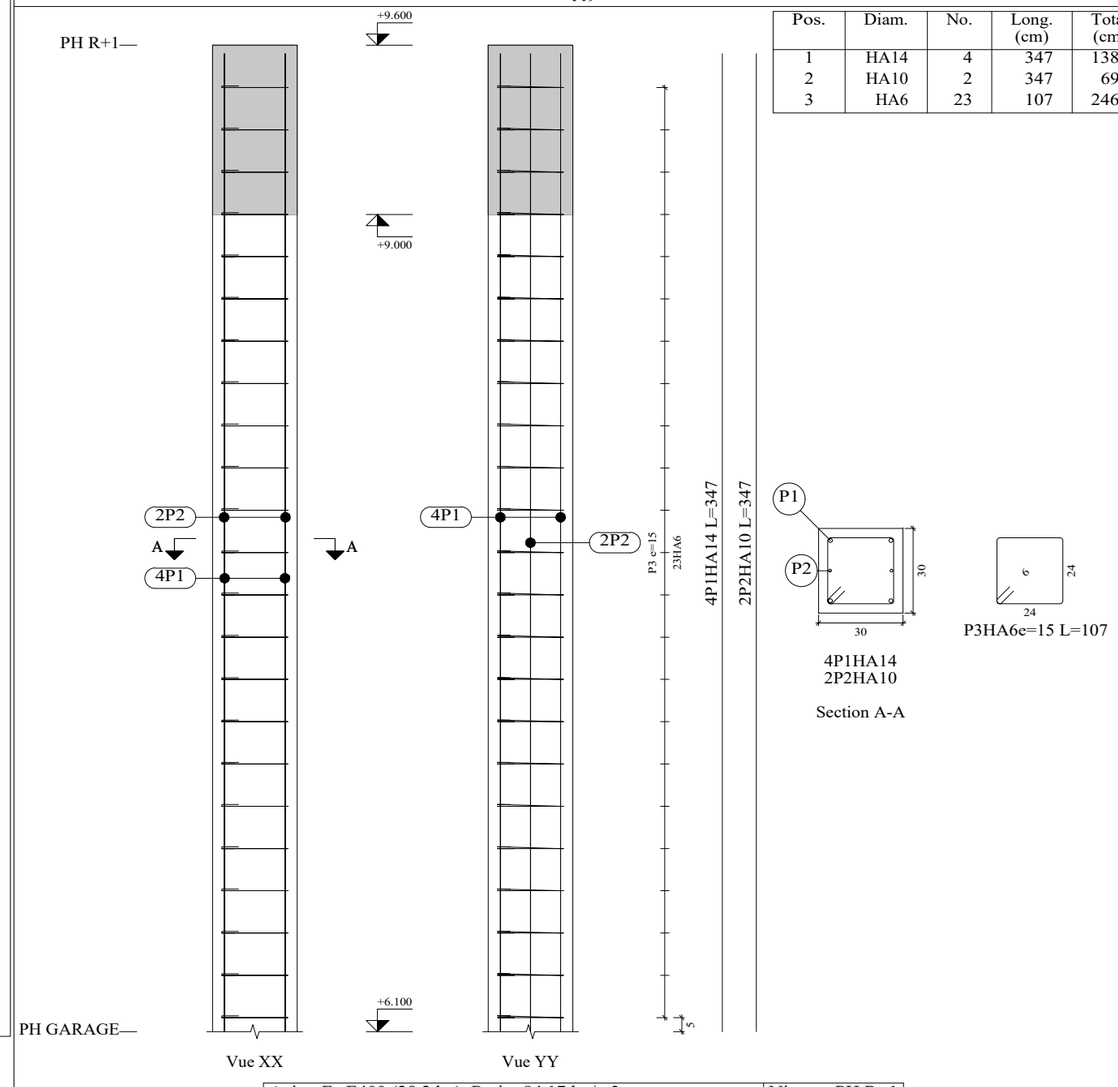
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	410	1640
2	HA10	2	395	790
3	HA6	23	107	2461



Acier: Fe E400 (33.2 kg). Ratio: 95.72 kg/m3		Niveau: PH R+1
Béton: B20 (0.32 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.20 m2	Enrobage géométrique: 3 cm	

P19

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	347	1388
2	HA10	2	347	694
3	HA6	23	107	2461



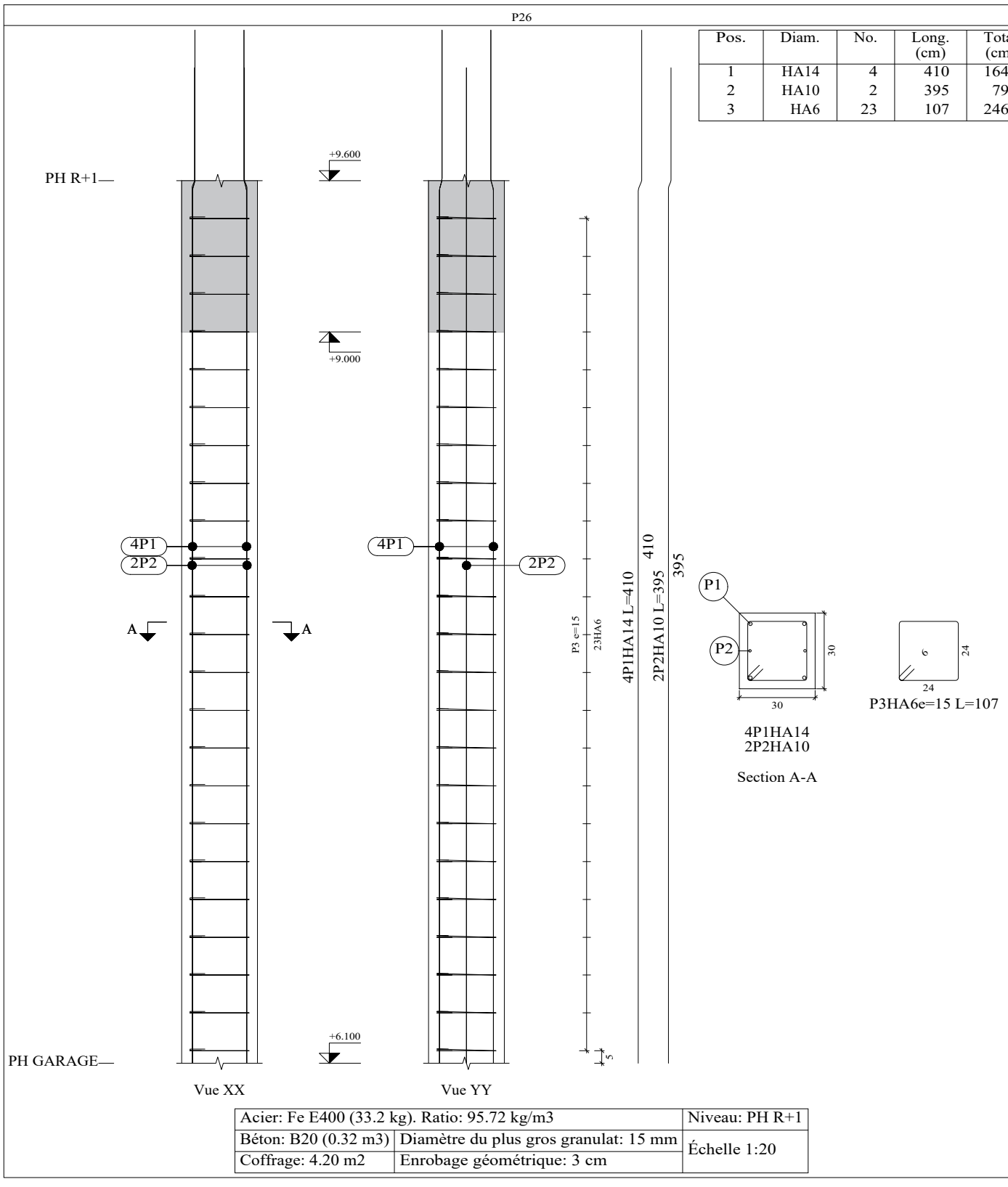
Acier: Fe E400 (29.2 kg). Ratio: 84.17 kg/m3		Niveau: PH R+1
Béton: B20 (0.32 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.20 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P26	1	HA14	4	347 60	410	1640	19.8
	2	HA10	2	348 45	395	790	4.9
	3	HA6	23	24 6 24	107	2461	5.5
						Total+10%:	33.2
						HA6:	6.0
						HA10:	5.4
						HA14:	21.8
						Total:	33.2

Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P26

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	410	1640
2	HA10	2	395	790
3	HA6	23	107	2461



Acier: Fe E400 (33.2 kg). Ratio: 95.72 kg/m3		Niveau: PH R+1
Béton: B20 (0.32 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.20 m2	Enrobage géométrique: 3 cm	

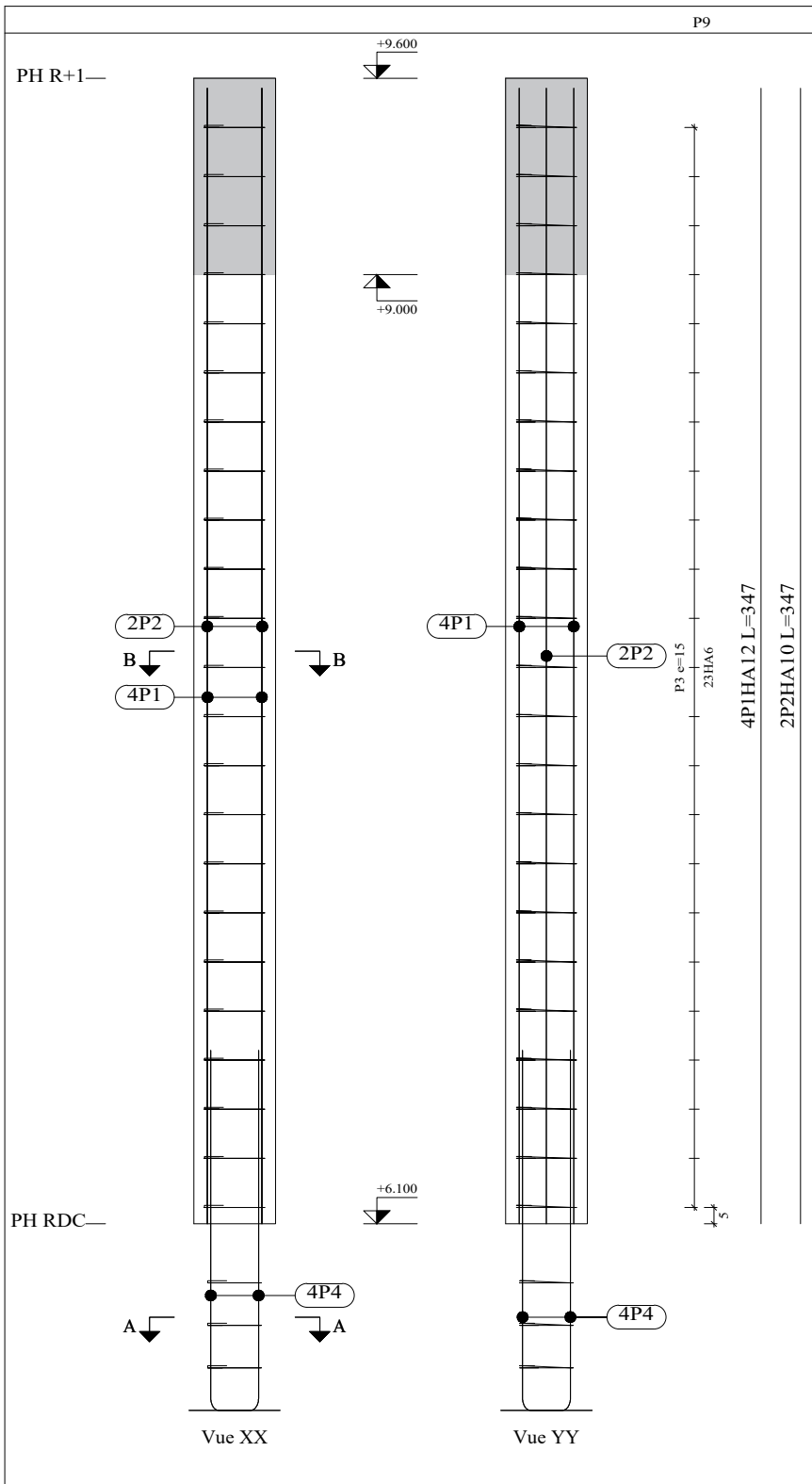
Niveau: PH R+1
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

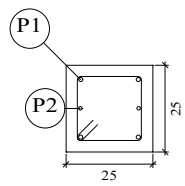
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P9	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	19 / 6 / 19	87	2001	4.4
	4	HA12	4	30 / 110	140	560	5.0
	5	HA6	3	17 / 6 / 17	79	237	0.5
				Total+10%:			29.2
						HA6:	5.4
						HA10:	4.8
						HA12:	19.0
						Total:	29.2

Récapitulatif Aciers Poteaux		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	720.5	176	
	HA10	178.3	121	
	HA12	275.4	269	
	HA14	118.6	158	724

P9

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	87	2001
4	HA12	4	140	560
5	HA6	3	79	237



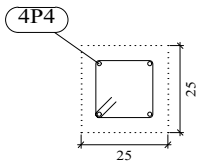


4P1HA12
2P2HA10

Section B-B

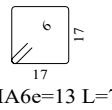


P3HA6e=15 L=87



4P4HA12

Section A-A



P5HA6e=13 L=79

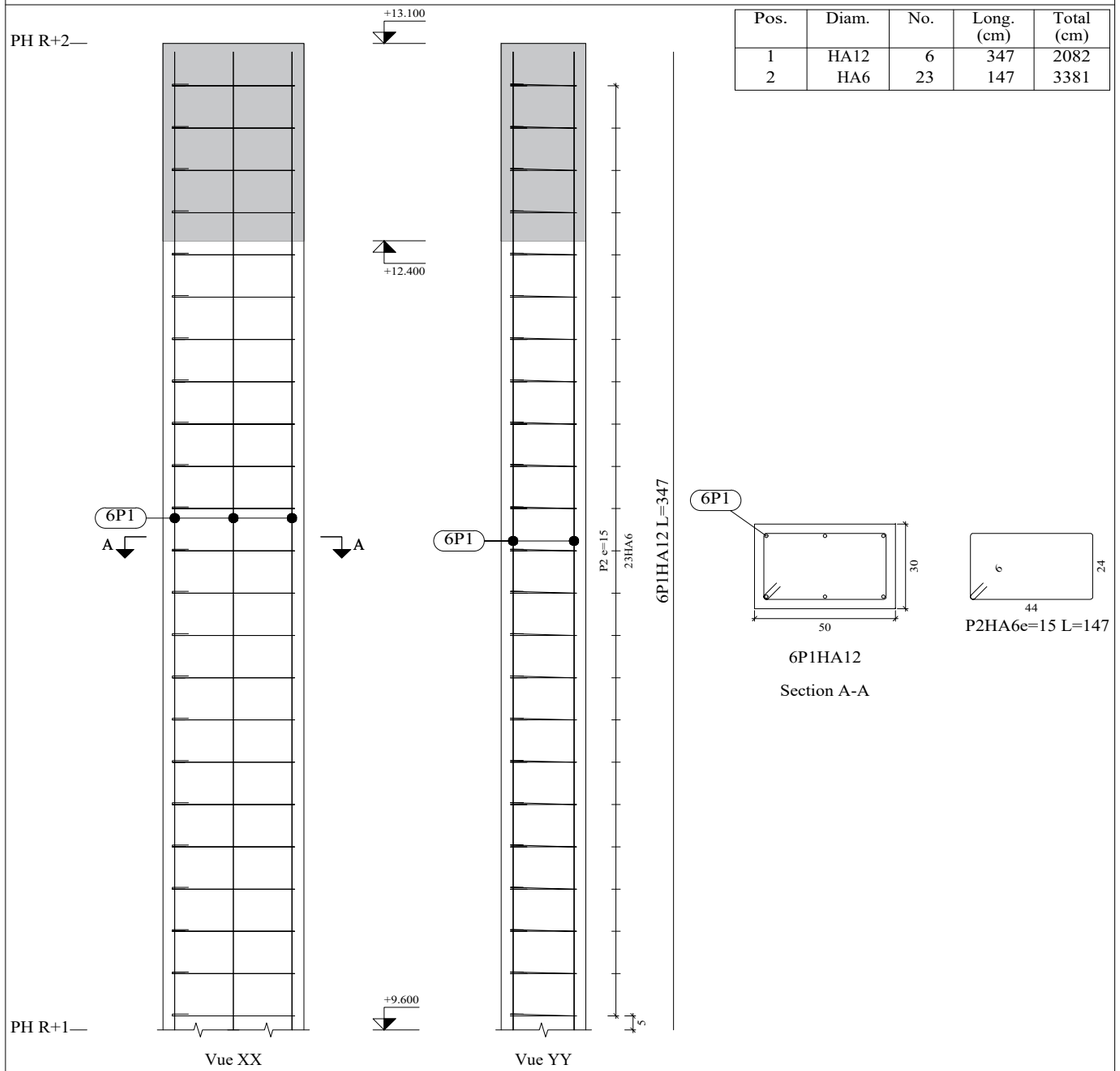
Acier: Fe E400 (23.1 kg). Ratio: 96.19 kg/m3		Niveau: PH R+1
Béton: B20 (0.22 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 3.50 m2	Enrobage géométrique: 3 cm	

Niveau: PH R+2
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P11	1	HA12	6	347	347	2082	18.5
	2	HA6	23	24 × 44, 6	147	3381	7.5
						Total+10%:	28.6
P17	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24 × 34, 6	127	2921	6.5
						Total+10%:	25.4
						HA6:	15.3
						HA10:	4.8
						HA12:	33.9
						Total:	54.0

P11

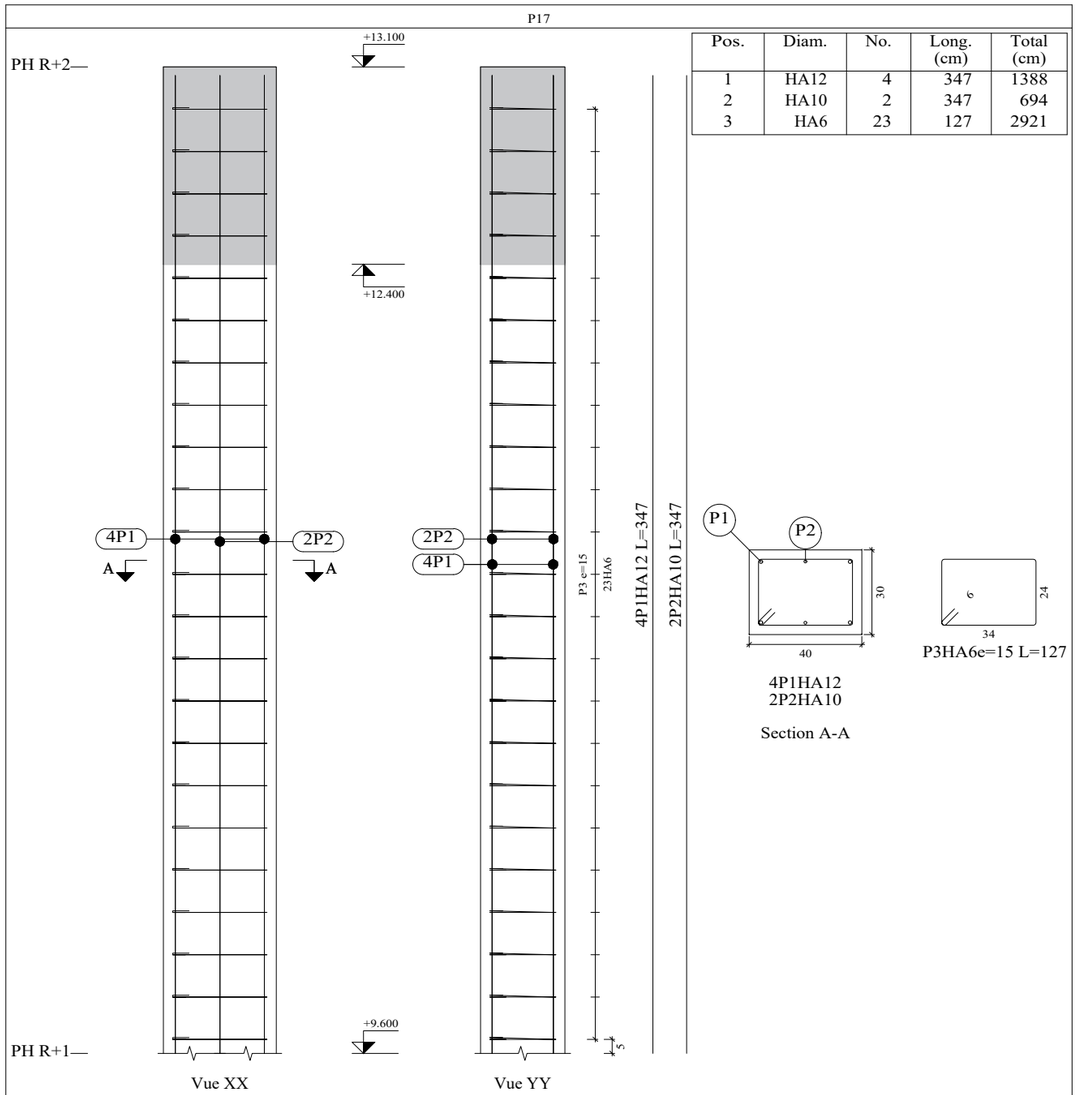
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	6	347	2082
2	HA6	23	147	3381



Acier: Fe E400 (28.6 kg). Ratio: 49.50 kg/m3		Niveau: PH R+2
Béton: B20 (0.53 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 5.60 m2	Enrobage géométrique: 3 cm	

P17

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	127	2921

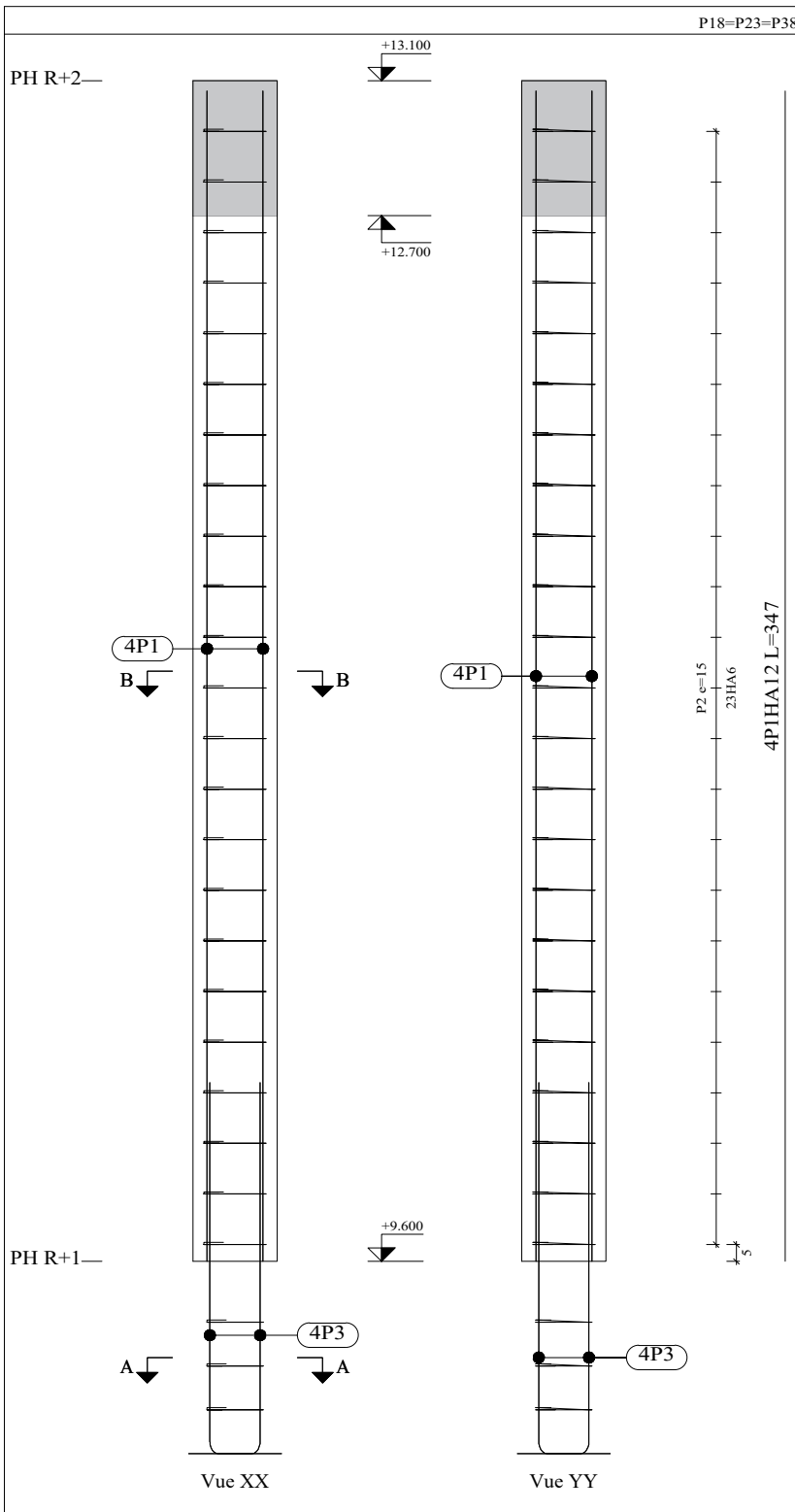


Acier: Fe E400 (25.4 kg). Ratio: 54.96 kg/m3		Niveau: PH R+2
Béton: B20 (0.42 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.90 m2	Enrobage géométrique: 3 cm	

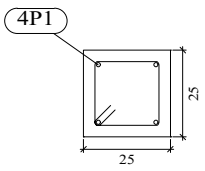
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P18=P23=P38	1	HA12	4	347	347	1388	12.3
	2	HA6	23	19 / 6 / 19	87	2001	4.4
	3	HA12	4	30 / 110	140	560	5.0
	4	HA6	3	17 / 6 / 17	80	240	0.5
				Total+10%:			24.4
				(x3):			73.2
				HA6:			16.2
				HA12:			57.0
				Total:			73.2

Niveau: PH R+2
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P18=P23=P38



Pos.	Diam.	No.	Long. (cm)	Total (cm)	x 3 (cm)
1	HA12	4	347	1388	4164
2	HA6	23	87	2001	6003
3	HA12	4	140	560	1680
4	HA6	3	80	240	720

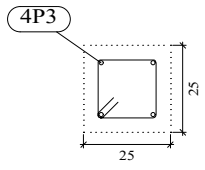


4P1HA12

Section B-B



P2HA6e=15 L=87



4P3HA12

Section A-A



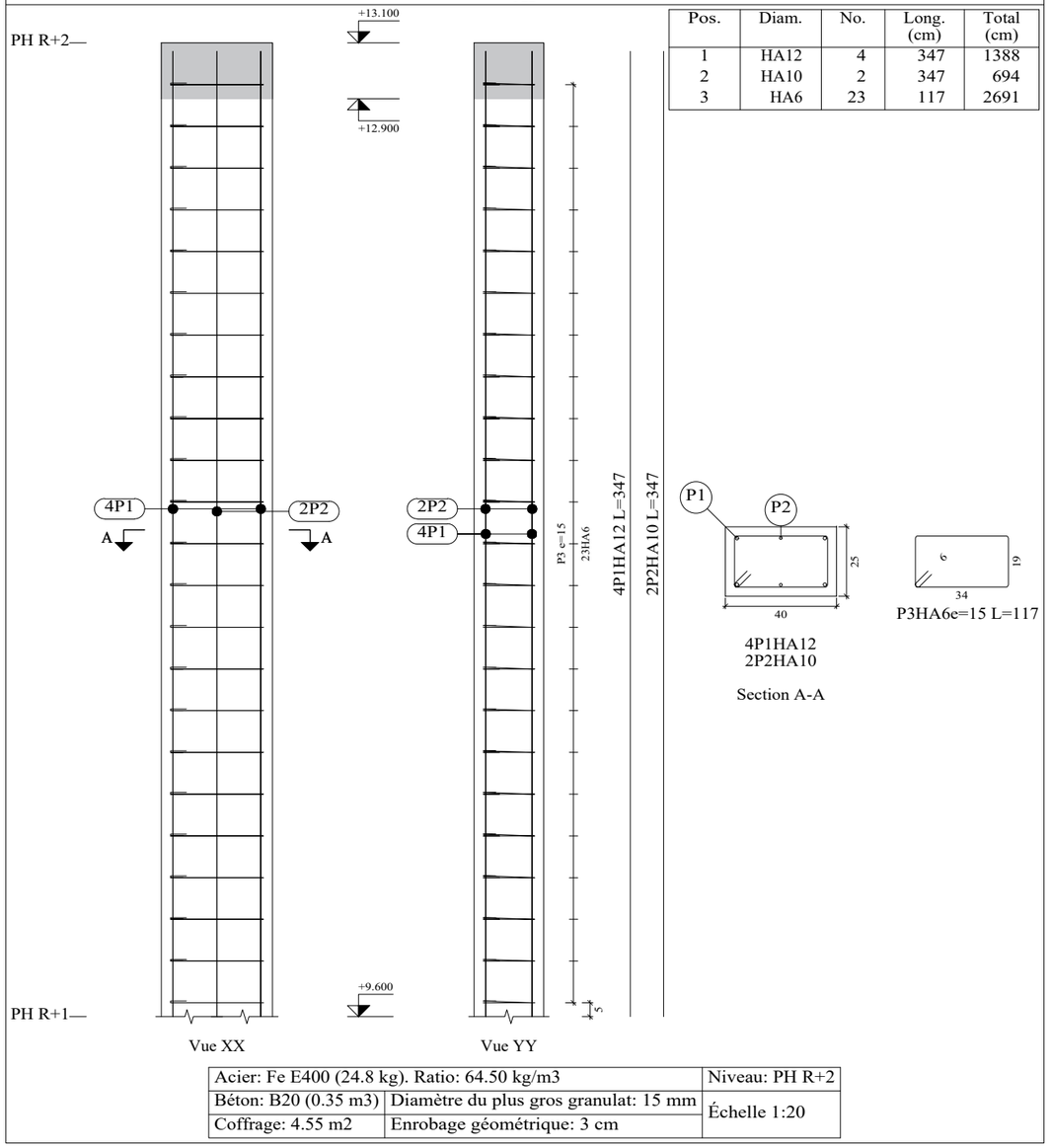
P4HA6e=13 L=80

Acier: Fe E400 (55.3 kg). Ratio: 76.63 kg/m3		Niveau: PH R+2
Béton: B20 (0.66 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 3.50 m2	Enrobage géométrique: 3 cm	

Niveau: PH R+2
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P22	1	HA12	4	347	347	1388	12.3
	2	HA10	2	347	347	694	4.3
	3	HA6	23	19 × 34, 6	117	2691	6.0
					Total+10%:		24.9
P7	1	HA14	4	346, 19	365	1460	17.6
	2	HA10	2	345, 15	360	720	4.4
	3	HA6	23	24 × 24, 6	107	2461	5.5
					Total+10%:		30.3
					HA6:		12.7
					HA10:		9.6
					HA12:		13.5
					HA14:		19.4
					Total:		55.2

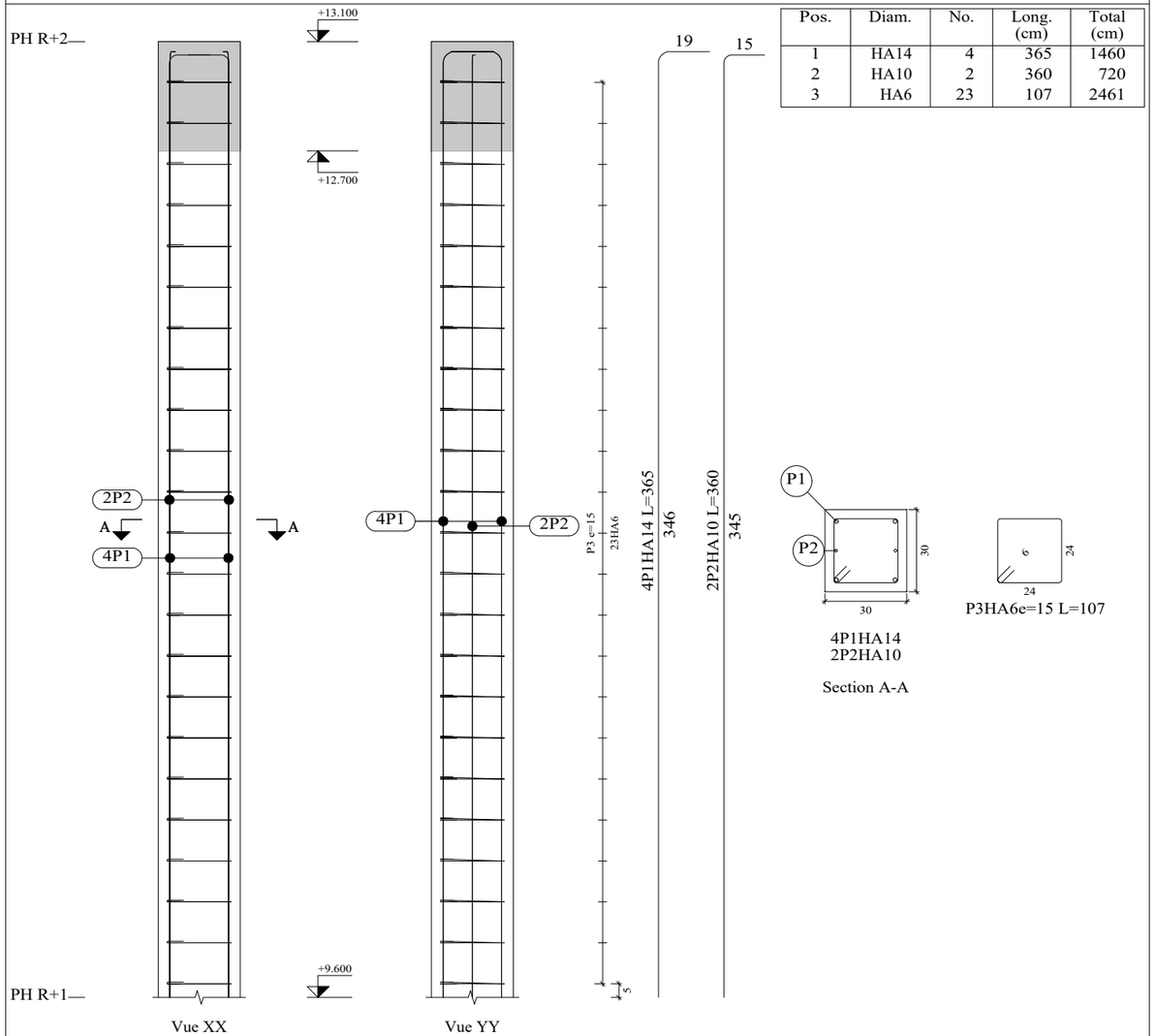
P22



Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	347	1388
2	HA10	2	347	694
3	HA6	23	117	2691

Acier: Fe E400 (24.8 kg). Ratio: 64.50 kg/m3		Niveau: PH R+2
Béton: B20 (0.35 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.55 m2	Enrobage géométrique: 3 cm	

P7



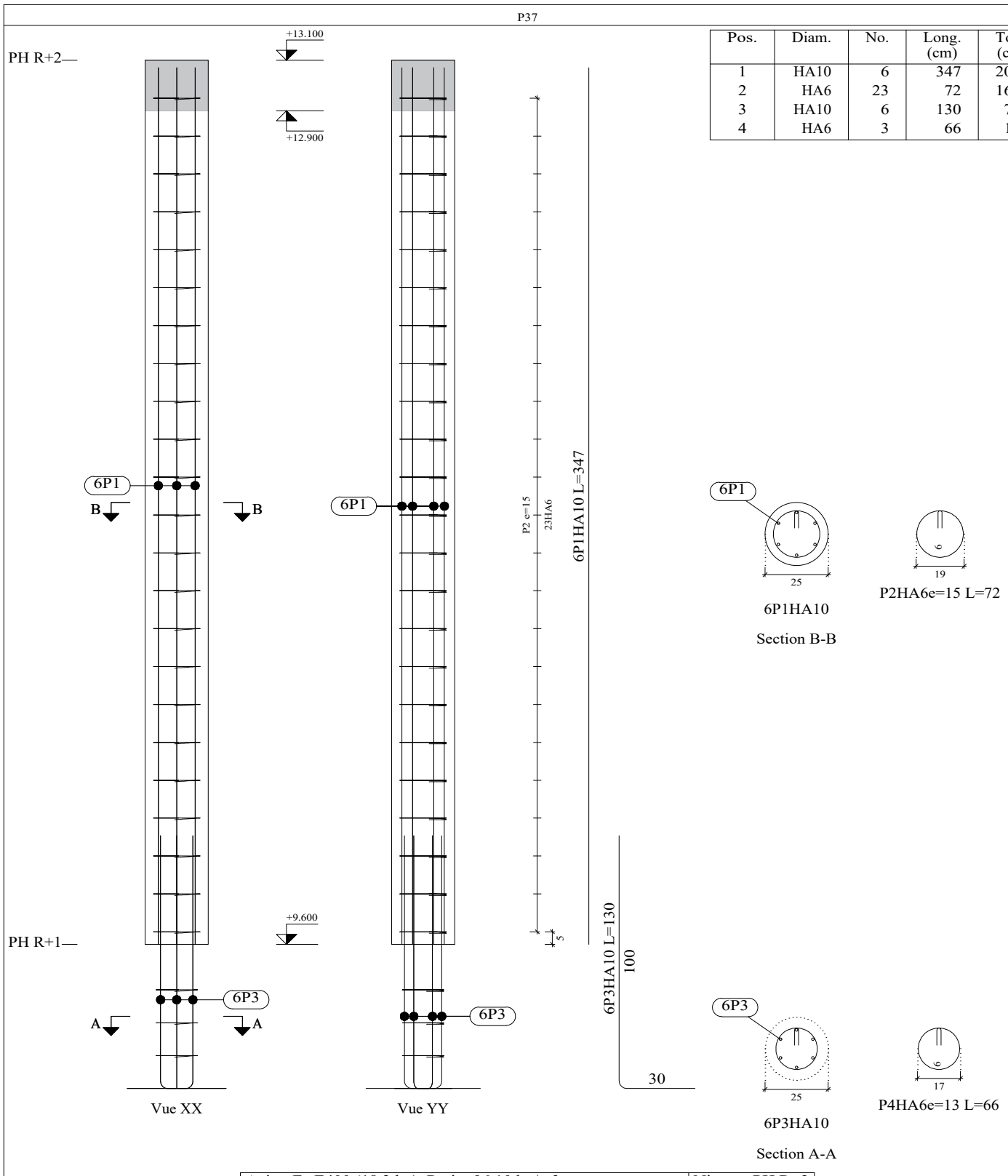
Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	365	1460
2	HA10	2	360	720
3	HA6	23	107	2461

Acier: Fe E400 (30.3 kg). Ratio: 87.44 kg/m3		Niveau: PH R+2
Béton: B20 (0.32 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.20 m2	Enrobage géométrique: 3 cm	

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P37	1	HA10	6	347	347	2082	12.8
	2	HA6	23	19 6	72	1656	3.7
	3	HA10	6	30 100	130	780	4.8
	4	HA6	3	17 6	66	198	0.4
						Total+10%:	23.9
						HA6:	4.5
						HA10:	19.4
						Total:	23.9

Niveau: PH R+2
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P37



Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA10	6	347	2082
2	HA6	23	72	1656
3	HA10	6	130	780
4	HA6	3	66	198

6P1HA10

Section B-B

P2HA6e=15 L=72

6P3HA10

Section A-A

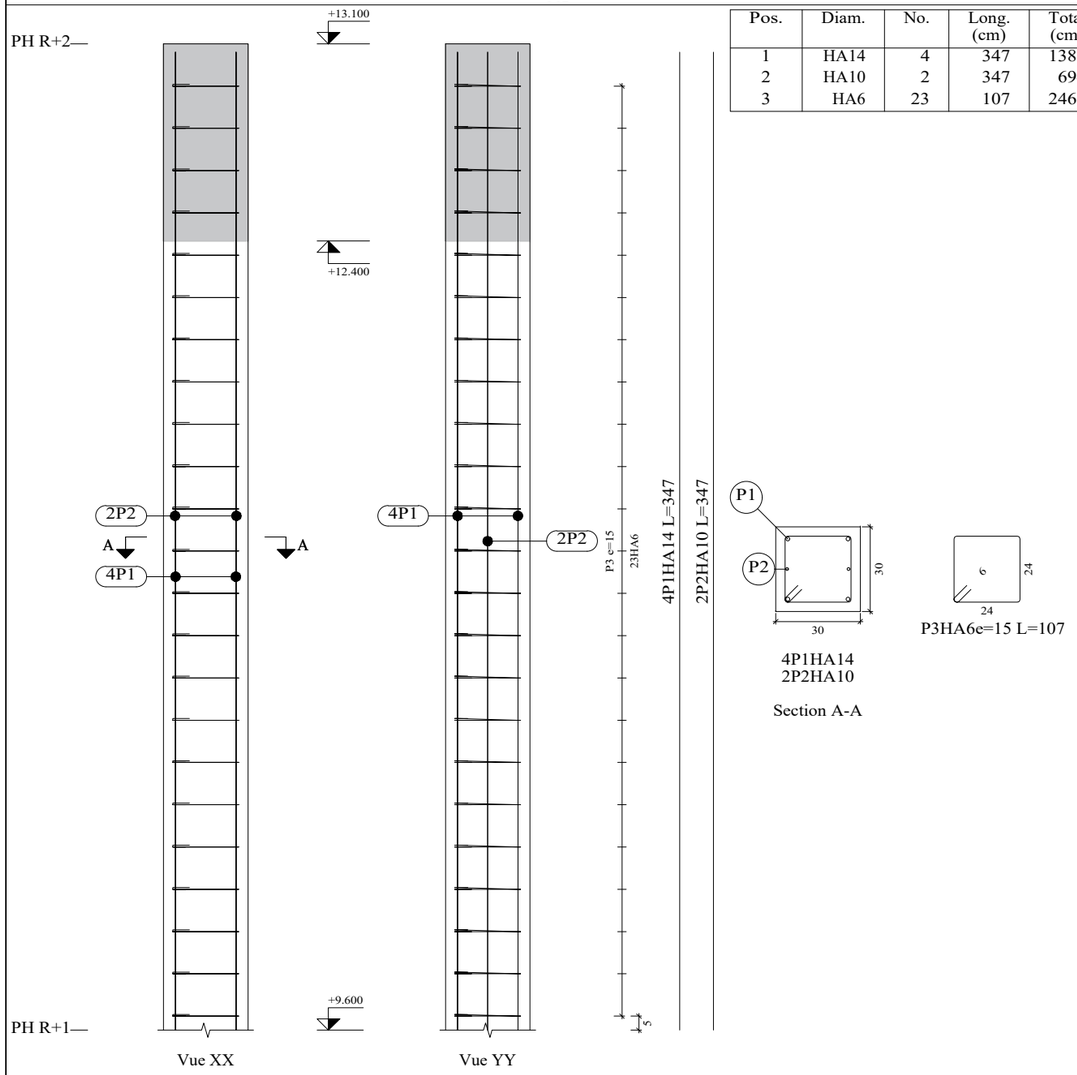
P4HA6e=13 L=66

Acier: Fe E400 (18.2 kg). Ratio: 96.10 kg/m3		Niveau: PH R+2
Béton: B20 (0.17 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 2.75 m2	Enrobage géométrique: 3 cm	

Niveau: PH R+2
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P12	1	HA14	4	347	347	1388	16.8
	2	HA10	2	347	347	694	4.3
	3	HA6	23	24 / 6 / 24	107	2461	5.5
						Total+10%:	29.3
P26	1	HA14	4	346 (17, 22)	385	1540	18.6
	2	HA10	2	345 (13, 22)	380	760	4.7
	3	HA6	23	24 / 6 / 24	107	2461	5.5
						Total+10%:	31.7
						HA6:	12.2
						HA10:	9.8
						HA14:	39.0
						Total:	61.0

P12

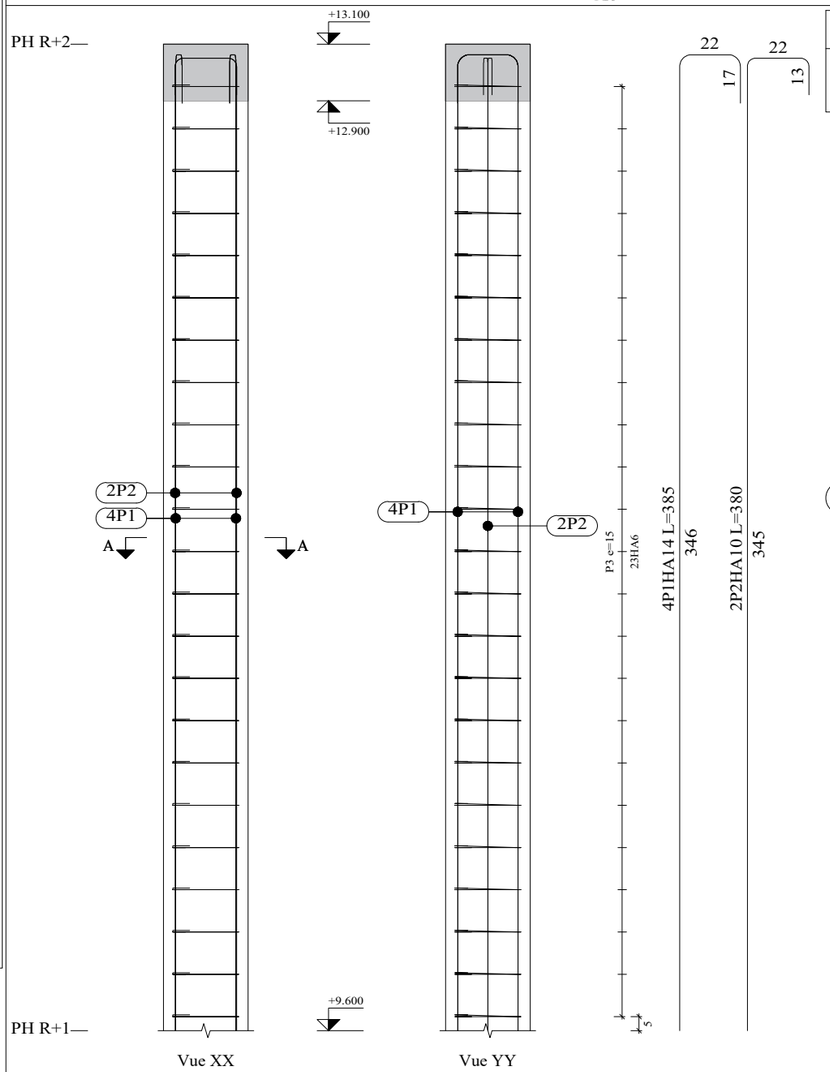


Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	347	1388
2	HA10	2	347	694
3	HA6	23	107	2461

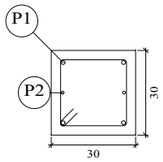
Section A-A

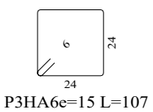
Acier: Fe E400 (29.2 kg). Ratio: 84.17 kg/m3		Niveau: PH R+2
Béton: B20 (0.32 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.20 m2	Enrobage géométrique: 3 cm	

P26



Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA14	4	385	1540
2	HA10	2	380	760
3	HA6	23	107	2461





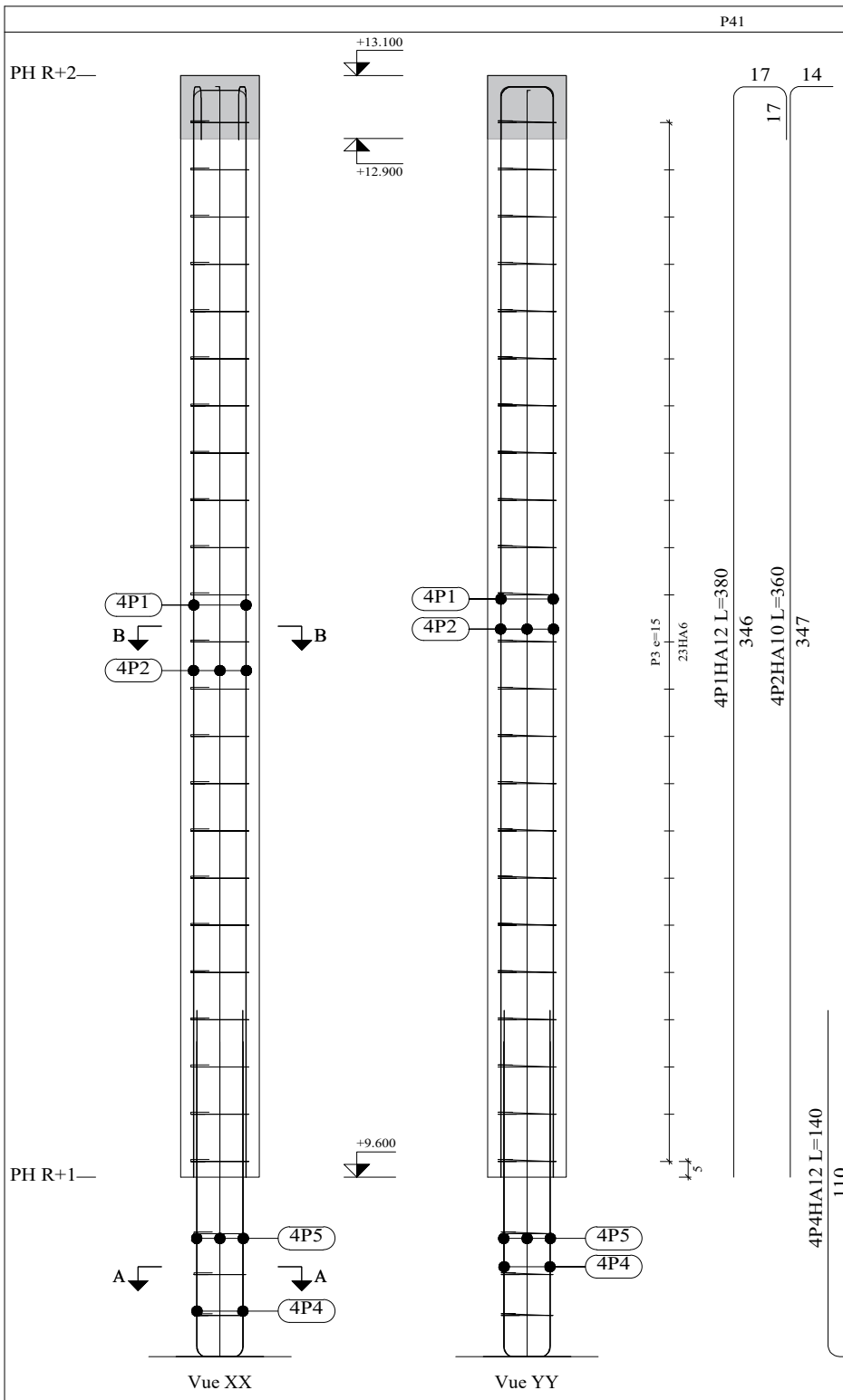
Section A-A

Acier: Fe E400 (31.6 kg). Ratio: 91.29 kg/m3		Niveau: PH R+2
Béton: B20 (0.32 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 4.20 m2	Enrobage géométrique: 3 cm	

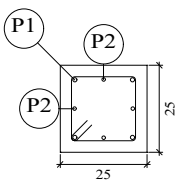
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P41	1	HA12	4	346, 17, 17	380	1520	13.5
	2	HA10	4	347, 14	360	1440	8.9
	3	HA6	23	19, 6, 19	87	2001	4.4
	4	HA12	4	30, 110	140	560	5.0
	5	HA10	4	30, 100	130	520	3.2
	6	HA6	3	17, 6, 17	79	237	0.5
					Total+10%:		39.1
					HA6:		5.4
					HA10:		13.3
					HA12:		20.4
					Total:		39.1

Niveau: PH R+2
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

P41

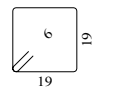


Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA12	4	380	1520
2	HA10	4	360	1440
3	HA6	23	87	2001
4	HA12	4	140	560
5	HA10	4	130	520
6	HA6	3	79	237

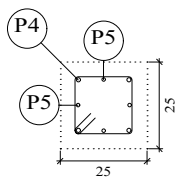


4P1HA12
4P2HA10

Section B-B

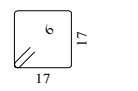


P3HA6e=15 L=87



4P4HA12
4P5HA10

Section A-A



P6HA6e=13 L=79

Acier: Fe E400 (29.5 kg). Ratio: 122.58 kg/m3		Niveau: PH R+2
Béton: B20 (0.22 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 3.50 m2	Enrobage géométrique: 3 cm	

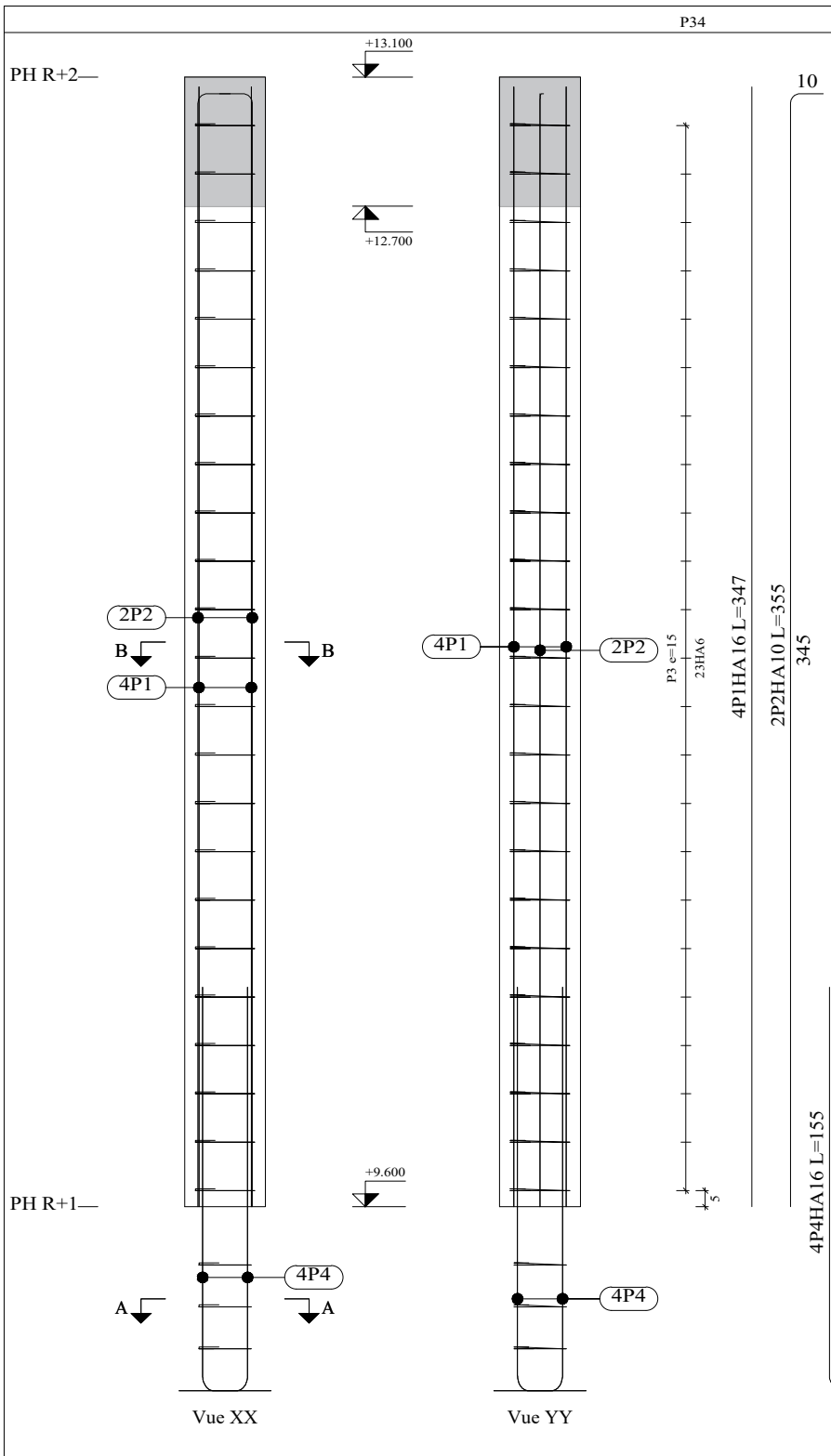
Niveau: PH R+2
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400

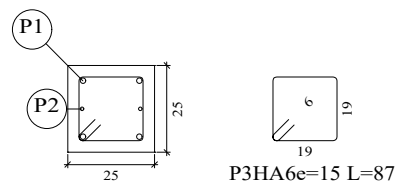
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
P34	1	HA16	4	347	347	1388	21.9
	2	HA10	2	345, 10	355	710	4.4
	3	HA6	23	19, 6, 19	87	2001	4.4
	4	HA16	4	30, 125	155	620	9.8
	5	HA6	3	17, 6, 17	78	234	0.5
					Total+10%:		45.1
					HA6:		5.4
					HA10:		4.8
					HA16:		34.9
					Total:		45.1

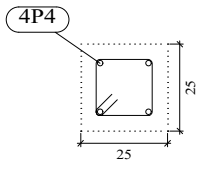
Récapitulatif Acier Poteaux		Long. total (m)	Poids+10% (kg)	Total
Fe E400	HA6	294.3	72	
	HA10	90.9	62	
	HA12	127.8	125	
	HA14	43.9	58	
	HA16	20.1	35	352

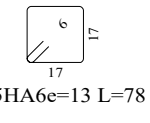
P34

Pos.	Diam.	No.	Long. (cm)	Total (cm)
1	HA16	4	347	1388
2	HA10	2	355	710
3	HA6	23	87	2001
4	HA16	4	155	620
5	HA6	3	78	234





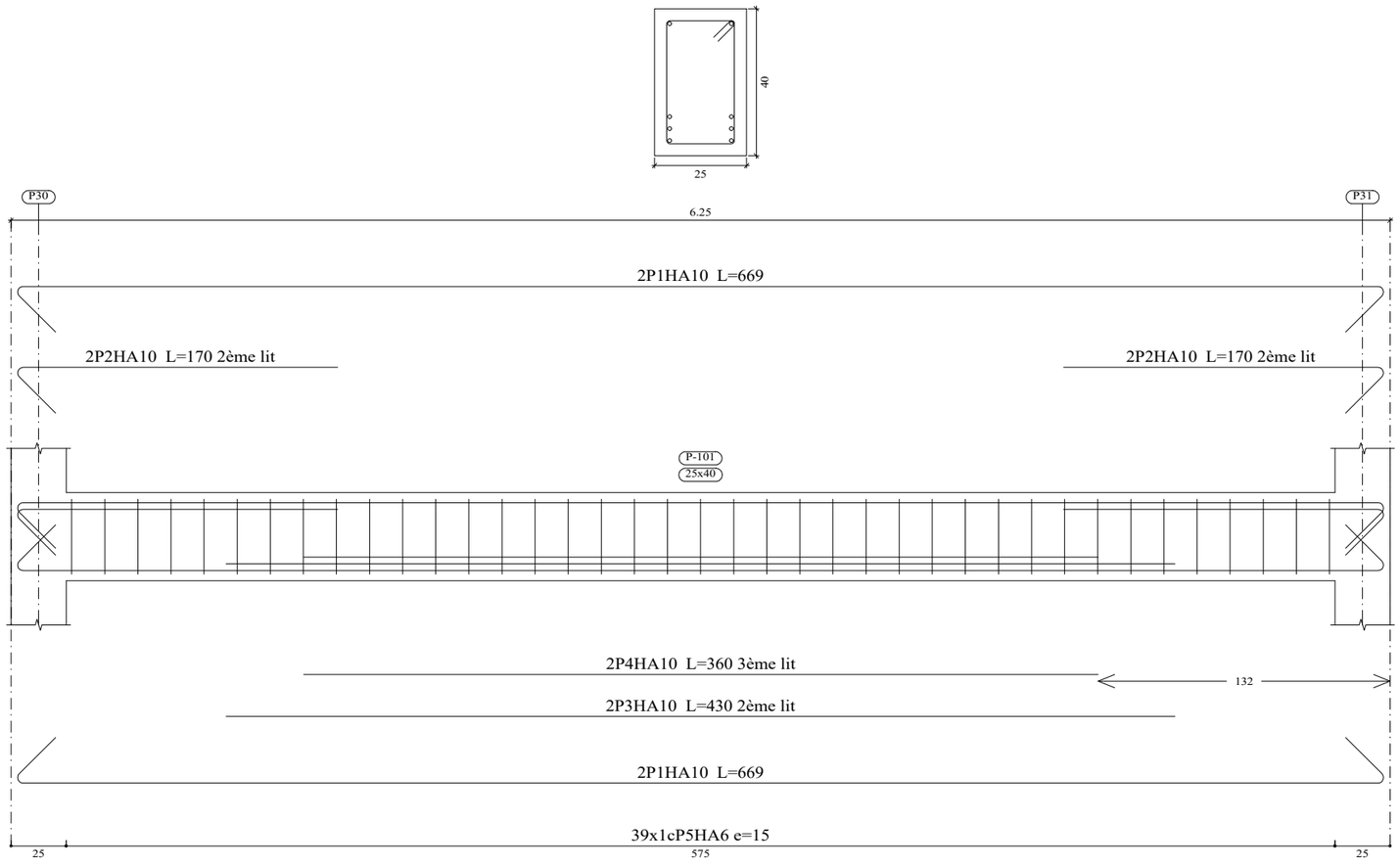




Acier: Fe E400 (33.8 kg). Ratio: 140.46 kg/m3		Niveau: PH R+2
Béton: B20 (0.22 m3)	Diamètre du plus gros granulat: 15 mm	Échelle 1:20
Coffrage: 3.50 m2	Enrobage géométrique: 3 cm	

LONGRINES

Portique 1



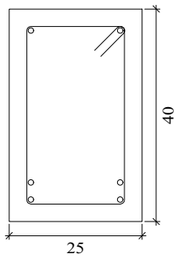
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 1	1	HA10	4	619, 25, 25	669	2676	16.5
	2	HA10	4	145, 25	170	680	4.2
	3	HA10	2	430	430	860	5.3
	4	HA10	2	360	360	720	4.4
	5	HA6	39	33, 6, 18	116	4524	10.0
						Total+10%:	44.4
						HA6:	11.0
						HA10:	33.4
						Total:	44.4

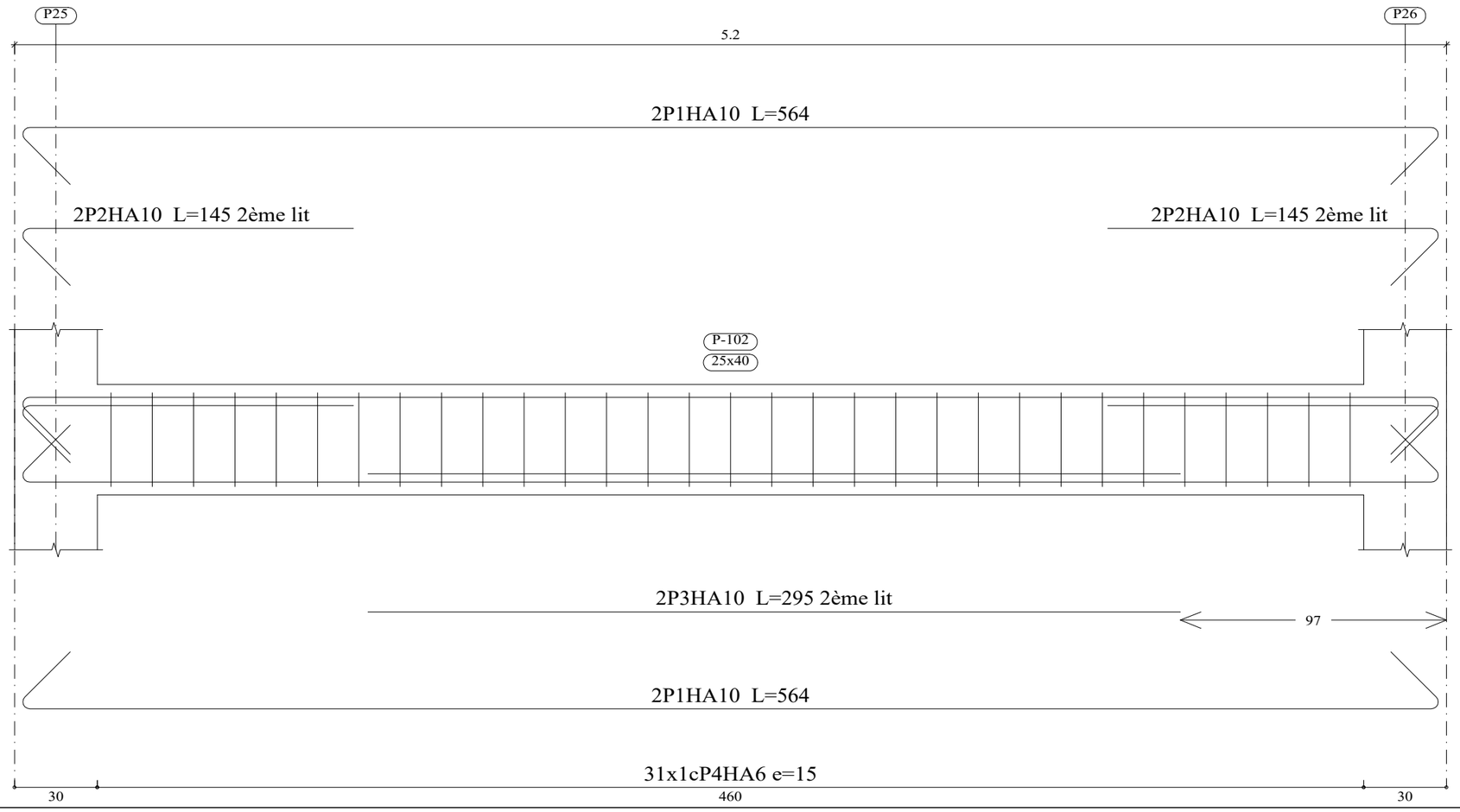
PB RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

PB RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

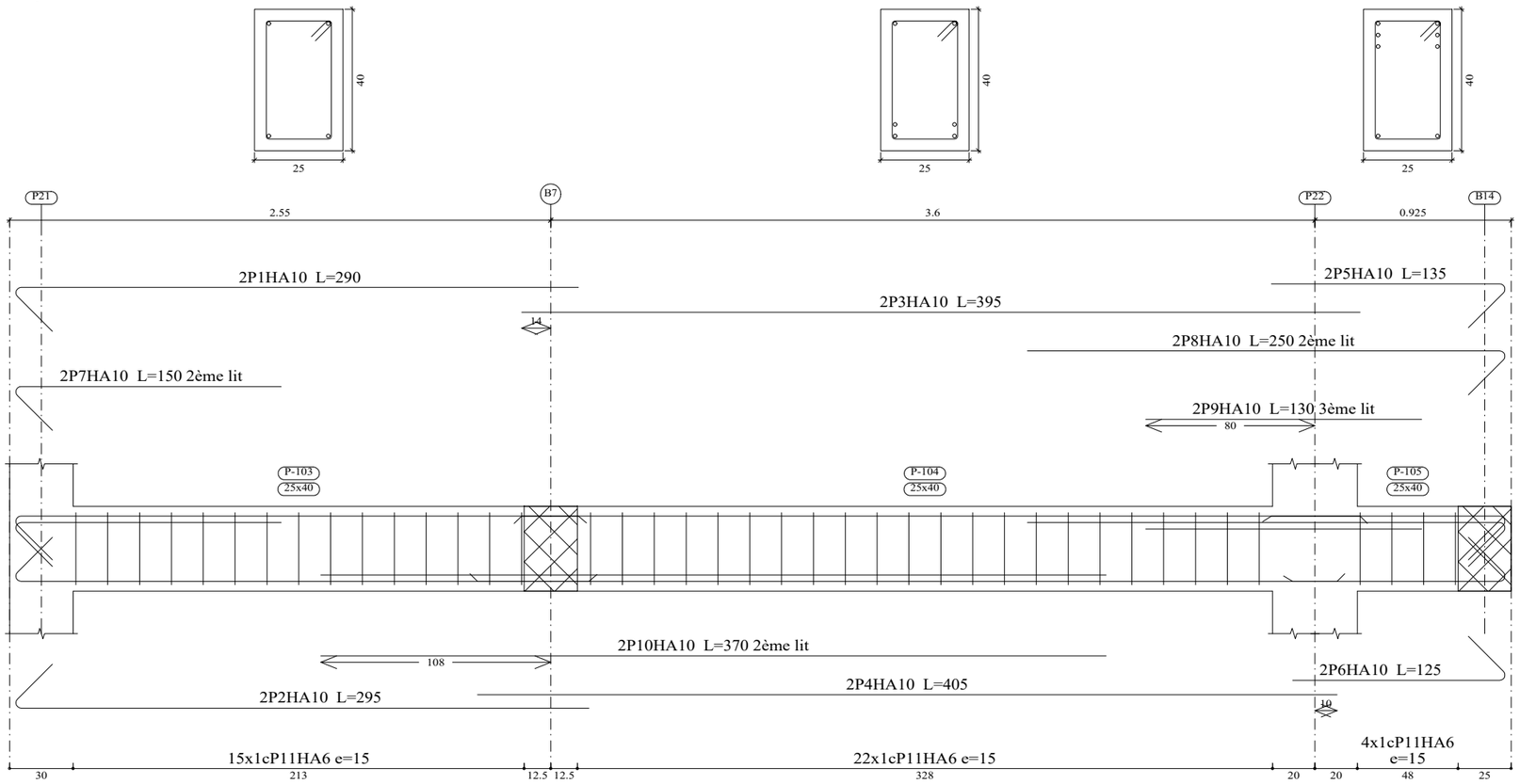
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 2	1	HA10	4	514, 25, 25	564	2256	13.9
	2	HA10	4	120, 25	145	580	3.6
	3	HA10	2	295	295	590	3.6
	4	HA6	31	33, 6, 18	116	3596	8.0
					Total+10%:		32.0
					HA6:		8.8
					HA10:		23.2
					Total:		32.0

Portique 2





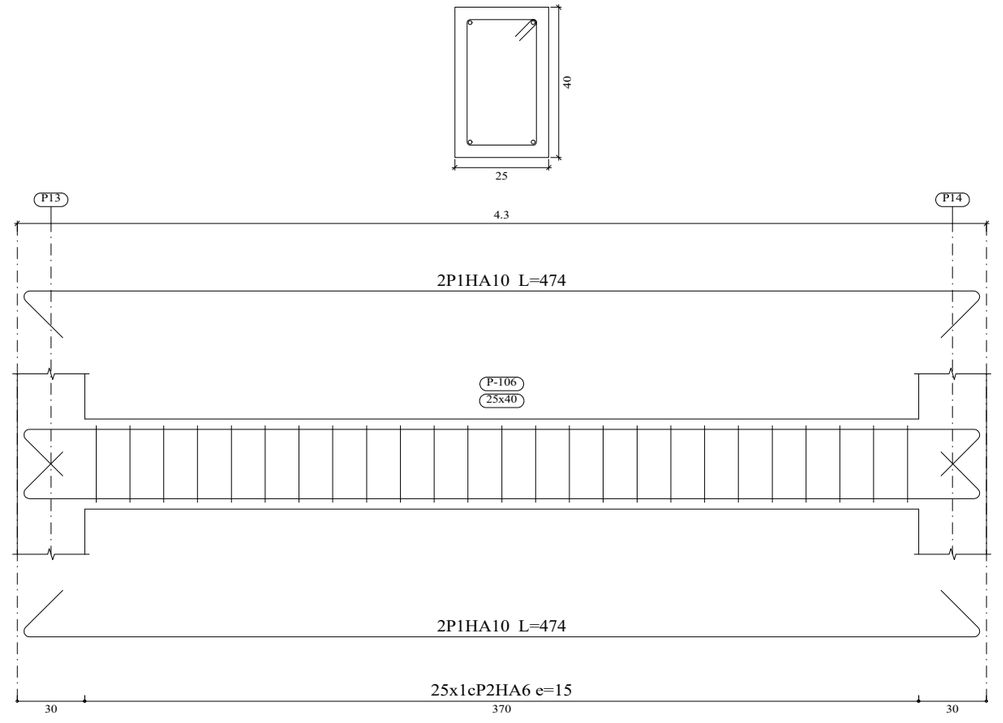
Portique 3



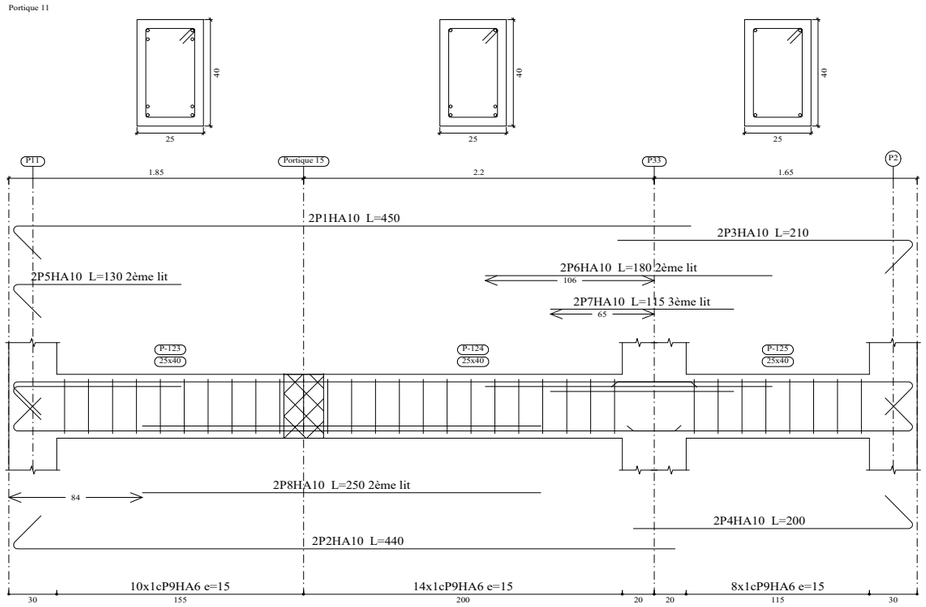
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 3	1	HA10	2	265, 25	290	580	3.6
	2	HA10	2	25, 270	295	590	3.6
	3	HA10	2	395	395	790	4.9
	4	HA10	2	405	405	810	5.0
	5	HA10	2	110, 25	135	270	1.7
	6	HA10	2	100, 25	125	250	1.5
	7	HA10	2	125, 25	150	300	1.8
	8	HA10	2	225, 25	250	500	3.1
	9	HA10	2	130	130	260	1.6
	10	HA10	2	370	370	740	4.6
	11	HA6	41	33, 6, 18	116	4756	10.6
						Total+10%:	46.2
Portique 4	1	HA10	4	424, 25	474	1896	11.7
	2	HA6	25	33, 6, 18	116	2900	6.4
						Total+10%:	19.9
						HA6:	18.7
						HA10:	47.4
						Total:	66.1

PB RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

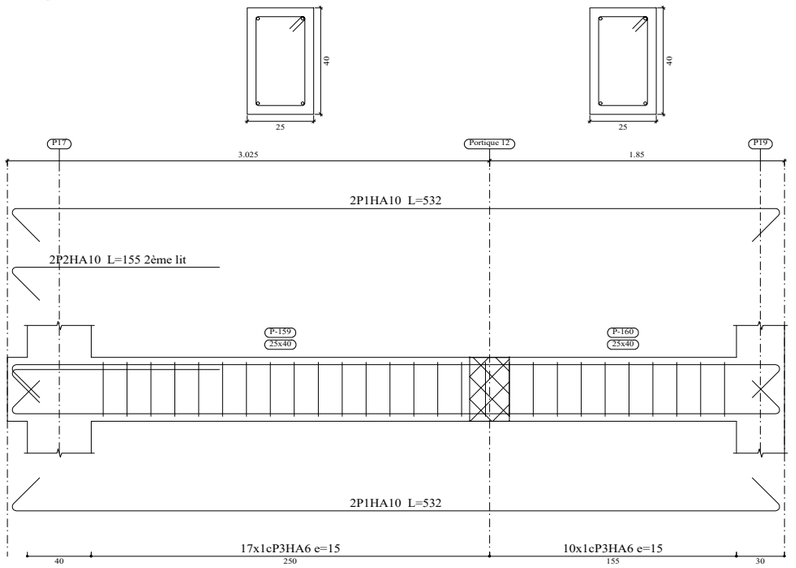
Portique 4



Portique 11

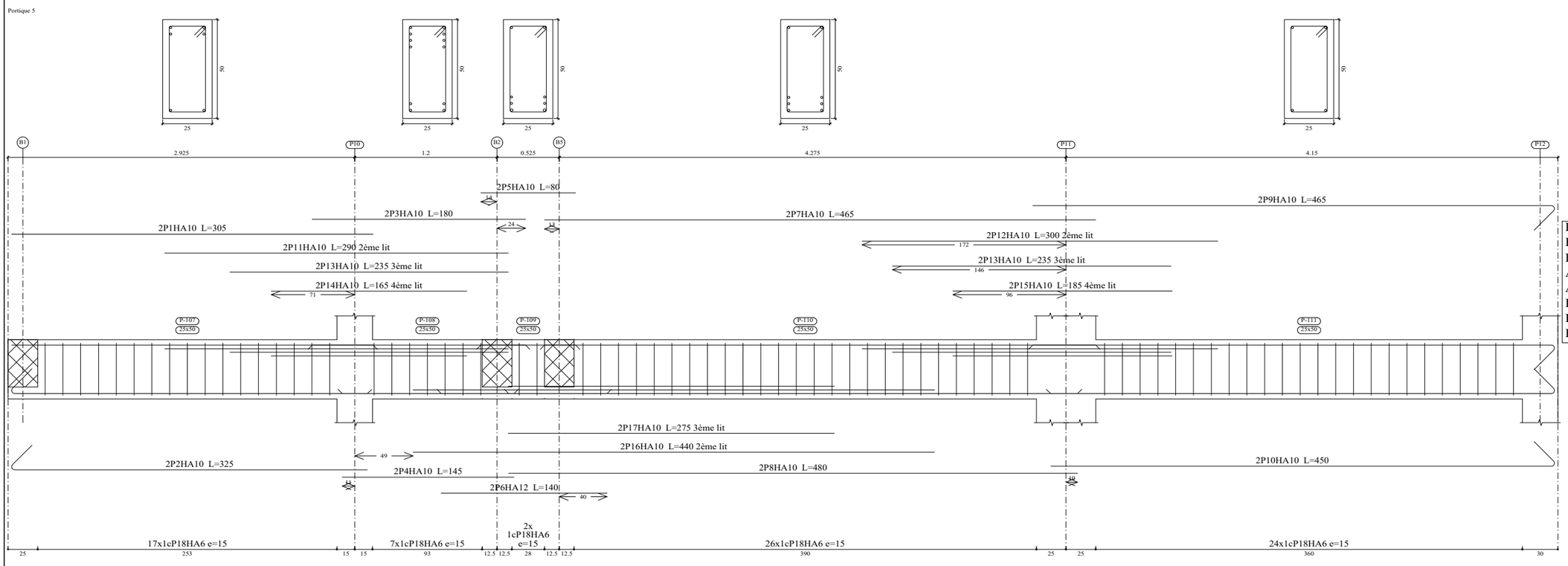


Portique 24



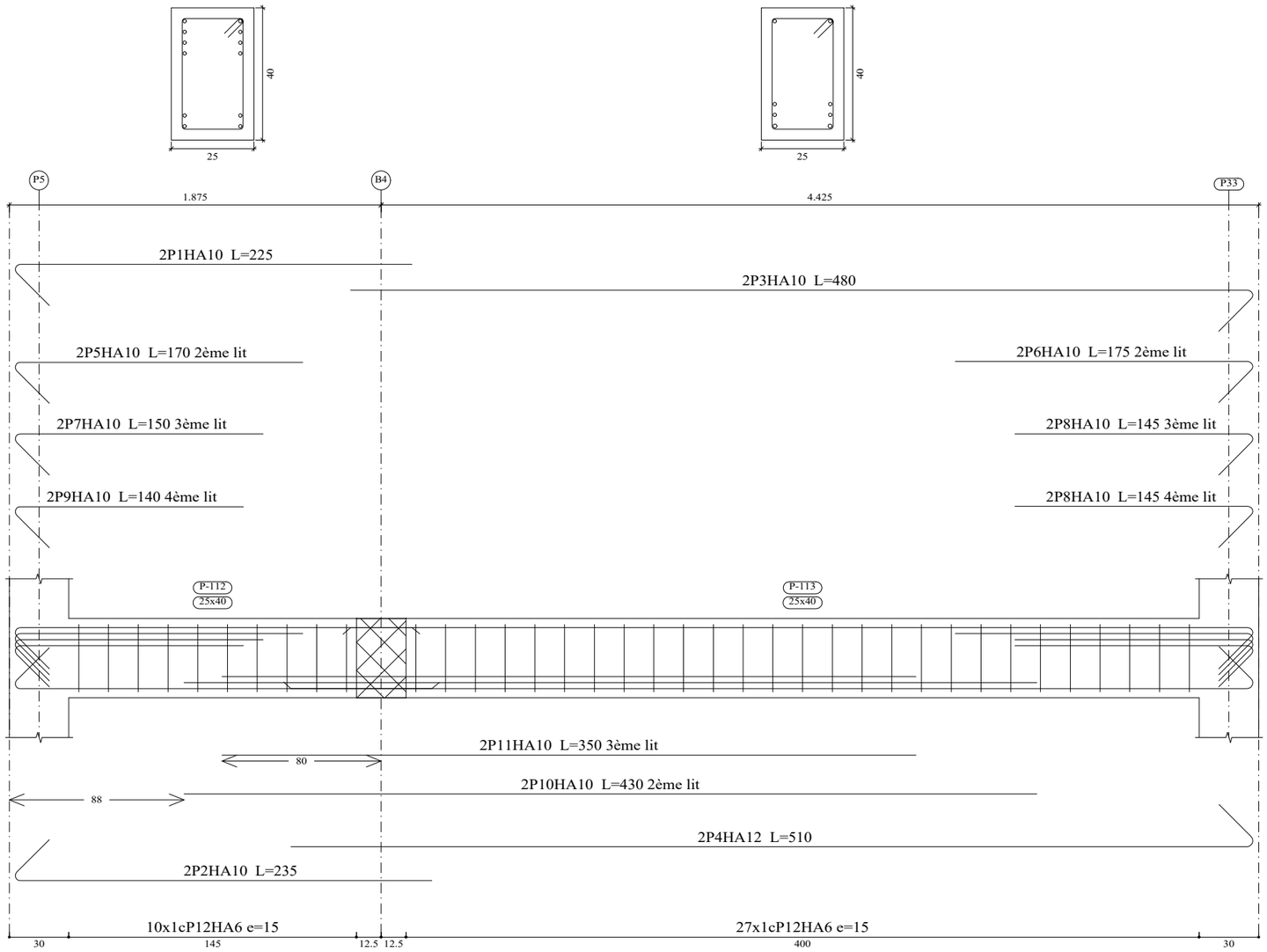
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 5	1	HA10	2	305	305	610	3.8
	2	HA10	2	25 300	325	650	4.0
	3	HA10	2	180	180	360	2.2
	4	HA10	2	145	145	290	1.8
	5	HA10	2	80	80	160	1.0
	6	HA12	2	140	140	280	2.5
	7	HA10	2	465	465	930	5.7
	8	HA10	2	480	480	960	5.9
	9	HA10	2	440 15	465	930	5.7
	10	HA10	2	425 25	450	900	5.5
	11	HA10	2	290	290	580	3.6
	12	HA10	2	300	300	600	3.7
	13	HA10	4	235	235	940	5.8
	14	HA10	2	165	165	330	2.0
	15	HA10	2	185	185	370	2.3
	16	HA10	2	440	440	880	5.4
	17	HA10	2	275	275	550	3.4
	18	HA6	76	6° 43 18	136	10336	22.9
						Total+10%:	95.9
Portique 11	1	HA10	2	25 425	450	900	5.5
	2	HA10	2	25 415	440	880	5.4
	3	HA10	2	185 25	210	420	2.6
	4	HA10	2	175 25	200	400	2.5
	5	HA10	2	25 105	130	260	1.6
	6	HA10	2	180	180	360	2.2
	7	HA10	2	115	115	230	1.4
	8	HA10	2	250	250	500	3.1
	9	HA6	32	6° 33 18	116	3712	8.2
						Total+10%:	35.8
Portique 24	1	HA10	4	25 482 25	532	2128	13.1
	2	HA10	2	25 130	155	310	1.9
	3	HA6	27	6° 33 18	116	3132	7.0
						Total+10%:	24.2
						HA6:	42.0
						HA10:	111.2
						HA12:	2.7
						Total:	155.9

Portique 5



PB RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Portique 6



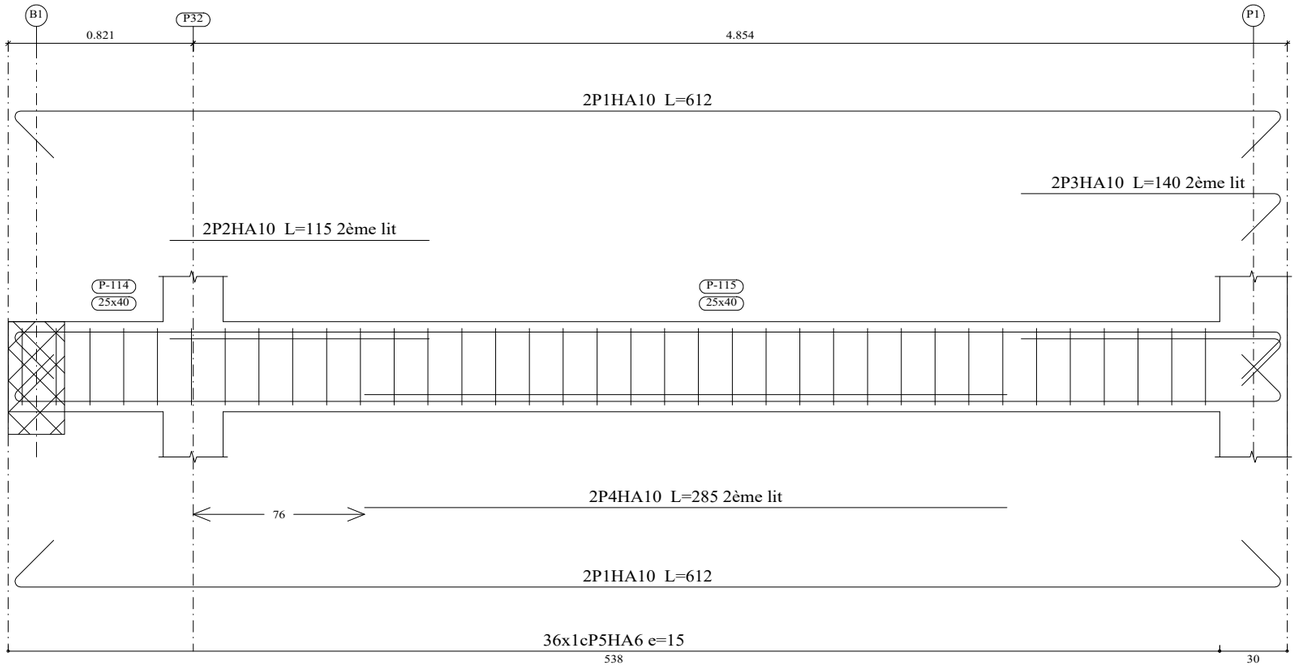
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 6	1	HA10	2	200, 2.5	225	450	2.8
	2	HA10	2	2.5, 210	235	470	2.9
	3	HA10	2	455, 2.5	480	960	5.9
	4	HA12	2	485, 2.5	510	1020	9.1
	5	HA10	2	145, 2.5	170	340	2.1
	6	HA10	2	150, 2.5	175	350	2.2
	7	HA10	2	125, 2.5	150	300	1.8
	8	HA10	4	120, 2.5	145	580	3.6
	9	HA10	2	115, 2.5	140	280	1.7
	10	HA10	2	430	430	860	5.3
	11	HA10	2	350	350	700	4.3
	12	HA6	37	33, 6, 18	116	4292	9.5
						Total+10%:	56.3
						HA6:	10.4
						HA10:	35.9
						HA12:	10.0
						Total:	56.3

PB RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 7	1	HA10	4	562, 25, 25	612	2448	15.1
	2	HA10	2	115	115	230	1.4
	3	HA10	2	115, 25	140	280	1.7
	4	HA10	2	285	285	570	3.5
	5	HA6	36	6, 33, 18	116	4176	9.3
				Total+10%:			34.1
				HA6:			10.2
				HA10:			23.9
				Total:			34.1

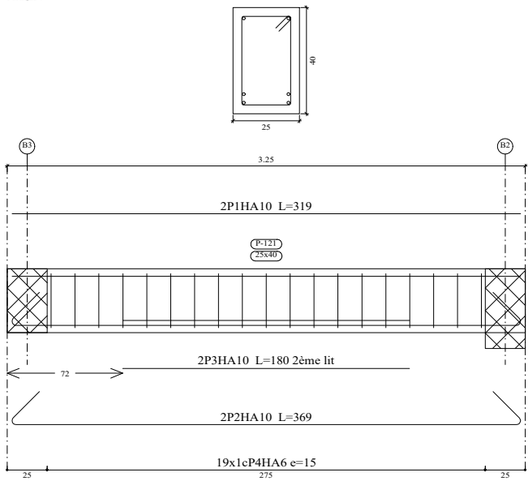
Portique 7



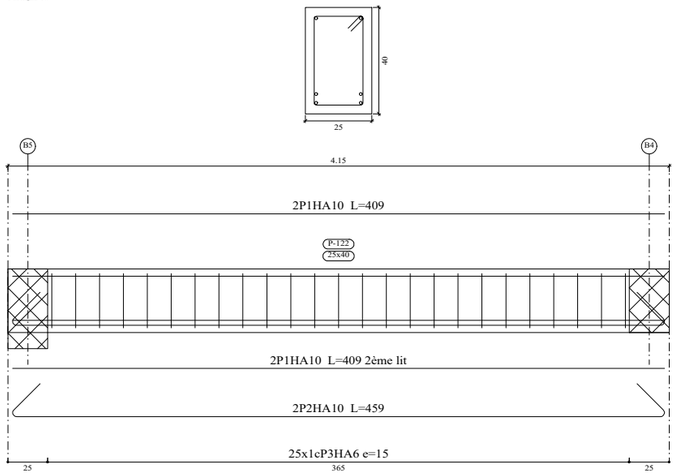


PB RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Portique 9

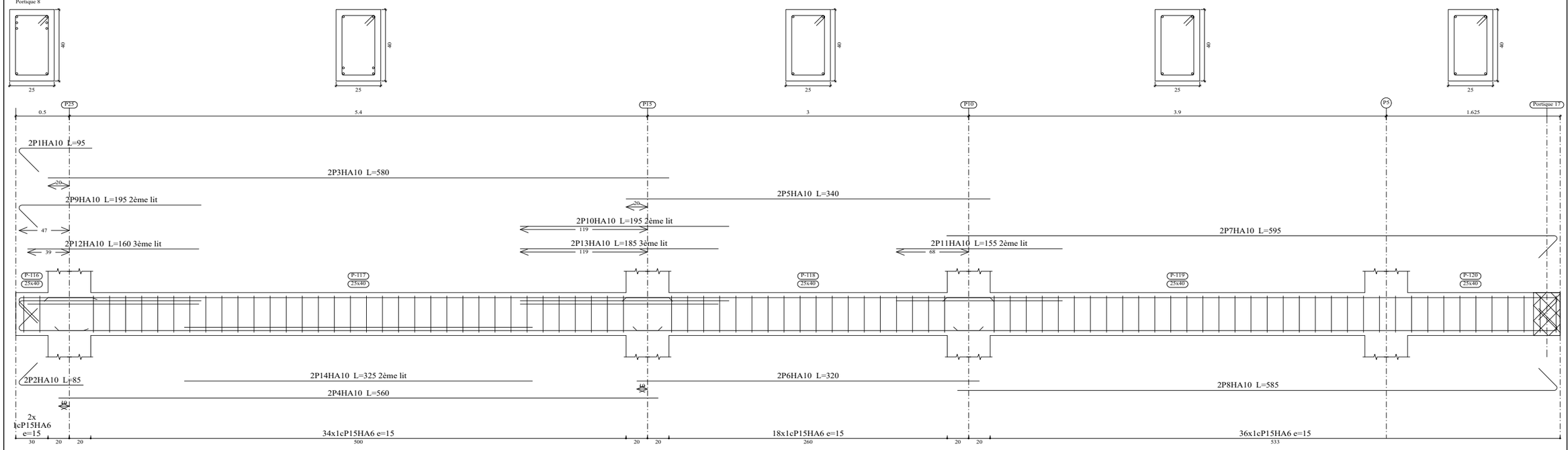


Portique 10



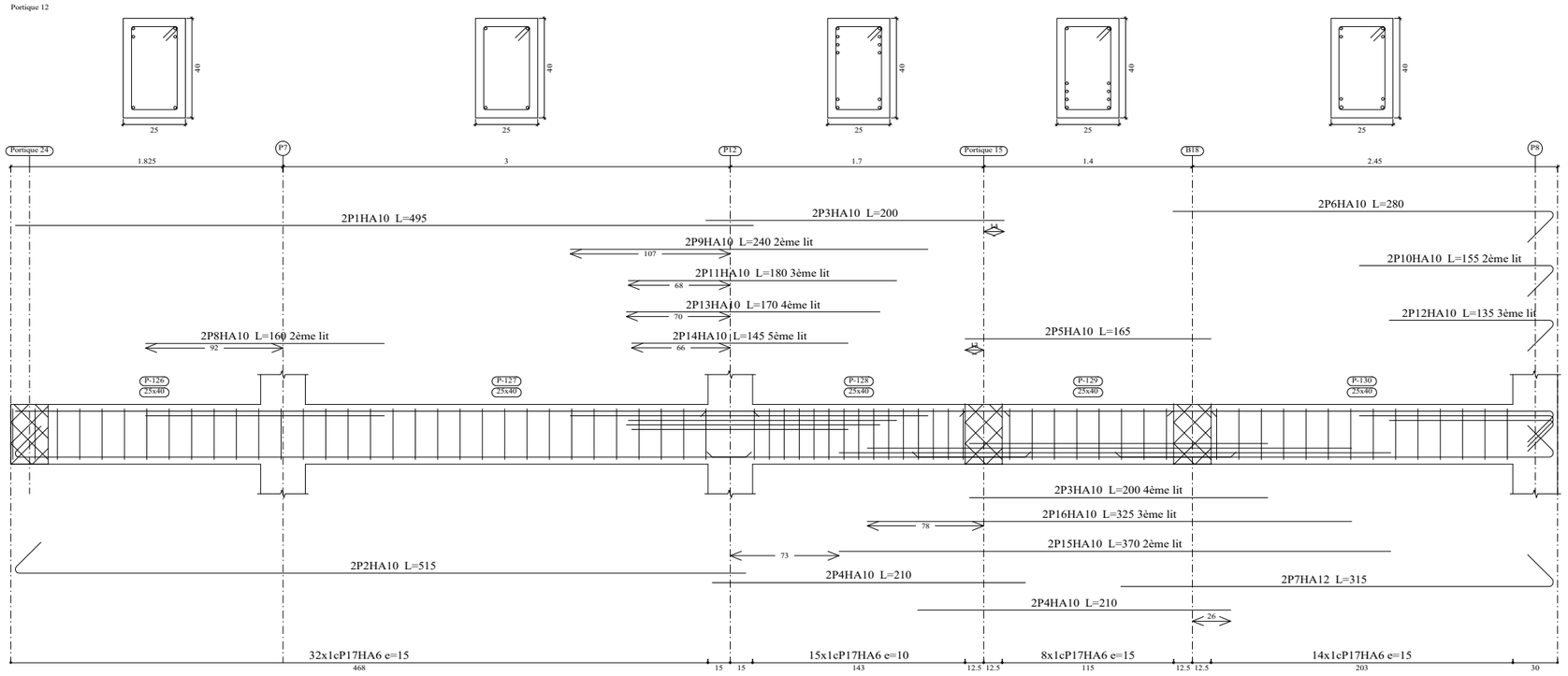
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 8	1	HA10	2	68	95	190	1.2
	2	HA10	2	60	85	170	1.0
	3	HA10	2	580	580	1160	7.2
	4	HA10	2	560	560	1120	6.9
	5	HA10	2	340	340	680	4.2
	6	HA10	2	320	320	640	3.9
	7	HA10	2	570	595	1190	7.3
	8	HA10	2	560	585	1170	7.2
	9	HA10	2	170	195	390	2.4
	10	HA10	2	195	195	390	2.4
	11	HA10	2	155	155	310	1.9
	12	HA10	2	160	160	320	2.0
	13	HA10	2	185	185	370	2.3
	14	HA10	2	325	325	650	4.0
	15	HA6	90	6 / 33 / 18	116	10440	23.2
						Total+10%:	84.8
Portique 9	1	HA10	2	319	319	638	3.9
	2	HA10	2	319	369	738	4.6
	3	HA10	2	180	180	360	2.2
	4	HA6	19	6 / 33 / 18	116	2204	4.9
						Total+10%:	17.2
Portique 10	1	HA10	4	409	409	1636	10.1
	2	HA10	2	409	459	918	5.7
	3	HA6	25	6 / 33 / 18	116	2900	6.4
						Total+10%:	24.4
						HA6:	37.9
						HA10:	88.5
						Total:	126.4

Portique 8



PB RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

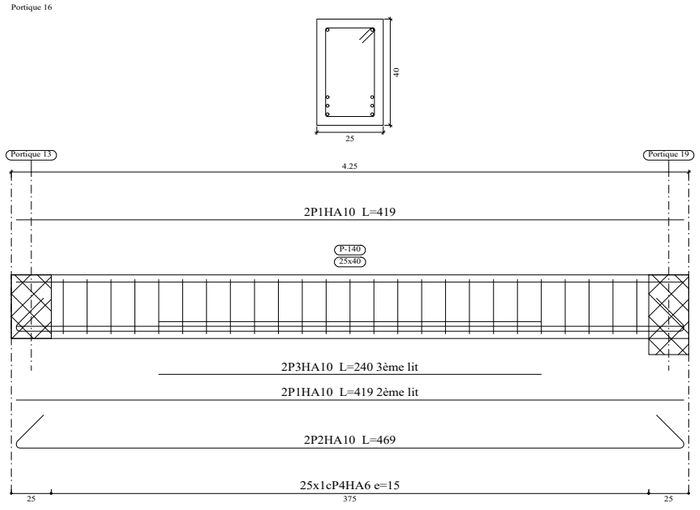
Portique 12



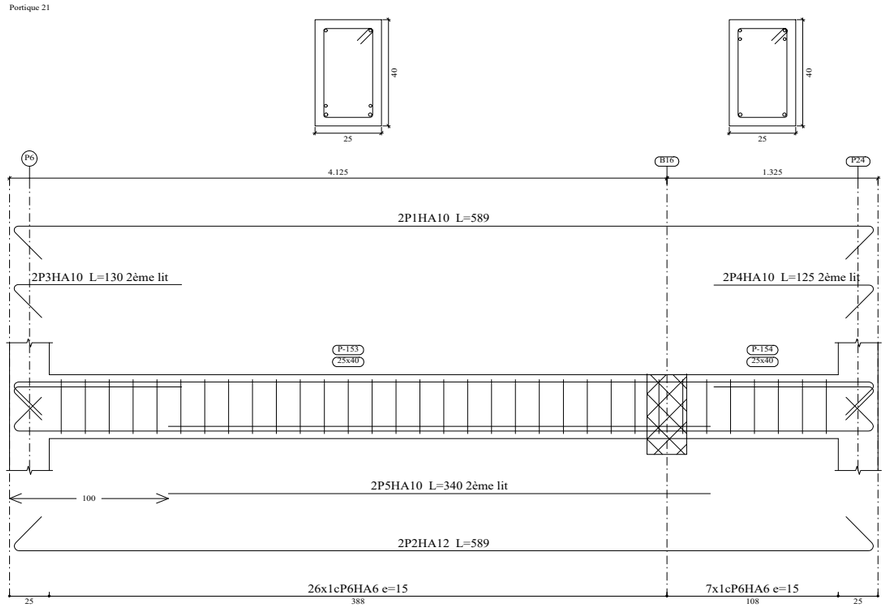
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 12	1	HA10	2	495	495	990	6.1
	2	HA10	2	25 490 25	515	1030	6.4
	3	HA10	4	200	200	800	4.9
	4	HA10	4	210	210	840	5.2
	5	HA10	2	165	165	330	2.0
	6	HA10	2	255 25	280	560	3.5
	7	HA12	2	290 25	315	630	5.6
	8	HA10	2	160	160	320	2.0
	9	HA10	2	240	240	480	3.0
	10	HA10	2	130 25	155	310	1.9
	11	HA10	2	180	180	360	2.2
	12	HA10	2	110 25	135	270	1.7
	13	HA10	2	170	170	340	2.1
	14	HA10	2	145	145	290	1.8
	15	HA10	2	370	370	740	4.6
	16	HA10	2	325	325	650	4.0
	17	HA6	69	6 33 18	116	8004	17.8
						Total+10%:	82.3
Portique 16	1	HA10	4	419	419	1676	10.3
	2	HA10	2	25 419 25	469	938	5.8
	3	HA10	2	240	240	480	3.0
	4	HA6	25	6 33 18	116	2900	6.4
						Total+10%:	28.1
Portique 21	1	HA10	2	25 539 25	589	1178	7.3
	2	HA12	2	25 539 25	589	1178	10.5
	3	HA10	2	25 105	130	260	1.6
	4	HA10	2	100 25	125	250	1.5
	5	HA10	2	340	340	680	4.2
	6	HA6	33	6 33 18	116	3828	8.5
						Total+10%:	37.0
Portique 23	1	HA10	2	397 25	422	844	5.2
	2	HA10	2	25 397 25	447	894	5.5
	3	HA10	2	205	205	410	2.5
	4	HA6	25	6 33 18	116	2900	6.4
						Total+10%:	21.6
						HA6:	43.2
						HA10:	108.1
						HA12:	17.7
						Total:	169.0

PB RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

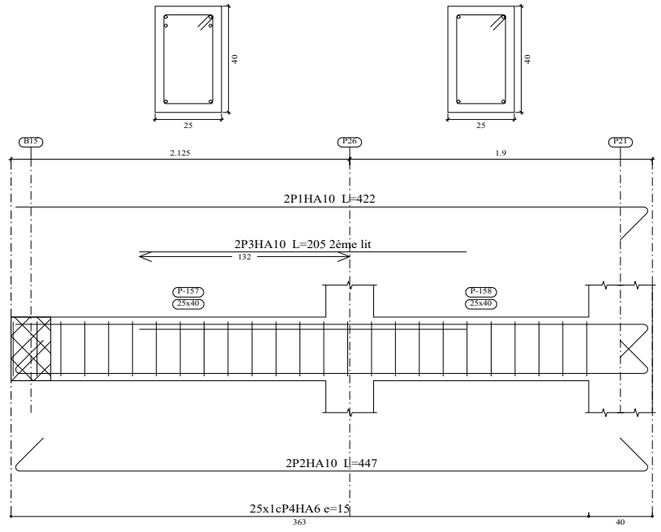
Portique 16



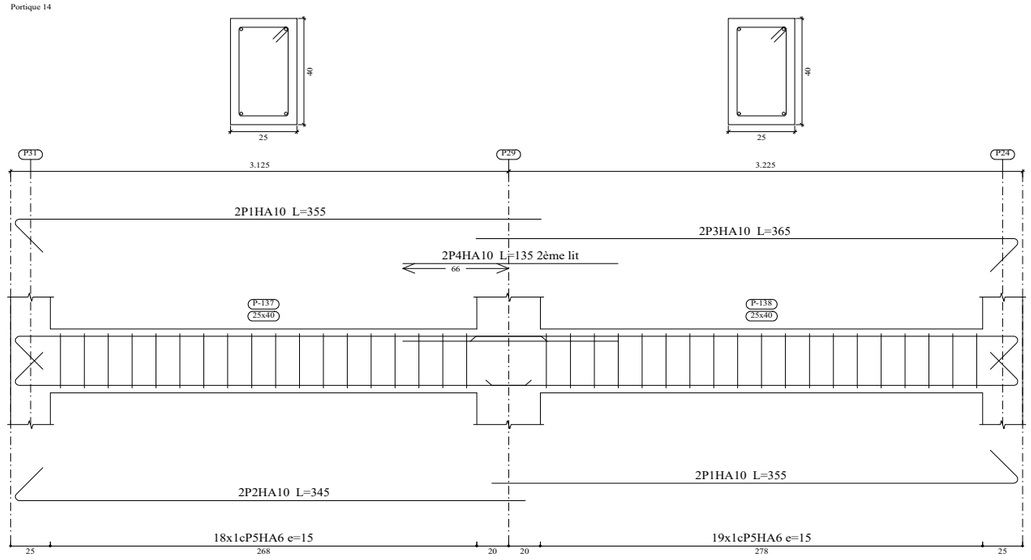
Portique 21



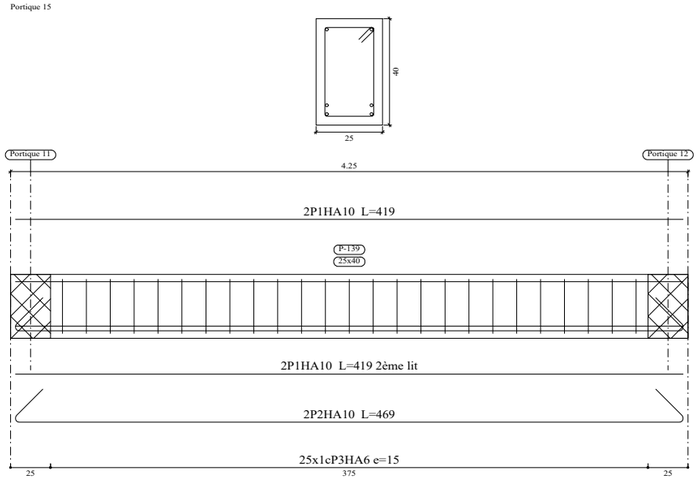
Portique 23



Portique 14

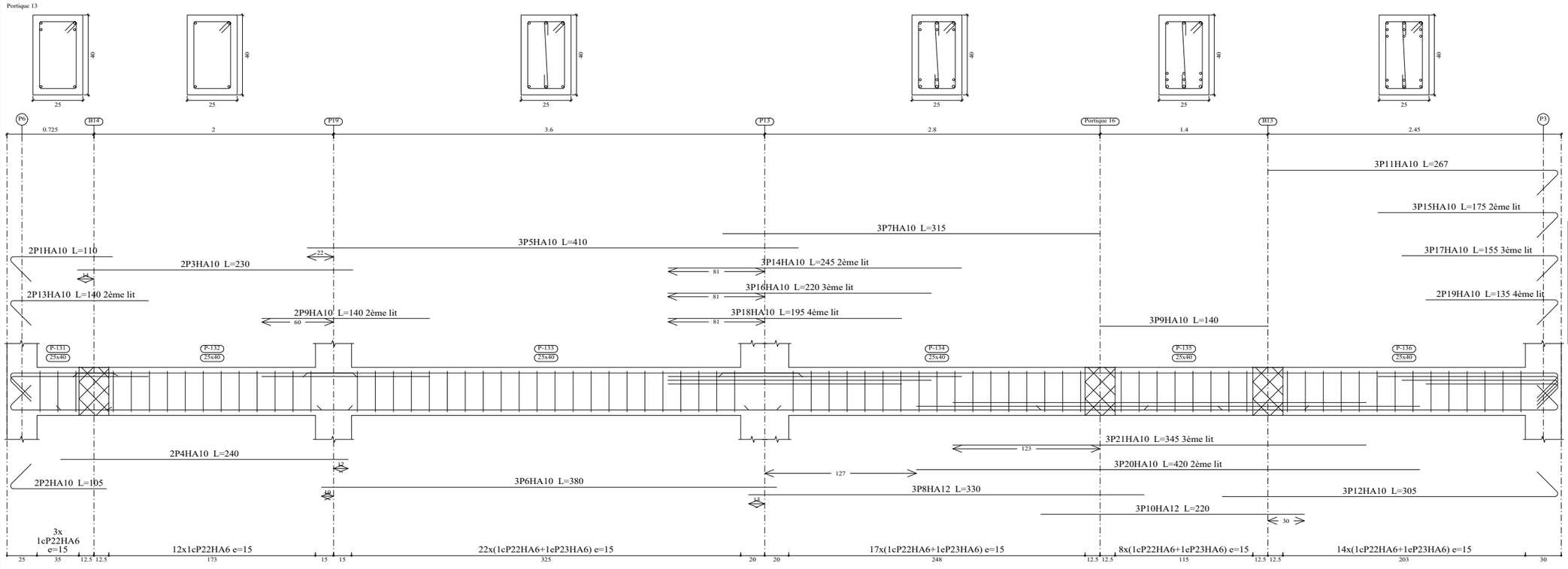


Portique 15

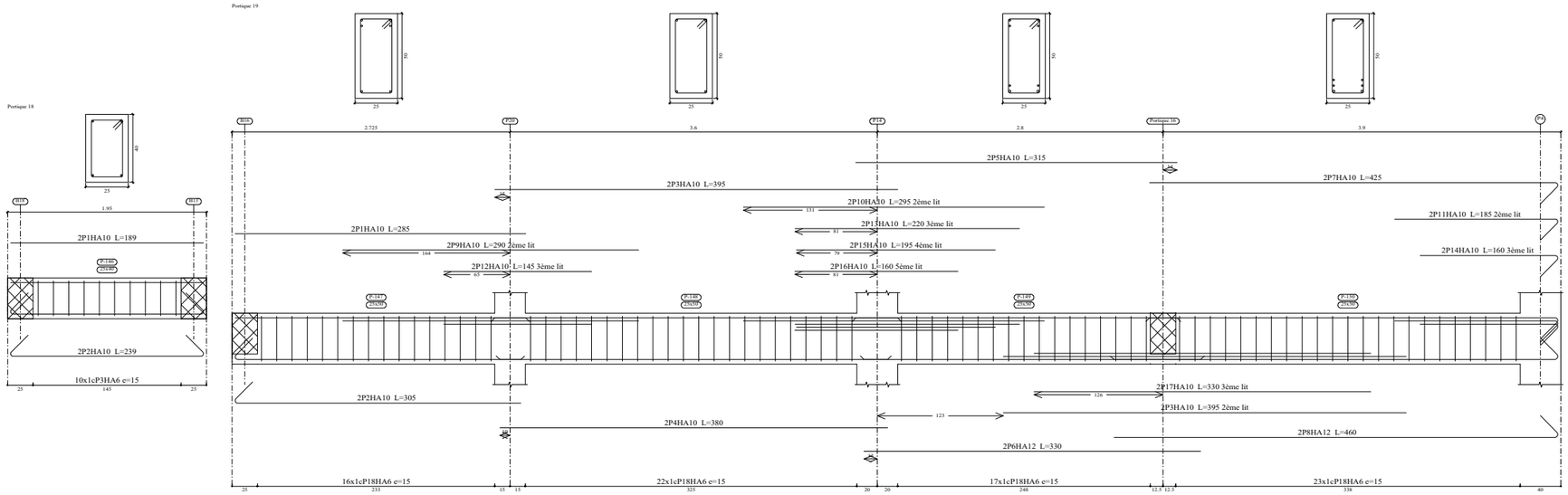


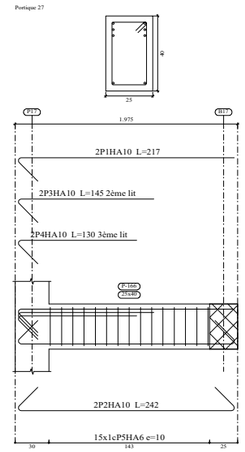
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 13	1	HA10	2	85	110	220	1.4
	2	HA10	2	80	105	210	1.3
	3	HA10	2	230	230	460	2.8
	4	HA10	2	240	240	480	3.0
	5	HA10	3	410	410	1230	7.6
	6	HA10	3	380	380	1140	7.0
	7	HA10	3	315	315	945	5.8
	8	HA12	3	330	330	990	8.8
	9	HA10	5	140	140	700	4.3
	10	HA12	3	220	220	660	5.9
	11	HA10	3	242 20	267	801	4.9
	12	HA10	3	280 20	305	915	5.6
	13	HA10	2	115	140	280	1.7
	14	HA10	3	245	245	735	4.5
	15	HA10	3	150 20	175	525	3.2
	16	HA10	3	220	220	660	4.1
	17	HA10	3	130 20	155	465	2.9
	18	HA10	3	195	195	585	3.6
	19	HA10	2	110 20	135	270	1.7
	20	HA10	3	420	420	1260	7.8
	21	HA10	3	345	345	1035	6.4
	22	HA6	76	6 33 18	116	8816	19.6
	23	HA6	61	6 33	45	2745	6.1
						Total+10%:	132.0
Portique 14	1	HA10	4	330	355	1420	8.8
	2	HA10	2	320	345	690	4.3
	3	HA10	2	340 20	365	730	4.5
	4	HA10	2	135	135	270	1.7
	5	HA6	37	6 33 18	116	4292	9.5
						Total+10%:	31.7
Portique 15	1	HA10	4	419	419	1676	10.3
	2	HA10	2	419	469	938	5.8
	3	HA6	25	6 33 18	116	2900	6.4
						Total+10%:	24.8
						HA6:	45.9
						HA10:	126.5
						HA12:	16.1
						Total:	188.5

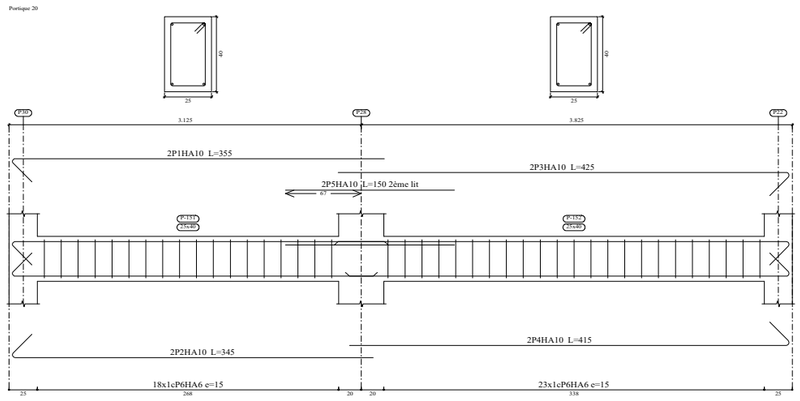
Portique 13

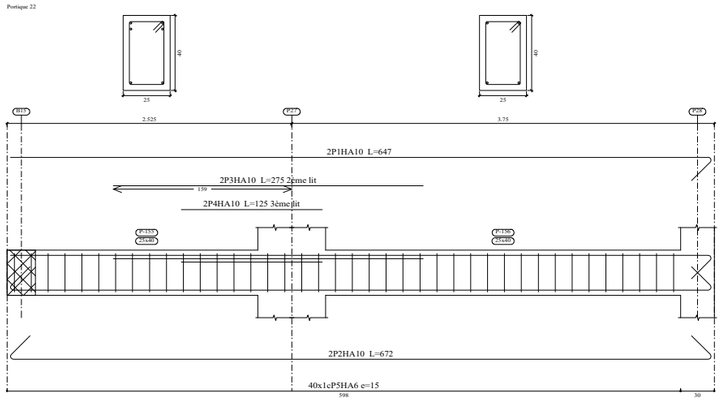


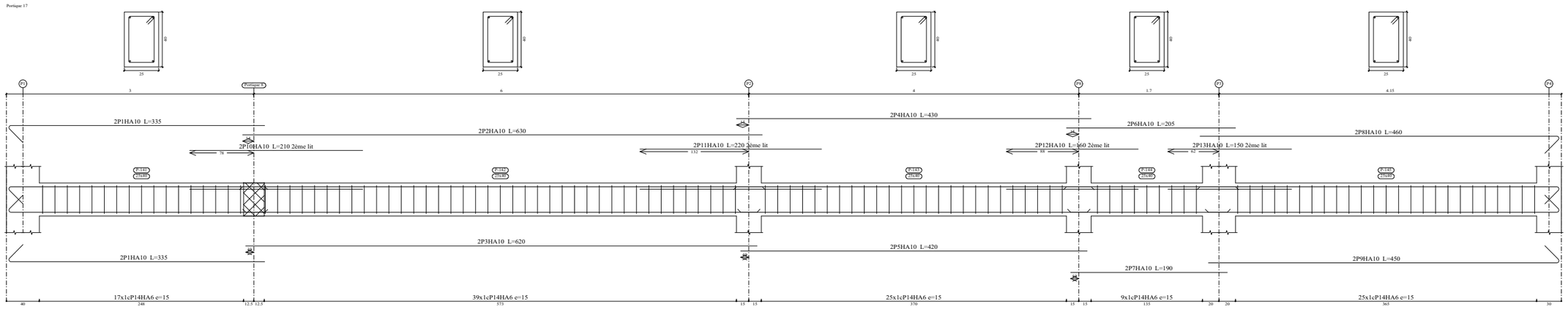
PB RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20











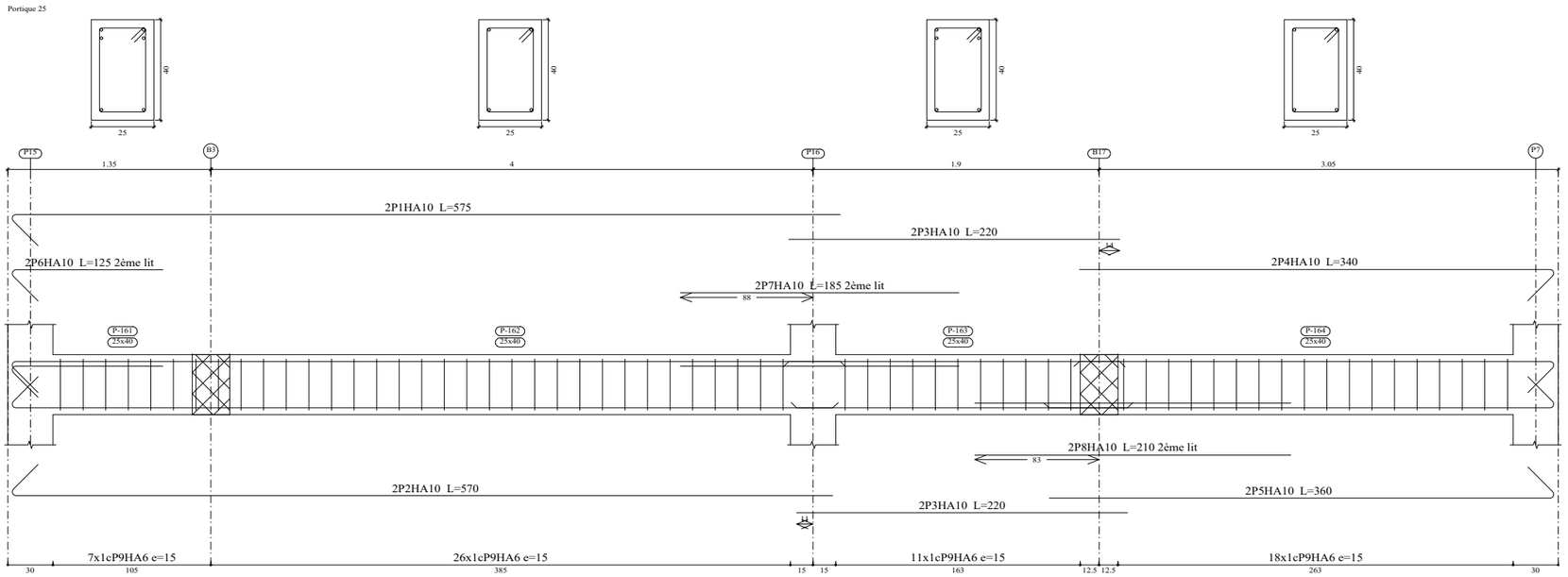
PB RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 17	1	HA10	4	310	335	1340	8.3
	2	HA10	2	630	630	1260	7.8
	3	HA10	2	620	620	1240	7.6
	4	HA10	2	430	430	860	5.3
	5	HA10	2	420	420	840	5.2
	6	HA10	2	205	205	410	2.5
	7	HA10	2	190	190	380	2.3
	8	HA10	2	435 / 25	460	920	5.7
	9	HA10	2	425 / 25	450	900	5.5
	10	HA10	2	210	210	420	2.6
	11	HA10	2	220	220	440	2.7
	12	HA10	2	160	160	320	2.0
	13	HA10	2	150	150	300	1.8
	14	HA6	115	25 / 18	116	13340	29.6
					Total+10%:		97.8
Portique 18	1	HA10	2	189	189	378	2.3
	2	HA10	2	189 / 25	239	478	2.9
	3	HA6	10	25 / 18	116	1160	2.6
					Total+10%:		8.6
Portique 19	1	HA10	2	285	285	570	3.5
	2	HA10	2	280 / 25	305	610	3.8
	3	HA10	4	395	395	1580	9.7
	4	HA10	2	380	380	760	4.7
	5	HA10	2	315	315	630	3.9
	6	HA12	2	320	330	660	5.9
	7	HA10	2	400 / 25	425	850	5.2
	8	HA12	2	435 / 25	460	920	8.2
	9	HA10	2	290	290	580	3.6
	10	HA10	2	295	295	590	3.6
	11	HA10	2	160 / 25	185	370	2.3
	12	HA10	2	145	145	290	1.8
	13	HA10	2	220	220	440	2.7
	14	HA10	2	135 / 25	160	320	2.0
	15	HA10	2	195	195	390	2.4
	16	HA10	2	160	160	320	2.0
	17	HA10	2	320	330	660	4.1
	18	HA6	78	25 / 18	136	10608	23.5
					Total+10%:		102.2
Portique 20	1	HA10	2	330	355	710	4.4
	2	HA10	2	320 / 25	345	690	4.3
	3	HA10	2	400 / 25	425	850	5.2
	4	HA10	2	390 / 25	415	830	5.1
	5	HA10	2	150	150	300	1.8
	6	HA6	41	25 / 18	116	4756	10.6
					Total+10%:		34.5
Portique 22	1	HA10	2	622	647	1294	8.0
	2	HA10	2	622 / 25	672	1344	8.3
	3	HA10	2	275	275	550	3.4
	4	HA10	2	125	125	250	1.5
	5	HA6	40	25 / 18	116	4640	10.3
					Total+10%:		34.7
Portique 27	1	HA10	2	192	217	434	2.7
	2	HA10	2	192 / 25	242	484	3.0
	3	HA10	2	120	145	290	1.8
	4	HA10	2	105	130	260	1.6
	5	HA6	15	25 / 18	116	1740	3.9
					Total+10%:		14.3
					HA6:		88.7
					HA10:		187.9
					HA12:		15.5
					Total:		292.1

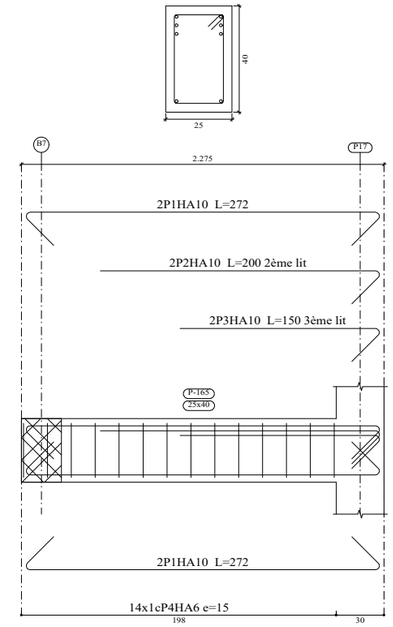
PB RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 25	1	HA10	2	550, 15	575	1150	7.1
	2	HA10	2	15, 545	570	1140	7.0
	3	HA10	4	220	220	880	5.4
	4	HA10	2	315, 25	340	680	4.2
	5	HA10	2	335, 25	360	720	4.4
	6	HA10	2	25, 100	125	250	1.5
	7	HA10	2	185	185	370	2.3
	8	HA10	2	210	210	420	2.6
	9	HA6	62	6, 33, 18	116	7192	16.0
					Total+10%:		55.6
Portique 26	1	HA10	4	25, 222, 25	272	1088	6.7
	2	HA10	2	25, 175, 25	200	400	2.5
	3	HA10	2	25, 125, 25	150	300	1.8
	4	HA6	14	6, 33, 18	116	1624	3.6
					Total+10%:		16.1
						HA6:	21.6
						HA10:	50.1
						Total:	71.7

Portique 25

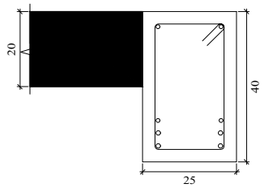


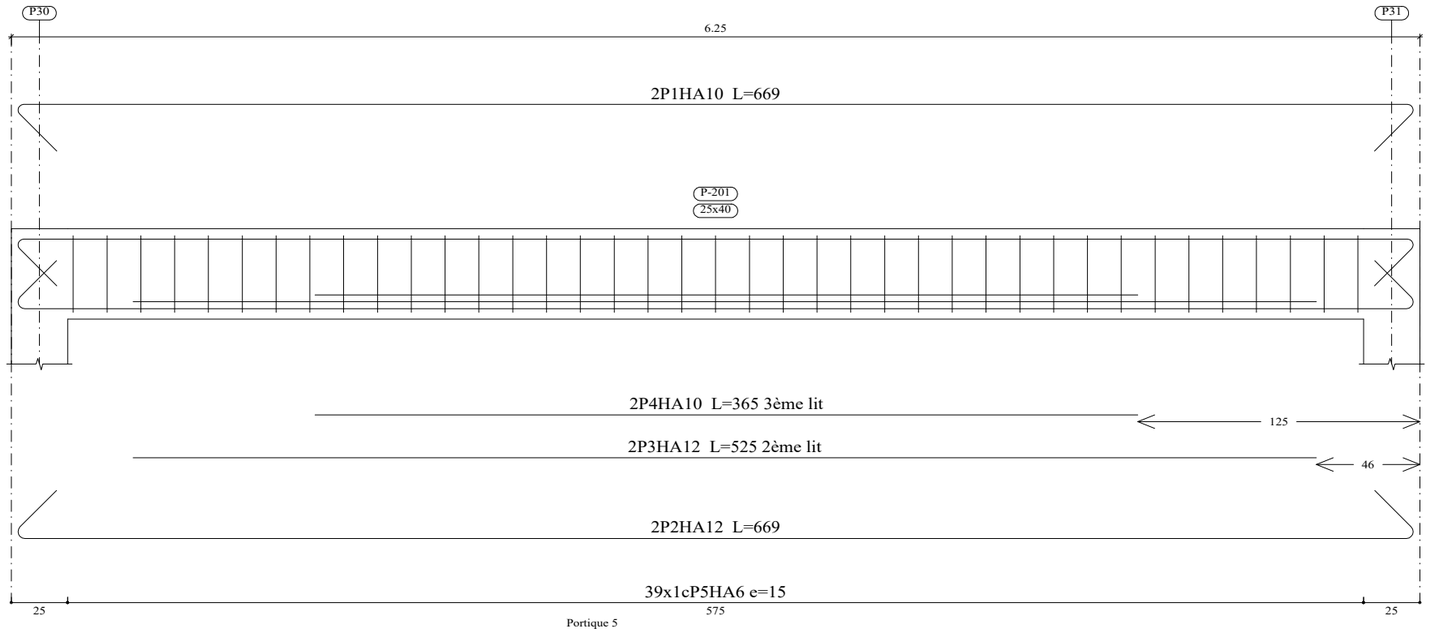
Portique 26



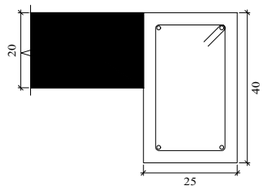
POUTRES PH GARAGE

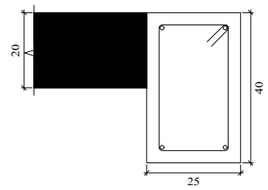
Portique 1





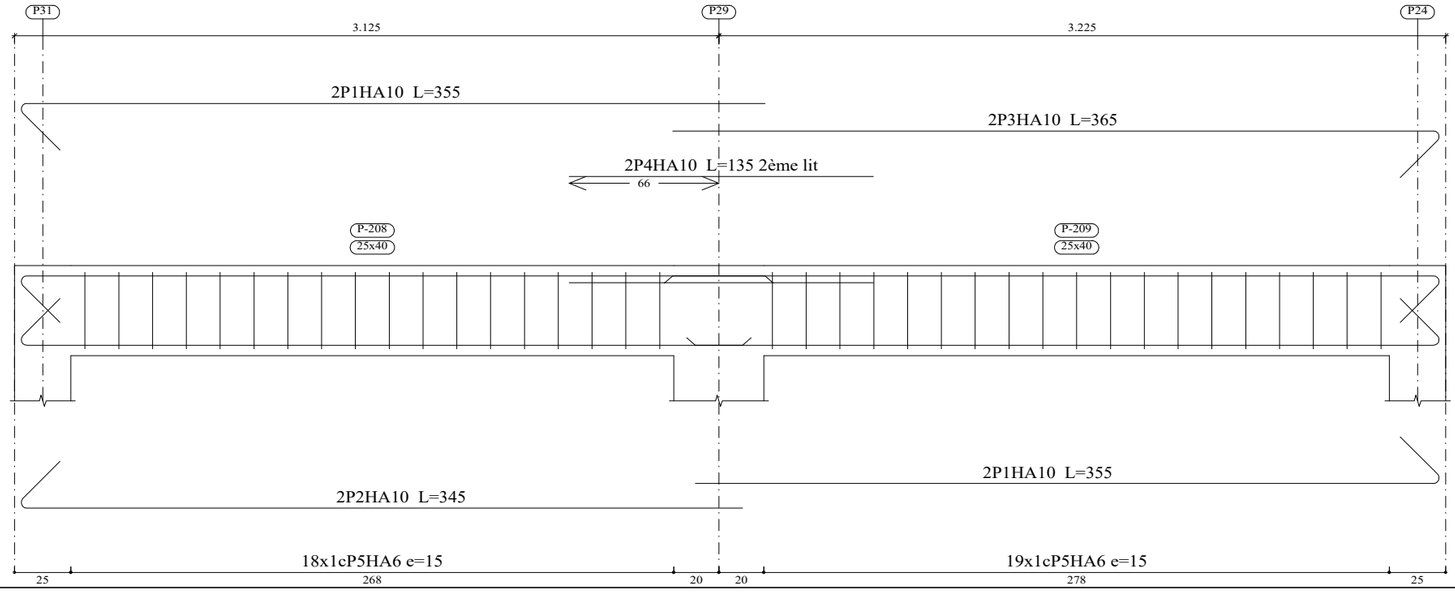
Portique 5



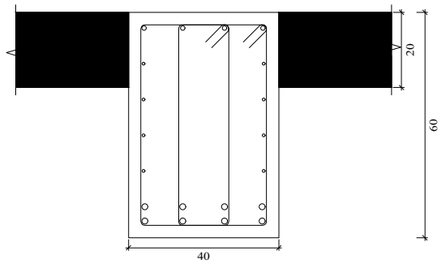


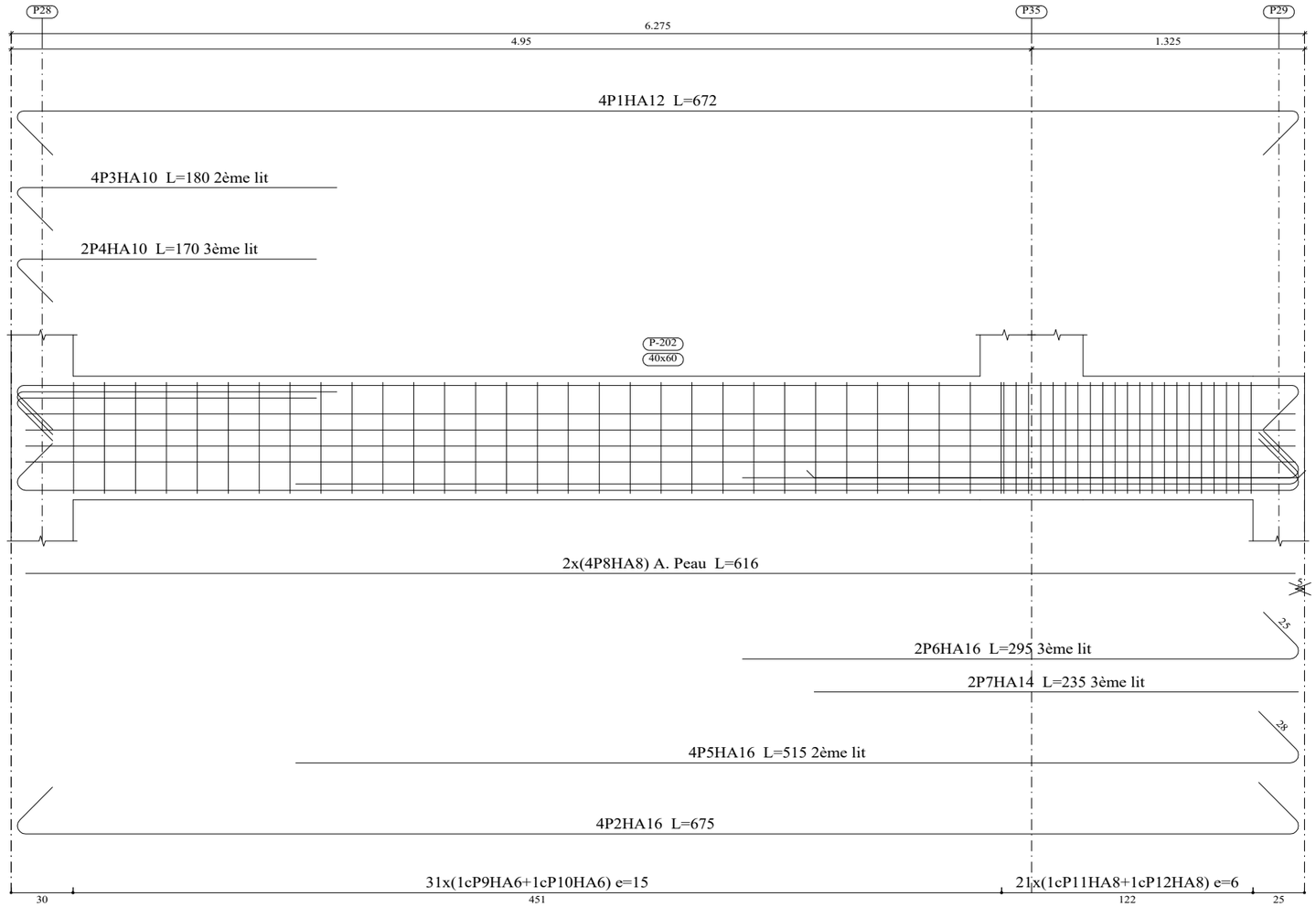
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 1	1	HA10	2	619, 25, 25	669	1338	8.2
	2	HA12	2	25, 619, 25	669	1338	11.9
	3	HA12	2	525	525	1050	9.3
	4	HA10	2	365	365	730	4.5
	5	HA6	39	33, 6, 18	116	4524	10.0
						Total+10%:	48.3
Portique 5	1	HA10	4	330, 25	355	1420	8.8
	2	HA10	2	25, 320	345	690	4.3
	3	HA10	2	340, 25	365	730	4.5
	4	HA10	2	135	135	270	1.7
	5	HA6	37	33, 6, 18	116	4292	9.5
						Total+10%:	31.7
						HA6:	21.5
						HA10:	35.2
						HA12:	23.3
						Total:	80.0

PH GARAGE
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20



Portique 2
Voir bases dans le détail des poteaux ou vue des murs



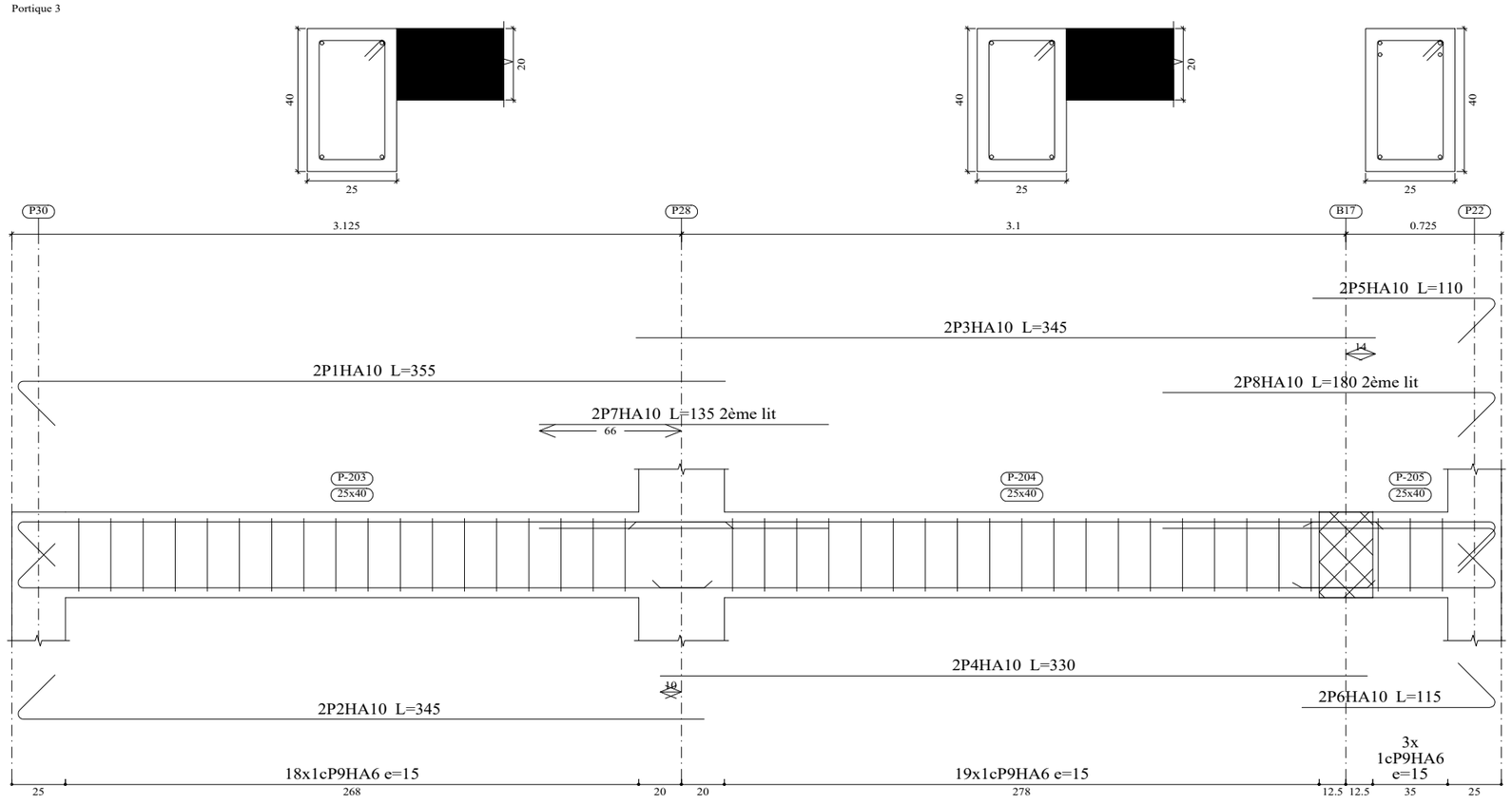


Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 2	1	HA12	4	622 (25, 25)	672	2688	23.9
	2	HA16	4	622 (25, 12)	675	2700	42.6
	3	HA10	4	155 (25)	180	720	4.4
	4	HA10	2	145 (25)	170	340	2.1
	5	HA16	4	487 (12)	515	2060	32.5
	6	HA16	2	270 (12)	295	590	9.3
	7	HA14	2	235	235	470	5.7
	8	HA8	8	616	616	4928	19.4
	9	HA6	31	53, 6, 33	186	5766	12.8
	10	HA6	31	53, 6, 11	141	4371	9.7
	11	HA8	21	53, 8, 33	189	3969	15.7
	12	HA8	21	53, 8, 11	145	3045	12.0
					Total+10%:		209.1

HA6:	24.7
HA8:	51.8
HA10:	7.2
HA12:	26.3
HA14:	6.3
HA16:	92.8
Total:	209.1

PH GARAGE
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Portique 3



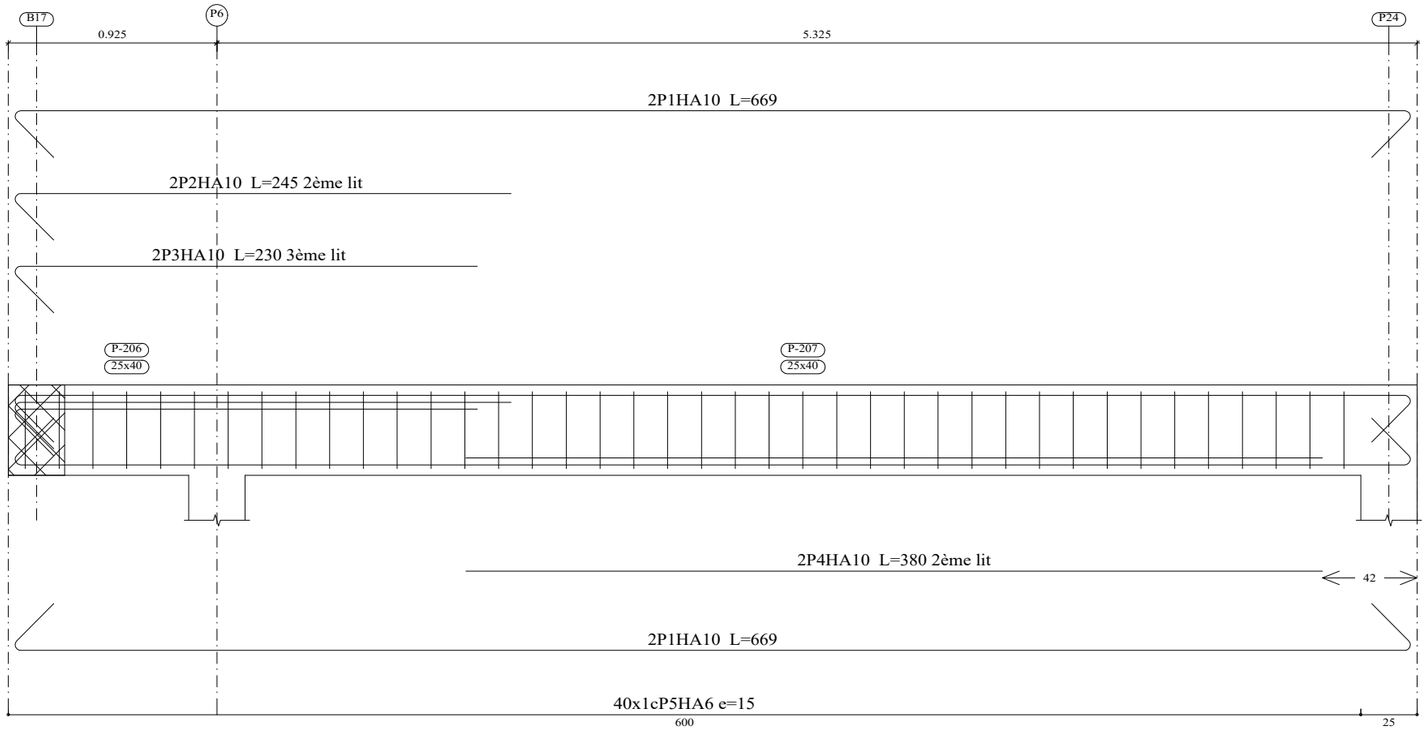
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 3	1	HA10	2	330, 25	355	710	4.4
	2	HA10	2	25, 320	345	690	4.3
	3	HA10	2	345	345	690	4.3
	4	HA10	2	330	330	660	4.1
	5	HA10	2	85, 25	110	220	1.4
	6	HA10	2	90, 25	115	230	1.4
	7	HA10	2	135	135	270	1.7
	8	HA10	2	155, 25	180	360	2.2
	9	HA6	40	33, 6, 18	116	4640	10.3
						Total+10%:	37.5
						HA6:	11.3
						HA10:	26.2
						Total:	37.5

PH GARAGE
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 4	1	HA10	4	619 (12.5 / 25)	669	2676	16.5
	2	HA10	2	220 (12.5)	245	490	3.0
	3	HA10	2	205 (12.5)	230	460	2.8
	4	HA10	2	380	380	760	4.7
	5	HA6	40	6 / 33 / 18	116	4640	10.3
					Total+10%:		41.0
					HA6:		11.3
					HA10:		29.7
					Total:		41.0

Portique 4





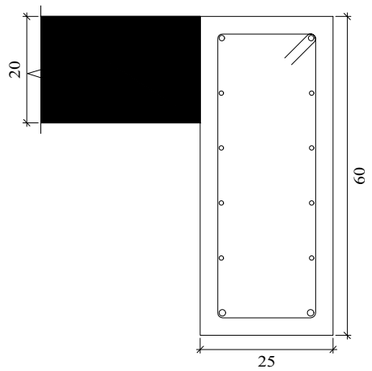
PH GARAGE
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

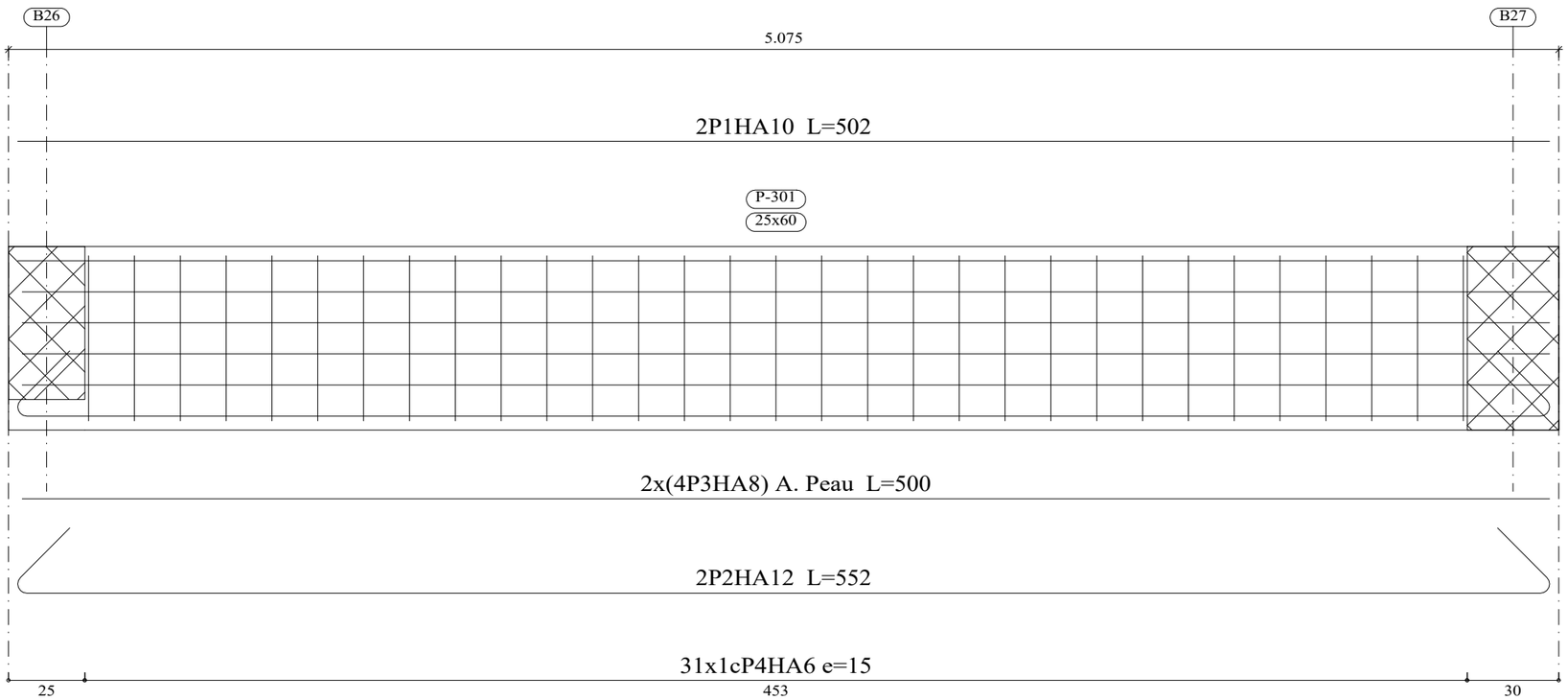
POUTRES PH RDC

PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

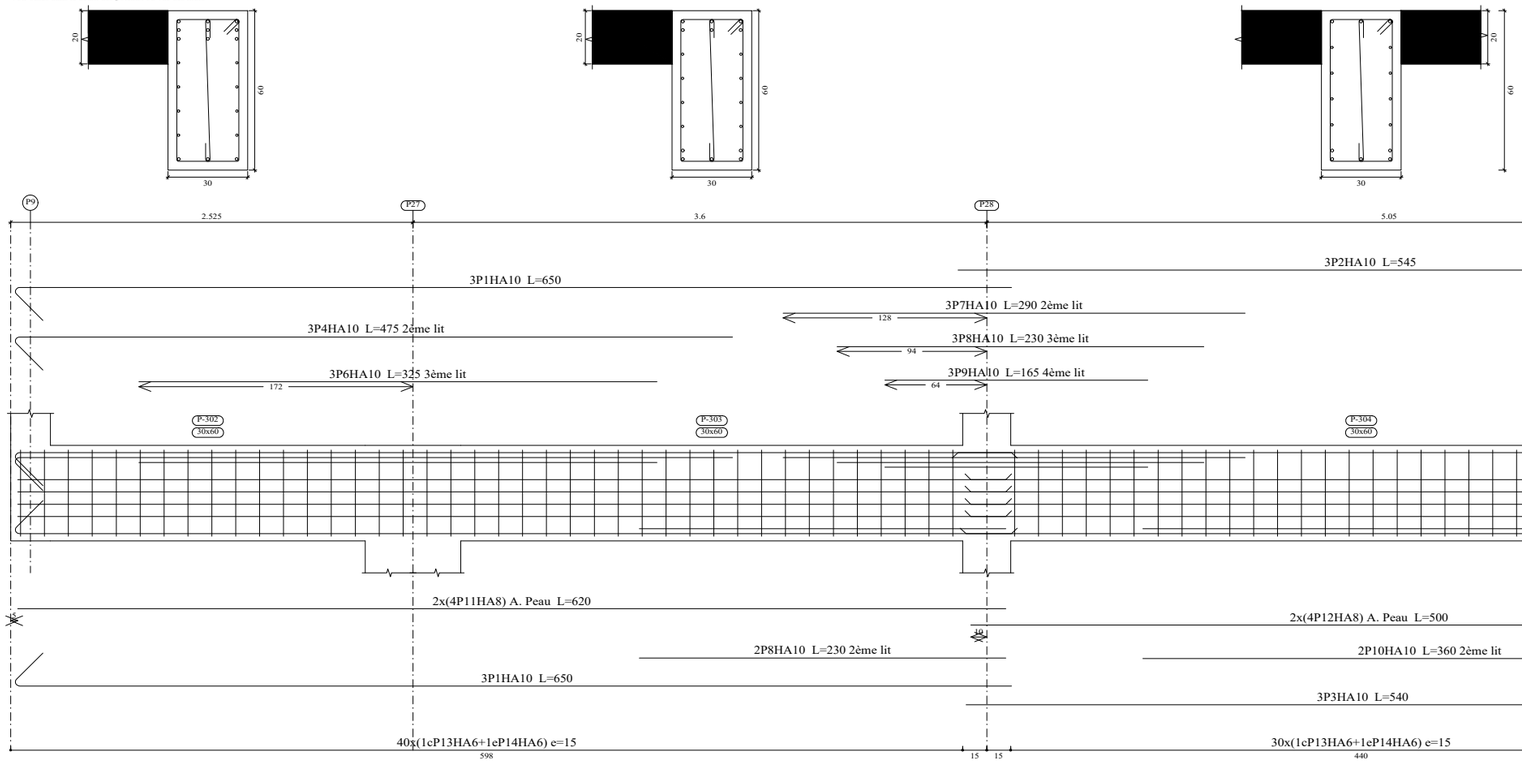
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 1	1	HA10	2	502	502	1004	6.2
	2	HA12	2	25 / 502 / 25	552	1104	9.8
	3	HA8	8	500	500	4000	15.8
	4	HA6	31	53 / 6 / 18	156	4836	10.7
					Total+10%:		46.8
					HA6:		11.8
					HA8:		17.4
					HA10:		6.8
					HA12:		10.8
					Total:		46.8

Portique 1





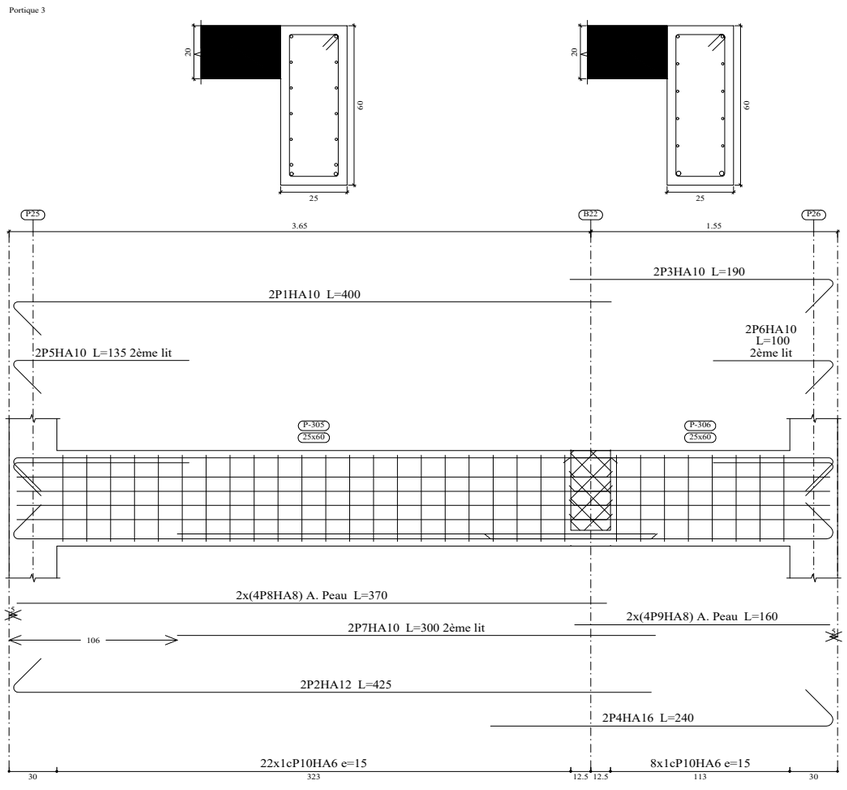
Portique 2
Voir bases dans le détail des poteaux ou vue des murs



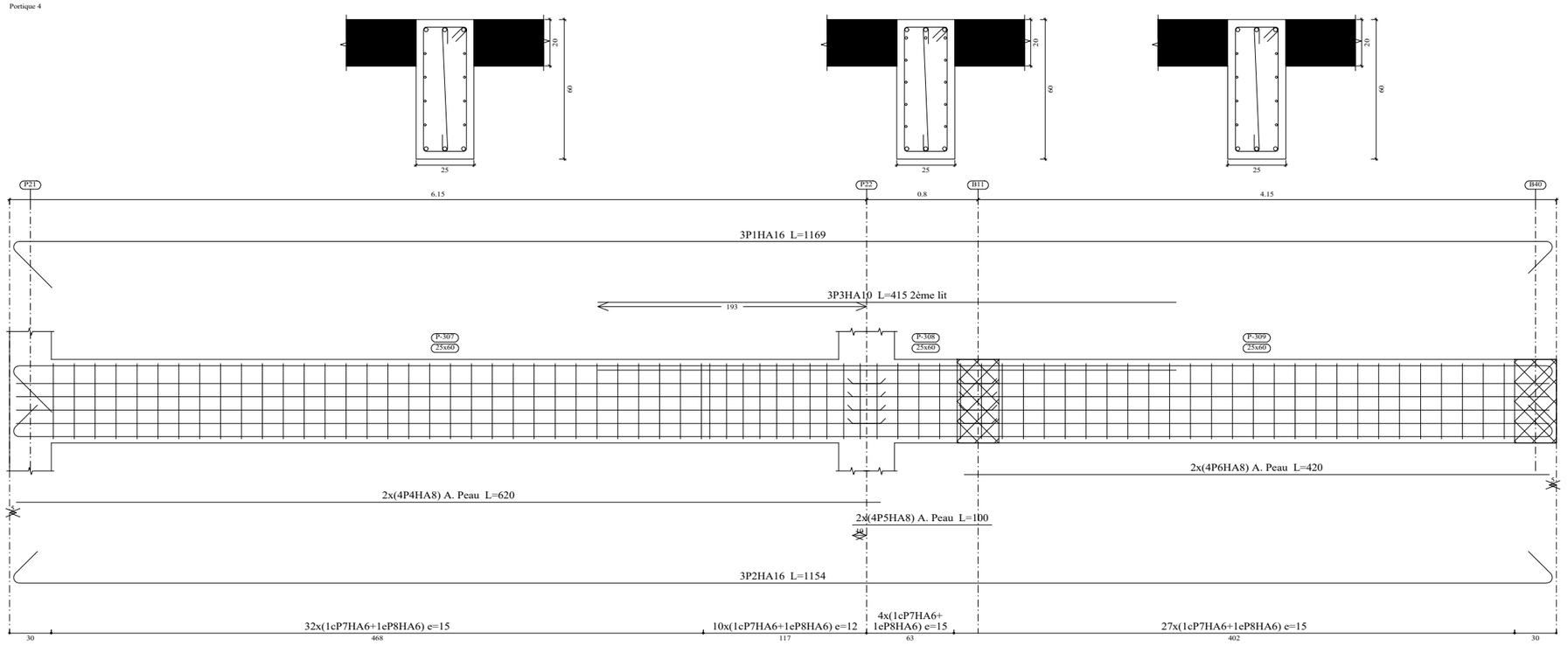
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 2	1	HA10	6	15 625	650	3900	24.0
	2	HA10	3	520 15	545	1635	10.1
	3	HA10	3	515 15	540	1620	10.0
	4	HA10	3	15 450	475	1425	8.8
	5	HA10	2	125 15	150	300	1.8
	6	HA10	3	325	325	975	6.0
	7	HA10	3	290	290	870	5.4
	8	HA10	5	230	230	1150	7.1
	9	HA10	3	165	165	495	3.1
	10	HA10	2	360	360	720	4.4
	11	HA8	8	620	620	4960	19.6
	12	HA8	8	500	500	4000	15.8
	13	HA6	70	53 6 23	166	11620	25.8
	14	HA6	70	6 53	65	4550	10.1
						Total+10%:	167.2
Portique 3	1	HA10	2	15 375	400	800	4.9
	2	HA12	2	25 400	425	850	7.5
	3	HA10	2	165 15	190	380	2.3
	4	HA16	2	215 25	240	480	7.6
	5	HA10	2	110 15	135	270	1.7
	6	HA10	2	75 15	100	200	1.2
	7	HA10	2	300	300	600	3.7
	8	HA8	8	370	370	2960	11.7
	9	HA8	8	160	160	1280	5.1
	10	HA6	30	53 6 18	156	4680	10.4
						Total+10%:	61.7
						HA6:	50.9
						HA8:	57.4
						HA10:	104.0
						HA12:	8.2
						HA16:	8.4
						Total:	228.9

PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Portique 3



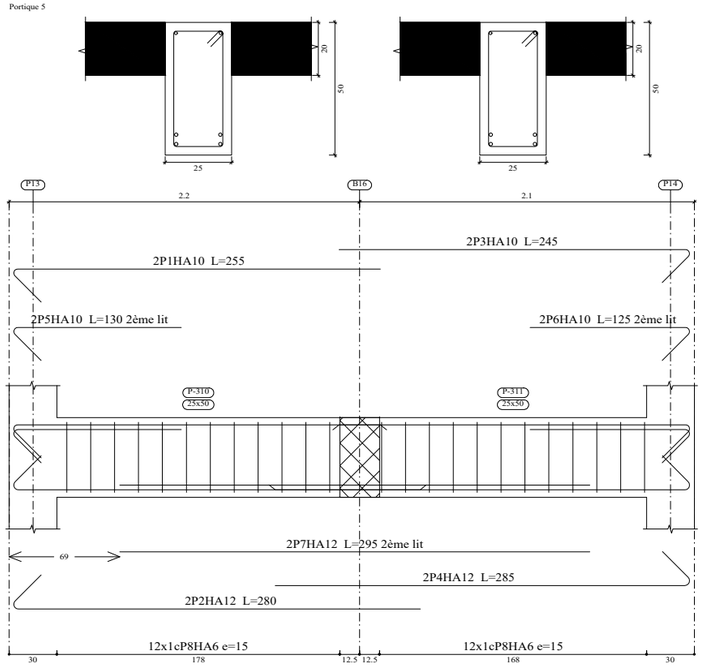
Portique 4



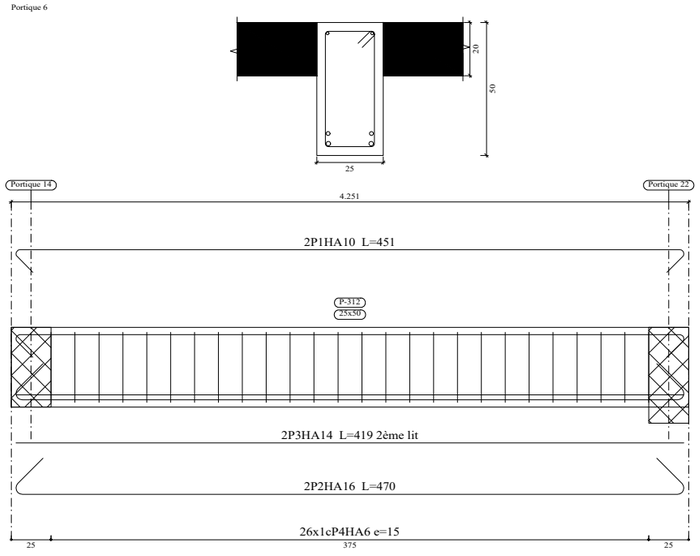
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 4	1	HA16	3	1104	1169	3507	55.4
	2	HA16	3	1104	1154	3462	54.6
	3	HA10	3	415	415	1245	7.7
	4	HA8	8	620	620	4960	19.6
	5	HA8	8	100	100	800	3.2
	6	HA8	8	420	420	3360	13.3
	7	HA6	73	43 6 18	156	11388	25.3
	8	HA6	73	6 53	65	4745	10.5
					Total+10%:		208.6
Portique 5	1	HA10	2	230	255	510	3.1
	2	HA12	2	255	280	560	5.0
	3	HA10	2	220	245	490	3.0
	4	HA12	2	260	285	570	5.1
	5	HA10	2	105	130	260	1.6
	6	HA10	2	100	125	250	1.5
	7	HA12	2	295	295	590	5.2
	8	HA6	24	43 6 18	136	3264	7.2
					Total+10%:		34.9
Portique 6	1	HA10	2	419	451	902	5.6
	2	HA16	2	419	470	940	14.8
	3	HA14	2	419	419	838	10.1
	4	HA6	26	43 6 18	136	3536	7.8
					Total+10%:		42.1
					HA6:		55.8
					HA8:		39.7
					HA10:		24.8
					HA12:		16.9
					HA14:		11.2
					HA16:		137.2
					Total:		285.6

PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Portique 5

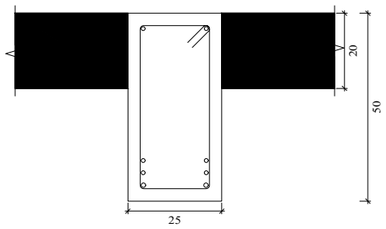


Portique 6

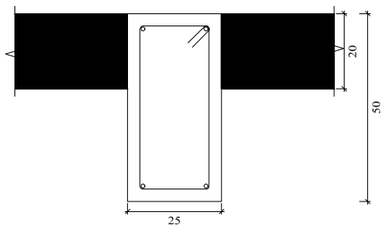


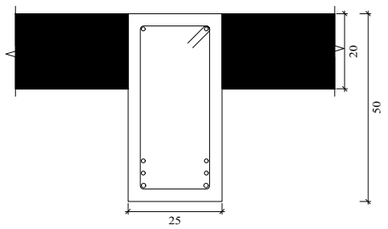
PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

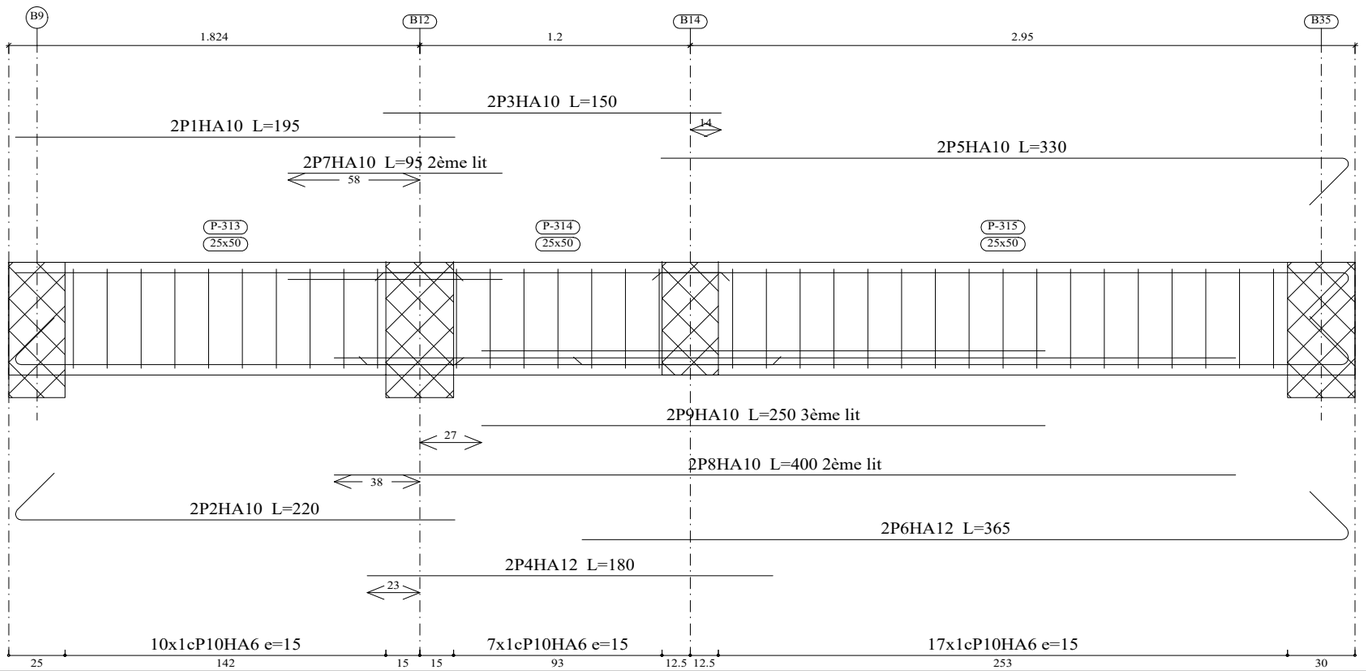
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 7	1	HA10	2	195	195	390	2.4
	2	HA10	2	25 195	220	440	2.7
	3	HA10	2	150	150	300	1.8
	4	HA12	2	180	180	360	3.2
	5	HA10	2	305 25	330	660	4.1
	6	HA12	2	340 12	365	730	6.5
	7	HA10	2	95	95	190	1.2
	8	HA10	2	400	400	800	4.9
	9	HA10	2	250	250	500	3.1
	10	HA6	34	43 6 18	136	4624	10.3
						Total+10%:	44.2
Portique 8	1	HA10	2	302 25	327	654	4.0
	2	HA12	2	25 302 12	352	704	6.3
	3	HA6	17	43 6 18	136	2312	5.1
						Total+10%:	16.9
						HA6:	16.9
						HA10:	26.6
						HA12:	17.6
						Total:	61.1



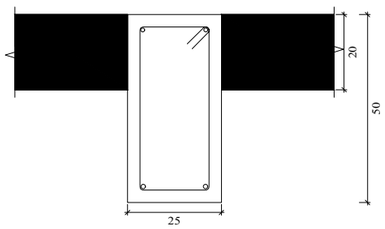
Portique 7

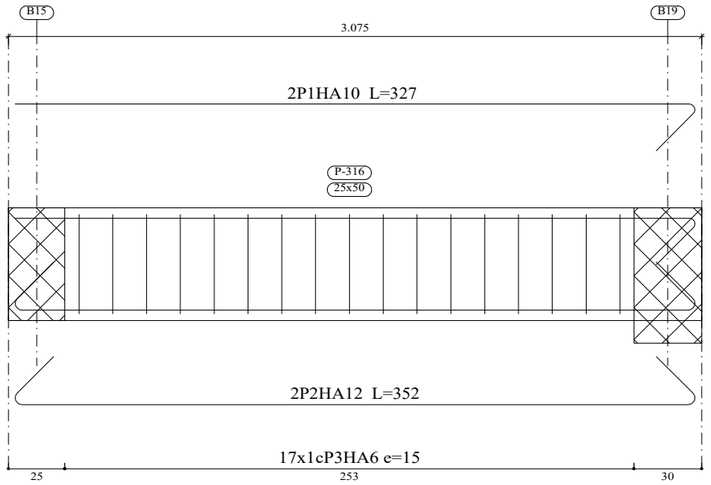




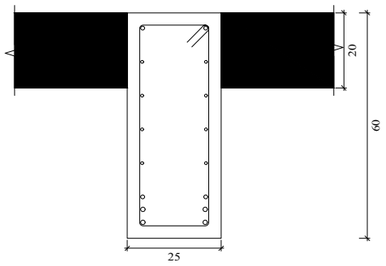


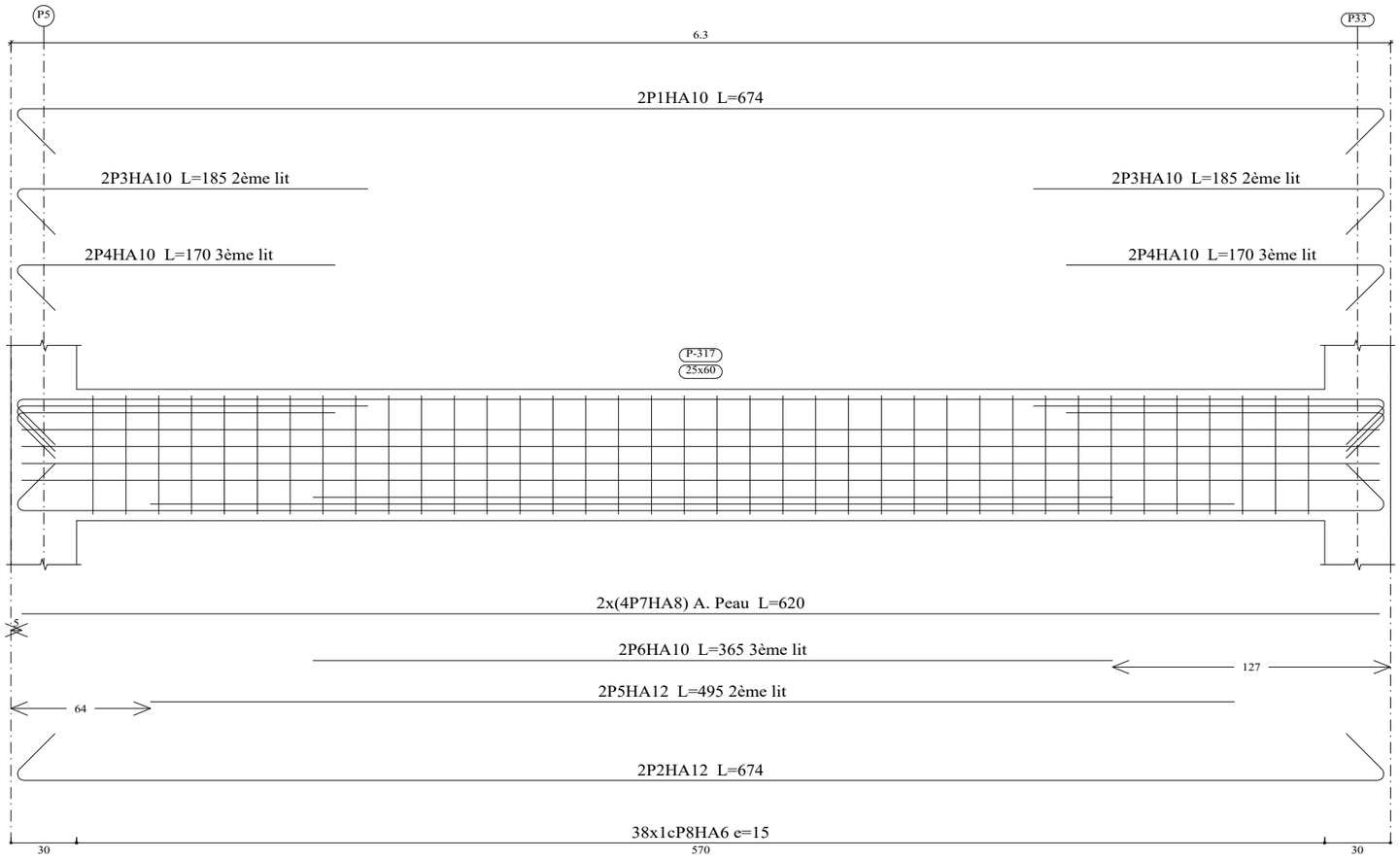
Portique 8



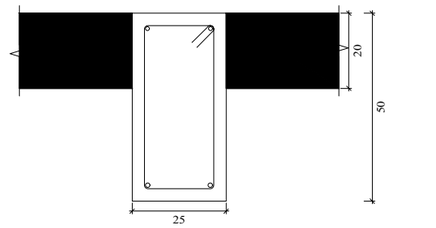


Portique 9

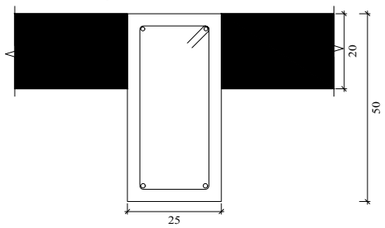


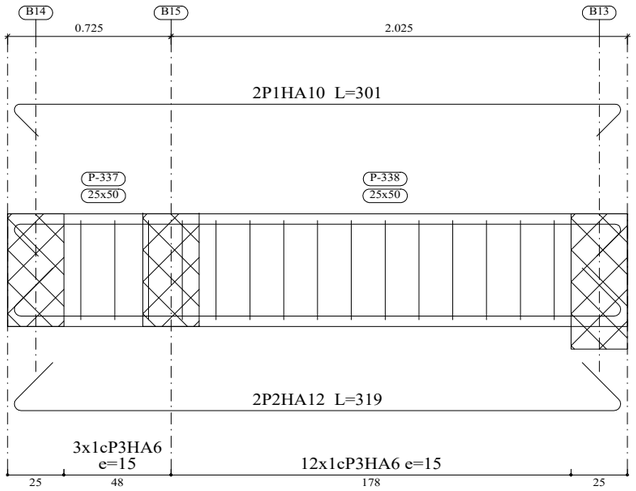


Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 9	1	HA10	2	624, 25, 25	674	1348	8.3
	2	HA12	2	25, 624, 25	674	1348	12.0
	3	HA10	4	25, 160	185	740	4.6
	4	HA10	4	25, 145	170	680	4.2
	5	HA12	2	495	495	990	8.8
	6	HA10	2	365	365	730	4.5
	7	HA8	8	620	620	4960	19.6
	8	HA6	38	53, 6, 18	156	5928	13.2
						Total+10%:	82.7
Portique 16	1	HA10	2	269, 16, 16	301	602	3.7
	2	HA12	2	25, 269, 25	319	638	5.7
	3	HA6	15	43, 6, 18	136	2040	4.5
						Total+10%:	15.3
						HA6:	19.5
						HA8:	21.6
						HA10:	27.9
						HA12:	29.0
						Total:	98.0



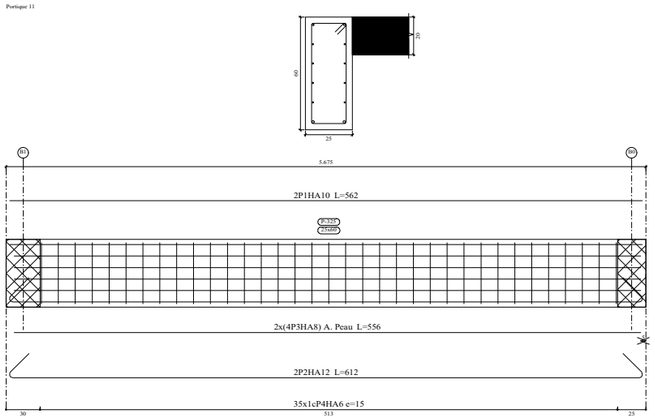
Portique 16



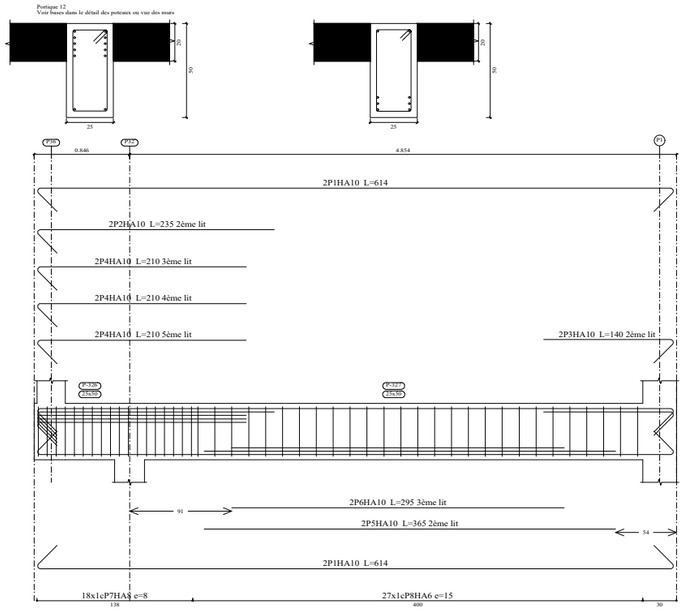


PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

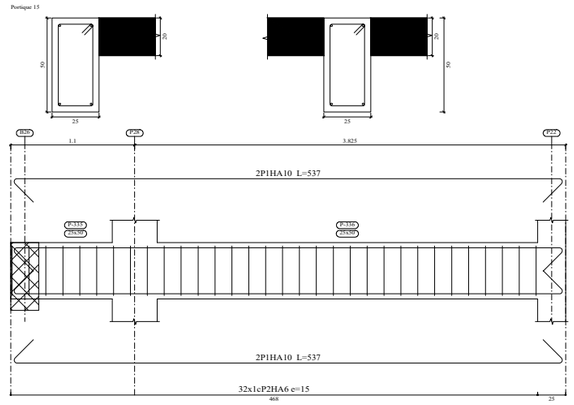
Portique 11



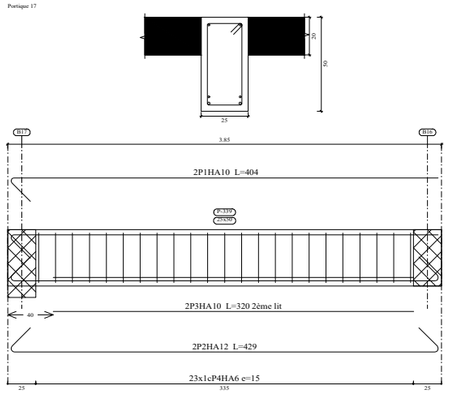
Portique 12
Voir bases dans le détail des poteaux ou voir des murs



Portique 15

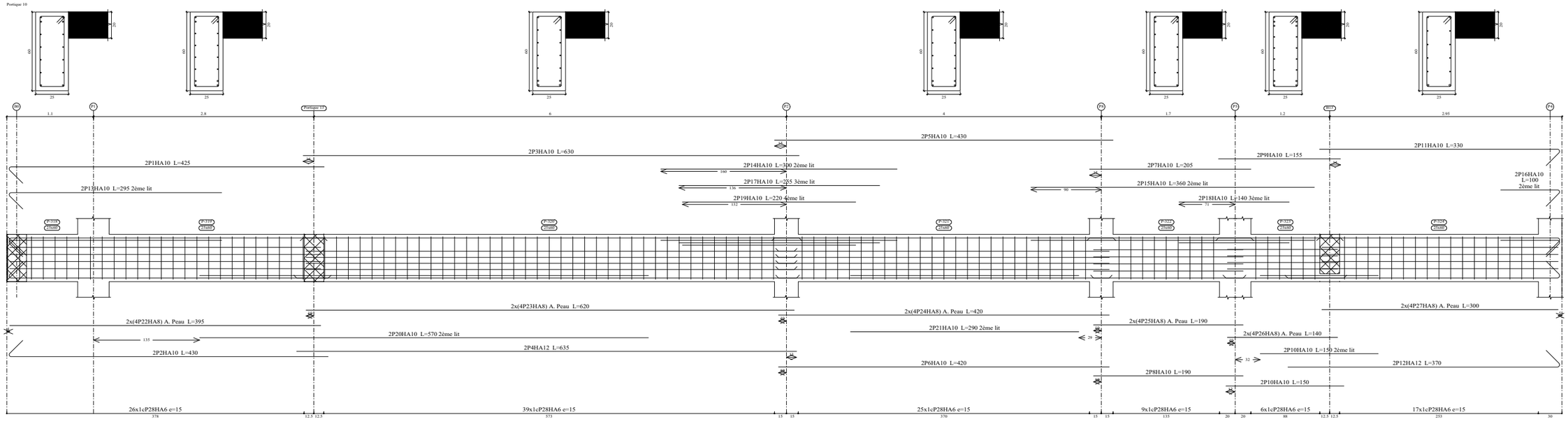


Portique 17



Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 10	1	HA10	2	400	425	850	5.2
	2	HA10	2	405	430	860	5.3
	3	HA10	2	620	630	1260	7.8
	4	HA12	2	635	635	1270	11.3
	5	HA10	2	420	430	860	5.3
	6	HA10	2	420	420	840	5.2
	7	HA10	2	205	205	410	2.5
	8	HA10	2	190	190	380	2.3
	9	HA10	2	155	155	310	1.9
	10	HA10	4	150	150	600	3.7
	11	HA10	2	105 / 225	330	660	4.1
	12	HA12	2	345 / 25	370	740	6.6
	13	HA10	2	270 / 25	295	590	3.6
	14	HA10	2	300	300	600	3.7
	15	HA10	2	360	360	720	4.4
	16	HA10	2	75 / 25	100	200	1.2
	17	HA10	2	255	255	510	3.1
	18	HA10	2	140	140	280	1.7
	19	HA10	2	220	220	440	2.7
	20	HA10	2	570	570	1140	7.0
	21	HA10	2	290	290	580	3.6
	22	HA8	8	395	395	3160	12.5
	23	HA8	8	620	620	4960	19.6
	24	HA8	8	420	420	3360	13.3
	25	HA8	8	190	190	1520	6.0
	26	HA8	8	140	140	1120	4.4
	27	HA8	8	300	300	2400	9.5
	28	HA6	122	25 / 18	156	19032	42.2
						Total+10%:	219.7
Portique 11	1	HA10	2	562	562	1124	6.9
	2	HA12	2	562 / 50	612	1224	10.9
	3	HA8	8	556	556	4448	17.6
	4	HA6	35	25 / 18	156	5460	12.1
						Total+10%:	52.3
Portique 12	1	HA10	4	564 / 50	614	2456	15.1
	2	HA10	2	210 / 25	235	470	2.9
	3	HA10	2	115 / 25	140	280	1.7
	4	HA10	6	185	210	1260	7.8
	5	HA10	2	365	365	730	4.5
	6	HA10	2	295	295	590	3.6
	7	HA8	18	25 / 18	139	2502	9.9
	8	HA6	27	25 / 18	136	3672	8.1
						Total+10%:	59.0
Portique 15	1	HA10	4	482 / 55	537	2148	13.2
	2	HA6	32	25 / 18	136	4352	9.7
						Total+10%:	25.2
Portique 17	1	HA10	2	379	404	808	5.0
	2	HA12	2	379 / 50	429	858	7.6
	3	HA10	2	320	320	640	3.9
	4	HA6	23	25 / 18	136	3128	6.9
						Total+10%:	25.7
						HA8:	86.9
						HA6:	100.3
						HA10:	152.1
						HA12:	80.1
						Total:	381.9

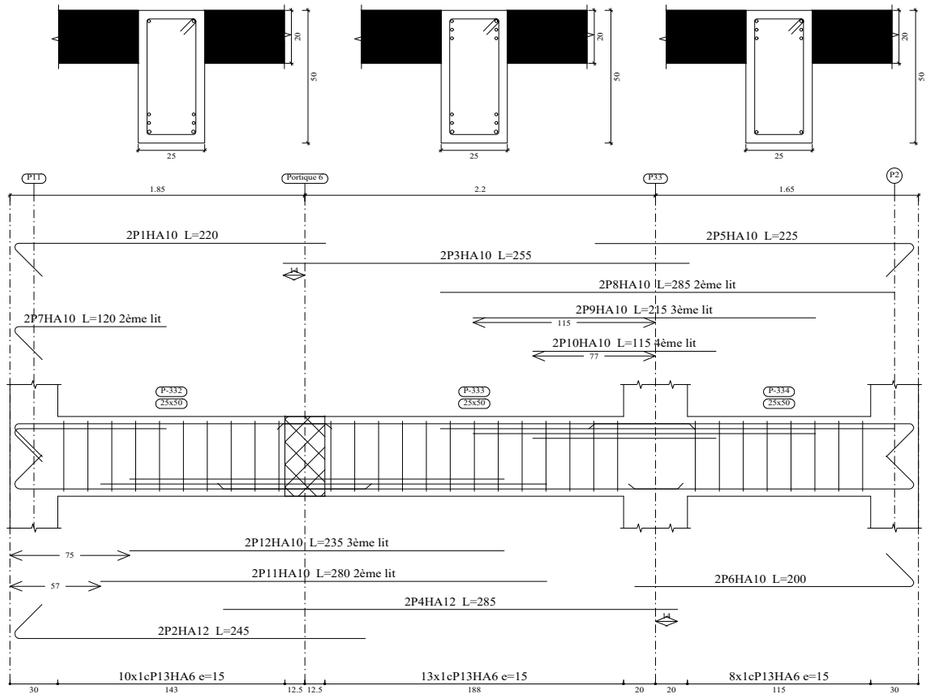
Portique 10



PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

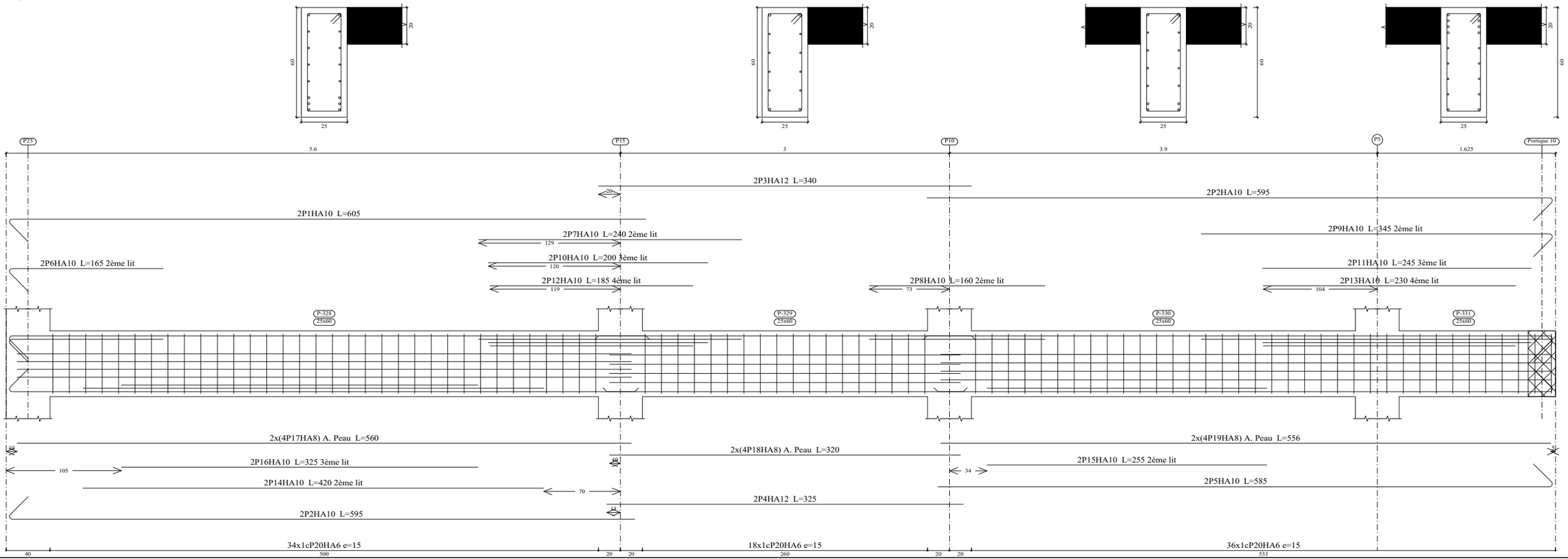
PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Portique 14



Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 13	1	HA10	2	580	605	1210	7.5
	2	HA10	4	570	595	2380	14.7
	3	HA12	2	340	340	680	6.0
	4	HA12	2	325	325	650	5.8
	5	HA10	2	560	585	1170	7.2
	6	HA10	2	140	165	330	2.0
	7	HA10	2	240	240	480	3.0
	8	HA10	2	160	160	320	2.0
	9	HA10	2	320	345	690	4.3
	10	HA10	2	200	200	400	2.5
	11	HA10	2	245	245	490	3.0
	12	HA10	2	185	185	370	2.3
	13	HA10	2	230	230	460	2.8
	14	HA10	2	420	420	840	5.2
	15	HA10	2	255	255	510	3.1
	16	HA10	2	325	325	650	4.0
	17	HA8	8	560	560	4480	17.7
	18	HA8	8	320	320	2560	10.1
	19	HA8	8	556	556	4448	17.6
	20	HA6	88	43 6 18	156	13728	30.5
						Total+10%:	166.4
Portique 14	1	HA10	2	195	220	440	2.7
	2	HA12	2	220	245	490	4.4
	3	HA10	2	255	255	510	3.1
	4	HA12	2	285	285	570	5.1
	5	HA10	2	200	225	450	2.8
	6	HA10	2	175	200	400	2.5
	7	HA10	2	95	120	240	1.5
	8	HA10	2	285	285	570	3.5
	9	HA10	2	215	215	430	2.7
	10	HA10	2	115	115	230	1.4
	11	HA10	2	280	280	560	3.5
	12	HA10	2	235	235	470	2.9
	13	HA6	31	43 6 18	136	4216	9.4
						Total+10%:	50.1
						HA6:	43.9
						HA8:	50.0
						HA10:	99.3
						HA12:	23.3
						Total:	216.5

Portique 13



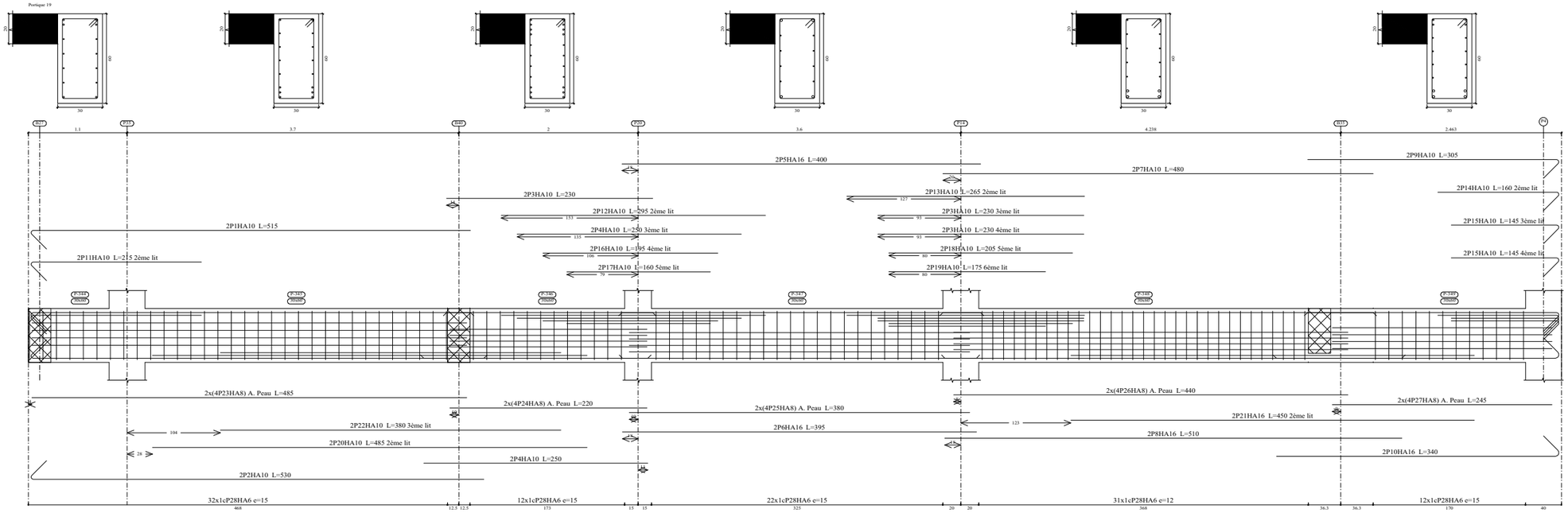
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 18	1	HA10	3	589	605	1815	11.2
	2	HA10	3	580	605	1815	11.2
	3	HA12	3	685	710	2130	18.9
	4	HA16	3	680	705	2115	33.4
	5	HA10	2	210	210	420	2.6
	6	HA10	2	155	180	360	2.2
	7	HA10	4	130	155	620	3.8
	8	HA12	3	450	450	1350	12.0
	9	HA12	3	315	315	945	8.4
	10	HA12	3	200	200	600	5.3
	11	HA10	2	150	150	300	1.8
	12	HA10	2	105	105	210	1.3
	13	HA14	2	490	490	980	11.8
	14	HA14	2	420	420	840	10.2
	15	HA14	2	345	345	690	8.3
	16	HA8	8	580	580	4640	18.3
	17	HA8	8	420	420	3360	13.3
	18	HA8	8	270	270	2160	8.5
	19	HA6	84	6 / 53 / 23	166	13944	30.9
	20	HA6	84	6 / 53	65	5460	12.1
					Total+10%:		248.1

HA6:	47.3
HA8:	44.2
HA10:	37.5
HA12:	49.1
HA14:	33.3
HA16:	36.7
Total:	248.1

Portique 18

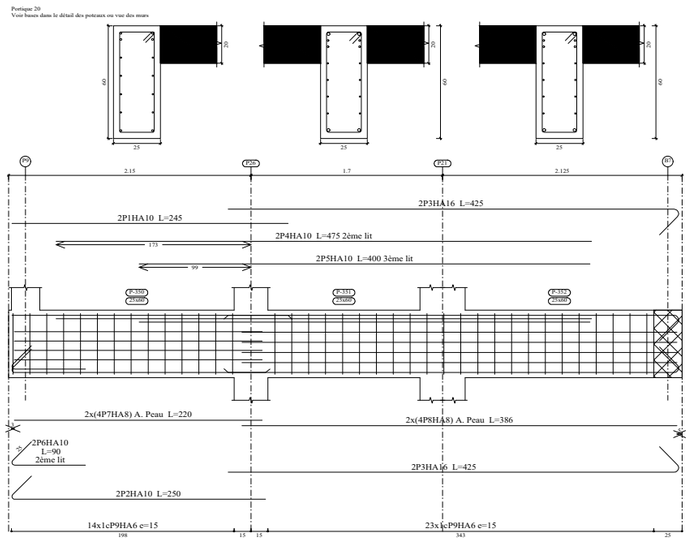


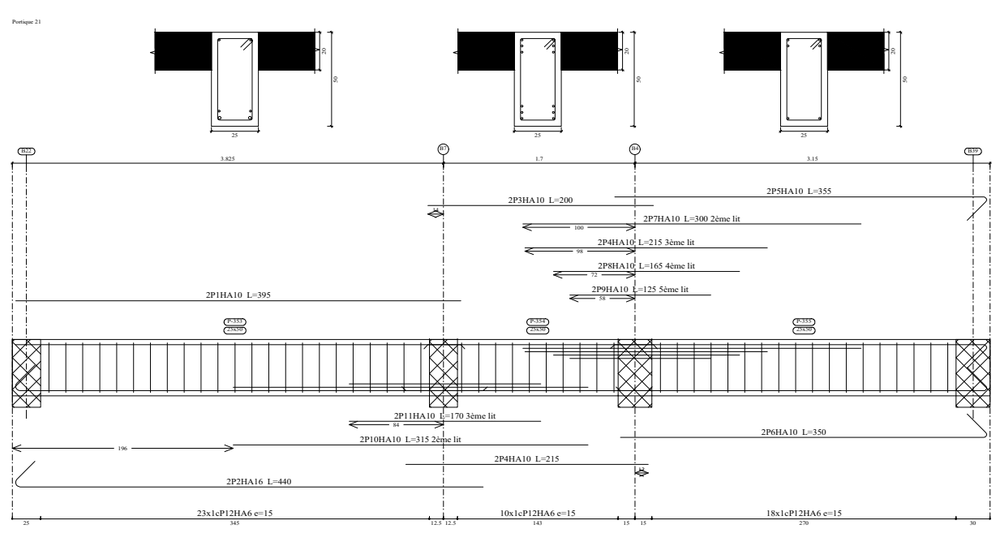
PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

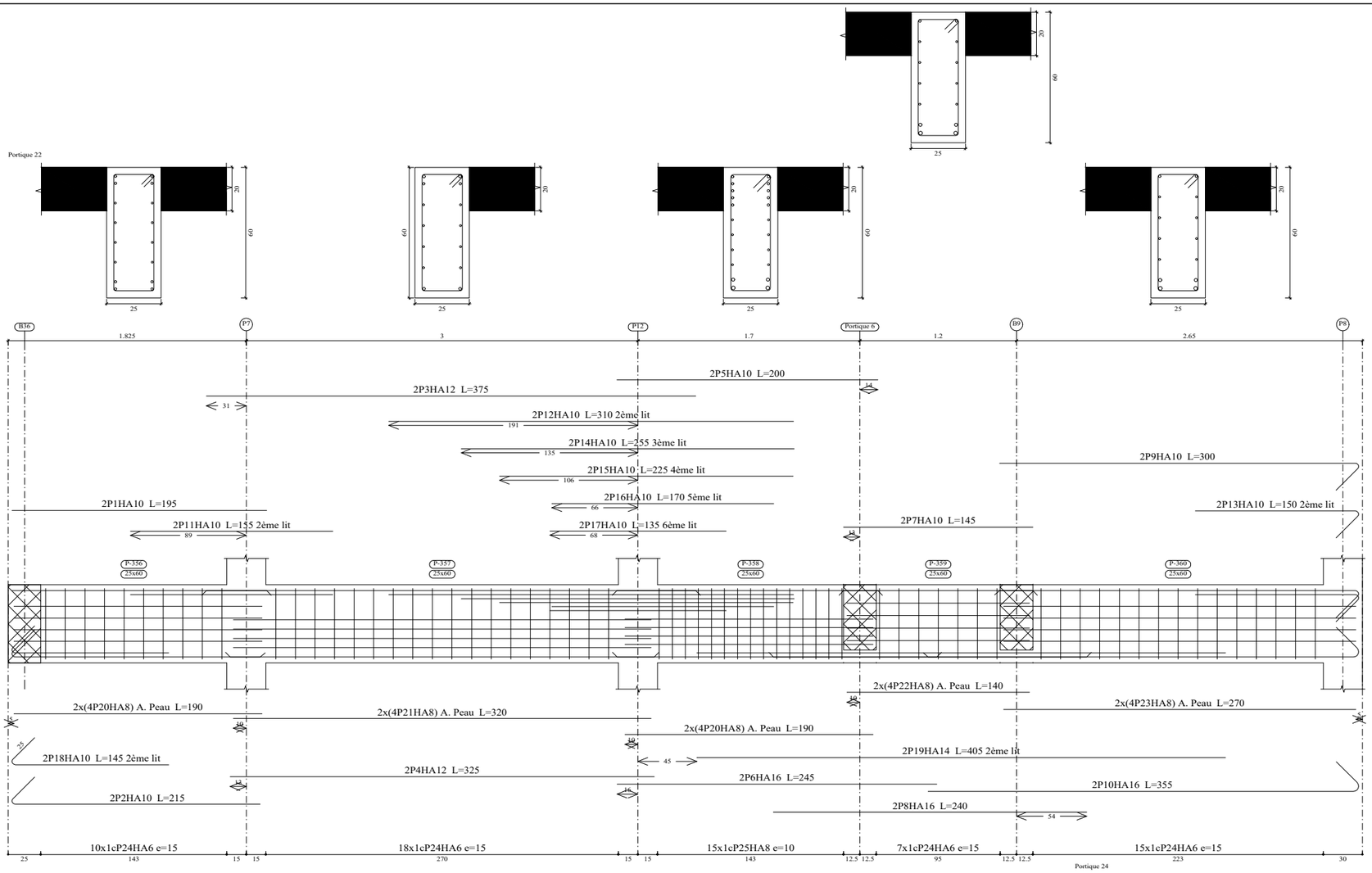


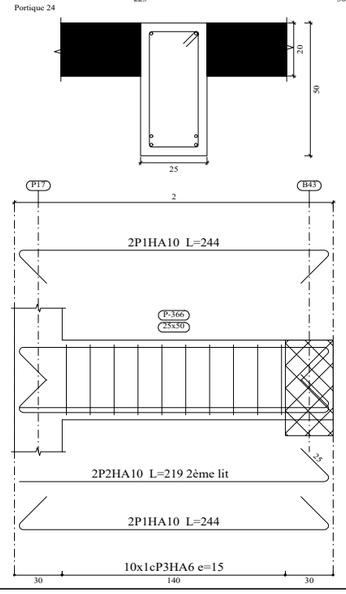
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 19	1	HA10	2	490	515	1030	6.4
	2	HA10	2	505	530	1060	6.5
	3	HA10	6	230	230	1380	8.5
	4	HA10	4	250	250	1000	6.2
	5	HA16	2	400	400	800	12.6
	6	HA16	2	395	395	790	12.5
	7	HA10	2	480	480	960	5.9
	8	HA16	2	510	510	1020	16.1
	9	HA10	2	280 / 25	305	610	3.8
	10	HA16	2	315 / 25	340	680	10.7
	11	HA10	2	190 / 25	215	430	2.7
	12	HA10	2	295	295	590	3.6
	13	HA10	2	265	265	530	3.3
	14	HA10	2	155 / 25	160	320	2.0
	15	HA10	4	120 / 25	145	580	3.6
	16	HA10	2	195	195	390	2.4
	17	HA10	2	160	160	320	2.0
	18	HA10	2	205	205	410	2.5
	19	HA10	2	175	175	350	2.2
	20	HA10	2	485	485	970	6.0
	21	HA16	2	450	450	900	14.2
	22	HA10	2	380	380	760	4.7
	23	HA8	8	485	485	3880	15.3
	24	HA8	8	220	220	1760	6.9
	25	HA8	8	380	380	3040	12.0
	26	HA8	8	440	440	3520	13.9
	27	HA8	8	245	245	1960	7.7
	28	HA6	109	25 / 23	166	18094	40.2
						Total+10%:	257.8
Portique 20	1	HA10	2	245	245	490	3.0
	2	HA10	2	225	250	500	3.1
	3	HA16	4	400 / 25	425	1700	26.8
	4	HA10	2	475	475	950	5.9
	5	HA10	2	400	400	800	4.9
	6	HA10	2	65	90	180	1.1
	7	HA8	8	220	220	1760	6.9
	8	HA8	8	386	386	3088	12.2
	9	HA6	37	25 / 18	156	5772	12.8
						Total+10%:	84.4
Portique 21	1	HA10	2	395	395	790	4.9
	2	HA16	2	415	440	880	13.9
	3	HA10	2	200	200	400	2.5
	4	HA10	4	215	215	860	5.3
	5	HA10	2	330 / 25	355	710	4.4
	6	HA10	2	325 / 25	350	700	4.3
	7	HA10	2	300	300	600	3.7
	8	HA10	2	165	165	330	2.0
	9	HA10	2	125	125	250	1.5
	10	HA10	2	315	315	630	3.9
	11	HA10	2	170	170	340	2.1
	12	HA6	51	25 / 18	136	6936	15.4
						Total+10%:	70.3
						HA6:	75.2
						HA8:	82.4
						HA10:	137.4
						HA16:	117.5
						Total:	412.5

PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20









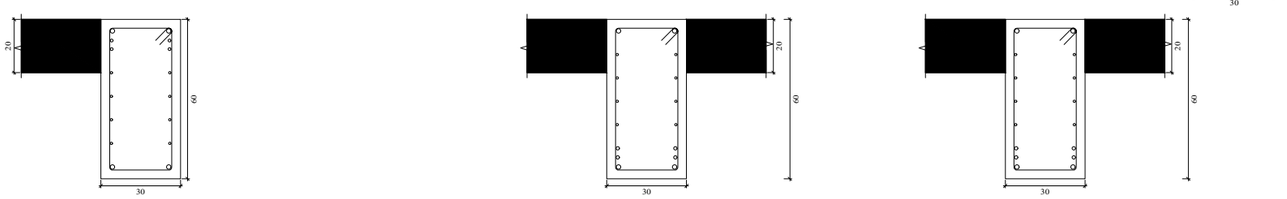
PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

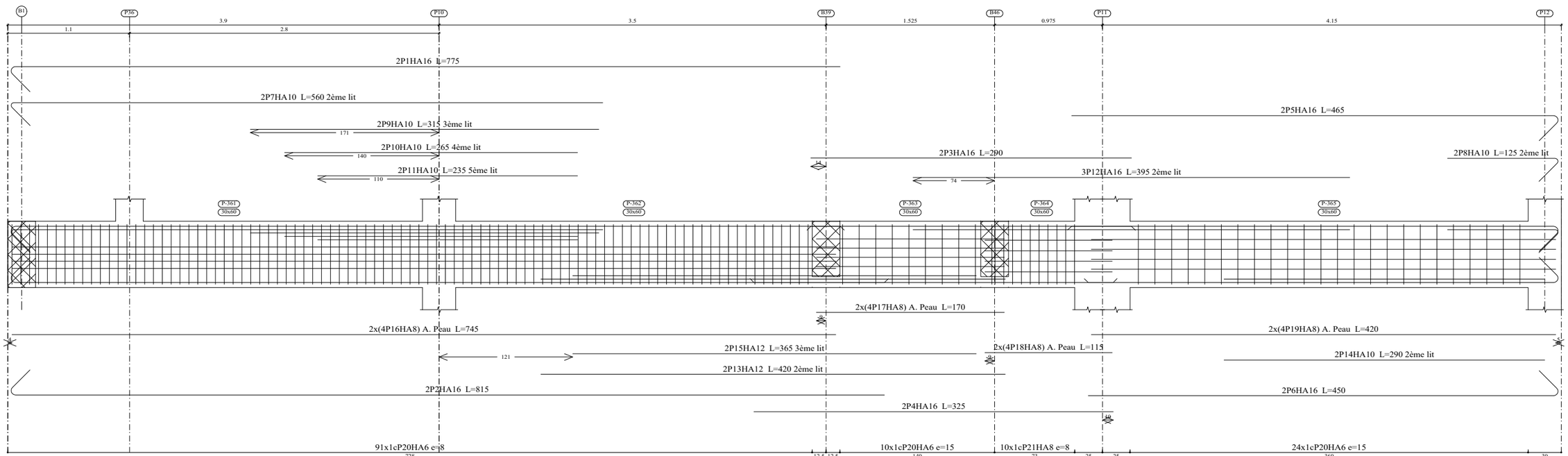
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 22	1	HA10	2	195	195	390	2.4
	2	HA10	2	5 190	215	430	2.7
	3	HA12	2	375	375	750	6.7
	4	HA12	2	325	325	650	5.8
	5	HA10	2	200	200	400	2.5
	6	HA16	2	245	245	490	7.7
	7	HA10	2	145	145	290	1.8
	8	HA16	2	240	240	480	7.6
	9	HA10	2	275 15	300	600	3.7
	10	HA16	2	330 5	355	710	11.2
	11	HA10	2	155	155	310	1.9
	12	HA10	2	310	310	620	3.8
	13	HA10	2	125 15	150	300	1.8
	14	HA10	2	255	255	510	3.1
	15	HA10	2	225	225	450	2.8
	16	HA10	2	170	170	340	2.1
	17	HA10	2	135	135	270	1.7
	18	HA10	2	5 120	145	290	1.8
	19	HA14	2	405	405	810	9.8
	20	HA8	16	190	190	3040	12.0
	21	HA8	8	320	320	2560	10.1
	22	HA8	8	140	140	1120	4.4
	23	HA8	8	270	270	2160	8.5
	24	HA6	50	57 6 18	156	7800	17.3
	25	HA8	15	57 8 18	159	2385	9.4
						Total+10%:	156.9
Portique 24	1	HA10	4	5 194 5	244	976	6.0
	2	HA10	2	194 5	219	438	2.7
	3	HA6	10	43 6 18	136	1360	3.0
						Total+10%:	12.9
						HA6:	22.4
						HA8:	48.8
						HA10:	44.9
						HA12:	13.8
						HA14:	10.8
						HA16:	29.1
						Total:	169.8

PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

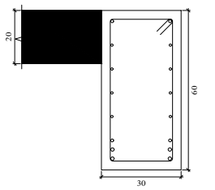
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 23	1	HA16	2	15 750	775	1550	24.5
	2	HA16	2	25 790	815	1630	25.7
	3	HA16	2	290	290	580	9.2
	4	HA16	2	325	325	650	10.3
	5	HA16	2	440 25	465	930	14.7
	6	HA16	2	425 25	450	900	14.2
	7	HA10	2	15 535	560	1120	6.9
	8	HA10	2	100 25	125	250	1.5
	9	HA10	2	315	315	630	3.9
	10	HA10	2	265	265	530	3.3
	11	HA10	2	235	235	470	2.9
	12	HA16	3	395	395	1185	18.7
	13	HA12	2	420	420	840	7.5
	14	HA10	2	290	290	580	3.6
	15	HA12	2	365	365	730	6.5
	16	HA8	8	745	745	5960	23.5
	17	HA8	8	170	170	1360	5.4
	18	HA8	8	115	115	920	3.6
	19	HA8	8	420	420	3360	13.3
	20	HA6	125	53 6 23	166	20750	46.0
	21	HA8	10	53 8 23	169	1690	6.7
						Total+10%:	277.1
						HA6:	50.6
						HA8:	57.8
						HA10:	24.3
						HA12:	15.4
						HA16:	129.0
						Total:	277.1

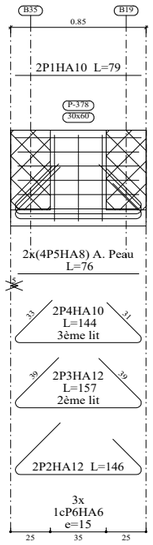
Portique 23
Voir bases dans le détail des poteaux ou vue des murs





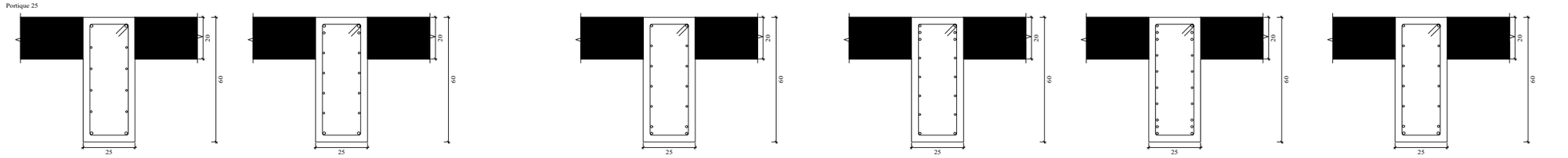
Portique 28

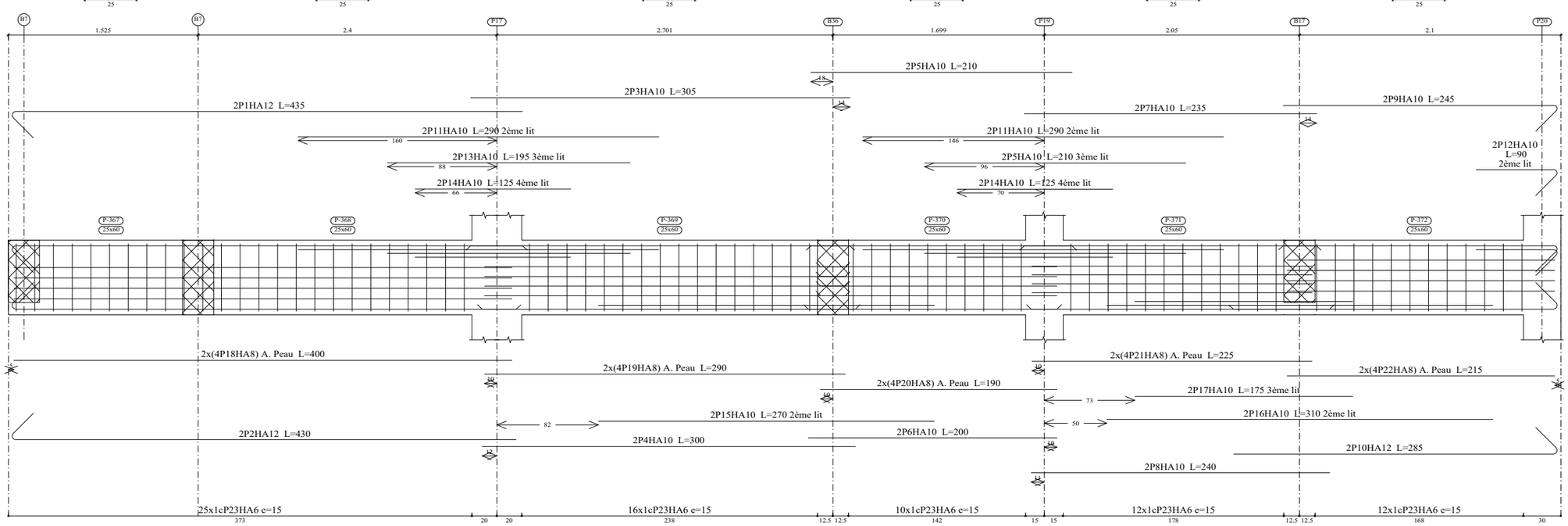




Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 25	1	HA12	2	410	435	870	7.7
	2	HA12	2	405	430	860	7.6
	3	HA10	2	305	305	610	3.8
	4	HA10	2	300	300	600	3.7
	5	HA10	4	210	210	840	5.2
	6	HA10	2	200	200	400	2.5
	7	HA10	2	235	235	470	2.9
	8	HA10	2	240	240	480	3.0
	9	HA10	2	220	245	490	3.0
	10	HA12	2	260	285	570	5.1
	11	HA10	4	290	290	1160	7.2
	12	HA10	2	65	90	180	1.1
	13	HA10	2	195	195	390	2.4
	14	HA10	4	125	125	500	3.1
	15	HA10	2	270	270	540	3.3
	16	HA10	2	310	310	620	3.8
	17	HA10	2	175	350	2.2	
	18	HA8	8	400	400	3200	12.6
	19	HA8	8	290	290	2320	9.2
	20	HA8	8	190	190	1520	6.0
	21	HA8	8	225	225	1800	7.1
	22	HA8	8	215	215	1720	6.8
	23	HA6	75	6 / 18	156	11700	26.0
						Total+10%:	148.8
Portique 28	1	HA10	2	79	79	158	1.0
	2	HA12	2	79	146	292	2.6
	3	HA12	2	79	157	314	2.8
	4	HA10	2	79	144	288	1.8
	5	HA8	8	76	76	608	2.4
	6	HA6	3	6 / 23	166	498	1.1
						Total+10%:	12.9
						HA6:	29.8
						HA8:	48.5
						HA10:	55.1
						HA12:	28.3
						Total:	161.7

Portique 25

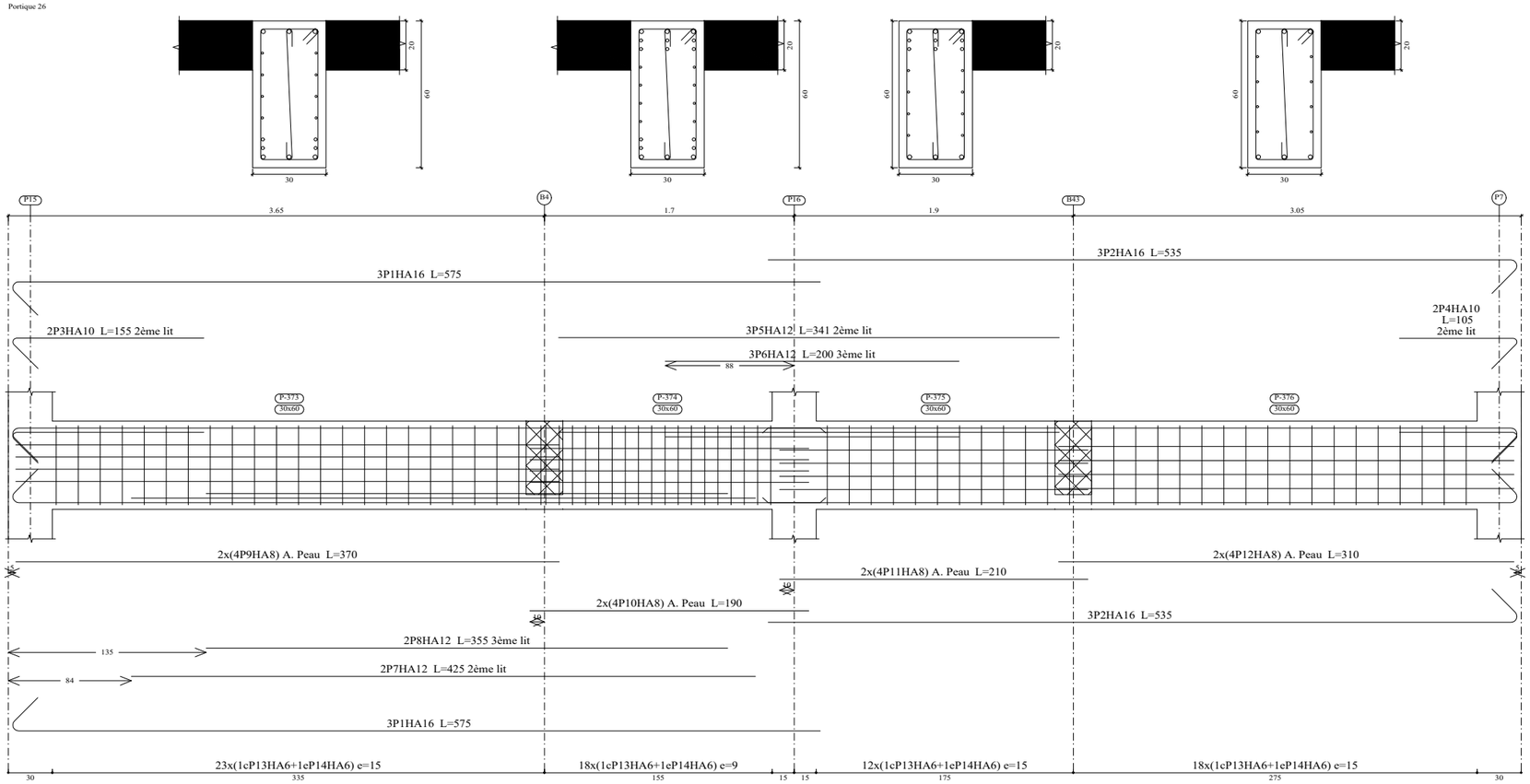




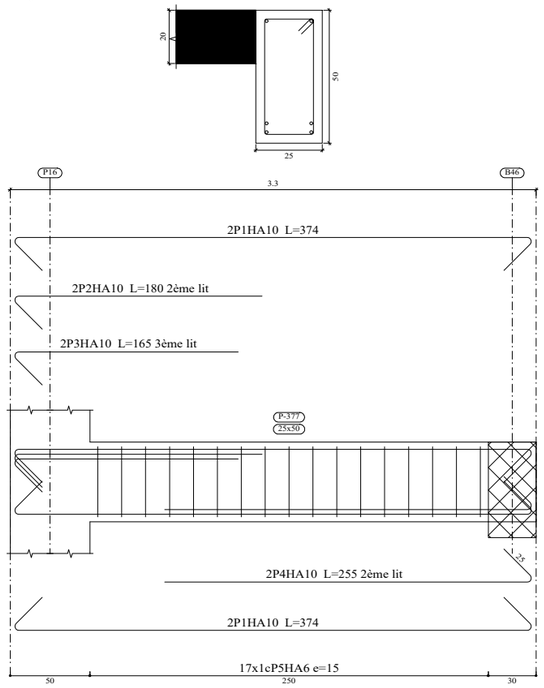
PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 26	1	HA16	6	15 550 25	575	3450	54.5
	2	HA16	6	510 25	535	3210	50.7
	3	HA10	2	15 130	155	310	1.9
	4	HA10	2	80 25	105	210	1.3
	5	HA12	3	341	341	1023	9.1
	6	HA12	3	200	200	600	5.3
	7	HA12	2	425	425	850	7.5
	8	HA12	2	355	355	710	6.3
	9	HA8	8	370	370	2960	11.7
	10	HA8	8	190	190	1520	6.0
	11	HA8	8	210	210	1680	6.6
	12	HA8	8	310	310	2480	9.8
	13	HA6	71	6 53 23	166	11786	26.2
	14	HA6	71	6 53	65	4615	10.2
						Total+10%:	227.8
Portique 27	1	HA10	4	15 324 25	374	1496	9.2
	2	HA10	2	15 155	180	360	2.2
	3	HA10	2	15 140	165	330	2.0
	4	HA10	2	230 25	255	510	3.1
	5	HA6	17	6 43 18	136	2312	5.1
						Total+10%:	23.8
						HA6:	45.6
						HA8:	37.5
						HA10:	21.7
						HA12:	31.1
						HA16:	115.7
						Total:	251.6

Portique 26



Portique 27

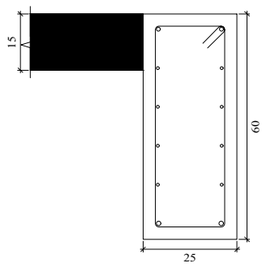


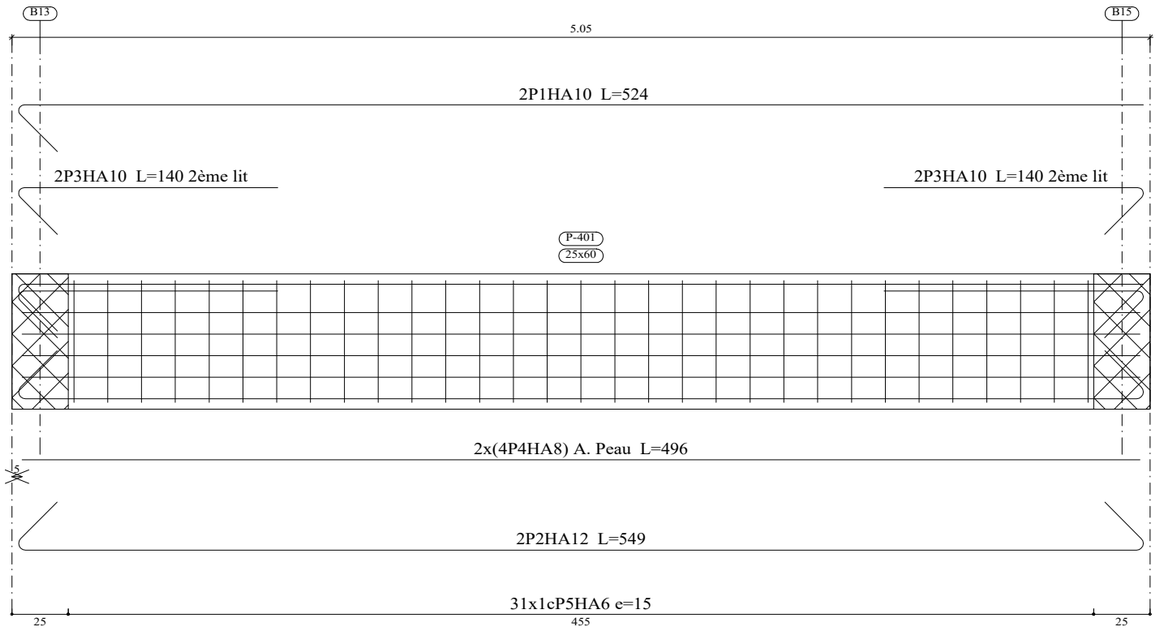
PH RDC
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

POUTRES PH R+1

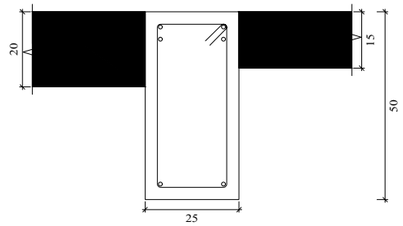
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 1	1	HA10	2	499, 13	524	1048	6.5
	2	HA12	2	25, 499, 25	549	1098	9.7
	3	HA10	4	115, 25	140	560	3.5
	4	HA8	8	496	496	3968	15.7
	5	HA6	31	53, 6, 18	156	4836	10.7
						Total+10%:	50.7
Portique 14	1	HA10	4	494, 13, 25	544	2176	13.4
	2	HA10	2	205	205	410	2.5
	3	HA6	32	43, 6, 18	136	4352	9.7
						Total+10%:	28.2
						HA6:	22.5
						HA8:	17.2
						HA10:	28.5
						HA12:	10.7
						Total:	78.9

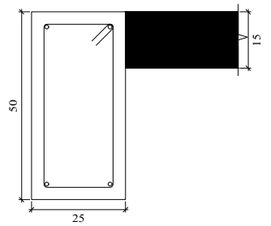
Portique 1

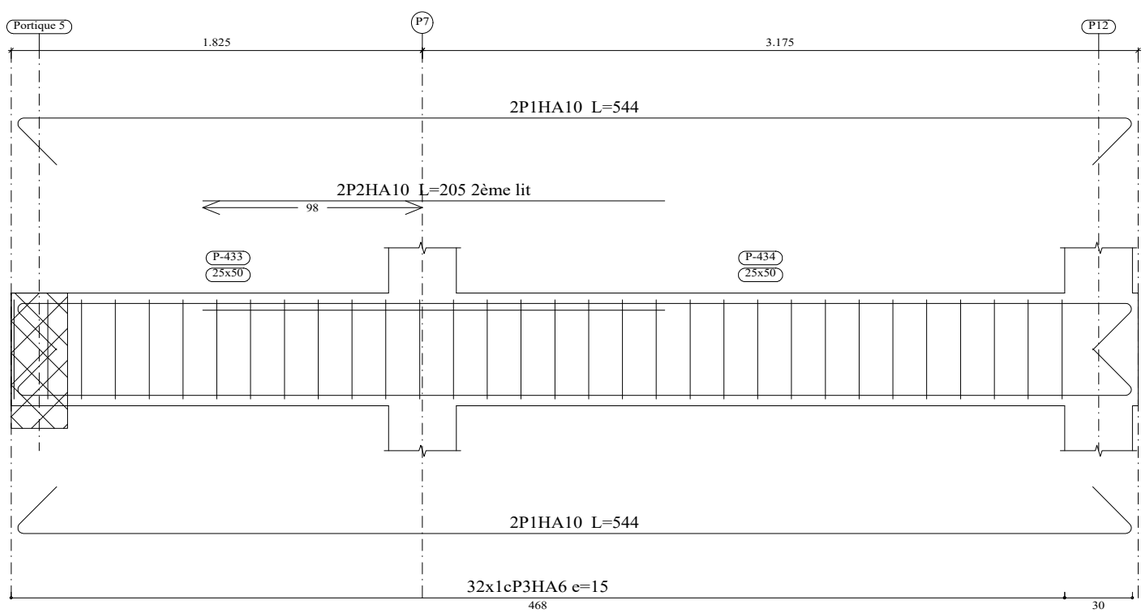




Portique 14



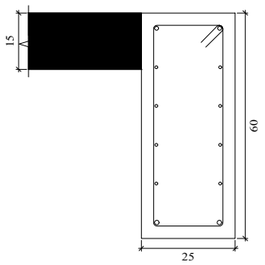


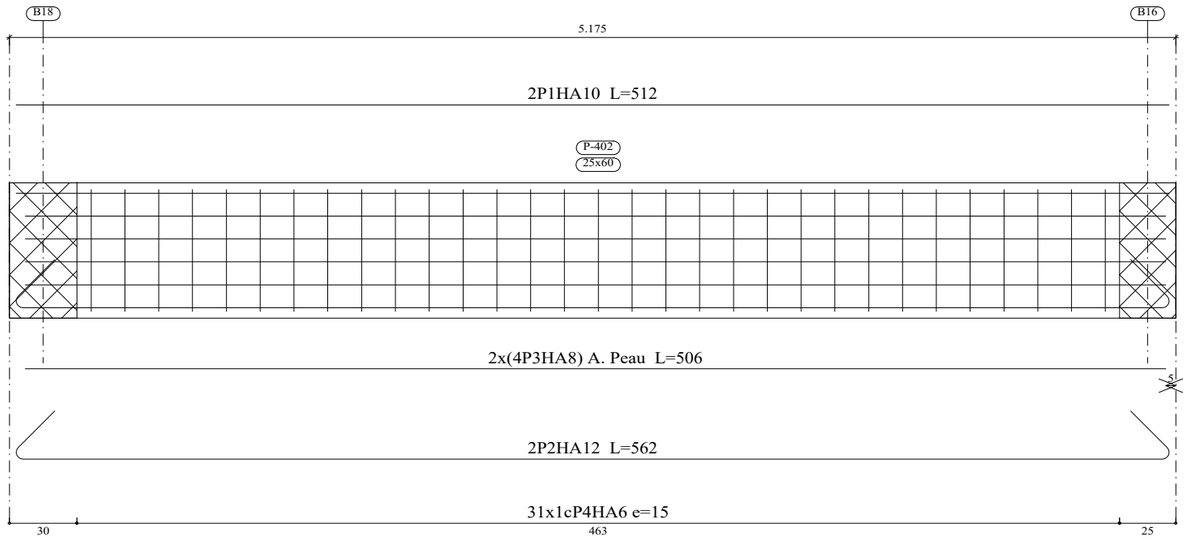


PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 2	1	HA10	2	512	512	1024	6.3
	2	HA12	2	25 512 12	562	1124	10.0
	3	HA8	8	506	506	4048	16.0
	4	HA6	31	6 53 18	156	4836	10.7
					Total+10%:		47.3
Portique 7	1	HA10	2	165 12	190	380	2.3
	2	HA16	2	25 210	235	470	7.4
	3	HA10	2	235	235	470	2.9
	4	HA12	2	269 12	300	600	5.3
	5	HA10	2	95 12	120	240	1.5
	6	HA12	2	273 12	305	610	5.4
	7	HA6	24	6 43 18	136	3264	7.2
					Total+10%:		35.2
					HA6:		19.7
					HA8:		17.6
					HA10:		14.3
					HA12:		22.8
					HA16:		8.1
					Total:		82.5

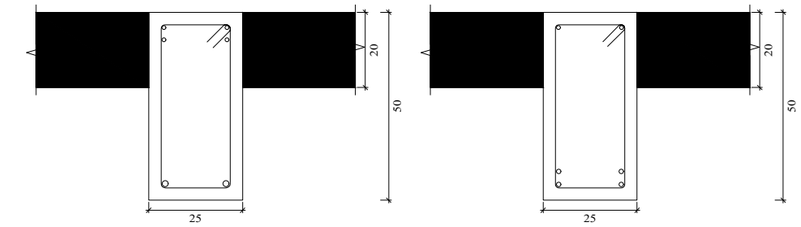
Portique 2

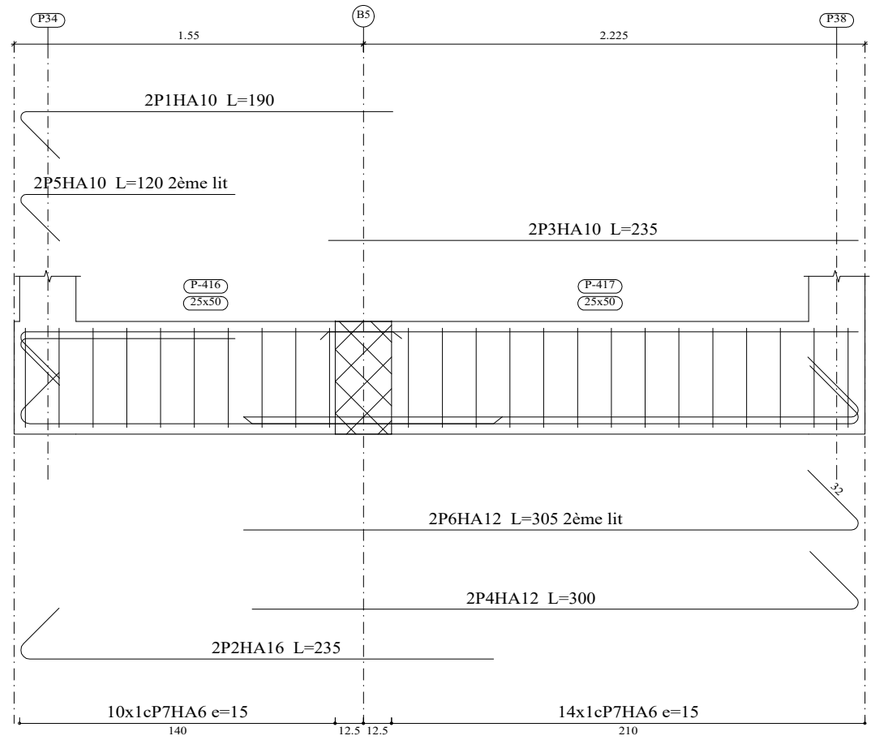




PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

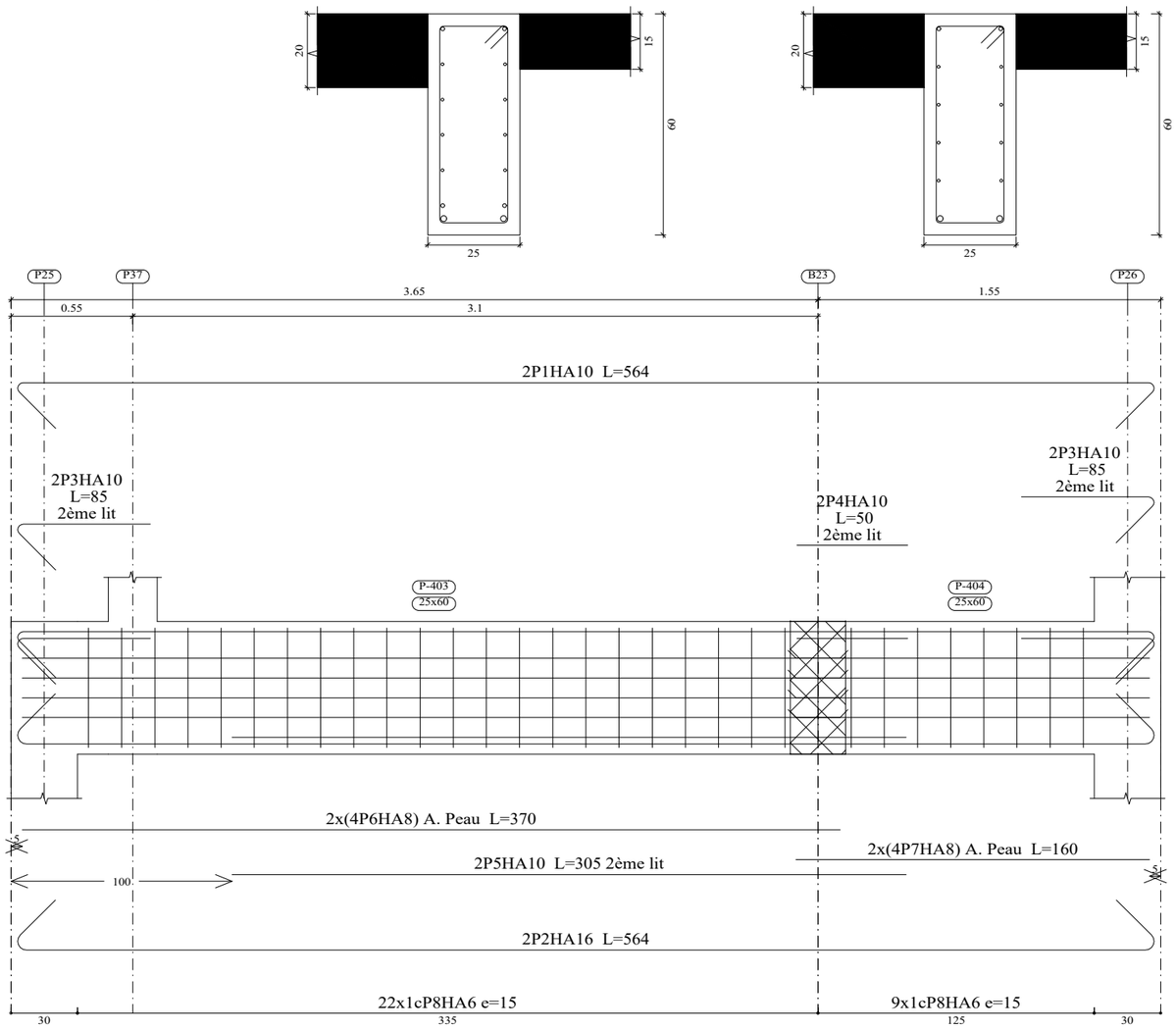
Portique 7
Voir bases dans le détail des poteaux ou vue des murs





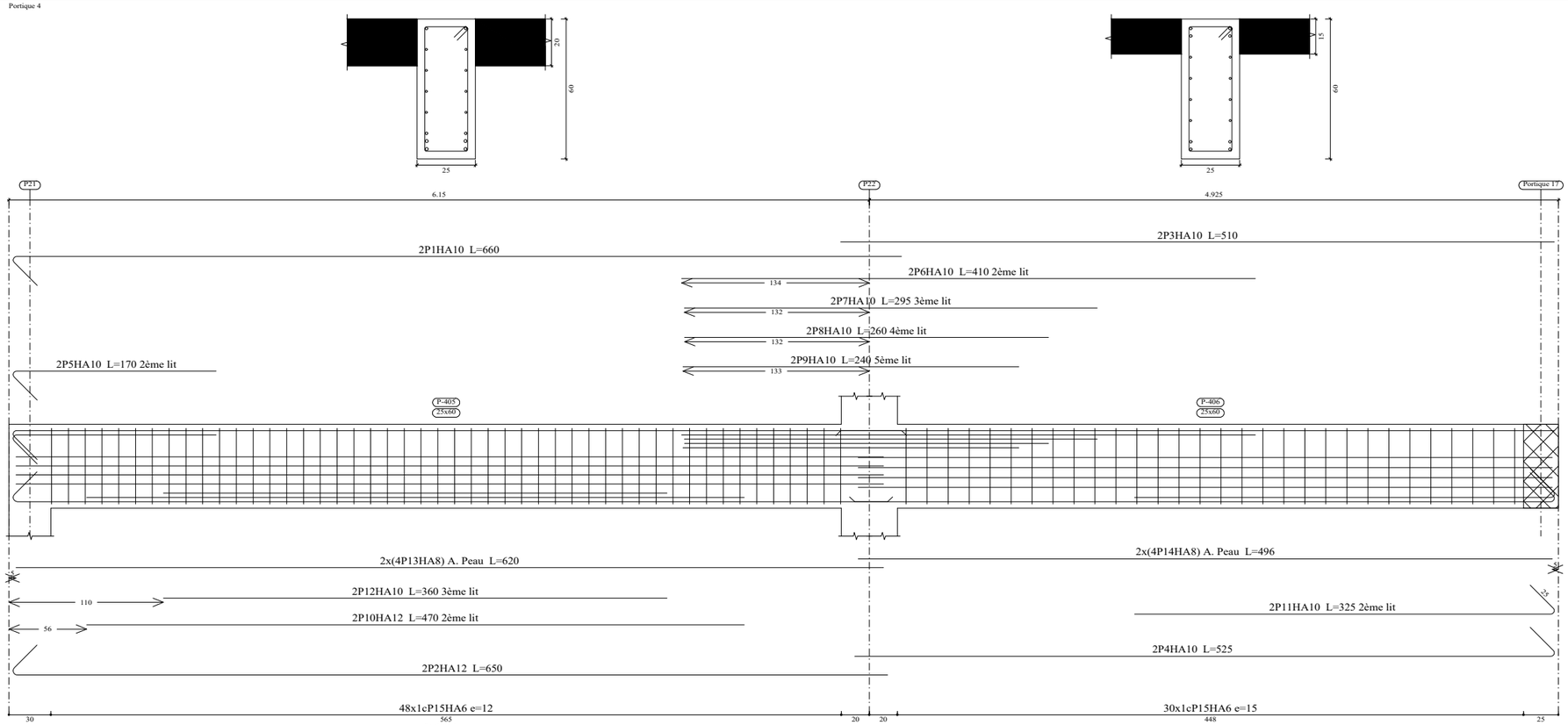
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 3	1	HA10	2	514, 25, 25	564	1128	7.0
	2	HA16	2	25, 514, 25	564	1128	17.8
	3	HA10	4	60, 25	85	340	2.1
	4	HA10	2	50	50	100	0.6
	5	HA10	2	305	305	610	3.8
	6	HA8	8	370	370	2960	11.7
	7	HA8	8	160	160	1280	5.1
	8	HA6	31	53, 6, 18	156	4836	10.7
					Total+10%:		64.7
					HA6:		11.8
					HA8:		18.5
					HA10:		14.9
					HA16:		19.5
					Total:		64.7

Portique 3
Voir bases dans le détail des poteaux ou vue des murs

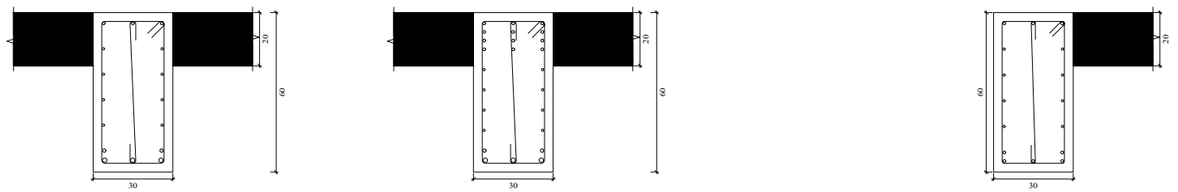


PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Portique 4



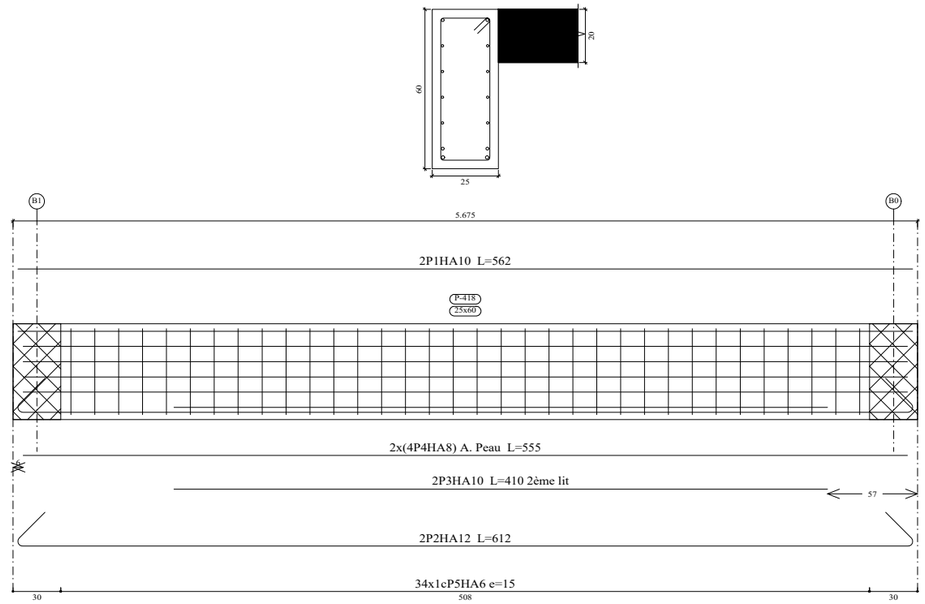
Portique 6
Voir bases dans le détail des poteaux ou vue des murs

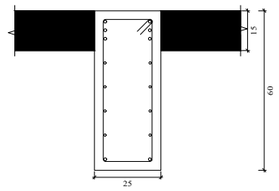


PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

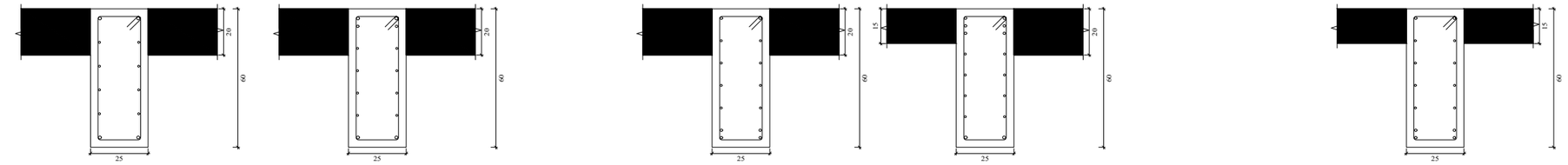
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 4	1	HA10	2	635	660	1320	8.1
	2	HA12	2	625	650	1300	11.5
	3	HA10	2	510	510	1020	6.3
	4	HA10	2	500	525	1050	6.5
	5	HA10	2	145	170	340	2.1
	6	HA10	2	410	410	820	5.1
	7	HA10	2	295	295	590	3.6
	8	HA10	2	260	260	520	3.2
	9	HA10	2	240	240	480	3.0
	10	HA12	2	470	470	940	8.3
	11	HA10	2	300	325	650	4.0
	12	HA10	2	360	360	720	4.4
	13	HA8	8	620	620	4960	19.6
	14	HA8	8	496	496	3968	15.7
	15	HA6	78	53 6 18	156	12168	27.0
					Total+10%:		141.2
Portique 6	1	HA10	3	550	575	1725	10.6
	2	HA16	3	550	575	1725	27.2
	3	HA10	3	510	535	1605	9.9
	4	HA10	3	505	530	1590	9.8
	5	HA10	3	395	395	1185	7.3
	6	HA10	3	285	285	855	5.3
	7	HA10	3	215	215	645	4.0
	8	HA12	2	460	460	920	8.2
	9	HA10	2	340	365	730	4.5
	10	HA8	8	370	370	2960	11.7
	11	HA8	8	190	190	1520	6.0
	12	HA8	8	500	500	4000	15.8
	13	HA6	74	53 6 23	166	12284	27.3
	14	HA6	74	6 53	65	4810	10.7
					Total+10%:		174.1
					HA6:		71.5
					HA8:		75.6
					HA10:		107.4
					HA12:		30.8
					HA16:		30.0
					Total:		315.3

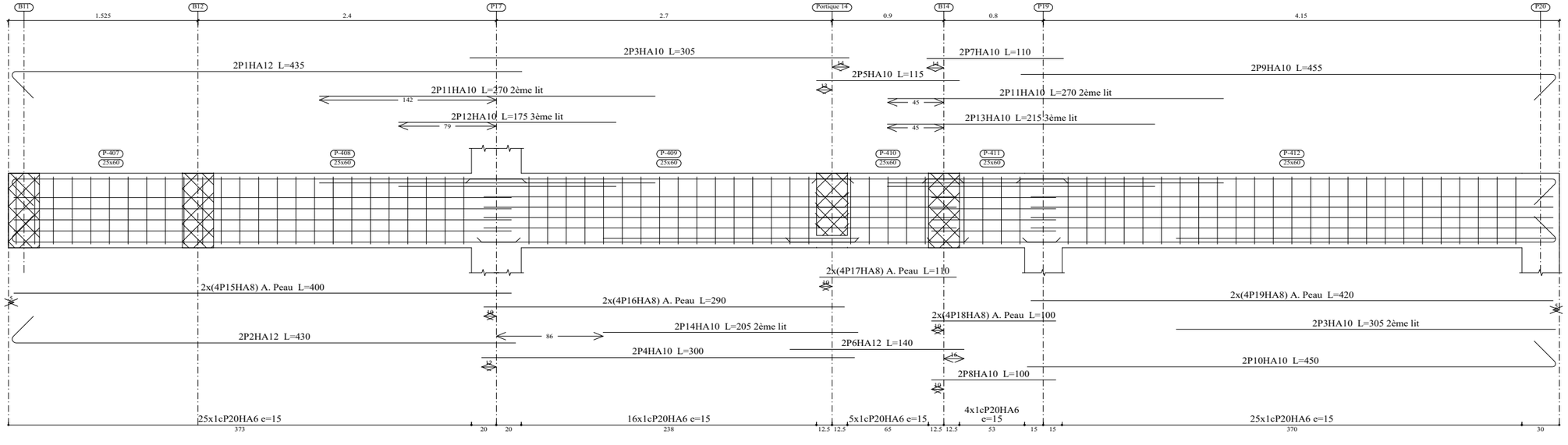
Portique 8





Portique 5



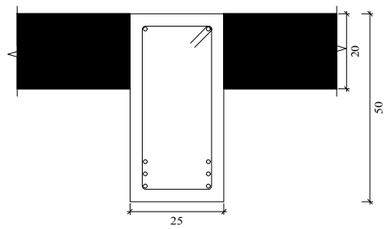


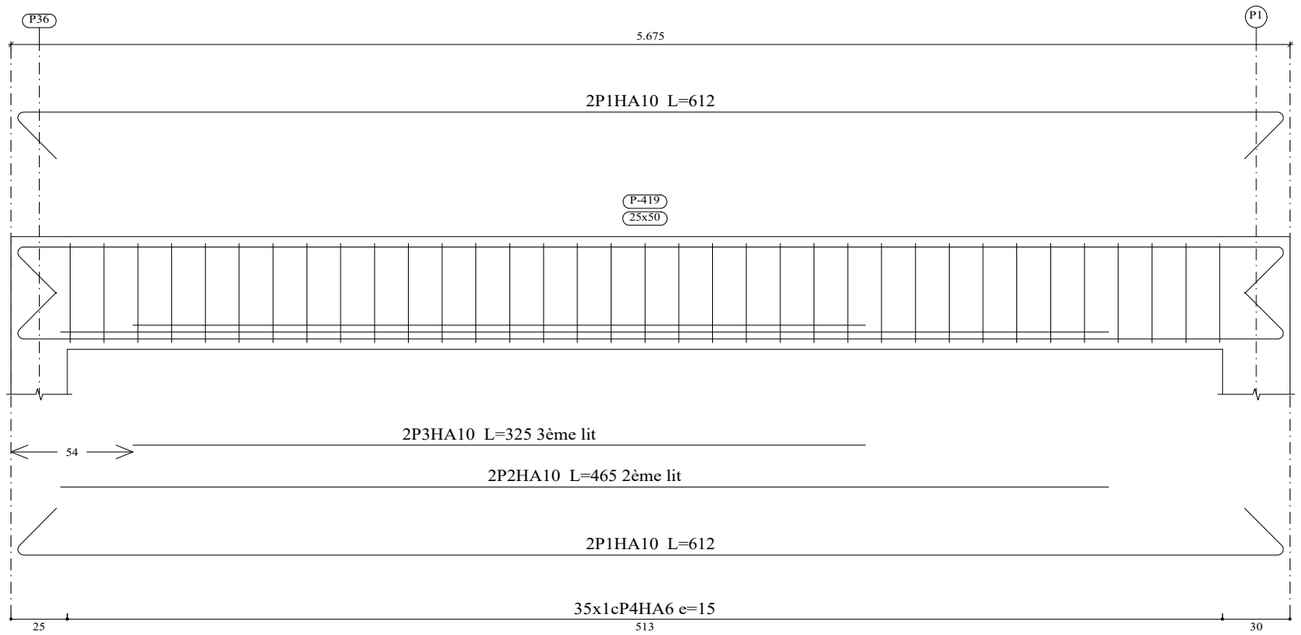
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 5	1	HA12	2	410	435	870	7.7
	2	HA12	2	405	430	860	7.6
	3	HA10	4	305	305	1220	7.5
	4	HA10	2	300	300	600	3.7
	5	HA10	2	115	115	230	1.4
	6	HA12	2	140	140	280	2.5
	7	HA10	2	110	110	220	1.4
	8	HA10	2	100	100	200	1.2
	9	HA10	2	430	455	910	5.6
	10	HA10	2	425	450	900	5.5
	11	HA10	4	270	270	1080	6.7
	12	HA10	2	175	175	350	2.2
	13	HA10	2	215	215	430	2.7
	14	HA10	2	205	205	410	2.5
	15	HA8	8	400	400	3200	12.6
	16	HA8	8	290	290	2320	9.2
	17	HA8	8	110	110	880	3.5
	18	HA8	8	100	100	800	3.2
	19	HA8	8	420	420	3360	13.3
	20	HA6	75	6 / 57 / 18	156	11700	26.0
						Total+10%:	138.6
Portique 8	1	HA10	2	562	562	1124	6.9
	2	HA12	2	562	612	1224	10.9
	3	HA10	2	410	410	820	5.1
	4	HA8	8	555	555	4440	17.5
	5	HA6	34	6 / 57 / 18	156	5304	11.8
						Total+10%:	57.4
						HA6:	41.6
						HA8:	65.2
						HA10:	57.6
						HA12:	31.6
						Total:	196.0

PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 9	1	HA10	4	562, 13, 25	612	2448	15.1
	2	HA10	2	465	465	930	5.7
	3	HA10	2	325	325	650	4.0
	4	HA6	35	43, 6, 18	136	4760	10.6
					Total+10%:		38.9
					HA6:		11.6
					HA10:		27.3
					Total:		38.9

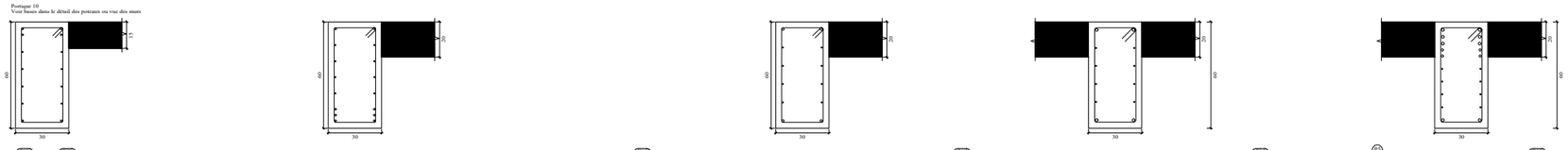
Portique 9

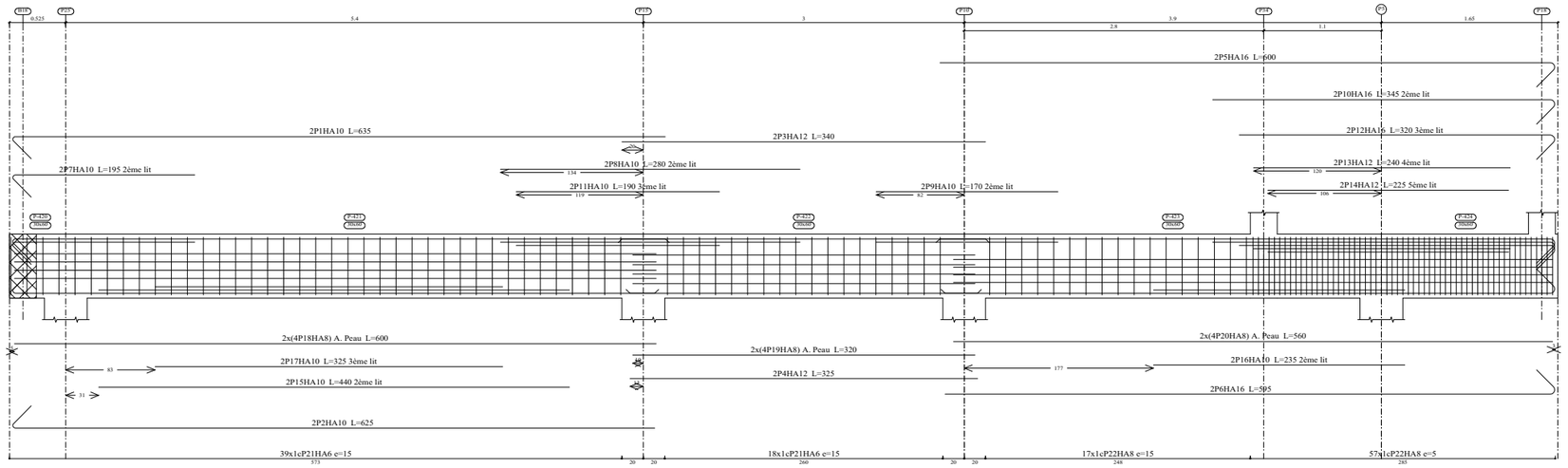




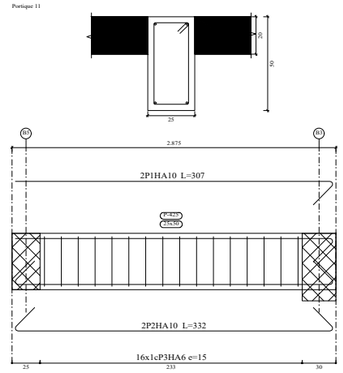
PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Portique 10
Voir bases dans le détail des poteaux ou vue des murs





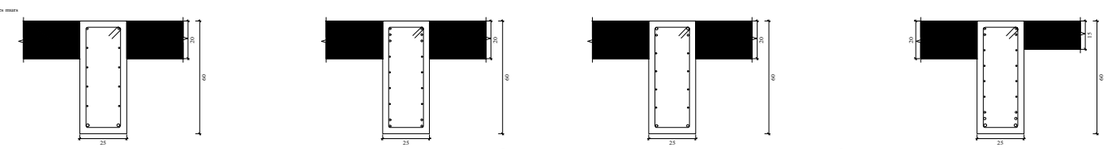
Portique 11

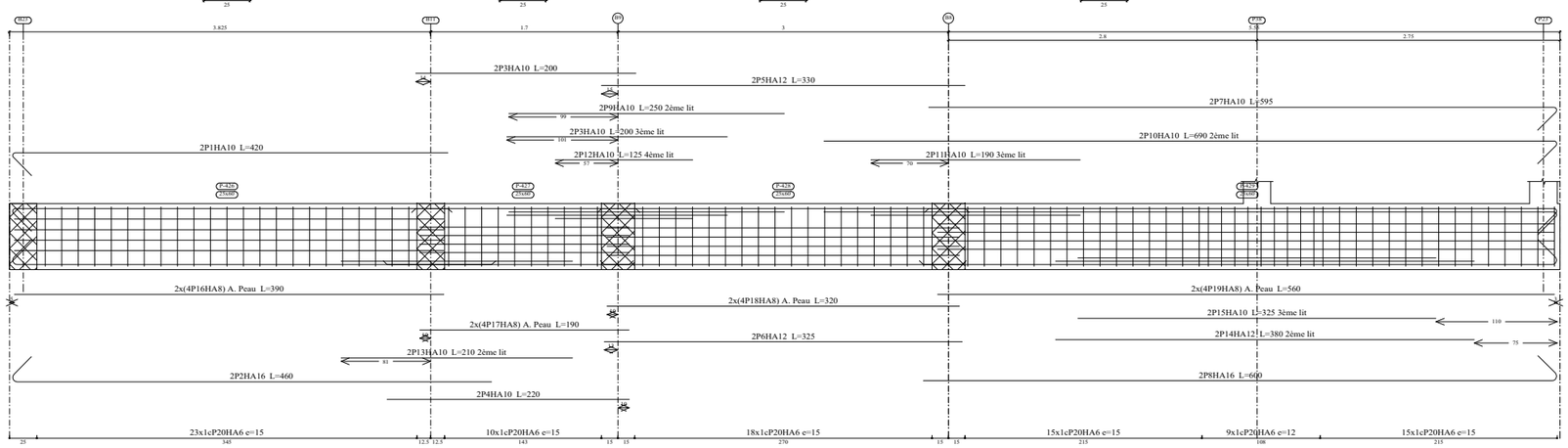


PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

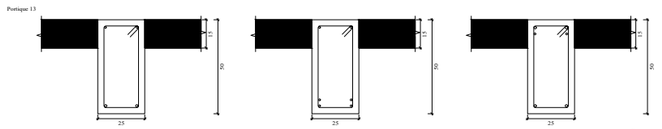
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 10	1	HA10	2	635	635	1270	7.8
	2	HA10	2	600	625	1250	7.7
	3	HA12	2	340	340	680	6.0
	4	HA12	2	325	325	650	5.8
	5	HA16	2	575 20	600	1200	18.9
	6	HA16	2	570 55	595	1190	18.8
	7	HA10	2	170	195	390	2.4
	8	HA10	2	280	280	560	3.5
	9	HA10	2	170	170	340	2.1
	10	HA16	2	320 25	345	690	10.9
	11	HA10	2	190	190	380	2.3
	12	HA16	2	295 25	320	640	10.1
	13	HA12	2	240	240	480	4.3
	14	HA12	2	225	225	450	4.0
	15	HA10	2	440	440	880	5.4
	16	HA10	2	235	235	470	2.9
	17	HA10	2	325	325	650	4.0
	18	HA8	8	600	600	4800	18.9
	19	HA8	8	320	320	2560	10.1
	20	HA8	8	560	560	4480	17.7
	21	HA6	57	25 15	166	9462	21.0
	22	HA8	74	25 15	169	12506	49.4
					Total+10%:		257.4
Portique 11	1	HA10	2	282	307	614	3.8
	2	HA10	2	282	332	664	4.1
	3	HA6	16	25 15	136	2176	4.8
					Total+10%:		14.0
Portique 12	1	HA10	2	395	420	840	5.2
	2	HA16	2	435	460	920	14.5
	3	HA10	4	200	200	800	4.9
	4	HA10	2	220	220	440	2.7
	5	HA12	2	330	330	660	5.9
	6	HA12	2	325	325	650	5.8
	7	HA10	2	570	595	1190	7.3
	8	HA16	2	575	600	1200	18.9
	9	HA10	2	250	250	500	3.1
	10	HA10	2	665	690	1380	8.5
	11	HA10	2	190	190	380	2.3
	12	HA10	2	125	125	250	1.5
	13	HA10	2	210	210	420	2.6
	14	HA12	2	380	380	760	6.7
	15	HA10	2	325	325	650	4.0
	16	HA8	8	390	390	3120	12.3
	17	HA8	8	190	190	1520	6.0
	18	HA8	8	320	320	2560	10.1
	19	HA8	8	560	560	4480	17.7
	20	HA6	90	25 15	156	14040	31.2
					Total+10%:		188.3
Portique 13	1	HA10	2	185	220	440	2.7
	2	HA12	2	210	235	470	4.2
	3	HA10	2	255	255	510	3.1
	4	HA10	2	260	260	520	3.2
	5	HA12	2	205	230	460	4.1
	6	HA12	2	175	200	400	3.6
	7	HA10	2	50	50	100	0.6
	8	HA10	2	225	225	450	2.8
	9	HA10	2	135	135	270	1.7
	10	HA10	2	120	120	240	1.5
	11	HA10	2	210	210	420	2.6
	12	HA6	31	25 15	136	4216	9.4
					Total+10%:		43.5
					HA6:		73.1
					HA8:		156.4
					HA10:		116.9
					HA12:		55.4
					HA16:		101.4
					Total:		503.2

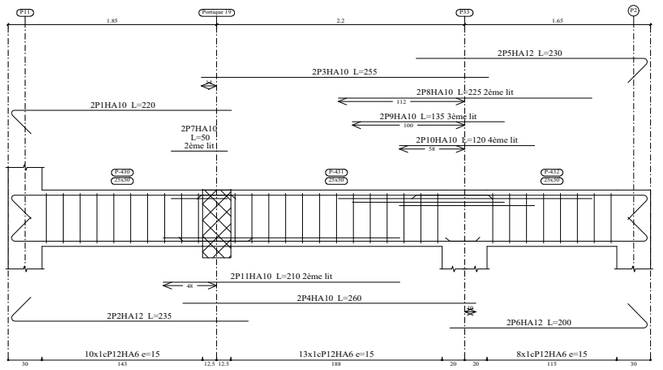
Portique 12
Voir bases dans le détail des poteaux ou vue des murs



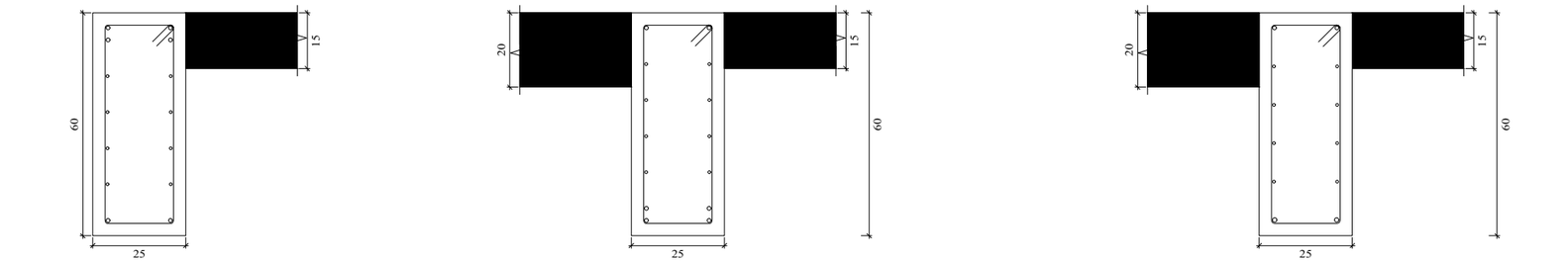


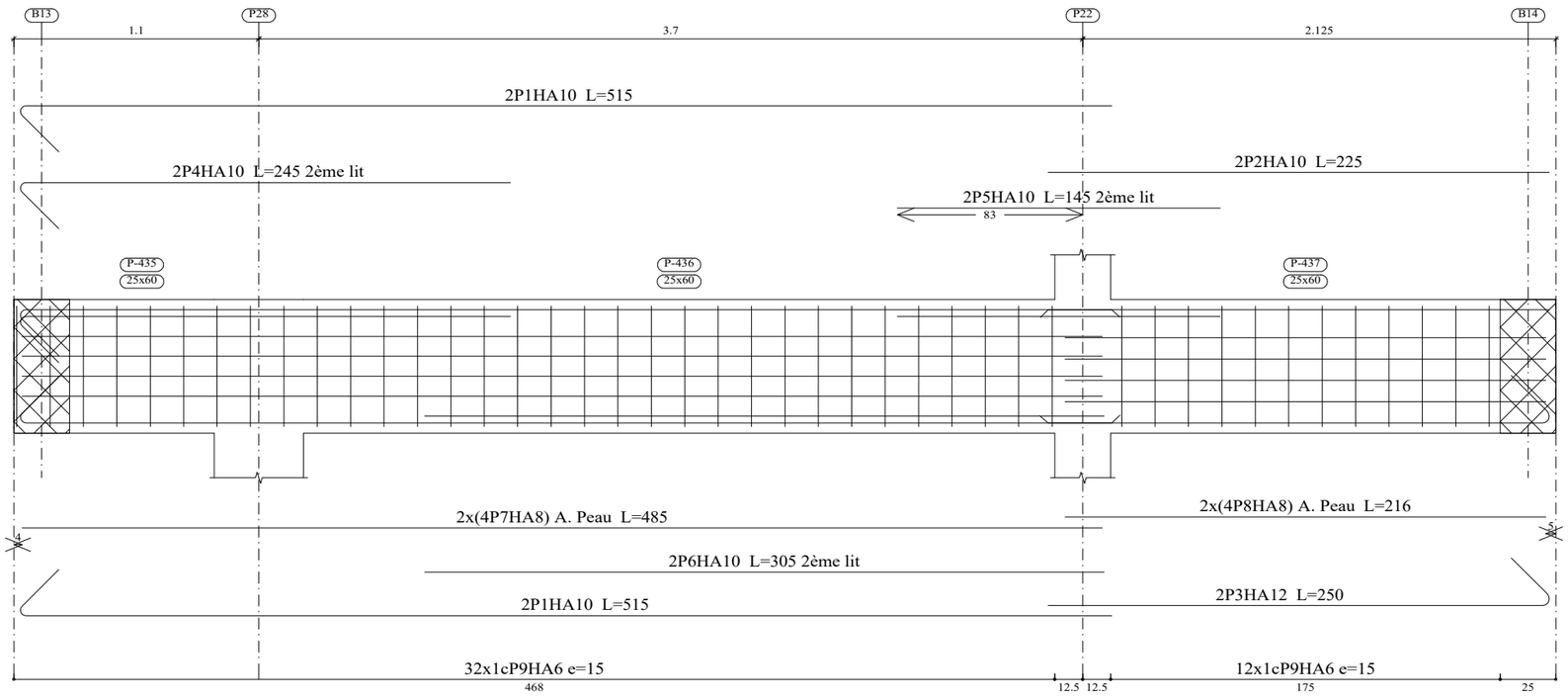
Portique 13





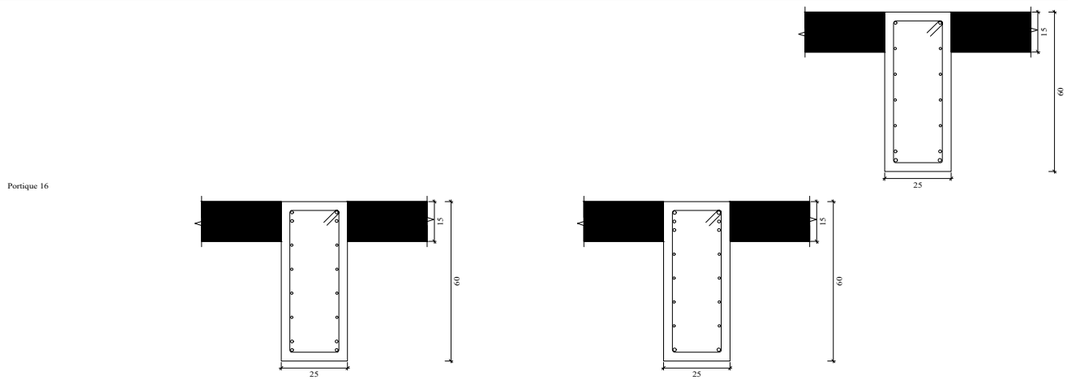
Portique 15

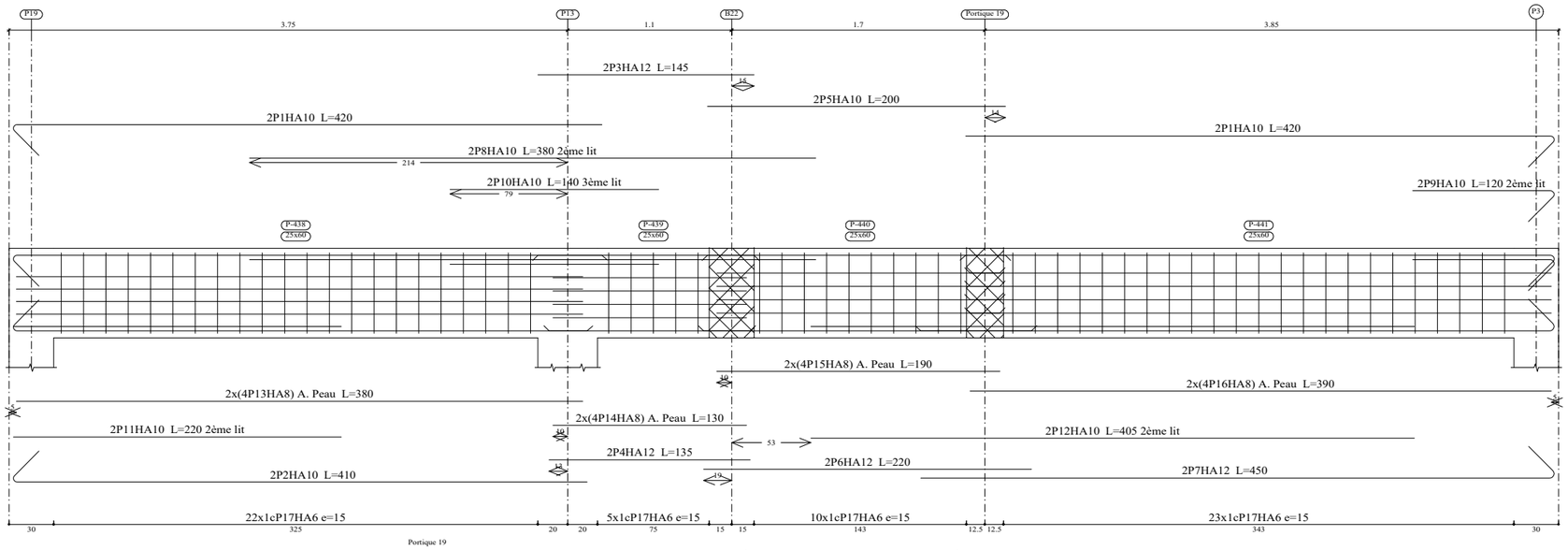




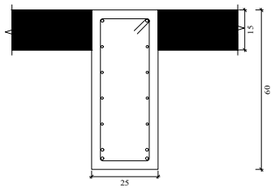
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 15	1	HA10	4	490, 15	515	2060	12.7
	2	HA10	2	225	225	450	2.8
	3	HA12	2	225, 12	250	500	4.4
	4	HA10	2	220, 15	245	490	3.0
	5	HA10	2	145	145	290	1.8
	6	HA10	2	305	305	610	3.8
	7	HA8	8	485	485	3880	15.3
	8	HA8	8	216	216	1728	6.8
	9	HA6	44	53, 6, 18	156	6864	15.2
					Total+10%:		72.4
					HA6:		16.7
					HA8:		24.3
					HA10:		26.5
					HA12:		4.9
					Total:		72.4

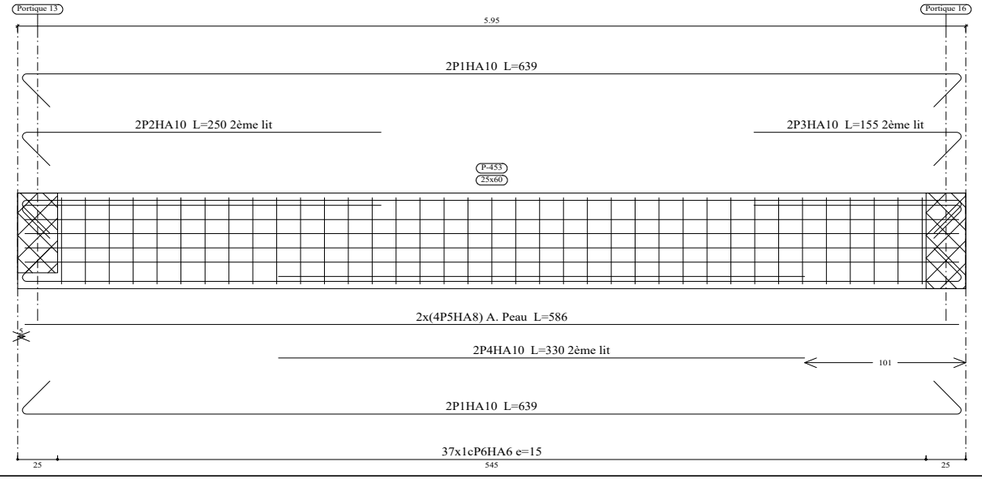
PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20



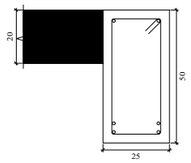


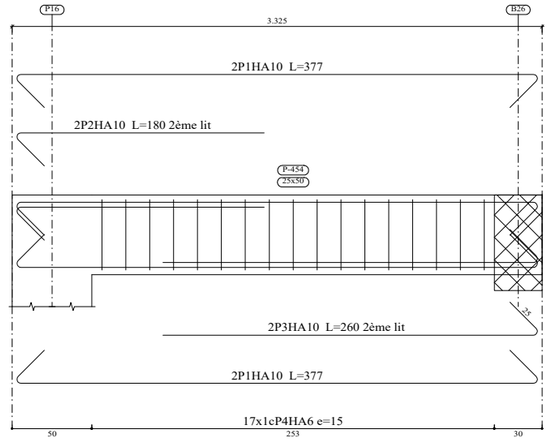
Portique 19





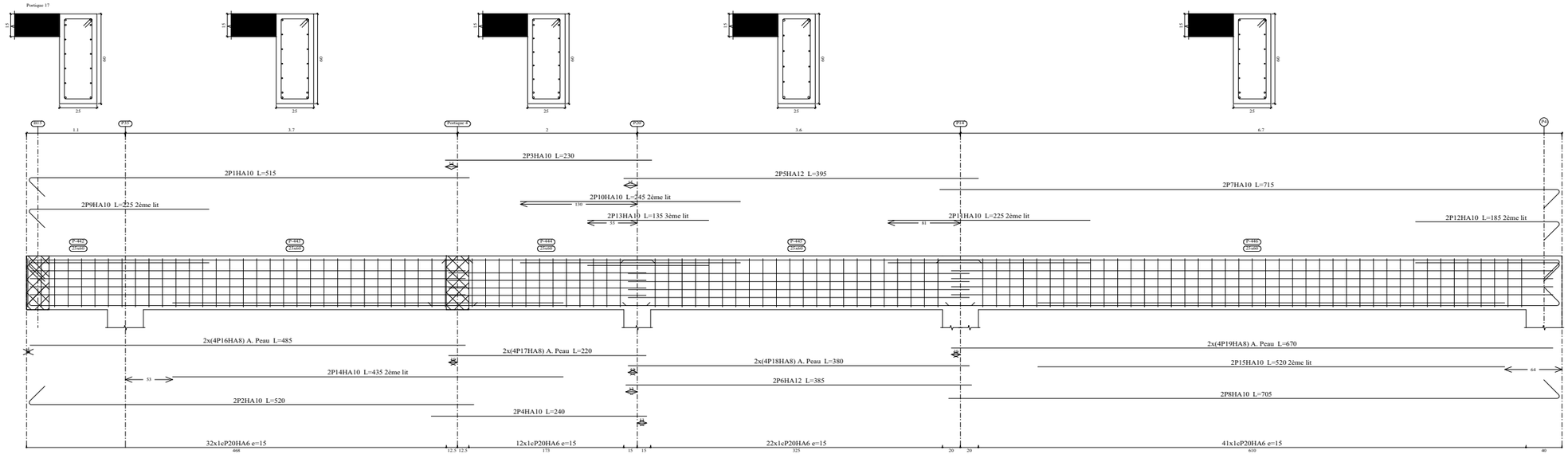
Portique 20





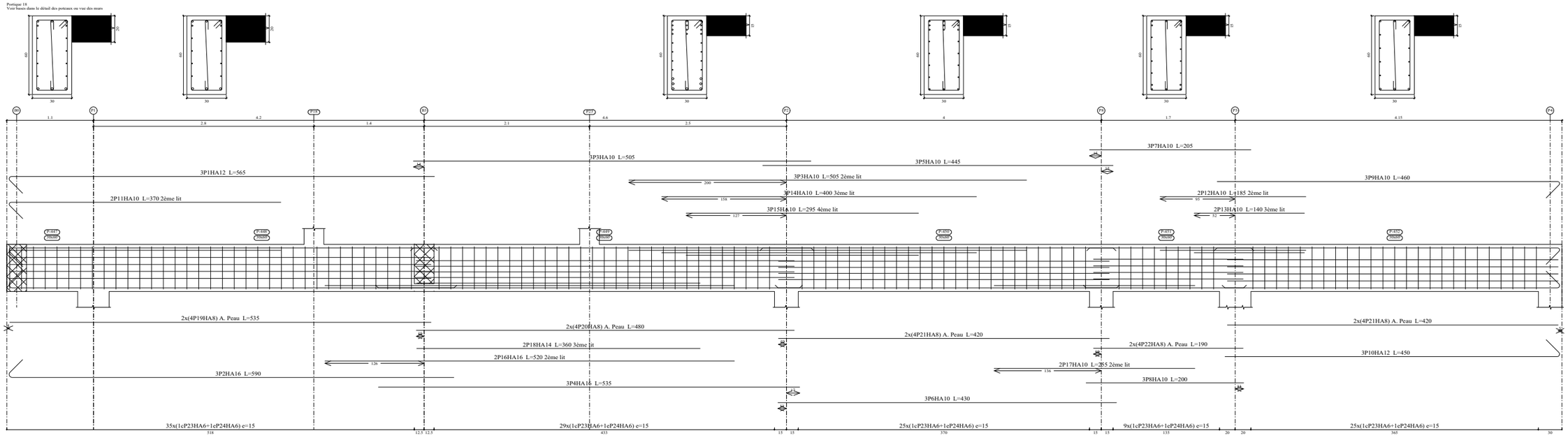
PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

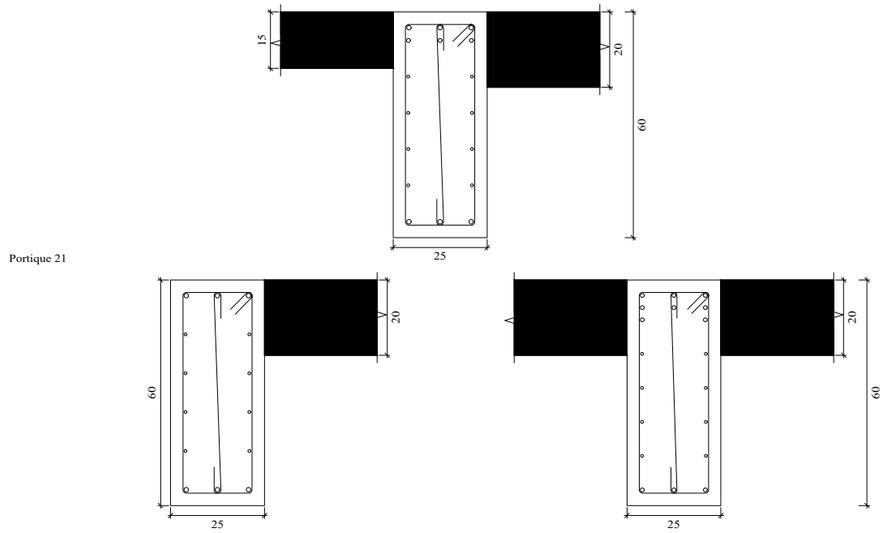
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 16	1	HA10	4	15 395	420	1680	10.4
	2	HA10	2	25 385	410	820	5.1
	3	HA12	2	145	145	290	2.6
	4	HA12	2	135	135	270	2.4
	5	HA10	2	200	200	400	2.5
	6	HA12	2	220	220	440	3.9
	7	HA12	2	425 15	450	900	8.0
	8	HA10	2	380	380	760	4.7
	9	HA10	2	95 25	120	240	1.5
	10	HA10	2	140	140	280	1.7
	11	HA10	2	220	220	440	2.7
	12	HA10	2	405	405	810	5.0
	13	HA8	8	380	380	3040	12.0
	14	HA8	8	130	130	1040	4.1
	15	HA8	8	190	190	1520	6.0
	16	HA8	8	390	390	3120	12.3
	17	HA6	60	53 6 18	156	9360	20.8
						Total+10%:	116.3
Portique 19	1	HA10	4	15 589 25	639	2556	15.8
	2	HA10	2	15 225	250	500	3.1
	3	HA10	2	130 25	155	310	1.9
	4	HA10	2	330	330	660	4.1
	5	HA8	8	586	586	4688	18.5
	6	HA6	37	53 6 18	156	5772	12.8
						Total+10%:	61.8
Portique 20	1	HA10	4	15 327 25	377	1508	9.3
	2	HA10	2	15 155	180	360	2.2
	3	HA10	2	235 25	260	520	3.2
	4	HA6	17	43 6 18	136	2312	5.1
						Total+10%:	21.8
						HA6:	42.6
						HA8:	58.1
						HA10:	80.6
						HA12:	18.6
						Total:	199.9



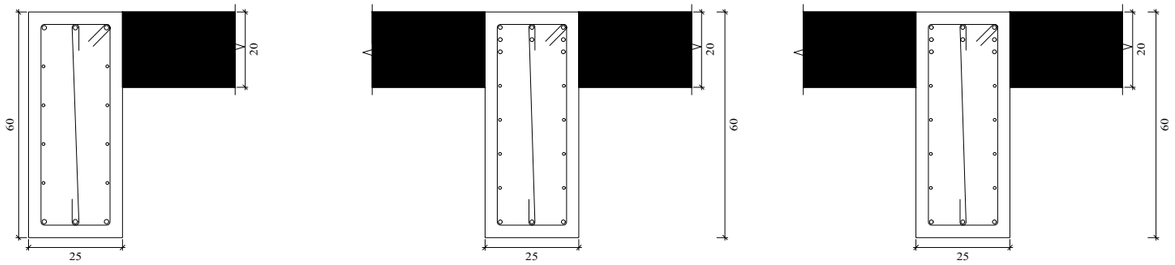
PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

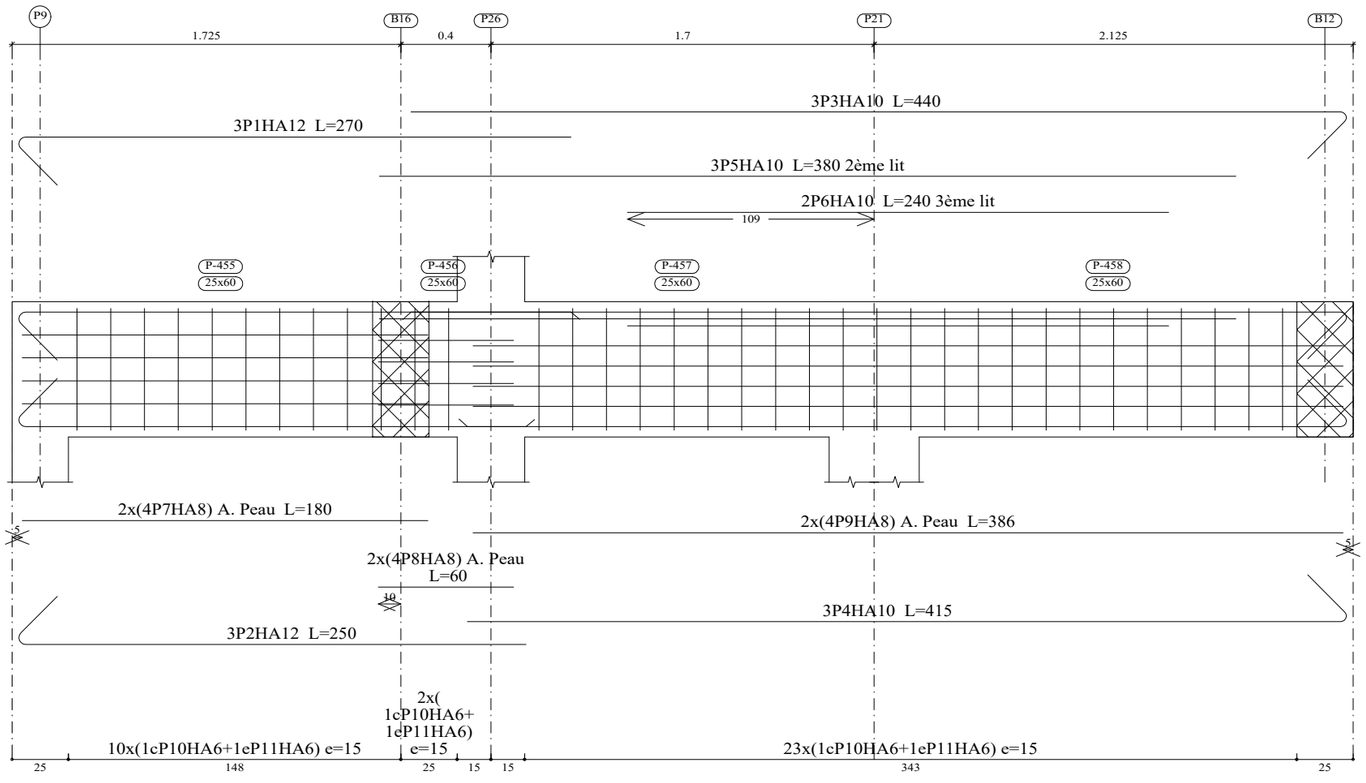
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 17	1	HA10	2	480	515	1030	6.4
	2	HA10	2	495	520	1040	6.4
	3	HA10	2	230	230	460	2.8
	4	HA10	2	240	240	480	3.0
	5	HA12	2	395	395	790	7.0
	6	HA12	2	385	385	770	6.8
	7	HA10	2	690 25	715	1430	8.8
	8	HA10	2	680 25	705	1410	8.7
	9	HA10	2	200 25	225	450	2.8
	10	HA10	2	245	245	490	3.0
	11	HA10	2	225	225	450	2.8
	12	HA10	2	100 25	185	370	2.3
	13	HA10	2	135	135	270	1.7
	14	HA10	2	435	435	870	5.4
	15	HA10	2	520	520	1040	6.4
	16	HA8	8	485	485	3880	15.3
	17	HA8	8	220	220	1760	6.9
	18	HA8	8	380	380	3040	12.0
	19	HA8	8	670	670	5360	21.2
	20	HA6	107	25 15	156	16692	37.0
						Total+10%:	183.4
Portique 18	1	HA12	3	540 25	565	1695	15.0
	2	HA16	3	565	590	1770	27.9
	3	HA10	6	505	505	3030	18.7
	4	HA16	3	535	535	1605	25.3
	5	HA10	3	445	445	1335	8.2
	6	HA10	3	430	430	1290	8.0
	7	HA10	3	205	205	615	3.8
	8	HA10	3	200	200	600	3.7
	9	HA10	3	435 25	460	1380	8.5
	10	HA12	3	425 25	450	1350	12.0
	11	HA10	2	345	370	740	4.6
	12	HA10	2	185	185	370	2.3
	13	HA10	2	140	140	280	1.7
	14	HA10	3	400	400	1200	7.4
	15	HA10	3	295	295	885	5.5
	16	HA16	2	520	520	1040	16.4
	17	HA10	2	255	255	510	3.1
	18	HA14	2	360	360	720	8.7
	19	HA8	8	535	535	4280	16.9
	20	HA8	8	480	480	3840	15.2
	21	HA8	16	420	420	6720	26.5
	22	HA8	8	190	190	1520	6.0
	23	HA6	123	25 25	166	20418	45.3
	24	HA6	123	53	65	7995	17.7
						Total+10%:	330.2
						HA6:	110.0
						HA8:	132.0
						HA10:	149.6
						HA12:	44.8
						HA14:	9.6
						HA16:	76.6
						Total:	522.6





Portique 21

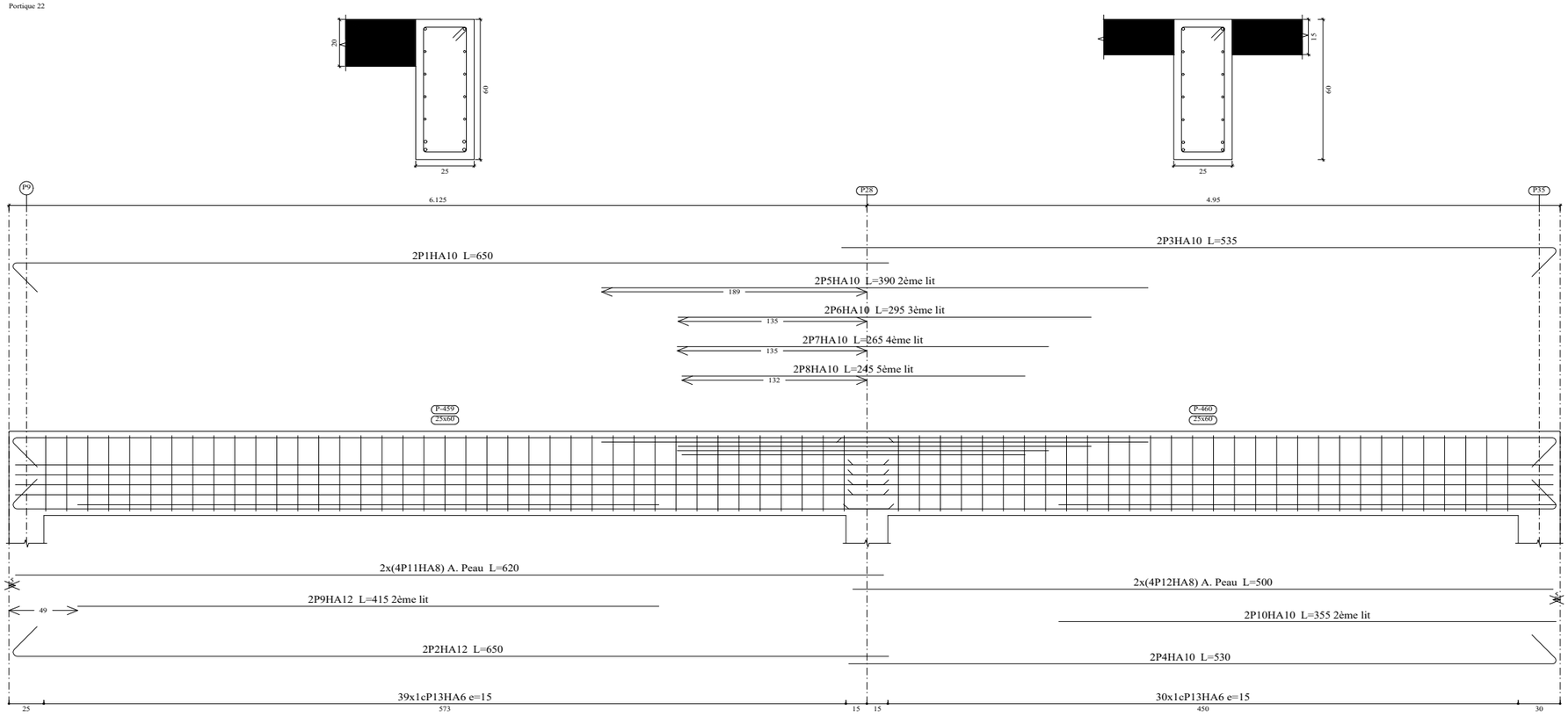




Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 21	1	HA12	3	25 245	270	810	7.2
	2	HA12	3	25 225	250	750	6.7
	3	HA10	3	415 25	440	1320	8.1
	4	HA10	3	390 25	415	1245	7.7
	5	HA10	3	380	380	1140	7.0
	6	HA10	2	240	240	480	3.0
	7	HA8	8	180	180	1440	5.7
	8	HA8	8	60	60	480	1.9
	9	HA8	8	386	386	3088	12.2
	10	HA6	35	53 6 18	156	5460	12.1
	11	HA6	35	6 53	65	2275	5.0
				Total+10%:			84.3
				HA6:			18.8
				HA8:			21.8
				HA10:			28.4
				HA12:			15.3
				Total:			84.3

PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Portique 22

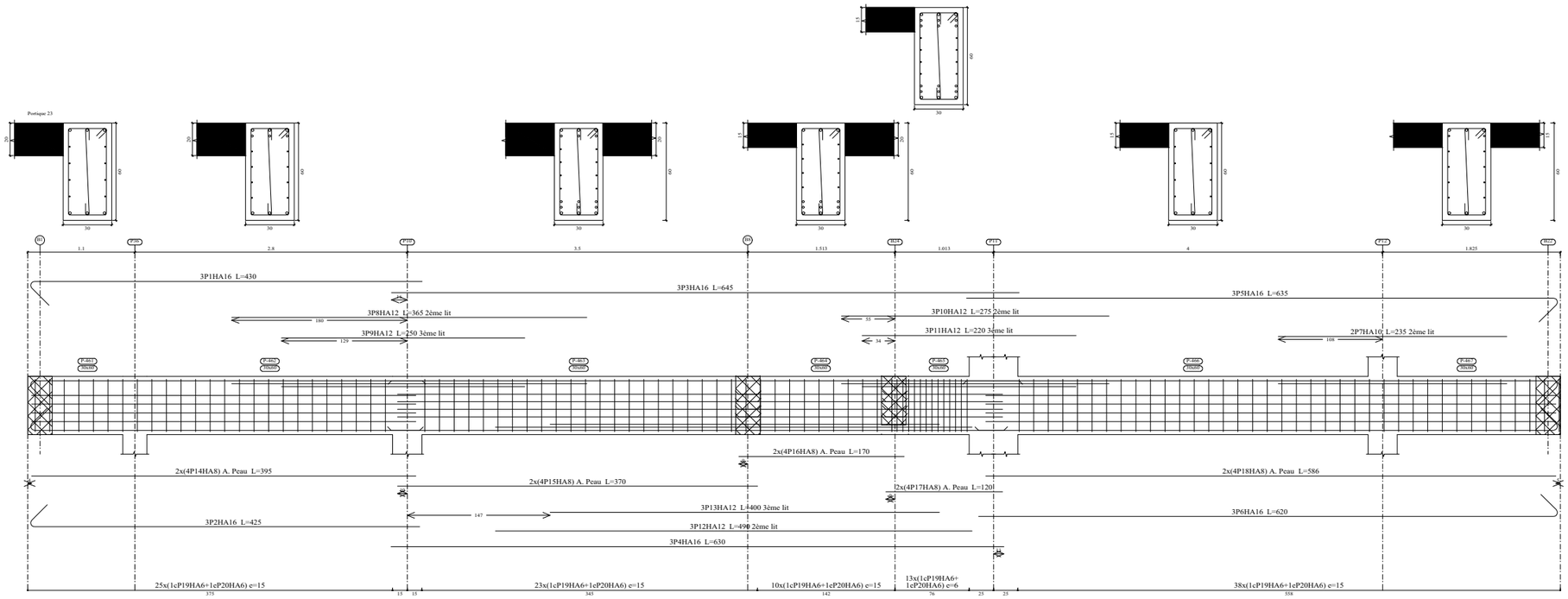


Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 22	1	HA10	2	625 15	650	1300	8.0
	2	HA12	2	15 625	650	1300	11.5
	3	HA10	2	510 15	535	1070	6.6
	4	HA10	2	505 15	530	1060	6.5
	5	HA10	2	390	390	780	4.8
	6	HA10	2	295	295	590	3.6
	7	HA10	2	265	265	530	3.3
	8	HA10	2	245	245	490	3.0
	9	HA12	2	415	415	830	7.4
	10	HA10	2	355	355	710	4.4
	11	HA8	8	620	620	4960	19.6
	12	HA8	8	500	500	4000	15.8
	13	HA6	69	6 53 18	156	10764	23.9
						Total+10%:	130.2
						HA6:	26.2
						HA8:	39.0
						HA10:	44.2
						HA12:	20.8
						Total:	130.2

PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 23	1	HA16	3	405	430	1290	20.4
	2	HA16	3	400	425	1275	20.1
	3	HA16	3	645	645	1935	30.5
	4	HA16	3	630	630	1890	29.8
	5	HA16	3	607	633	1905	30.1
	6	HA16	3	595	620	1860	29.4
	7	HA10	2	235	235	470	2.9
	8	HA12	3	365	365	1095	9.7
	9	HA12	3	250	250	750	6.7
	10	HA12	3	275	275	825	7.3
	11	HA12	3	220	220	660	5.9
	12	HA12	3	490	490	1470	13.1
	13	HA12	3	400	400	1200	10.7
	14	HA8	8	395	395	3160	12.5
	15	HA8	8	370	370	2960	11.7
	16	HA8	8	170	170	1360	5.4
	17	HA8	8	120	120	960	3.8
	18	HA8	8	586	586	4688	18.5
	19	HA6	109	25 / 53	166	18094	40.2
	20	HA6	109	53	65	7085	15.7
						Total+10%:	356.8

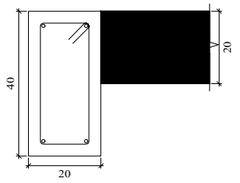
HA6:	61.4
HA8:	57.1
HA10:	3.2
HA12:	58.8
HA16:	176.3
Total:	356.8

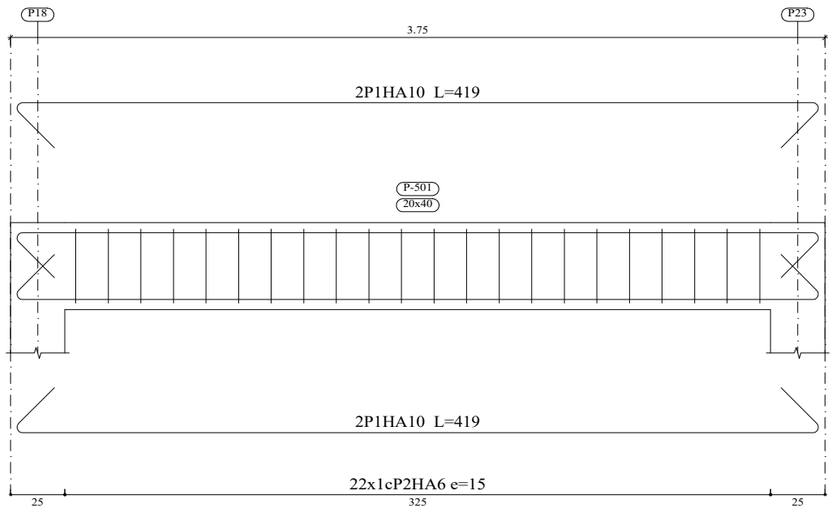


PH R+1
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

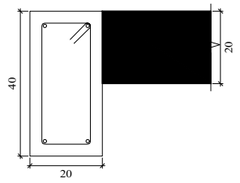
POUTRES PH R+2

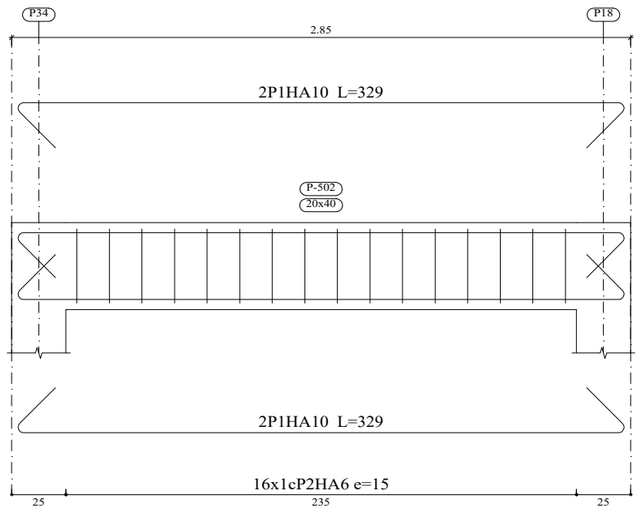
Portique 1





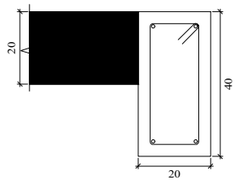
Portique 2

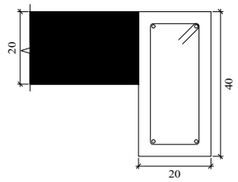




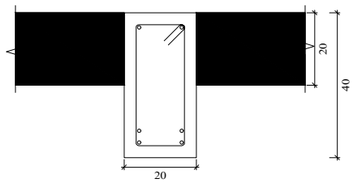
Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 1	1	HA10	4	369, 25, 25	419	1676	10.3
	2	HA6	22	6, 33, 13	106	2332	5.2
						Total+10%:	17.1
Portique 2	1	HA10	4	279, 25, 25	329	1316	8.1
	2	HA6	16	6, 33, 13	106	1696	3.8
						Total+10%:	13.1
Portique 6	1	HA10	2	199, 31	230	460	2.8
	2	HA10	2	25, 200	225	450	2.8
	3	HA10	4	330, 25	355	1420	8.8
	4	HA10	2	115	115	230	1.4
	5	HA6	28	6, 33, 13	106	2968	6.6
						Total+10%:	24.6
Portique 7	1	HA10	4	369, 25, 25	419	1676	10.3
	2	HA10	2	260	260	520	3.2
	3	HA6	22	6, 33, 13	106	2332	5.2
						Total+10%:	20.6
						HA6:	22.9
						HA10:	52.5
						Total:	75.4

Portique 6

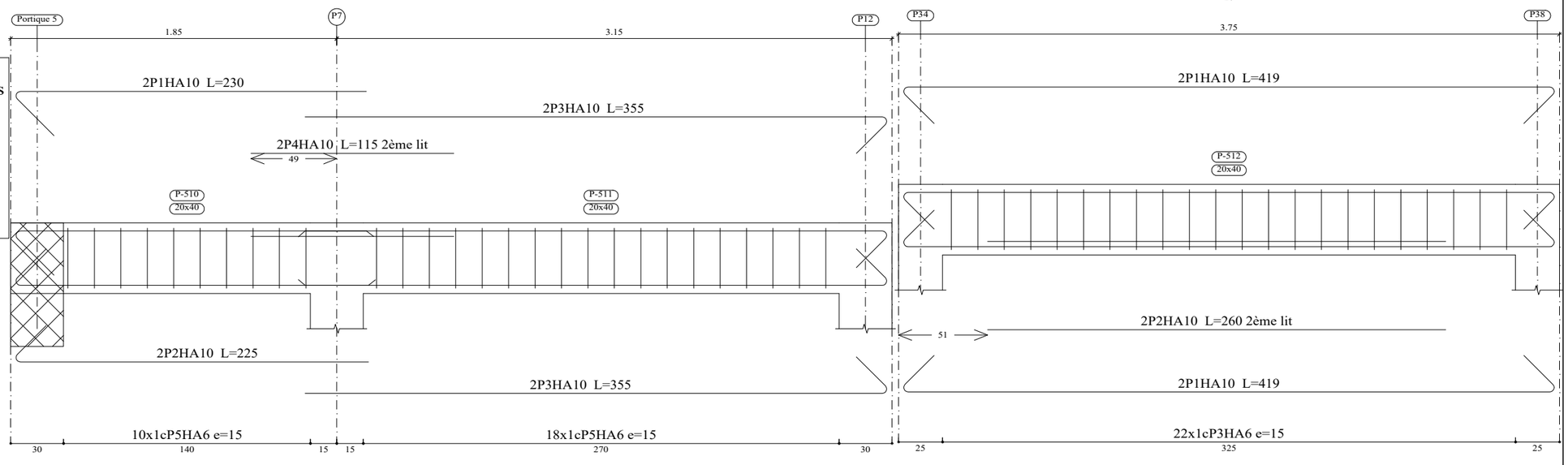




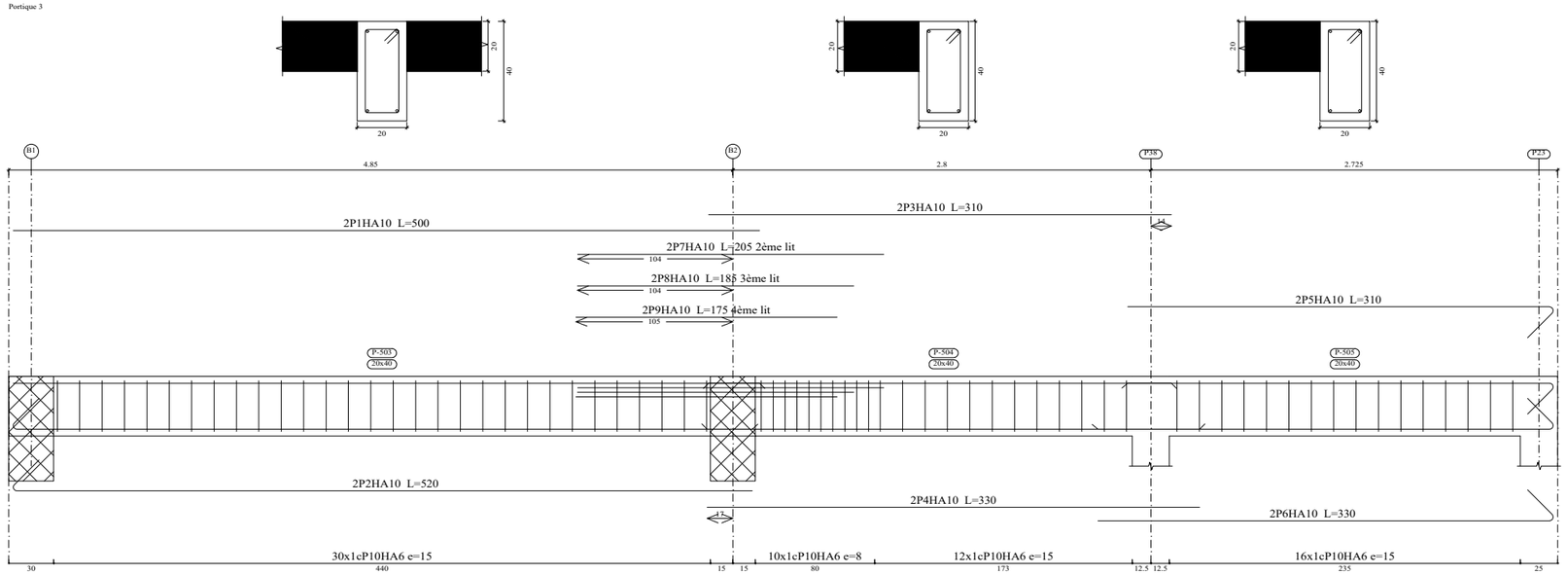
Portique 7



PH R+2
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20



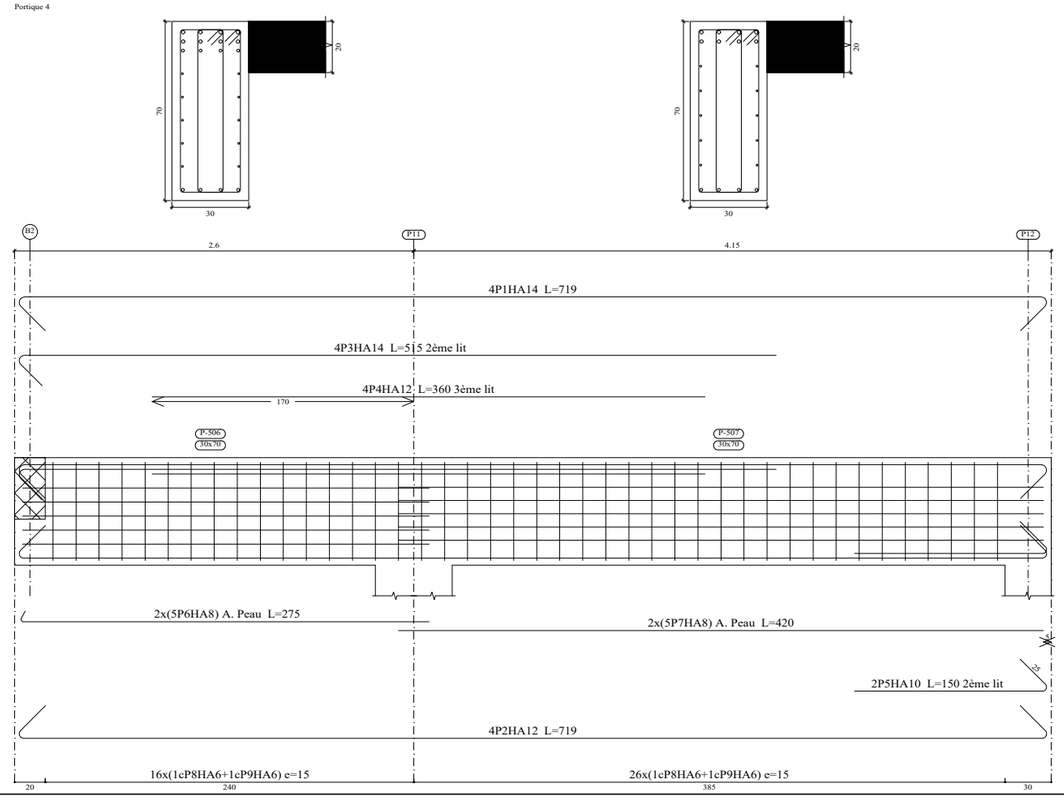
Portique 3



PH R+2
Plan de ferraillage des poutres
Béton: B20
Acier des barres: Fe E400
Acier des cadres: Fe E400
Échelle portiques 1:25
Échelle sections 1:15
Échelle réservation 1:20

Élément	Pos.	Diam.	No.	Schéma (cm)	Long. (cm)	Total (cm)	Fe E400 (kg)
Portique 3	1	HA10	2	500	500	1000	6.2
	2	HA10	2	25 495	520	1040	6.4
	3	HA10	2	310	310	620	3.8
	4	HA10	2	330	330	660	4.1
	5	HA10	2	285 25	310	620	3.8
	6	HA10	2	305 25	330	660	4.1
	7	HA10	2	205	205	410	2.5
	8	HA10	2	185	185	370	2.3
	9	HA10	2	175	175	350	2.2
	10	HA6	68	6 13 13	106	7208	16.0
						Total+10%:	56.5
Portique 4	1	HA14	4	669 25 25	719	2876	34.8
	2	HA12	4	25 669 25	719	2876	25.5
	3	HA14	4	493 25	515	2060	24.9
	4	HA12	4	360	360	1440	12.8
	5	HA10	2	125 25	150	300	1.8
	6	HA8	10	25 267	275	2750	10.9
	7	HA8	10	420	420	4200	16.6
	8	HA6	42	6 63 23	186	7812	17.3
	9	HA6	42	6 63 8	154	6468	14.4
						Total+10%:	174.9
Portique 5	1	HA14	4	664 25 25	711	2844	34.4
	2	HA12	4	25 664 25	714	2856	25.4
	3	HA14	4	415	415	1660	20.1
	4	HA12	4	315	315	1260	11.2
	5	HA10	2	25 190	215	430	2.7
	6	HA8	10	25 397	405	4050	16.0
	7	HA8	10	287 6	295	2950	11.6
	8	HA6	42	6 63 23	186	7812	17.3
	9	HA6	84	6 63	75	6300	14.0
						Total+10%:	168.0
						HA6:	87.0
						HA8:	60.5
						HA10:	43.9
						HA12:	82.3
						HA14:	125.7
						Total:	399.4

Portique 4



Portique 5

